Literature Review Draft

1. Deepfake Detection Basics

What are the main deepfake detection approaches?

There are two main ways to detect deepfakes: deep learning methods and digital forensics.

Deep learning is the most popular approach (about 75% of all detection work). It uses CNNs, neural networks that analyze face images frame-by-frame to spot fakes. Common models include XceptionNet, VGG19, ResNet, and MesoNet. These can achieve 95-98% accuracy on test videos.

For video (not just single frames), researchers add temporal methods like CNN-LSTM, 3D CNN, and Transformers to track motion and artifacts across multiple frames. This captures patterns that change over time.

Attention-based models highlight the parts of a face most likely to be fake, making the detection more interpretable that you can see where the model is looking.

Digital forensics analyzes the traces left by deepfake generation blending boundaries, frequency artifacts, and convolutional patterns. These methods are slower but more robust to unseen manipulations.

Approach Category	Key Methods	Strengths	Limitations
CNN-based	Xception, VGG19, ResNet, MesoNet	High accuracy (95–98%)	Overfits to dataset-specific artifacts
Spatio-temporal	CNN-LSTM, 3D CNN, TimeSformer	Captures motion artifacts across frames	Needs longer clips for peak performance
Forensic	Frequency analysis, blending-boundary detection	Robust to unseen manipulations	Slower; requires manual review
Attention-based	ADD, multi-head self-attention CNN	Focuses on forgery regions; interpretable	Generalization across datasets remains challenging

What datasets exist for deepfake detection?

Early datasets were small and limited. Now there are larger, more diverse benchmark datasets:

Dataset	Scale	Manipulation Types	Key Feature
FaceForensics++ (FF++)	~4K fake videos	4 methods	Most widely used benchmark
Celeb-DF	~5,639 fake videos	1 method (face swap)	Large-scale, high visual quality

Dataset	Scale	Manipulation Types	Key Feature
DFDC	Large-scale	8 deepfake techniques	Hidden test set for evaluation
DF40	40 techniques	Face swap, reenactment, synthesis, editing	Includes recent diffusion-based methods
ExDDV	~5,400 videos	Multiple deepfake types	First explainability benchmark with text descriptions and visual annotations

FaceForensics++ is still the most popular. Newer datasets like DF40 and DDL include many more deepfake generation methods. ExDDV is special because it's the first dataset designed for explainable detection videos come with text explanations and click annotations showing where the artifacts are, not just whether a video is fake or real.

2. Explainability in ML Detection

How do explainability methods work in detection systems?

Explainability methods help us understand *why* a detection model made a decision. There are two main types:

Saliency methods work by backpropagating gradients from the model's output back to the input, creating heatmaps that highlight which pixels or regions were most important for the classification decision. Grad-CAM is the most popular variant, it highlights pixels from the last convolutional layer that increased the probability of the predicted class. Other saliency methods like Integrated Gradients and Layer-wise Relevance Propagation reduce noise in these attributions.

Perturbation-based methods (LIME, SHAP) take a different approach: they modify small regions of the input image and measure how much the model's prediction changes. If changing a region drastically changes the prediction, that region is important.

Attention mechanisms are built directly into model architectures, so they provide explainability by design so that you can see where the model is focusing without extra analysis. Advanced versions like HAG-XAI learn from human attention patterns to improve explanations.

Method Type	Examples	How It Works	Strength	Weakness
Gradient-based saliency	Grad-CAM, SmoothGrad, Integrated Gradients	Backpropagation creates heatmaps	Good spatial coherence	Can be noisy from unstable gradients

Method Type	Examples	How It Works	Strength	Weakness
Perturbation-based	LIME, SHAP, occlusion	Modify image regions, measure prediction change	Competitive across deepfake types	Weaker localization, less temporal stability
Attention-based	HAG-XAI, Concept Bottleneck Models	Built into architecture; highlights focus regions	Inherent interpretability by design	Requires task-specific retraining

What existing work adds explanations to deepfake detection?

Several frameworks now combine detection with explainability:

Framework	Explanation Type	How It Works	Key Limitation
DF-P2E	Visual + text narrative	Three-stage pipeline: Grad-CAM → image captioning → LLM refinement into natural language	Post-hoc; depends on captioning quality
DD-VQA	Text (common-sense reasoning)	Frames detection as Visual Question Answering; generates textual explanations about unnatural facial features	Requires annotated dataset
ExDDV	Spatial clicks + text descriptions	First explainable video deepfake benchmark with manual click annotations and text descriptions	Manual annotation overhead
Neurosymbolic	Text (emotion-based reasoning)	Combines neural networks with symbolic logic to detect emotional inconsistencies	Limited to emotion-injective deepfakes
DeepDect	Visual (Grad-CAM) + text	Human-centered platform integrating expert and non-expert needs	81% accuracy; moderate detection ceiling

DF-P2E is the most advanced. It integrates three components: a detection model that generates heatmaps via Grad-CAM, a captioning module that describes the heatmap, and an LLM that refines the description into natural language explanations for users. Unlike older methods that add explanations after the fact, DF-P2E makes explanations part of the main detection pipeline.

ExDDV is unique because it's the first dataset designed specifically for explainable deepfake *video* detection. Videos come with both spatial click annotations (pointing to artifacts) and text descriptions explaining them.

Remaining challenges: Spatio-temporal methods often highlight the entire face instead of specific artifacts. High-quality explanations require additional training data or fine-tuning. Saliency maps are hard for non-experts to understand without help, which is why newer work focuses on text-based explanations instead.

3. Real-World Video Degradation

What types of degradation affect videos?

Real-world videos suffer from many types of quality loss. These degradations happen during recording, transmission, or storage:

Degradation Category	Specific Types	Source / What Causes It
Compression	Blockiness, ringing, mosquito noise	Video encoding (H.264, H.265), JPEG compression
Noise	Gaussian noise, Poisson noise, sensor noise	Low-light cameras, camera sensor electronics
Blur	Defocus blur, motion blur, Gaussian blur	Camera shake, moving subjects, focus issues
Environmental	Haze, rain, low light, color distortion	Weather, atmospheric effects, poor lighting
Resolution & Temporal	Downsampling, low frame rate, frame dropping	Bandwidth limits, hardware constraints, live streaming

Compression creates spatial artifacts (blurring, blocking, color bleeding) and temporal artifacts (flickering, floating artifacts across frames).

Blur and noise are fundamental degradation factors. Blur can be from camera movement or subjects moving in the shot. Noise comes from low-light sensors or the camera's internal electronics.

Environmental degradations like haze, rain, and low-light often happen together in real-world videos and make restoration harder.

Temporal degradations like reduced frame rate happen when videos are transmitted over limited bandwidth or recorded on hardware with constraints.

How does degradation impact detection accuracy in general?

Degradation seriously hurts deepfake detection accuracy:

- **Significant accuracy drops**: Deepfake detectors show consistent and major accuracy losses when tested on degraded videos. Models that overfit (trained only on clean videos) can lose over 30% accuracy when facing real-world compression and resizing.
- **Complete failure in extreme cases**: In heavily degraded or live-streaming scenarios, most deepfake detectors fail completely, none achieve accuracy higher than 50%.
- **Not correlated with storage size**: Detection performance doesn't depend on how much data the degradation takes up. A platform with small storage changes can actually have the lowest detection accuracy.

- **Generalization matters**: Models that generalize well to different domains also resist degradation better. This shows that robustness to degradation and robustness to domain changes are linked.
- **Type of degradation matters**: At the same frame rate, accuracy can vary by over 40% depending on which type of temporal degradation is applied. Different degradations affect detection differently.
- **Training matters**: If you include degraded videos in training, models perform better on degraded test videos. But low resolution can still significantly hurt performance if not handled properly.

4. The Gap

When videos are degraded, do explanations from detection models remain reliable/valid?

Research from adjacent domains indicates that explanations may not remain reliable under video degradation, though direct evidence in deepfake detection is limited.

In medical imaging, explanation maps generated by neural networks degrade under various corruptions (noise, compression, blur). Critically, this degradation occurs independently of model accuracy. Explanations can become unstable or unreliable even when the model continues to make correct predictions.

In general XAI research, saliency-based methods such as Grad-CAM, LIME, and SHAP demonstrate inconsistent robustness across degradation conditions. Whether an explanation is considered "robust" depends on the evaluation metric used pixel consistency, spatial agreement, or prediction fidelity under distribution shift all yield different rankings of method reliability.

ROPE is one framework explicitly optimized for explanation robustness under real distribution shifts, achieving fidelity drop of 0–7% compared to 13–19% for LIME and SHAP baselines. However, this evidence derives from tabular-style black-box settings rather than from degraded video environments.

The central concern is that prediction stability does not guarantee explanation stability. A deepfake detector may correctly identify a manipulated video in compressed or noisy conditions, but the accompanying explanation whether a saliency map, attention heatmap, or click annotation may become misaligned, unreliable, or inconsistent.

Has anyone tested explainability robustness under real-world conditions?

Testing of explanation robustness under degradation exists, but predominantly outside deepfake detection.

The most comprehensive evidence comes from medical imaging and XAI benchmarking studies, which evaluate post-hoc methods including Grad-CAM, Integrated Gradients, SmoothGrad, LIME, and SHAP under natural noise, JPEG compression, and data distribution shifts. These studies confirm that explanations degrade under corruption even when model accuracy remains relatively stable.

For video systems, related work from video-LLMs and structured reasoning tasks provides relevant findings. INFACT evaluates explanation reliability under visual degradation, evidence corruption, and temporal intervention, demonstrating that accuracy in clean conditions does not predict reliability under induced degradations. Similarly, predicate-level manipulation understanding studies show that fine-grained spatial predicates are more susceptible to degradation than coarse-level predicates across blur, illumination changes, occlusion, frame dropping, and detection noise.

Domain	Evidence Type	Key Finding
Medical Imaging	Direct testing of explanation stability under corruption	Explanations degrade independently of model accuracy
Generic XAI Benchmarking	Robustness evaluation across methods and metrics	No saliency method exhibits uniform robustness
Video-LLMs & Reasoning	Temporal and degradation sensitivity analysis	Clean-setting accuracy does not predict degraded-setting reliability

5. Experiments

This chapter records every detection test run for the thesis. Older results are never removed: each new run is added as a new test. Every test is written in the same format — overview, code pipeline diagram, setup, degradations, data, training, results (original vs degraded, by modality, per degradation), explainability, findings, limitations, a suggested thesis sentence and output files — so results can be compared directly and quoted in the thesis. Source code is not reproduced; each script is summarised as a pipeline diagram. Section 5.0 describes the tool used to make all degraded videos.

5.0 Degradation Tool Used to Create the Degraded Videos

All degraded videos used in these experiments were produced with my own tool, the **DeepFake Bulk Video Degradation Tool**. I built it to convert dataset videos in bulk into degraded versions that imitate real-world quality loss. The tool runs FFmpeg locally, so no database, cloud or upload is involved. It never modifies, renames, moves or deletes the original videos. Every output goes into a separate *degraded_output/* folder inside each dataset, together with a *manifest.json* that links each degraded video to its original. The experiment scripts read this manifest to pair every original with its degraded versions.

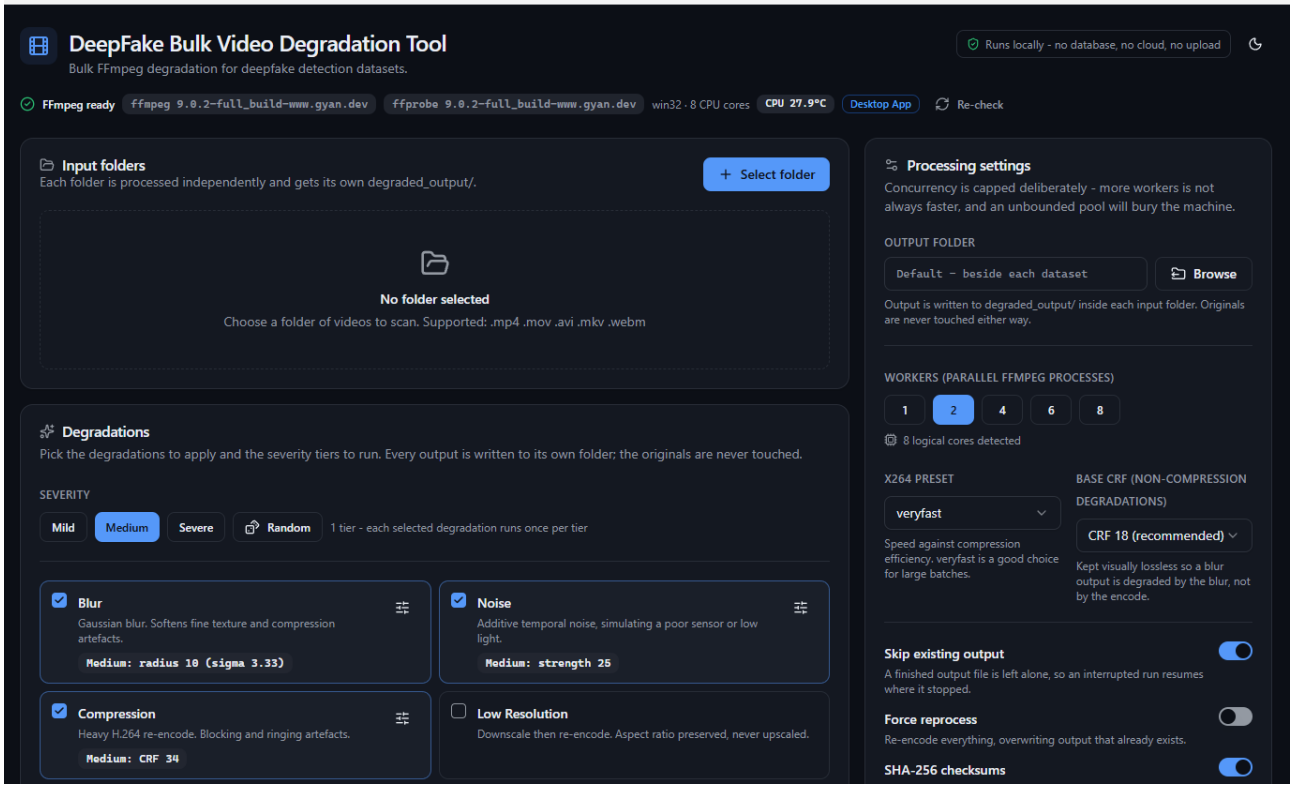

Figure 5.1: DeepFake Bulk Video Degradation Tool: input folders, severity selection, degradation cards and processing settings (FFmpeg 9.0.2, 8 CPU cores).

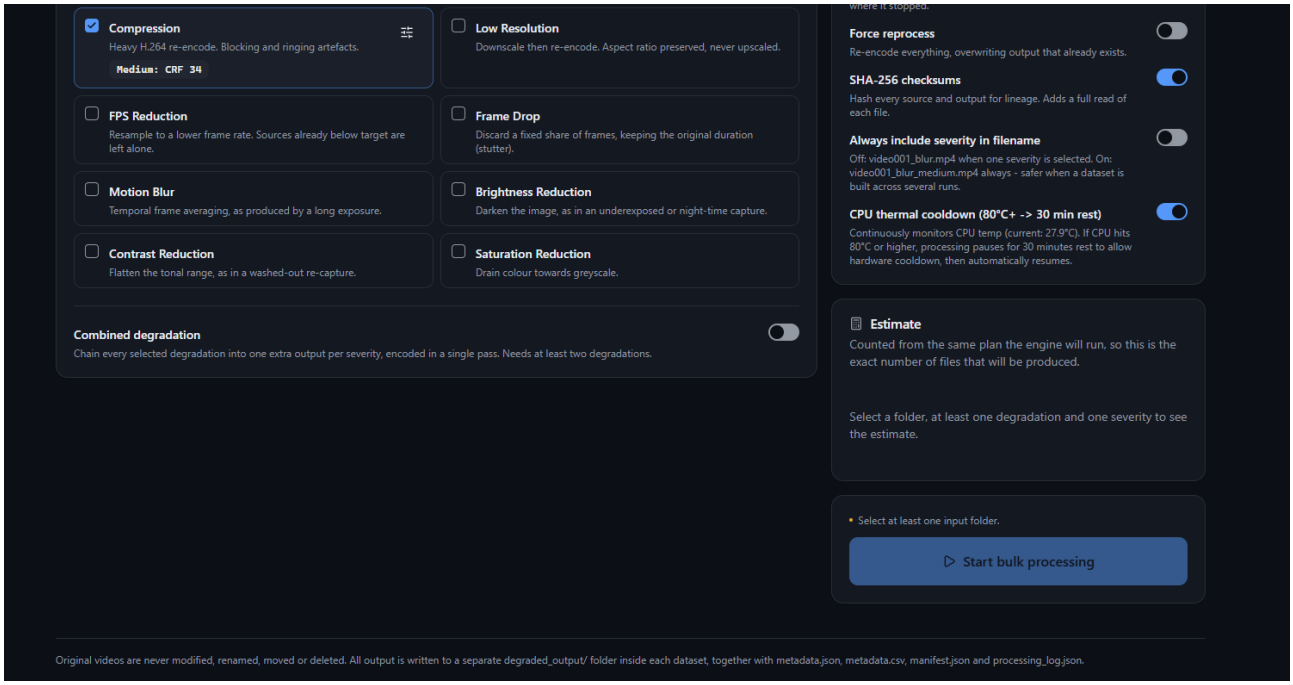

Figure 5.2: Remaining degradation types, combined-degradation option, safety settings, file estimate and start button.

Figure 5.3: How the tool fits into the experiments.

Feature	What it does
Input folders	Each selected folder is processed on its own and gets its own degraded_output/ folder (.mp4, .mov, .avi, .mkv, .webm).
Degradations (10)	Blur (Gaussian), noise (temporal sensor noise), compression (heavy H.264), low resolution (downscale, never upscale), FPS reduction, frame drop (stutter, same duration), motion blur (frame averaging), brightness, contrast and saturation reduction. Settings per level are in Section 5.2.4.
Severity	Mild, medium, severe, or random; each selected degradation runs once per chosen tier. In random mode each output draws one of the selected tiers; the number of random sets per video and a seed are set by the user (same seed = same picks, so a resumed run stays consistent).
Combined degradation	Optional: chains all selected degradations into one extra output per severity in a single encode.
Processing	1, 2, 4, 6 or 8 parallel FFmpeg workers; x264 preset (e.g. veryfast); base CRF 18 for non-compression degradations, so the encode itself adds almost no damage.
Safety and resuming	Skip existing output (an interrupted run resumes), force reprocess, CPU thermal cooldown (pause 30 min at 80 °C or more).
Traceability	SHA-256 checksums of every source and output; optional severity in the file name; estimate of the exact number of files before starting.
Outputs	Degraded videos + metadata.json, metadata.csv, manifest.json and processing_log.json.

Table 5.1: Main features of the degradation tool.

Severity settings. Every degradation has three strength levels. In random mode the user chooses which tiers may be drawn, how many random degraded outputs each video gets, and a seed; the tool then picks the degradation type and tier for each output.

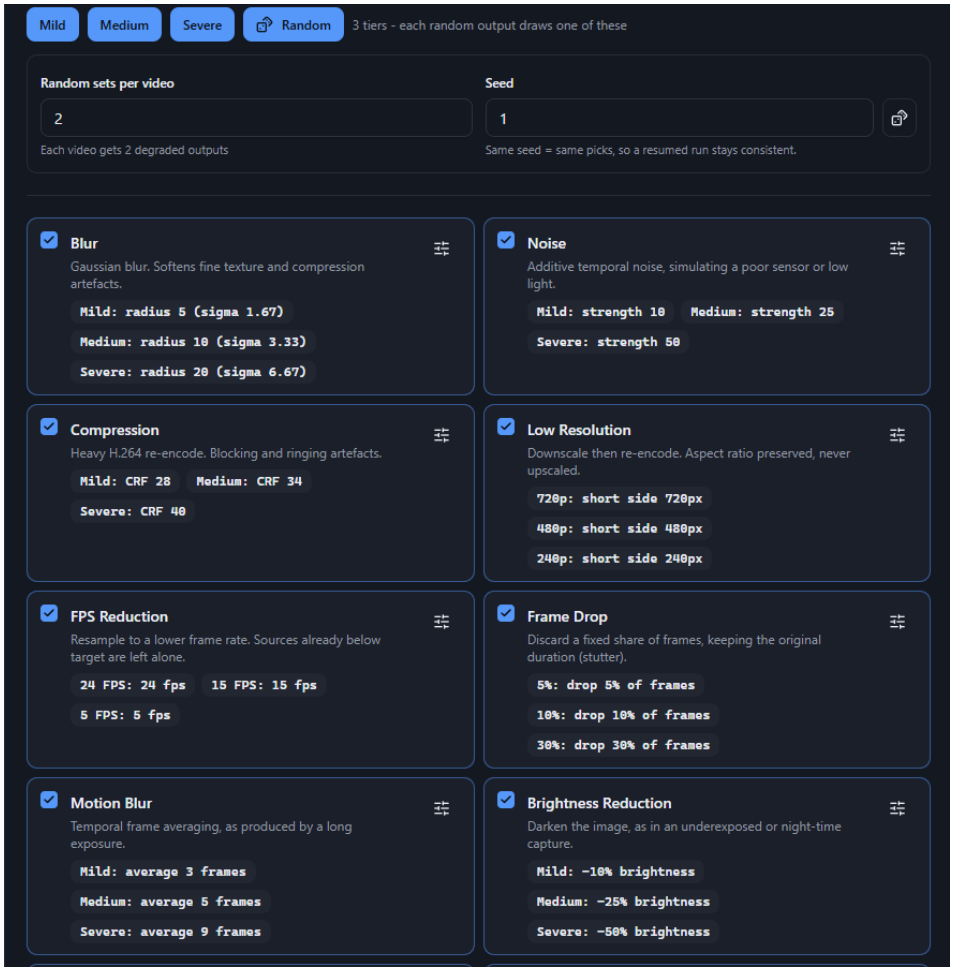

Figure 5.4: Random mode (random sets per video and seed) and the three strength levels of each degradation.

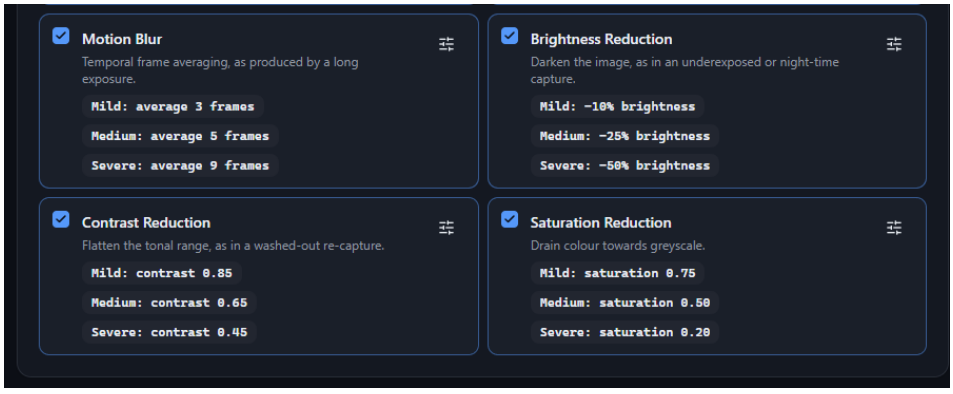

Figure 5.5: Strength levels of motion blur, brightness, contrast and saturation reduction.

Degradation	FFmpeg filter	Mild	Medium	Severe	Effect
Blur	gblur	radius 5 (σ 1.67)	radius 10 (σ 3.33)	radius 20 (σ 6.67)	Softens fine texture and compression artefacts
Noise	noise (temporal)	strength 10	strength 25	strength 50	Poor sensor or low light
Compression	libx264 CRF	CRF 28	CRF 34	CRF 40	Blocking and ringing
Low resolution	scale	short side 720 px	480 px	240 px	Downscale, aspect kept, never upscaled
FPS reduction	fps	24 fps	15 fps	5 fps	Lower frame rate; slower sources left alone
Frame drop	select	drop 5%	drop 10%	drop 30%	Stutter, original duration kept
Motion blur	tmix	average 3 frames	5 frames	9 frames	Long-exposure smear
Brightness reduction	eq=brightness	−10%	−25%	−50%	Under-exposed / night-time capture
Contrast reduction	eq=contrast	0.85	0.65	0.45	Washed-out re-capture
Saturation reduction	eq=saturation	0.75	0.50	0.20	Colour drained towards greyscale

Table 5.2: Degradation strength levels in the tool (all outputs except compression re-encoded at base CRF 18).

How each test used the tool. Test 1 used tool outputs at the medium level for all 9 degradation types. Tests 2 and 3 used tool outputs (random mode, mild / medium / severe) for BioDeepAV and FF++ DeepFakeDetection, and added code-generated degradations only for FF++ folders that had no tool outputs yet. Test 4 used **only** tool outputs: 79,582 degraded videos for all 9 folders, of which 19,980 were selected (Section 5.5.4). Note that tool outputs are re-encoded with libx264 while the originals keep their own encoding.

5.1 Overview of All Tests

	Test 1 — Mixed degradation (av_mixed.py)	**Test 2 — Two-stage (30-9-26test.py)**	**Test 3 — Face crop, balanced (1-10-26.py)**	**Test 4 — Shortcut-resistant, consistency (6-10-26.py)**
Training data	BioDeepAV	BioDeepAV + FF++	BioDeepAV + FF++	One dataset per model (BioDeepAV; FF++), tested on both
Input	Full frame 96×96	Full frame 96×96	Face crop 128×128	Face crop 128×128, resolution-normalised (80 px face)
Degradations	9 video types, medium	8 video types, 3 levels	8 video types, 3 levels	Only the provided files: 8 (BioDeepAV) / 10 (FF++) types, 3 levels + 6 audio types
Training	Single stage, 12 ep., ~90% degraded	Stage 1 → Stage 2	As Test 2 + balanced sampling	Stage 2 paired + feature & Grad-CAM consistency, curriculum, mixed audio
Threshold	0.5	0.5	0.31 (validation)	Validation (both final models 0.50)
Test set	738	2,141	2,141 (= Test 2)	2,141 originals (same split) + 3,993 degraded
BioDeepAV AUC orig. | deg.	0.954 | 0.959	0.992 | 0.990	0.996 | 0.992	0.992 | 0.988 (BioDeepAV-trained)
FF++ AUC orig. | deg.	—	0.584 | 0.584	0.804 | 0.747	0.747 | 0.680 (FF++-trained)
Cross-dataset AUC	—	—	0.457 / 0.165*	0.527 (Bio→FF++) / 0.500 (FF++→Bio)
Explanation changed while prediction stable	Not measured	43.2%	35.8%	19.6% (Bio-trained) / 11.1% (FF++-trained)

*Table 5.3: Summary of all tests (audio-visual model; Tests 2–4: Stage 2 model). *Test 3 single-dataset video-only models.*

Implementation status

Component	T1	T2	T3	T4	Notes
Detection & pre-processing					
Audio-visual detection (video + audio CNN)	Yes	Yes	Yes	Yes	Same network in all tests
Face detection and cropping	No	No	Yes	Yes	YuNet, 99% of frames
Pixel-aligned original/degraded crops	No	No	Yes	Yes	
Resolution normalisation	No	No	No	Yes	360 px frame, 80 px face bottleneck
Degradations					
Blur, brightness, compression, contrast, frame drop, motion blur, noise, saturation	Yes	Yes	Yes	Yes	
FPS reduction / low resolution	Yes	No	No	Partial	T4: FF++ only (both classes)
Several strength levels	No	Yes	Yes	Yes	
Synthetic (code-generated) degradations	No	Yes	Yes	No	T4 uses only the provided files
Audio degradation	No	No	No	Yes	Noise, volume, low-pass, resample, dropout, silence
Training					
Two-stage training (original → degraded)	No	Yes	Yes	Yes	
Balanced sampling	No	No	Yes	Yes	dataset × class
Validation-selected threshold / macro-AUC selection	No	No	Yes	Yes	
Paired training + feature consistency loss	No	No	No	Yes	$\lambda = 0.1$
Grad-CAM consistency loss	No	No	No	Yes	$\lambda = 0.05$
Degradation curriculum	No	No	No	Yes	mild → medium → all
Mixed audio policy	No	No	No	Yes	50% clean / 25% degraded / 25% none
Joint training on both datasets	No	Yes	Yes	No	T4 trains one model per dataset
Evaluation					
Original vs degraded, flips, per-degradation	Yes	Yes	Yes	Yes	
Degradation × severity	No	No	Yes	Yes	
Video-only / audio-only ablation	Yes	Yes	Partial	Yes	T4: full audio × video grid
Cross-dataset evaluation	No	No	Yes	Yes	
Shortcut diagnostics	No	No	Yes	Yes	
Enforced leakage check	No	Partial	Yes	Yes	0 crossings
Explainability					
Grad-CAM heatmaps / stability	No	Yes	Yes	Yes	Pearson r, IoU, centre shift
Temporal Grad-CAM	No	No	Yes	Yes	
LIME / SHAP	No	No	No	No	
Frozen final model for XAI phase	No	No	Yes	Yes	SHA-256 recorded

Table 5.4: Implementation status of each test.

Shared model architecture

All four tests use the same AudioVisualDeepFakeCNN (584,978 parameters; adaptive pooling makes it work at both 96×96 and 128×128). The video and audio branches produce 256 and 128 features, which are joined and classified. Video-only and audio-only results come from this same trained model with one branch replaced by zeros (a modality-removal ablation), not from separately trained models.

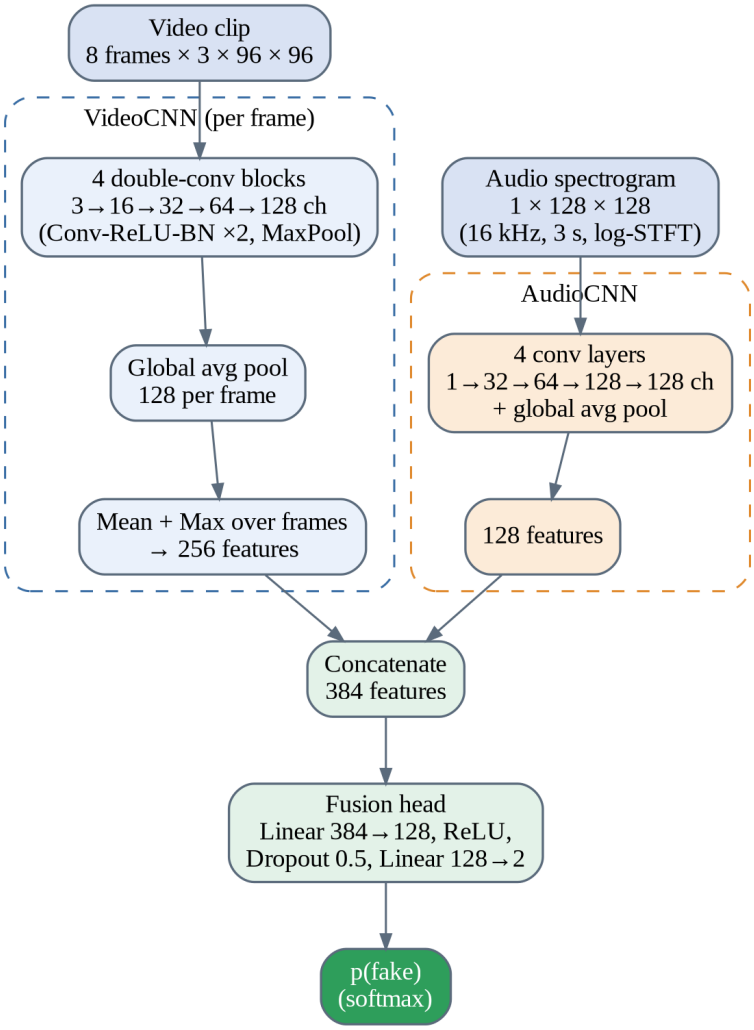

Figure 5.6: AudioVisualDeepFakeCNN, used in all tests.

5.2 Test 1 — Mixed Degradation Training on BioDeepAV

5.2.1 Overview

The audio-visual detector was trained for 12 epochs on BioDeepAV using the EXPANDED regime: every training video in its clean form and all nine degraded versions in each epoch. The best checkpoint (epoch 11) was tested on 738 identity-disjoint held-out videos in clean form, with one random degradation per video, and with all ten variants pooled. Script: *av_mixed.py*.

Key result. ROC-AUC stayed between 0.954 and 0.978 in every condition; no degradation lowered AUC below the clean baseline (0.9541). Accuracy at the 0.5 threshold was limited by real videos being called fake (specificity 62.6% on clean video). Audio added less than 0.005 AUC over video alone.

Note on the earlier Mid report. The Mid report described the same run (same epochs, training time and 738-video test set). Its headline figures (clean AUC 0.9503, accuracy 84.69%; degraded 0.9550, 90.24%; pooled 0.9568, 88.73%) and its per-degradation table match the **video-only** column of this run, not the audio-visual model. Both are kept below; the audio-visual values are the main Test 1 result.

5.2.2 Code pipeline

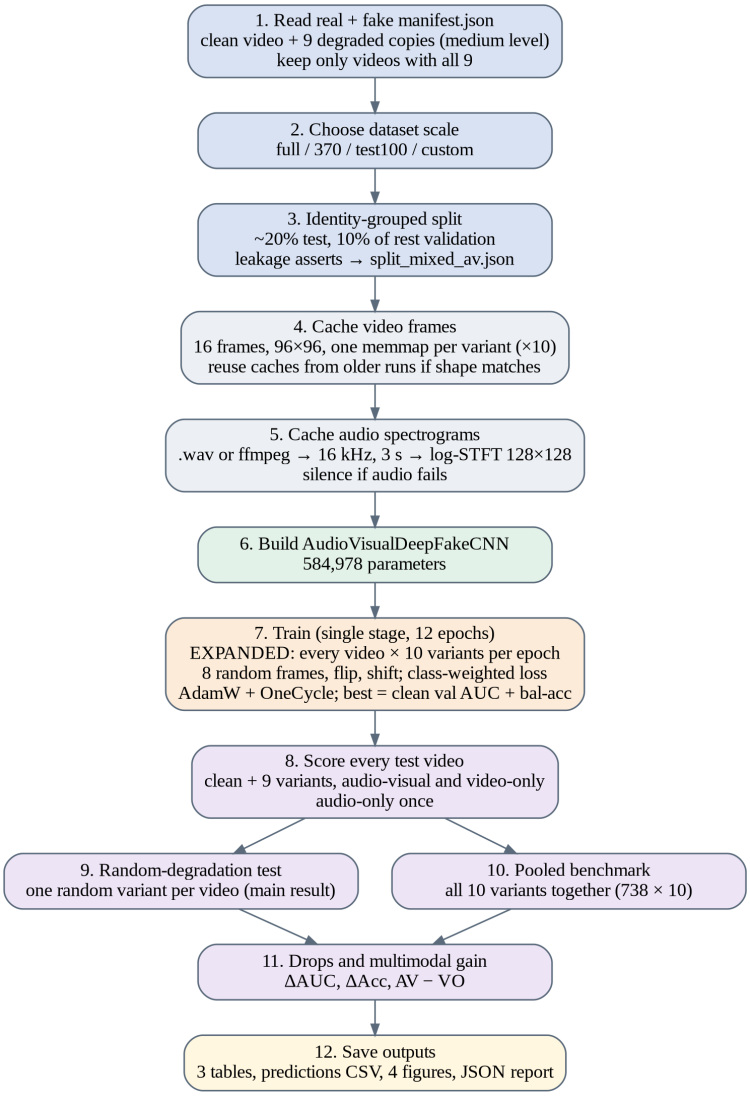

Figure 5.7: What av_mixed.py does, step by step (1,495 lines of Python). Modes: train_test (default), train, test.

5.2.3 Setup

Item	Setting
Mode / hardware	train_test; CUDA — NVIDIA GeForce RTX 3050; GPU stayed at 60–66°C, thermal guard never paused
Dataset scale	full (options: full, 370, test100, custom)
Training regime	EXPANDED (options: expanded, random_mix, compound_mix)
Input	16 frames at 96×96 cached, 8 random frames per training clip; audio 16 kHz, 3 s (48,000 samples), STFT n_fft 512
Optimiser	AdamW (lr 1e-3, weight decay 1e-4) + OneCycleLR; class-weighted cross-entropy; batch 32
Epochs / steps	12 epochs, 821 steps/epoch, 9,852 steps; training time 2:34:41
Checkpoint	Best clean validation AUC + balanced accuracy
Augmentation	Random horizontal flip, ±4 px shift
Caches	All 10 video caches reused from another cache_degraded folder; audio cache built in this run

Table 5.5: Test 1 configuration.

5.2.4 Degradations

Degraded copies were made with the Team01 FFmpeg degradation tool. Each degradation has three levels; every file in Test 1 used the **medium** level. Outputs are re-encoded with libx264 at CRF 18 (compression uses its own CRF).

Degradation	FFmpeg filter	Mild	Medium (used)	Severe	What it simulates
Blur	gblur	radius 5 (σ 1.67)	radius 10 (σ 3.33)	radius 20 (σ 6.67)	Out-of-focus camera
Noise	noise=alls:allf=t+u	10	25	50	Low-light sensor grain
Compression	libx264 CRF	28	34	40	Social-media re-encoding
FPS reduction	fps	24	15	5	Low frame rate
Frame drop	select	5%	10%	30%	Bad connection (frame held)
Motion blur	tmix	3 frames	5 frames	9 frames	Slow shutter
Brightness	eq=brightness	−10%	−25%	−50%	Under-exposure
Contrast	eq=contrast	0.85	0.65	0.45	Flat lighting
Saturation	eq=saturation	0.75	0.50	0.20	Faded colour
Low resolution*	scale	720p	480p	240p	Downscaled video

*Table 5.6: Degradation strength levels. *Low resolution exists in the tool but was not in the manifests, so it was not used.*

How these settings affect the results

- **96×96 input hides damage.** Shrinking full-HD frames about 11 times removes most of a 10 px blur, fine noise and compression blocks before the model sees them.
- **Only 16 sampled frames.** FPS reduction and frame drop barely change the 16 evenly spaced frames, so they behave almost like clean video.
- **Re-encoding confound.** Degraded files are re-encoded with libx264; clean originals are not, so "clean vs degraded" is also "original vs re-encoded".
- **Skipped videos.** FPS reduction is skipped for sources at ≤ 15 fps; such videos lack a variant and are silently dropped.
- **One strength level** and **clean audio**: robustness is shown at medium strength only, and the audio is never degraded.

5.2.5 Data

Partition	Clean	Degraded	Total
Training pool (80%) — Real 1,641 / Fake 1,312	2,953	26,577	29,530
Active train	2,625	23,625	26,250
Validation	328	2,952	3,280
Held-out test (20%) — Real 369 / Fake 369	738	6,642	7,380
All — Real 2,010 / Fake 1,681	**3,691**	**33,219**	**36,910**

Table 5.7: Test 1 data. Real videos are grouped by person/channel so no identity is in two splits; the test set is balanced, so accuracy equals balanced accuracy.

5.2.6 Training

Epoch	Loss	Val AUC	Val Bal-Acc	Epoch time	Best
1	0.2139	0.9870	0.9421	15:09	Yes
2	0.0818	0.9905	0.9451	15:45	Yes
3	0.0593	0.9880	0.9604	16:11	Yes
4	0.0431	0.9837	0.8902	14:43	
5	0.0296	0.9856	0.8171	13:14	
6	0.0254	0.9913	0.9360	12:05	
7	0.0253	0.9907	0.9604	11:33	Yes
8	0.0096	0.9874	0.9482	11:20	
9	0.0071	0.9881	0.9177	11:13	
10	0.0051	0.9882	0.9512	11:09	
11	0.0032	0.9893	0.9665	11:07	Yes
12	0.0022	0.9895	0.9634	11:07	

Table 5.8: Test 1 training (clean validation, 328 videos). Best epoch 11: AUC 0.9893, balanced accuracy 0.9665.

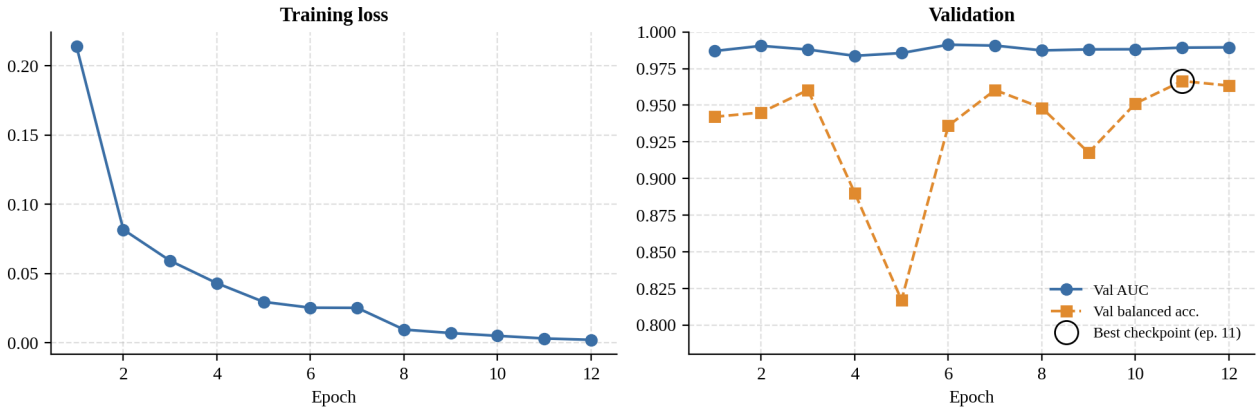

Figure 5.8: Test 1 training loss and validation metrics.

5.2.7 Results: original vs degraded

Metric	Original (clean)	Random degradation	Change
Accuracy	80.89%	85.37%	+4.47 pts
Balanced accuracy	80.89%	85.37%	+4.47 pts
ROC-AUC	0.9541	0.9593	+0.0052
Precision (fake)	72.62%	77.59%	+4.97 pts
Recall / sensitivity (fake)	99.19%	99.46%	+0.27 pts
Specificity (real)	62.60%	71.27%	+8.67 pts
F1-score	0.8385	0.8717	+0.0332
Confusion [TN, FP]	[231, 138]	[263, 106]	FP −32
Confusion [FN, TP]	[3, 366]	[2, 367]	FN −1

Table 5.9: Test 1 audio-visual model on 738 test videos.

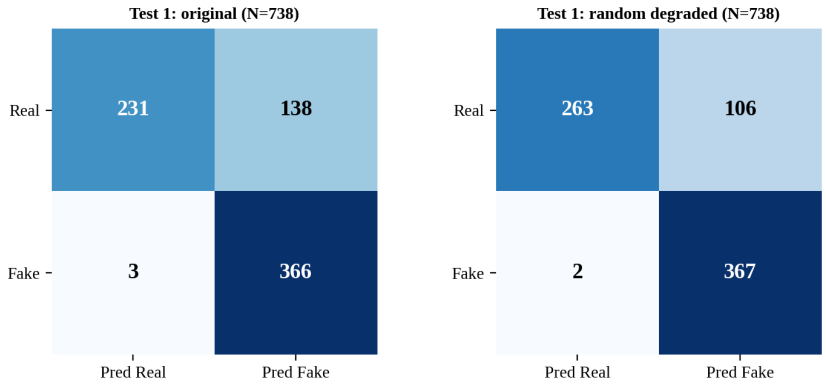

Figure 5.9: Test 1 confusion matrices.

5.2.8 Results by modality

Benchmark	N	AV AUC	AV Acc	AV F1	VO AUC	VO Acc	AO AUC	AV − VO
Clean	738	0.9541	80.89%	0.8385	0.9503	84.69%	0.8194	+0.0038
Random degradation	738	0.9593	85.37%	0.8717	0.9550	90.24%	0.8194	+0.0044
Pooled (all 10)	7,380	0.9589	83.75%	0.8593	0.9568	88.73%	0.8194	+0.0021

Table 5.10: Test 1 by modality. AV = audio-visual, VO = video-only (audio zeroed), AO = audio-only (accuracy 69.38%). VO values equal the Mid report figures.

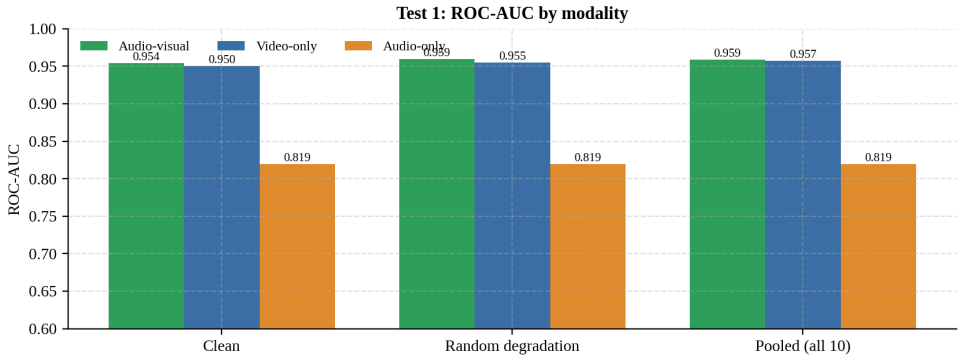

Figure 5.10: Test 1 ROC-AUC by modality.

5.2.9 Results per degradation

Variant	AV AUC	AV Acc	AV F1	ΔAUC vs clean	ΔAcc vs clean	VO AUC	VO Acc	AV − VO
Clean	0.9541	80.89%	0.8385	+0.0000	+0.00 pts	0.9503	84.69%	+0.0038
Blur	0.9778	90.24%	0.9102	+0.0237	+9.35 pts	0.9791	93.63%	−0.0013
Brightness	0.9614	87.13%	0.8851	+0.0073	+6.23 pts	0.9607	92.95%	+0.0007
Compression	0.9595	84.28%	0.8635	+0.0054	+3.39 pts	0.9600	89.43%	−0.0005
Contrast	0.9617	83.88%	0.8602	+0.0076	+2.98 pts	0.9598	89.97%	+0.0019
FPS reduction	0.9555	81.84%	0.8453	+0.0014	+0.95 pts	0.9522	85.77%	+0.0033
Frame drop	0.9558	82.11%	0.8472	+0.0017	+1.22 pts	0.9526	85.50%	+0.0032
Motion blur	0.9546	82.52%	0.8502	+0.0005	+1.63 pts	0.9504	86.18%	+0.0042
Noise	0.9574	80.89%	0.8389	+0.0033	+0.00 pts	0.9542	85.50%	+0.0032
Saturation	0.9597	83.74%	0.8588	+0.0055	+2.85 pts	0.9666	93.63%	−0.0069
Random degradation	0.9593	85.37%	0.8717	+0.0052	+4.47 pts	0.9550	90.24%	+0.0043
Pooled (all 10)	0.9589	83.75%	0.8593	+0.0048	+2.86 pts	0.9568	88.73%	+0.0021

Table 5.11: Test 1 per-degradation results (738 test videos each).

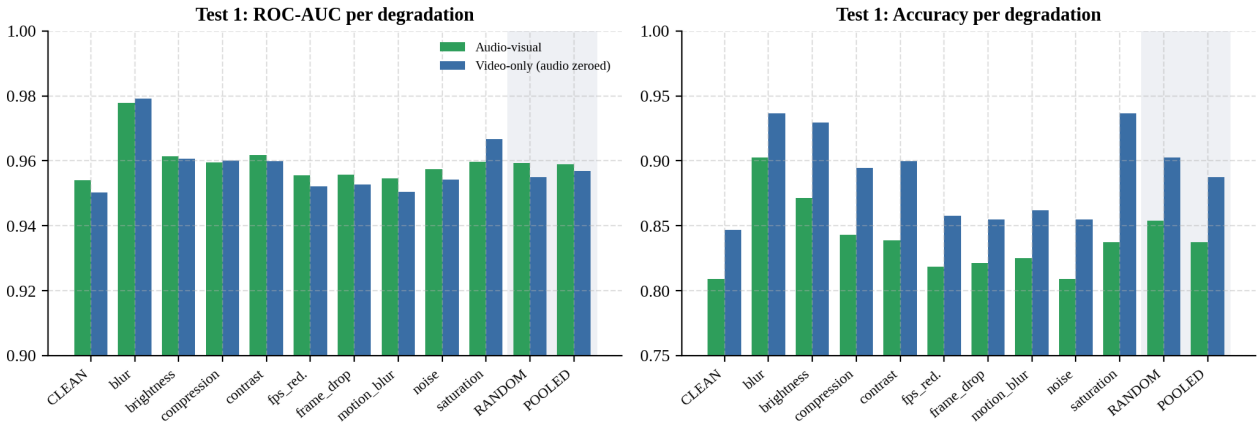

Figure 5.11: Test 1 per-degradation ROC-AUC and accuracy, audio-visual vs video-only (dashed line: audio-only AUC 0.819; shaded: random and pooled).

5.2.10 Explainability

Not implemented in Test 1 (Grad-CAM, heatmaps and LIME/SHAP were planned; explanation robustness was listed as future work). The team's separate Explainable CNN-GRU script already had Grad-CAM, temporal attention and explanation-robustness measures, which informed Test 2.

5.2.11 Key findings

- **Ranking is strong and stable:** AUC 0.954–0.978 in every condition; no degradation lowered AUC below clean.
- **Degraded videos scored higher than clean.** Nine of every ten training examples were degraded, so clean video was the model's least familiar input. Report this as "no loss under degradation", not "degradation improves detection".

- **Main weakness: real videos called fake.** Fake recall was 99.2% but only 62.6% of clean real videos were recognised (138 of 369 flagged fake). With AUC 0.954, this points to the fixed 0.5 threshold; choosing the threshold on validation should raise accuracy without retraining.
- **Audio adds very little:** AV beat VO by at most 0.004 AUC and was lower for blur, compression and saturation. VO was more accurate at 0.5 in every row, most likely because zeroing audio shifts scores toward "real" and partly corrects the threshold bias.
- **Audio-only AUC never changes (0.8194)** because audio is never degraded.
- **Validation was optimistic:** about 0.99 AUC on validation vs 0.954 on test, with training loss near zero — some overfitting and a small (328-video) validation set.

5.2.12 Limitations and things to check

- **Video-only and audio-only are not separate models** — describe them as a modality-removal ablation.
- **Reused caches** were matched only by shape; confirm once that split_mixed_av.json and split_mixed_cnngru.json list the same videos in the same order (the high AUC suggests they do).
- **Silent-audio fallback**: count how many videos fell back to silence, by class, to rule out a "silence means real/fake" shortcut.
- **Test mode rebuilds the split** instead of loading the saved one; testing a checkpoint trained with another scale or seed could leak.
- **Random-degradation test is not balanced by type**; use the per-degradation table for per-type claims.
- **Adding datasets**: identity grouping depends on BioDeepAV file names, caches are keyed only by video count, and dataset shortcuts must be checked — all of which became relevant in Test 2.

5.2.13 Suggested thesis sentence

> "On 738 held-out, identity-disjoint test videos from the BioDeepAV dataset, the audio-visual model reached a ROC-AUC of 0.954 on clean video and 0.959 under randomly assigned degradations, with no drop in AUC for any of the nine degradation types. Accuracy at a fixed 0.5 threshold was limited by a high false-positive rate on real videos (specificity 62.6% on clean video), and the audio stream added less than 0.005 AUC over video alone."

5.2.14 Output files

File	Contents
best_model_mixed_av.pt	Best audio-visual model weights
split_mixed_av.json	Train / validation / test video names
cache_degraded/cache_*.npy, cache_audio_*.npy	Cached frames for every variant and audio spectrograms
test_predictions_degraded_av.csv	Per test video: probabilities for every variant and modality, assigned degradation
report_degraded_av_date_*.json, latest_report_degraded_av.json	All metrics, drops, per-degradation results, summary tables
degraded_av_roc_curves.png, multimodal_degraded_ablation.png, confusion_matrix_clean_vs_degraded.png, training_curves_mixed_av.png	Figures

Table 5.12: Test 1 output files.

5.3 Test 2 — Two-Stage Audio-Visual Training on BioDeepAV + FaceForensics++

5.3.1 Overview

The same detector was trained in two stages — first on original videos only, then on degraded + original videos — on BioDeepAV and FaceForensics++ C23. Both stage checkpoints were tested on 2,141 identity-disjoint held-out videos in original form and with one random degraded version each. For the first time, Grad-CAM explanations of the original and degraded version of the same video were compared. Script: *30-9-26test.py* (run 30-9-26).

Key result. Degraded training made predictions robust: the decision flip rate under degradation fell from 12.4% (Stage 1) to 1.4% (Stage 2), and on BioDeepAV degraded accuracy rose from 88.5% to 94.9%. But explanations did not follow: 43.2% of Stage 2 predictions that stayed the same had a substantially different Grad-CAM map. FaceForensics++ real videos were almost all called fake, so overall FF++ numbers do not reflect detection ability.

5.3.2 Code pipeline

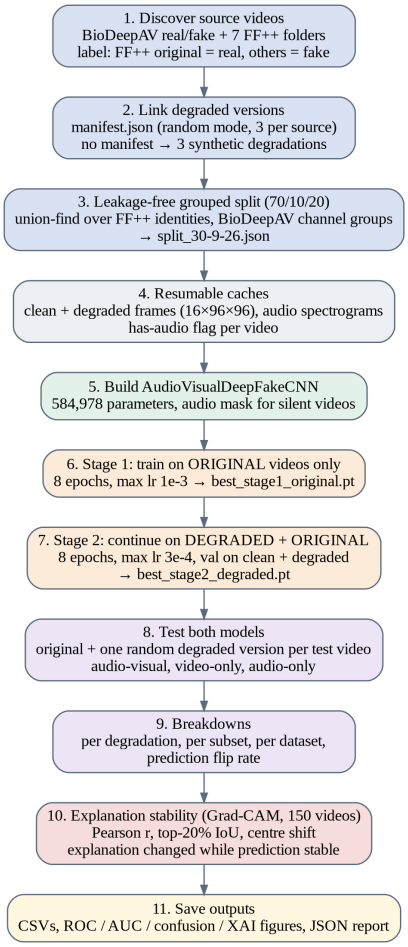

Figure 5.12: What 30-9-26test.py does, step by step. Modes: full (default), stage1, stage2, test.

5.3.3 Setup

Item	Setting
Mode / hardware	FULL (Stage 1 → Stage 2 → Test); CUDA — NVIDIA GeForce RTX 3050; thermal guard at 72°C
Datasets	BioDeepAV (with audio) + FaceForensics++ C23: original, Deepfakes, Face2Face, FaceShifter, FaceSwap, NeuralTextures, DeepFakeDetection (no audio)
Input	16 frames at 96×96 cached, 8-frame clips; audio as in Test 1; audio dropout 0.2 in training
Stage 1	Original videos only; 8 epochs, 235 steps/epoch; AdamW + OneCycle max lr 1e-3; class-weighted loss
Stage 2	Initialised from Stage 1; degraded + original; 8 epochs, 937 steps/epoch; max lr 3e-4
Checkpoint	Best mean of (AUC + balanced accuracy) over the validation sets
Augmentation	Random horizontal flip, ±4 px shift; new random synthetic degradation every epoch
Caching	Clean 4.73 GB (54:38), degraded 6.24 GB (1:20:18), audio for 10,703 videos; 0 failed videos
Training time	Stage 1: 26:31; Stage 2: 1:44:46

Table 5.13: Test 2 configuration.

5.3.4 Degradations

Two sources of degraded video were used. **Manifest videos** (BioDeepAV real, BioDeepAV fake, FF++ DeepFakeDetection) come from the degradation tool in random mode: each source has 3 random degradations, each at a mild, medium or severe level. **Synthetic degradations** are applied in code to the other FF++ folders (including FF++ original), so that “degraded” can never become a shortcut for “fake”. Their settings, applied to the 96×96 frames, are:

Degradation	Mild	Medium	Severe	Method
Noise	σ 5	σ 10	σ 18	Gaussian pixel noise
Blur	σ 0.8	σ 1.4	σ 2.2	Gaussian blur
Brightness	±25	±45	±70	Added to pixel values (random sign)
Contrast	×0.75 / 1.25	×0.55 / 1.45	×0.35 / 1.7	Around the mean (lower 70% of the time)
Saturation	×0.6 / 1.5	×0.3 / 2.0	×0.05 / 2.5	HSV saturation (lower 70% of the time)
Compression	JPEG q 45	q 25	q 10	Per-frame JPEG
Motion blur	3 px	5 px	9 px	Line kernel, random direction
Frame drop	5%	10%	30%	Dropped frame replaced by previous

Table 5.14: Test 2 synthetic degradation settings (fps_reduction not used).

5.3.5 Data

Folder	Label	Original	Degraded (manifest)	Audio	Train	Val	Test
BioDeepAV/real	REAL	2,010	6,030	2,009	1,407	201	402
BioDeepAV/fake	FAKE	1,693	5,079	1,693	1,185	169	339
FF++/original	REAL	1,000	0	0	700	100	200
FF++/Deepfakes	FAKE	1,000	0	0	700	100	200
FF++/Face2Face	FAKE	1,000	0	0	700	100	200
FF++/FaceShifter	FAKE	1,000	0	0	700	100	200
FF++/FaceSwap	FAKE	1,000	0	0	700	100	200
FF++/NeuralTextures	FAKE	1,000	0	0	700	100	200
FF++/DeepFakeDetection	FAKE	1,000	3,000	0	700	100	200
Total		**10,703**	**14,109**	**3,702**	**7,492**	**1,070**	**2,141**

Table 5.15: Test 2 data (Real 3,010 | Fake 7,693). Split by identity groups (FF++ source/target identities, BioDeepAV channels); test videos have 6,423 degraded versions (2,823 manifest + 3,600 synthetic).

5.3.6 Training

Epoch	Loss	Val AUC (clean)	Val Bal-Acc (clean)	Val AUC (degraded)	Val Bal-Acc (degraded)	Best
S1-1	0.5623	0.5561	0.4993	—	—	Yes
S1-2	0.5043	0.8268	0.7722	—	—	Yes
S1-3	0.4704	0.8586	0.8105	—	—	Yes
S1-4	0.4417	0.8612	0.8150	—	—	Yes
S1-5	0.4213	0.8437	0.7714	—	—	
S1-6	0.4047	0.8772	0.8205	—	—	Yes
S1-7	0.3775	0.8820	0.8120	—	—	
S1-8	0.3725	0.8805	0.8234	—	—	Yes
S2-1	0.4227	0.8748	0.8224	0.8708	0.8043	Yes
S2-2	0.4014	0.8676	0.8165	0.8610	0.8061	
S2-3	0.3910	0.8802	0.8247	0.8805	0.8183	Yes
S2-4	0.3799	0.8847	0.8250	0.8819	0.8067	
S2-5	0.3655	0.8828	0.8272	0.8745	0.8246	Yes
S2-6	0.3537	0.8812	0.8243	0.8817	0.8227	Yes
S2-7	0.3467	0.8792	0.8249	0.8826	0.8233	Yes
S2-8	0.3397	0.8807	0.8236	0.8854	0.8244	Yes

Table 5.16: Test 2 training (validation: 1,070 videos clean, 1,070 degraded). Stage 2 instances per epoch: 12,600 synthetic + 9,876 manifest-degraded + 7,492 clean.

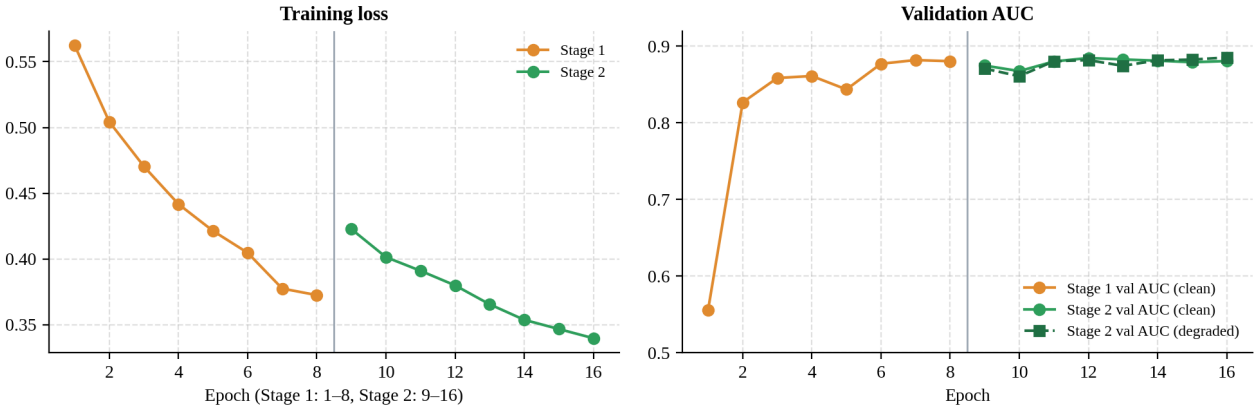

Figure 5.13: Test 2 training loss and validation AUC across both stages.

5.3.7 Results: original vs degraded

Metric	S1 original	S1 degraded	S1 change	S2 original	S2 degraded	S2 change
Accuracy	88.28%	79.68%	−8.59 pts	89.02%	88.14%	−0.89 pts
ROC-AUC	0.8772	0.8207	−0.0565	0.8700	0.8656	−0.0043
Precision (fake)*	87.93%	86.27%	−1.66 pts	87.21%	86.90%	−0.31 pts
Recall (fake)*	97.01%	85.32%	−11.70 pts	99.29%	98.31%	−0.97 pts
Specificity (real)*	65.95%	65.28%	−0.66 pts	62.79%	62.13%	−0.66 pts
F1-score	0.9225	0.8579	−0.0646	0.9286	0.9226	−0.0060
Confusion [TN, FP]	[397, 205]	[393, 209]	FP +4	[378, 224]	[374, 228]	FP +4
Confusion [FN, TP]	[46, 1493]	[226, 1313]	FN +180	[11, 1528]	[26, 1513]	FN +15
Pooled AUC (original + all degraded)	0.8338			0.8688		
Prediction flip rate	12.42%			1.45%		

*Table 5.17: Test 2 on 2,141 test videos (S1 = Stage 1, S2 = Stage 2). *Computed from the confusion matrices.*

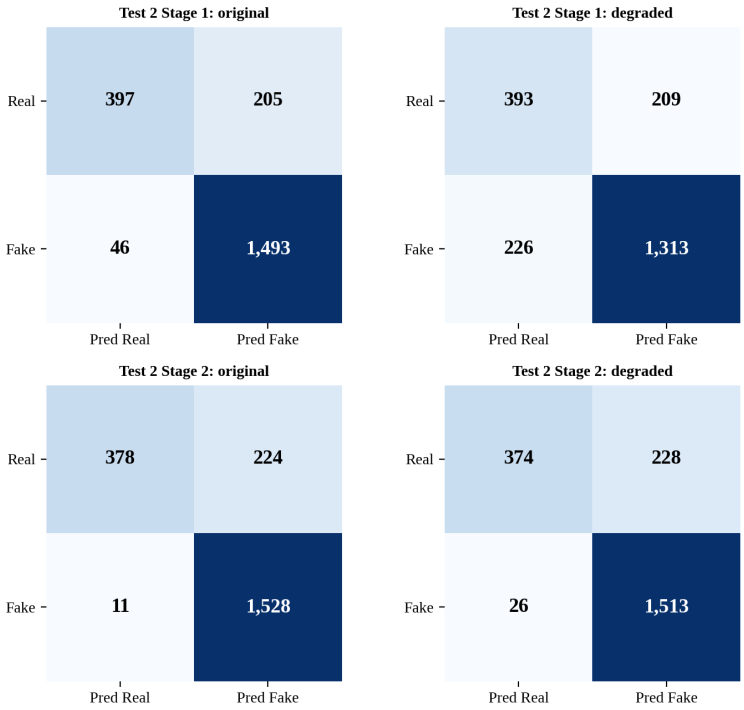

Figure 5.14: Test 2 confusion matrices.

Dataset	Model	Original AUC	Original Acc	Degraded AUC	Degraded Acc
BioDeepAV	Stage 1	0.9931	96.09%	0.9507	88.53%
BioDeepAV	Stage 2	0.9923	95.82%	0.9900	94.87%
FF++	Stage 1	0.5844	84.14%	0.5469	75.00%
FF++	Stage 2	0.5838	85.43%	0.5842	84.57%

Table 5.18: Test 2 results per dataset.

5.3.8 Results by modality

Benchmark	AV AUC	VO AUC	AO AUC	AV − VO
Stage 1 original	0.8772	0.8691	0.7204	+0.0081
Stage 1 degraded	0.8207	0.8004	0.7204	+0.0203
Stage 2 original	0.8700	0.8560	0.8137	+0.0140
Stage 2 degraded	0.8656	0.8480	0.8137	+0.0176

Table 5.19: Test 2 by modality (AO measured on BioDeepAV, the only dataset with audio; VO accuracy was not logged).

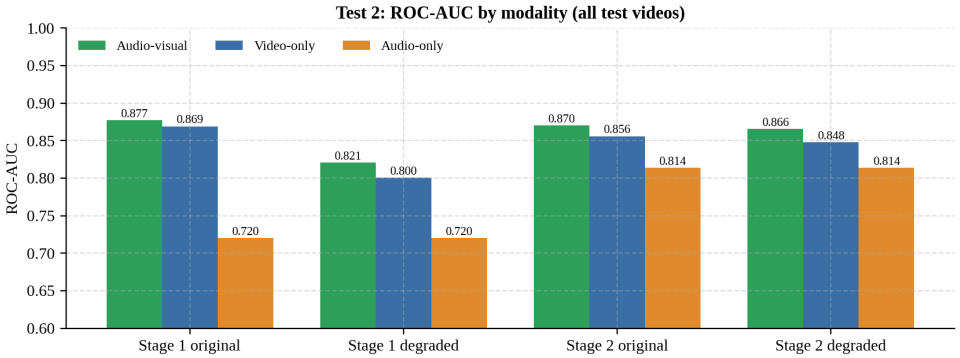

Figure 5.15: Test 2 ROC-AUC by modality.

5.3.9 Results per degradation

Degradation	S1 AUC	S2 AUC	S1 Acc	S2 Acc	ΔAUC S2 vs original	ΔAcc S2 vs original
Blur	0.8452	0.8593	64.39%	88.34%	−0.0107	−0.68 pts
Brightness	0.8045	0.8351	76.17%	86.51%	−0.0349	−2.51 pts
Compression	0.8834	0.8711	82.09%	88.34%	+0.0011	−0.68 pts
Contrast	0.8519	0.8839	84.80%	89.57%	+0.0139	+0.55 pts
Frame drop	0.8636	0.8589	88.23%	89.06%	−0.0111	+0.04 pts
Motion blur	0.8885	0.8766	79.46%	88.76%	+0.0066	−0.26 pts
Noise	0.6125	0.8810	79.03%	89.83%	+0.0110	+0.81 pts
Saturation	0.8434	0.8736	81.33%	86.77%	+0.0036	−2.25 pts

Table 5.20: Test 2 per-degradation results (all degraded versions of the test videos).

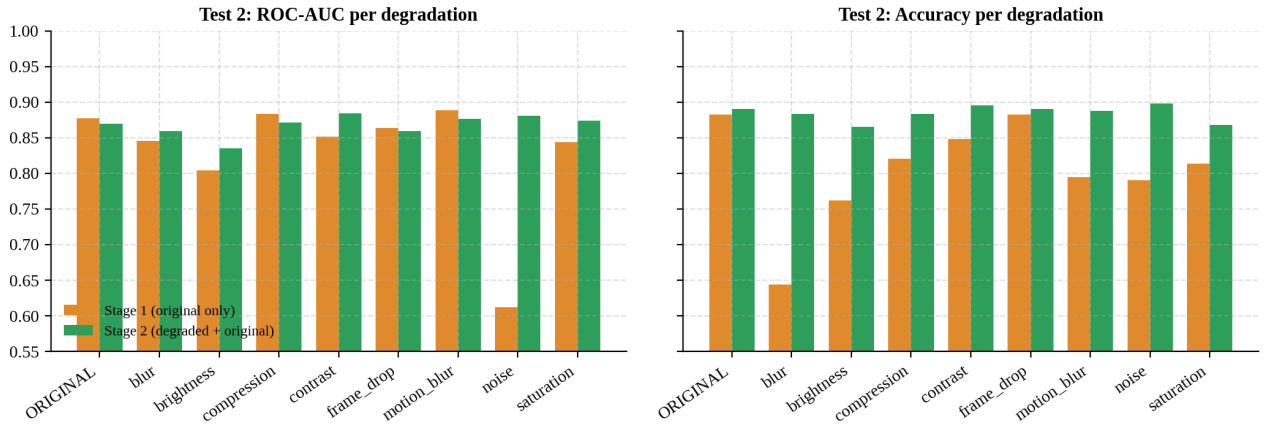

Figure 5.16: Test 2 per-degradation ROC-AUC and accuracy, Stage 1 vs Stage 2.

5.3.10 Results per subset

Subset	N	S1 Orig.	S1 Deg.	S1 Δ	S2 Orig.	S2 Deg.	S2 Δ
BioDeepAV/fake	339	94.4%	86.1%	−8.3	98.2%	97.0%	−1.2
BioDeepAV/real	402	97.5%	90.5%	−7.0	93.8%	93.0%	−0.7
FF++/DeepFakeDetection	200	100.0%	95.0%	−5.0	100.0%	100.0%	+0.0
FF++/Deepfakes	200	96.5%	82.5%	−14.0	99.5%	99.0%	−0.5
FF++/Face2Face	200	98.0%	81.0%	−17.0	99.5%	98.5%	−1.0
FF++/FaceShifter	200	98.0%	88.0%	−10.0	99.5%	97.0%	−2.5
FF++/FaceSwap	200	97.0%	85.5%	−11.5	99.5%	99.0%	−0.5
FF++/NeuralTextures	200	97.0%	78.5%	−18.5	99.5%	98.5%	−1.0
FF++/original (REAL)	**200**	**2.5%**	**14.5%**	**+12.0**	**0.5%**	**0.0%**	**−0.5**

Table 5.21: Test 2 accuracy per subset (Δ in points). Red: FF++ real videos are almost all called fake.

Sample predictions (Stage 2, 20 random test videos)

Video	True	p(fake) orig.	Result	Degradation	p(fake) deg.	Result
Bio/real/-axp0KRQJYc_0004	REAL	0.000	OK	blur/mild	0.000	OK
Bio/real/-BNR218UFX4_0007	REAL	0.000	OK	contrast/severe	0.001	OK
Bio/real/-BNR218UFX4_0041	REAL	0.000	OK	noise/medium	0.000	OK
Bio/fake/b3a27b80…	FAKE	0.999	OK	noise/mild	1.000	OK
Bio/fake/f4722d30…	FAKE	0.999	OK	frame_drop/medium	0.999	OK
FF++/original/095	REAL	0.706	**ERR**	compression/medium	0.768	**ERR**
FF++/original/911	REAL	0.673	**ERR**	frame_drop/30pct	0.673	**ERR**
FF++/Deepfakes/175_159	FAKE	0.647	OK	frame_drop/5pct	0.648	OK
FF++/Deepfakes/824_419	FAKE	0.648	OK	motion_blur/medium	0.735	OK
FF++/Deepfakes/834_852	FAKE	0.682	OK	contrast/severe	0.645	OK
FF++/FaceShifter/301_262	FAKE	0.647	OK	compression/severe	0.646	OK
FF++/FaceShifter/346_351	FAKE	0.652	OK	frame_drop/5pct	0.652	OK
FF++/FaceShifter/430_459	FAKE	0.671	OK	compression/medium	0.673	OK
FF++/FaceShifter/657_644	FAKE	0.658	OK	noise/severe	0.661	OK
FF++/FaceSwap/737_719	FAKE	0.653	OK	contrast/medium	0.656	OK
FF++/NeuralTextures/784_769	FAKE	0.648	OK	blur/mild	0.664	OK
FF++/DFD/01_15__meeting_serious	FAKE	0.997	OK	saturation/medium	0.992	OK
FF++/DFD/06_25__outside_talking_pan	FAKE	0.998	OK	contrast/mild	0.998	OK
FF++/DFD/21_11__walking_down_indoor	FAKE	0.996	OK	saturation/severe	0.955	OK
FF++/DFD/22_23__outside_talking_still	FAKE	1.000	OK	compression/severe	1.000	OK

Table 5.22: Test 2 sample predictions. Most FF++ videos (except DFD) sit in a narrow 0.65–0.77 band, whether real or fake.

5.3.11 Explainability: Grad-CAM stability

For 150 test videos, the Grad-CAM map (last video conv block, averaged over frames) of the original clip was compared with the map of its degraded version using Pearson correlation, the overlap (IoU) of the top-20% most salient pixels, and the shift of the map's centre of mass. An explanation counts as changed when IoU < 0.5.

Model	Pearson r	Top-20% IoU	Centre shift	Prediction stable	Explanation changed while prediction stable
Stage 1	0.678	0.557	0.081	90.7%	39.7%
Stage 2	0.741	0.567	0.053	98.7%	**43.2%**

Table 5.23: Test 2 Grad-CAM stability, original vs degraded version of the same video.

Degradation	IoU Stage 1	IoU Stage 2	Change
Blur	0.4146	0.5685	+0.1539
Brightness	0.3857	0.2801	−0.1056
Compression	0.6077	0.6089	+0.0012
Contrast	0.4663	0.4543	−0.0120
Frame drop	0.9395	0.8978	−0.0417
Motion blur	0.6999	0.6736	−0.0263
Noise	0.4512	0.5683	+0.1171
Saturation	0.4071	0.4366	+0.0295

Table 5.24: Mean top-20% Grad-CAM IoU per degradation type.

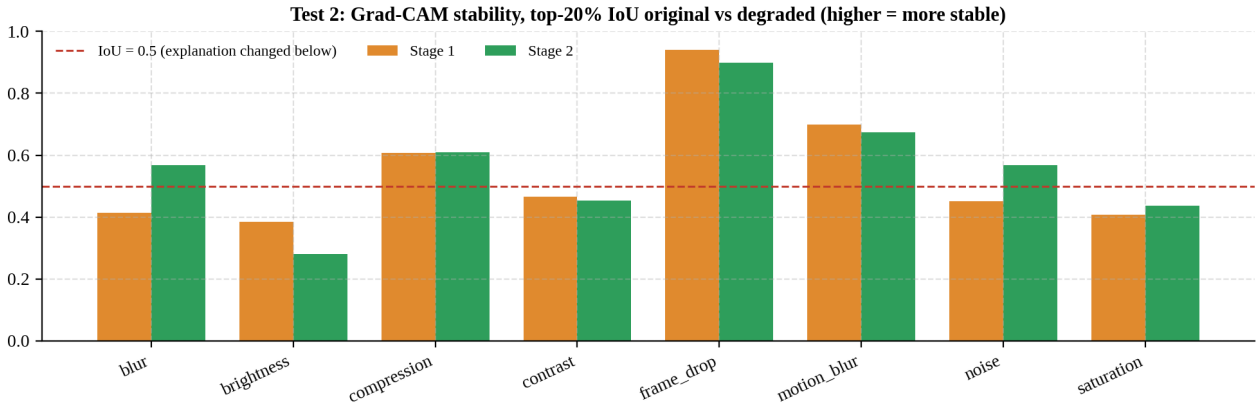

Figure 5.17: Test 2 Grad-CAM stability per degradation type (higher = more stable; red line: IoU 0.5).

5.3.12 Key findings

- **Degraded training makes predictions robust.** Stage 2 cut the AUC drop under degradation from −0.0565 to −0.0043, the accuracy drop from −8.6 to −0.9 points and the flip rate from 12.4% to 1.4%, with almost no clean loss. On BioDeepAV, degraded AUC rose from 0.9507 to 0.9900.
- **Stable predictions do not mean stable explanations.** 98.7% of Stage 2 predictions were unchanged, yet 43.2% of those had a substantially different Grad-CAM map — direct evidence for the gap in Section 4.
- **Explanation stability depends on the degradation type.** Frame drop barely changes the maps (IoU ≈ 0.90); brightness, saturation and contrast change them most. Stage 2 improved blur and noise but made brightness worse (0.386 → 0.280).
- **Without degraded training, noise and blur hurt most** (Stage 1 noise AUC 0.6125, blur accuracy 64.4%).
- **Audio helps more than in Test 1** (AV − VO up to +0.020 AUC) and the audio branch improved in Stage 2 (audio-only AUC 0.7204 → 0.8137).
- **BioDeepAV real videos are now well recognised** (93.8% in Stage 2, vs 62.6% specificity in Test 1).

5.3.13 Limitations and things to check

- **FF++ real videos are almost all called fake** (199 of 200 in Stage 2, accounting for 199 of its 224 false positives); FF++ AUC is about 0.58, close to chance.
- **Likely dataset shortcut:** FF++ has no audio and 6 of its 7 folders are fake, so the model separates the dataset rather than the manipulation (see the narrow probability band in the sample predictions).
- **Grad-CAM on FF++** may explain the shortcut rather than a forgery; the stability numbers are most meaningful for BioDeepAV.
- **Video-only and audio-only are modality-removal ablations**, as in Test 1; audio is still never degraded.
- **Next steps:** report per-dataset balanced accuracy, rebalance FF++ originals, crop faces before resizing, tune the threshold on validation, and repeat the Grad-CAM test per dataset on correctly classified videos.

5.3.14 Suggested thesis sentence

> "When the audio-visual detector was first trained on original videos only, a randomly degraded version changed its decision for 12.4% of 2,141 held-out videos; after a second training stage on degraded and original videos this fell to 1.4%, and on BioDeepAV the degraded-video AUC rose from 0.951 to 0.990. However, the Grad-CAM explanations did not stabilise to the same degree: for 43.2% of the videos whose prediction stayed the same, the most salient regions overlapped by less than 50% between the original and degraded version."

5.3.15 Output files

File	Contents
best_stage1_original.pt, best_stage2_degraded.pt	Stage 1 and Stage 2 model weights
split_30-9-26.json	Train / validation / test video names
cache/cache_clean_*, cache_degraded_*, cache_audio_*	Cached frames and spectrograms (resumable)
test_predictions_30-9-26.csv, test_predictions_all_degraded_30-9-26.csv	Per-video probabilities, original and degraded
summary_original_vs_degraded.csv, summary_per_degradation.csv, summary_per_subset.csv	Summary tables
xai_stability_30-9-26.csv	Grad-CAM stability per video
report_30-9-26_20261001_0313.json, latest_report_30-9-26.json	All metrics
roc_original_vs_degraded.png, auc_per_degradation.png, confusion_original_vs_degraded.png, xai_stability_by_degradation.png, xai_examples_original_vs_degraded.png, training_curves_30-9-26.png	Figures

Table 5.25: Test 2 output files (F:\Code\run_30-9-26).

5.4 Test 3 — Face-Crop, Balanced, Threshold-Calibrated Detector on BioDeepAV + FaceForensics++

5.4.1 Overview

Test 3 keeps the Test 2 network, two-stage training, degradations and test split, and changes how the data reach the model: faces are detected and cropped to 128×128, every training batch is balanced across BioDeepAV/FF++ and real/fake, models are selected on the macro per-dataset validation AUC, and the decision threshold is chosen on validation only. It adds cross-dataset training, shortcut diagnostics and Grad-CAM on pixel-aligned original/degraded crops, and freezes the final model for the XAI phase. Script: *1-10-26.py* (run 1-10-26, finished 2 Oct 2026).

Key result. On the same 2,141 test videos as Test 2, the Stage 2 model reached AUC 0.936 on original and 0.913 on degraded video. FF++ AUC rose from 0.58 to 0.80 and FF++ real videos recognised rose from 0.5% to 40.0%, while BioDeepAV stayed at AUC 0.996 / 0.992. Grad-CAM maps were more stable (IoU 0.605, Pearson r 0.788), but 35.8% of predictions that did not change still had a substantially different explanation. Diagnostics show BioDeepAV real and fake videos differ in resolution, so part of BioDeepAV's near-perfect score may come from that cue.

5.4.2 Code pipeline

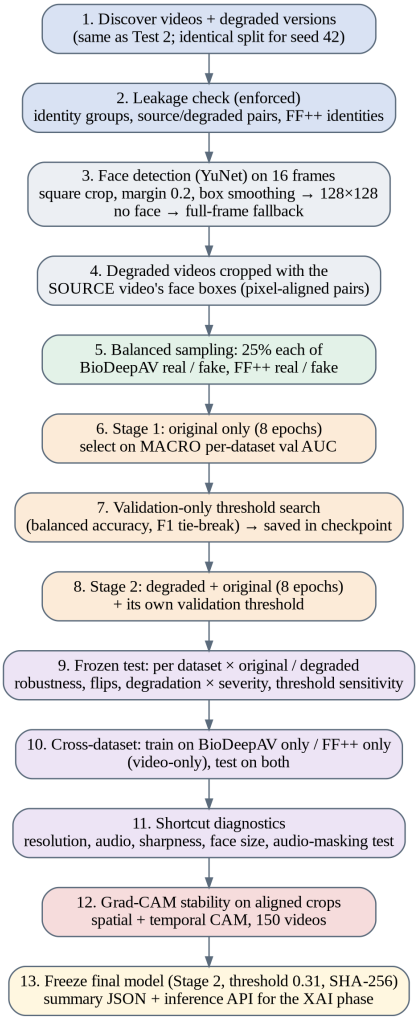

Figure 5.18: What 1-10-26.py does, step by step. Modes: full (default), stage1, stage2, evaluate, threshold, cross, selftest, predict.

5.4.3 Setup

Item	Test 2	Test 3
Input	Full frame resized to 96×96	YuNet face crop (margin 0.2, no rotation alignment, 3-frame box smoothing) → 128×128
Degraded inputs	Own full frame	Cropped with the source video's face boxes (aligned pairs)
Sampling	Natural mix, class-weighted loss	25% each: BioDeepAV real / fake, FF++ real / fake (oversampling cap 4×), plain loss
Model selection	Mean of AUC + balanced accuracy	Macro per-dataset validation AUC
Threshold	Fixed 0.5	Searched 0.01–0.99 on validation (balanced accuracy, F1 tie-break), stored in checkpoint
Augmentation	Flip, ±4 px shift	Flip, ±5 px shift (same relative jitter)
Unchanged	—	Network, 16 frames / 8-frame clips, batch 32, AdamW + OneCycle, lr 1e-3 / 3e-4, 8 + 8 epochs, seed 42, audio dropout 0.2, split
Hardware / time	RTX 3050	RTX 3050; Stage 1 47:06, Stage 2 3:06:27, cross-dataset 16:32 + 30:59
Caches	4.73 + 6.24 GB	Face crops 8.42 GB (original) + 11.10 GB (degraded); 0 failed videos

Table 5.26: Test 3 configuration compared with Test 2.

5.4.4 Degradations

Same sources and settings as Test 2 (Section 5.3.4): manifest videos with mild / medium / severe levels for BioDeepAV and FF++ DeepFakeDetection, and synthetic degradations of the same 8 types for the other FF++ folders. The difference is that the degraded video is cropped with exactly the same face box as its source, so original and degraded crops show the same pixels and Grad-CAM maps can be compared position by position. Test 3 also reports every degradation at every severity (Section 5.4.9).

5.4.5 Data

The inventory and split are identical to Test 2 (Section 5.3.5): 10,703 original videos (Real 3,010 | Fake 7,693), 14,109 manifest-degraded videos; train 7,492 | validation 1,070 | test 2,141. The enforced leakage check found 2,523 identity groups, 0 groups crossing splits, 0 degraded/source pairs crossing splits and 0 FF++ identities crossing splits.

Subset	Videos	Frames	Face found	Detection rate	Full-frame fallback	Mean crop size*
BioDeepAV/fake	1,693	27,088	27,080	99.97%	0	0.877
BioDeepAV/real	2,010	32,160	30,534	94.94%	56	0.871
FF++/DeepFakeDetection	1,000	16,000	15,995	99.97%	0	0.319
FF++/Deepfakes	1,000	16,000	15,999	99.99%	0	0.491
FF++/Face2Face	1,000	16,000	16,000	100.00%	0	0.493
FF++/FaceShifter	1,000	16,000	15,999	99.99%	0	0.496
FF++/FaceSwap	1,000	16,000	15,999	99.99%	0	0.492
FF++/NeuralTextures	1,000	16,000	15,999	99.99%	0	0.493
FF++/original	1,000	16,000	15,999	99.99%	0	0.494
All	**10,703**	**171,248**	**169,604**	**99.04%**	**56**	

*Table 5.27: Test 3 face detection on original videos. *Crop side relative to the shorter frame side (BioDeepAV videos are already tight face crops).*

Cell	Training items (Stage 1)	Natural share	Sampled share	Oversampling
BioDeepAV fake	1,185	15.82%	25.0%	1.58×
BioDeepAV real	1,407	18.78%	25.0%	1.33×
FF++ fake	4,200	56.06%	25.0%	0.45×
FF++ real	700	9.34%	25.0%	2.68×

Table 5.28: Test 3 balanced sampling (Stage 2 uses the same shares over 29,968 items: 12,600 synthetic + 9,876 manifest-degraded + 7,492 original).

5.4.6 Training and threshold selection

Epoch	Train loss	Val loss	Val AUC	Val BAcc@0.5	AUC Bio	AUC FF++	Val deg. AUC	Select	Best
S1-1	0.5849	0.7765	0.4406	0.4492	0.763	0.500	—	0.6316	Yes
S1-2	0.4973	0.5575	0.6913	0.6245	0.976	0.569	—	0.7729	Yes
S1-3	0.4664	0.6610	0.7604	0.6594	0.965	0.602	—	0.7834	Yes
S1-4	0.4163	0.5066	0.8202	0.7325	0.994	0.656	—	0.8246	Yes
S1-5	0.3871	0.4639	0.8450	0.7455	0.998	0.682	—	0.8398	Yes
S1-6	0.3644	0.5431	0.9198	0.7534	0.999	0.716	—	0.8574	Yes
S1-7	0.3566	0.4421	0.9195	0.7735	0.999	0.729	—	0.8641	Yes
S1-8	0.3444	0.4324	0.9243	0.7763	0.999	0.730	—	0.8646	Yes
S2-1	0.4104	0.5074	0.9059	0.7302	0.998	0.718	0.8844	0.8297	Yes
S2-2	0.3925	0.4786	0.9201	0.7542	0.999	0.717	0.8901	0.8360	Yes
S2-3	0.3796	0.5158	0.9267	0.7237	1.000	0.738	0.8932	0.8425	Yes
S2-4	0.3658	0.7887	0.5999	0.5476	0.996	0.732	0.6208	0.8357	
S2-5	0.3544	0.3829	0.9020	0.8119	1.000	0.729	0.8826	0.8457	Yes
S2-6	0.3360	0.4364	0.9186	0.7753	0.999	0.782	0.9014	0.8652	Yes
S2-7	0.3258	0.4487	0.9178	0.7706	1.000	0.777	0.9012	0.8651	
S2-8	0.3183	0.4036	0.9165	0.7955	1.000	0.779	0.9003	0.8669	Yes

Table 5.29: Test 3 training (S1 = Stage 1, 235 steps/epoch; S2 = Stage 2, 937 steps/epoch). Select = macro per-dataset validation AUC; best epoch 8 in both stages.

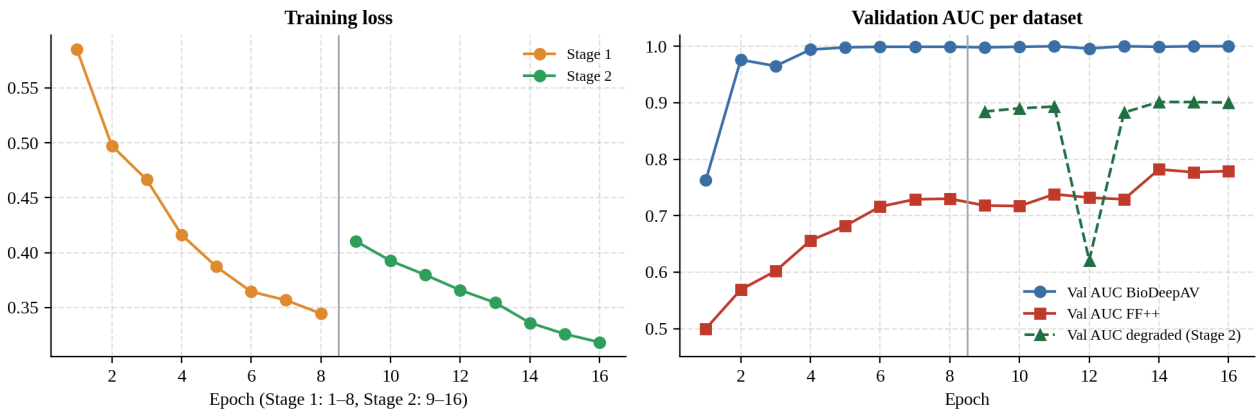

Figure 5.19: Test 3 training loss and validation AUC per dataset.

Stage	Selected threshold	Selected on	Val balanced accuracy	Val F1	BioDeepAV optimum*	FF++ optimum*
Stage 1	0.31	Validation original	0.8529	0.9213	0.49	0.44
Stage 2	0.31	Validation original + degraded	0.8300	0.9151	0.83	0.54

*Table 5.30: Validation-selected thresholds. *Per-dataset optima are diagnostic only and were not used.*

5.4.7 Results: original vs degraded

Model	Dataset	Cond.	Acc	BAcc	Prec.	Recall	Spec.	F1	AUC	TN / FP / FN / TP
Stage 1	Combined	original	85.24%	0.8265	0.9069	0.8856	0.7674	0.8961	0.9143	462 / 140 / 176 / 1363
		degraded	80.01%	0.7674	0.8751	0.8421	0.6927	0.8583	0.8316	417 / 185 / 243 / 1296
	BioDeepAV	original	97.44%	0.9750	0.9624	0.9823	0.9677	0.9723	0.9960	389 / 13 / 6 / 333
		degraded	88.12%	0.8792	0.8815	0.8555	0.9030	0.8683	0.9149	363 / 39 / 49 / 290
	FF++	original	78.79%	0.6117	0.8902	0.8583	0.3650	0.8740	0.7278	73 / 127 / 170 / 1030
		degraded	75.71%	0.5542	0.8733	0.8383	0.2700	0.8554	0.6310	54 / 146 / 194 / 1006
Stage 2	Combined	original	88.93%	0.8563	0.9157	0.9318	0.7807	0.9237	0.9362	470 / 132 / 105 / 1434
		degraded	88.04%	0.8324	0.8967	0.9422	0.7226	0.9189	0.9127	435 / 167 / 89 / 1450
	BioDeepAV	original	97.71%	0.9777	0.9653	0.9853	0.9701	0.9752	0.9958	390 / 12 / 5 / 334
		degraded	96.09%	0.9623	0.9379	0.9794	0.9453	0.9582	0.9915	380 / 22 / 7 / 332
	FF++	original	84.29%	0.6583	0.9016	0.9167	0.4000	0.9091	0.8042	80 / 120 / 100 / 1100
		degraded	83.79%	0.6033	0.8852	0.9317	0.2750	0.9078	0.7468	55 / 145 / 82 / 1118

Table 5.31: Test 3 test metrics at the validation threshold 0.31 (n = 2,141 Combined, 741 BioDeepAV, 1,400 FF++).

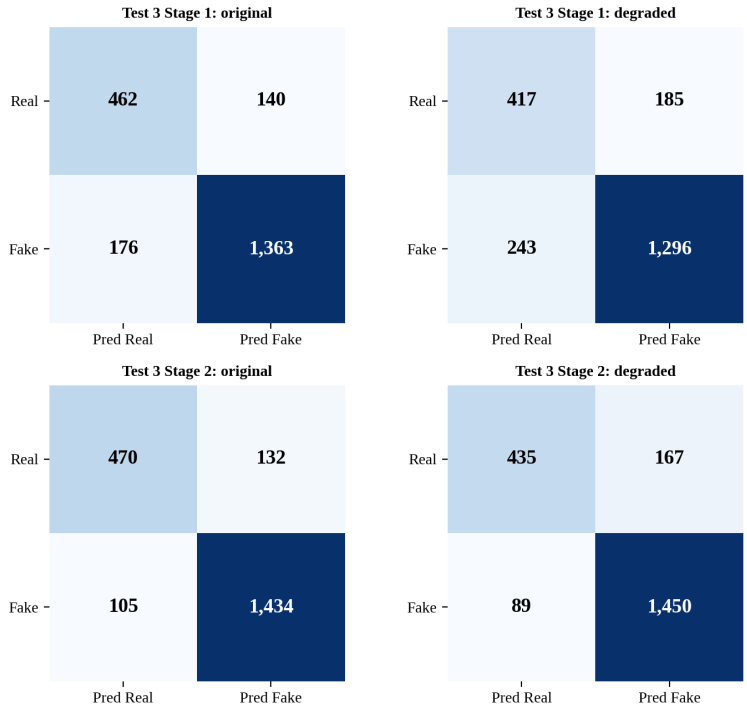

Figure 5.20: Test 3 confusion matrices (Combined test set, threshold 0.31).

Model	Dataset	AUC orig.	AUC deg.	AUC drop	BAcc drop	F1 drop	Flip rate	Mean |Δp|
Stage 1	Combined	0.9143	0.8316	0.0828	0.0591	0.0378	13.55%	0.139
	BioDeepAV	0.9960	0.9149	0.0811	0.0958	0.1040	10.66%	0.114
	FF++	0.7278	0.6310	0.0969	0.0575	0.0186	15.07%	0.153
Stage 2	Combined	0.9362	0.9127	0.0235	0.0239	0.0048	6.21%	0.067
	BioDeepAV	0.9958	0.9915	0.0043	0.0154	0.0170	2.16%	0.021
	FF++	0.8042	0.7468	0.0575	0.0550	0.0013	8.36%	0.091

Table 5.32: Test 3 robustness: same videos, frozen thresholds. Flip rate = decisions that change after degradation; |Δp| = mean change in p(fake).

Model	Dataset	BAcc at 0.31	BAcc at 0.5	Specificity at 0.31	Specificity at 0.5	Recall at 0.31	Recall at 0.5
Stage 2	Combined	0.8563	0.8260	0.7807	0.9269	0.9318	0.7251
	BioDeepAV	0.9777	0.9789	0.9701	0.9726	0.9853	0.9853
	FF++	0.6583	0.7433	0.4000	0.8350	0.9167	0.6517

Table 5.33: Validation threshold (0.31) vs fixed 0.5 on original test videos (0.5 shown for analysis only).

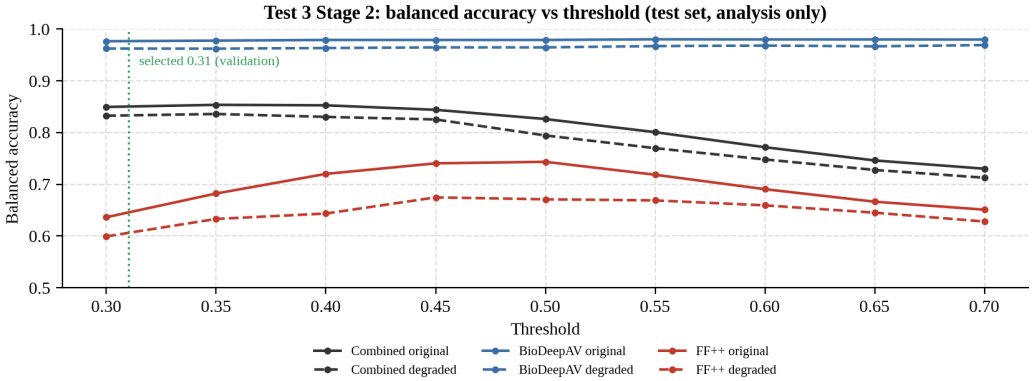

Figure 5.21: Test 3 Stage 2 balanced accuracy vs threshold on the test set (analysis only; the threshold was chosen on validation).

5.4.8 Results by modality

Test 3 did not repeat the video-only/audio-only ablation. Instead it tested how much the decision depends on audio being present: the 741 BioDeepAV test videos (the only ones with audio) were scored with and without their audio.

Measure	With audio	Audio masked	Change
Mean p(fake)	0.468	0.512	+0.044
AUC	0.9958	0.9877	−0.0082
Decisions that flip	—	—	12.1%

Table 5.34: Test 3 audio-masking test (Stage 2, BioDeepAV).

5.4.9 Results per degradation and severity

Degradation	Severity	n	AUC S1	AUC S2	BAcc S1	BAcc S2	F1 S1	F1 S2	Flip S1	Flip S2
blur	mild	268	0.873	0.875	0.789	0.773	0.889	0.913	13.4%	11.2%
	medium	278	0.828	0.868	0.678	0.822	0.697	0.929	33.5%	10.8%
	severe	243	0.741	0.862	**0.547**	0.780	0.365	0.915	52.7%	9.9%
brightness	mild	292	0.923	0.927	0.839	0.844	0.906	0.920	7.2%	5.5%
	medium	236	0.885	0.912	0.798	0.807	0.889	0.903	9.7%	9.3%
	severe	265	0.771	0.881	0.710	0.787	0.789	0.914	31.3%	12.5%
compression	mild	270	0.928	0.936	0.839	0.852	0.919	0.927	5.2%	4.8%
	medium	285	0.929	0.936	0.843	0.837	0.926	0.903	10.2%	6.0%
	severe	260	0.890	0.904	0.785	0.802	0.912	0.919	9.2%	4.2%
contrast	mild	291	0.893	0.914	0.823	0.833	0.900	0.916	5.5%	4.5%
	medium	279	0.916	0.934	0.822	0.850	0.870	0.919	15.8%	6.8%
	severe	226	0.868	0.889	0.743	0.836	0.732	0.903	31.4%	9.3%
frame_drop	mild	288	0.913	0.932	0.819	0.863	0.912	0.938	1.0%	1.0%
	medium	297	0.901	0.943	0.841	0.869	0.907	0.928	0.0%	1.7%
	severe	256	0.914	0.940	0.843	0.848	0.919	0.925	1.2%	1.2%
motion_blur	mild	256	0.896	0.918	0.827	0.820	0.904	0.915	9.0%	5.1%
	medium	250	0.901	0.932	0.824	0.841	0.887	0.933	12.8%	6.8%
	severe	268	0.886	0.882	0.768	0.789	0.826	0.911	20.5%	9.3%
noise	mild	289	0.930	0.939	0.850	0.862	0.934	0.922	8.3%	2.4%
	medium	262	**0.557**	0.936	**0.504**	0.870	0.822	0.937	25.6%	3.8%
	severe	255	**0.471**	0.944	**0.507**	0.865	0.827	0.935	29.8%	6.7%
saturation	mild	275	0.922	0.931	0.812	0.836	0.869	0.902	4.0%	5.5%
	medium	262	0.890	0.906	0.792	0.832	0.857	0.898	9.2%	5.3%
	severe	272	0.872	0.892	0.761	0.813	0.857	0.919	18.0%	8.5%

Table 5.35: Test 3 per degradation × severity (Combined test set, all degraded versions; flip = vs the same video's original prediction). Red: near or below chance.

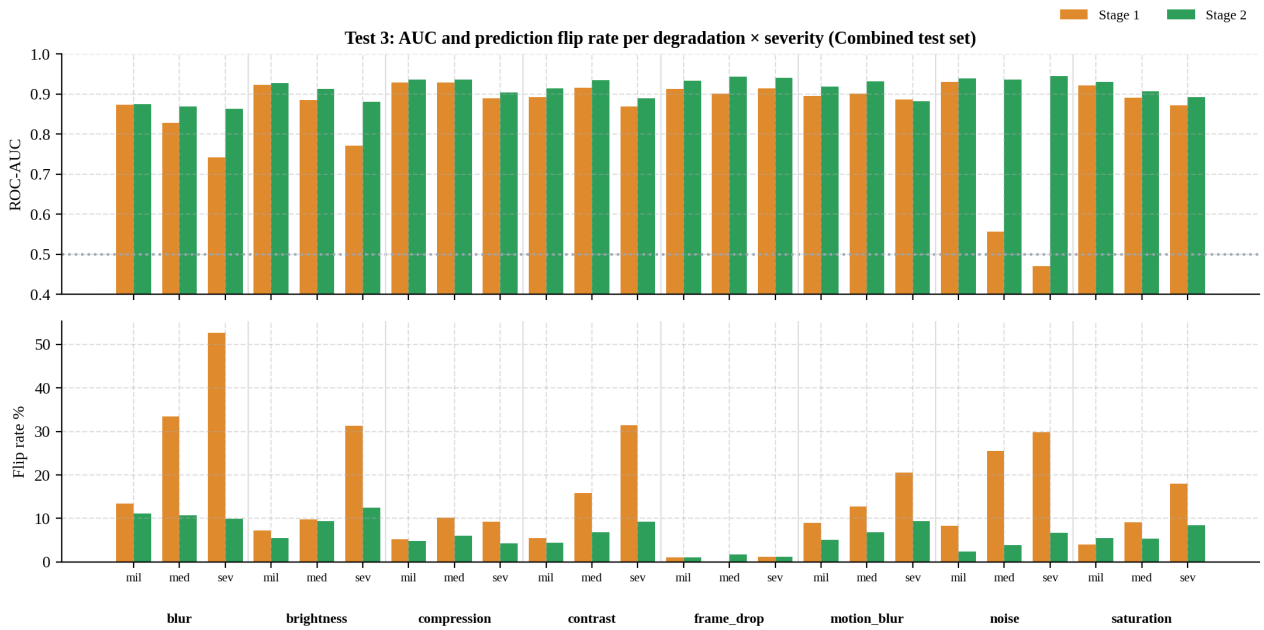

Figure 5.22: Test 3 AUC and flip rate per degradation and severity, Stage 1 vs Stage 2.

Degradation	S1 BioDeepAV	S1 FF++	S2 BioDeepAV	S2 FF++
blur	28.2%	30.4%	2.4%	13.1%
brightness	11.5%	14.0%	12.6%	7.3%
compression	3.9%	9.9%	1.9%	8.2%
contrast	5.3%	18.6%	0.0%	9.3%
frame_drop	0.0%	0.6%	0.0%	2.2%
motion_blur	0.0%	21.0%	0.0%	10.2%
noise	29.1%	11.9%	0.0%	5.0%
saturation	6.1%	13.4%	1.0%	10.8%

Table 5.36: Test 3 prediction flip rate per dataset and degradation (one random degraded version per test video).

5.4.10 Cross-dataset results

To see whether the detector learns manipulation cues that transfer, video-only models were trained on one dataset’s original videos (8 epochs, class-balanced, own validation threshold) and tested on both. The combined rows are the main Test 3 models.

Setting	Train on	Test on	Condition	Threshold	AUC	BAcc	F1
in-domain	BioDeepAV	BioDeepAV	original	0.40	0.9898	0.9227	0.9161
in-domain	BioDeepAV	BioDeepAV	degraded	0.40	0.9098	0.8375	0.8259
cross-dataset	BioDeepAV	FF++	original	0.40	**0.4574**	0.4746	0.6471
cross-dataset	BioDeepAV	FF++	degraded	0.40	**0.4640**	0.4688	0.5346
cross-dataset	Both	BioDeepAV	original	0.46	**0.1654**	0.2458	0.3266
cross-dataset	Both	BioDeepAV	degraded	0.46	**0.3015**	0.3259	0.4478
in-domain	Both	FF++	original	0.46	0.7353	0.6654	0.7087
in-domain	Both	FF++	degraded	0.46	0.6671	0.6142	0.8379
combined (stage1)	Both	BioDeepAV	original	0.31	0.9960	0.9750	0.9723
combined (stage1)	Both	BioDeepAV	degraded	0.31	0.9149	0.8792	0.8683
combined (stage1)	Both	FF++	original	0.31	0.7278	0.6117	0.8740
combined (stage1)	Both	FF++	degraded	0.31	0.6310	0.5542	0.8554
combined (stage2)	Both	BioDeepAV	original	0.31	0.9958	0.9777	0.9752
combined (stage2)	Both	BioDeepAV	degraded	0.31	0.9915	0.9623	0.9582
combined (stage2)	Both	FF++	original	0.31	0.8042	0.6583	0.9091
combined (stage2)	Both	FF++	degraded	0.31	0.7468	0.6033	0.9078

Table 5.37: Test 3 cross-dataset results. Red: chance level or below. Best epochs: BioDeepAV-only 8 (val AUC 0.998), FF++-only 6 (val AUC 0.740).

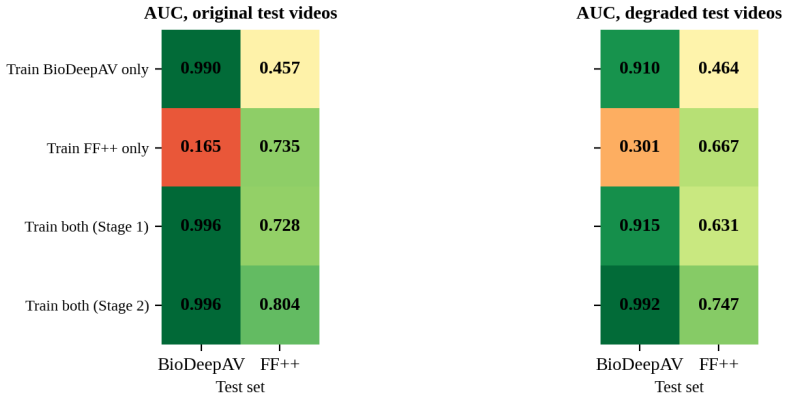

Figure 5.23: Test 3 cross-dataset AUC (rows: training data; columns: test dataset).

5.4.11 Shortcut diagnostics

The script checks whether predictions follow properties that are not the manipulation itself. These are **potential** shortcuts, not proof.

Subset	Label	n	Predicted fake	Audio	Median size (px)	Face crop size	Sharpness
BioDeepAV/fake	FAKE	339	98.5%	Yes	512 × 512	0.880	592.1
BioDeepAV/real	REAL	402	3.0%	Yes	168 × 168	0.848	252.5
FF++/DeepFakeDetection	FAKE	200	100.0%	No	1920 × 1080	0.322	311.4
FF++/Deepfakes	FAKE	200	95.0%	No	854 × 480	0.499	443.9
FF++/Face2Face	FAKE	200	90.0%	No	832 × 480	0.500	464.9
FF++/FaceShifter	FAKE	200	99.0%	No	854 × 480	0.503	386.2
FF++/FaceSwap	FAKE	200	83.5%	No	854 × 480	0.500	468.7
FF++/NeuralTextures	FAKE	200	82.5%	No	832 × 480	0.500	400.7
FF++/original	REAL	200	**60.0%**	No	854 × 480	0.501	487.9

Table 5.38: Test 3 prediction rate by subset (Stage 2, original test videos) with nuisance properties.

Scope	Feature	AUC feature vs label	ρ within real	ρ within fake	Flag
Combined	has_audio	0.276	−0.759	0.652	prediction follows feature
Combined	resolution_px	0.827	0.699	−0.032	separates classes; prediction follows feature
Combined	total_frames	0.725	0.624	−0.078	prediction follows feature
Combined	face_crop_rel_size	0.315	−0.573	0.309	prediction follows feature
Combined	fps / brightness / sharpness	0.563 / 0.580 / 0.643	0.281 / 0.409 / 0.510	−0.486 / −0.053 / 0.163	prediction follows feature
BioDeepAV	**resolution_px**	**0.994**	0.208	−0.062	**feature nearly separates the classes**
BioDeepAV	crop_sharpness	0.792	0.402	0.050	prediction follows feature
BioDeepAV	fps / brightness	0.524 / 0.565	−0.316 / 0.439	0.269 / 0.225	prediction follows feature
FF++	resolution_px / fps	0.566 / 0.419	0.077 / −0.023	0.482 / −0.441	prediction follows feature

Table 5.39: Test 3 flagged features (ρ = Spearman correlation between the feature and p(fake) with the label held fixed).

5.4.12 Explainability: Grad-CAM stability

Grad-CAM (last video conv block) was computed for the same 150 test videos in original and degraded form. Because the crops are pixel-aligned, differences in the maps come from the degradation, not from a shifted face. Test 3 also measures the stability of the temporal CAM (how attention is spread over the 8 frames).

Model	Pearson r	Top-20% IoU	Centre shift	Temporal CAM r	Prediction stable	Explanation changed while prediction stable
Stage 1	0.685	0.531	0.062	0.536	89.3%	40.3%
Stage 2	0.788	0.605	0.053	0.530	98.7%	**35.8%**

Table 5.40: Test 3 Grad-CAM stability, original vs degraded version of the same video.

Degradation	S1 r	S1 IoU	S1 shift	S2 r	S2 IoU	S2 shift
blur	0.3148	0.2722	0.1102	0.5974	0.4724	0.0859
brightness	0.4969	0.3947	0.0956	0.6927	0.4885	0.0595
compression	0.8397	0.5747	0.0353	0.8194	0.5818	0.0518
contrast	0.6718	0.4953	0.0734	0.8739	0.6805	0.0488
frame_drop	0.9794	0.8793	0.0114	0.9849	0.8884	0.0113
motion_blur	0.7538	0.5175	0.0524	0.8225	0.5871	0.0489
noise	0.5159	0.4447	0.0627	0.7027	0.5229	0.0745
saturation	0.7878	0.5495	0.0646	0.7468	0.5247	0.0590
BioDeepAV	0.6868	0.5334	0.0811	0.7674	0.5841	0.0731
FF++	0.6835	0.5299	0.0514	0.7994	0.6160	0.0420

Table 5.41: Test 3 Grad-CAM stability per degradation type and per dataset.

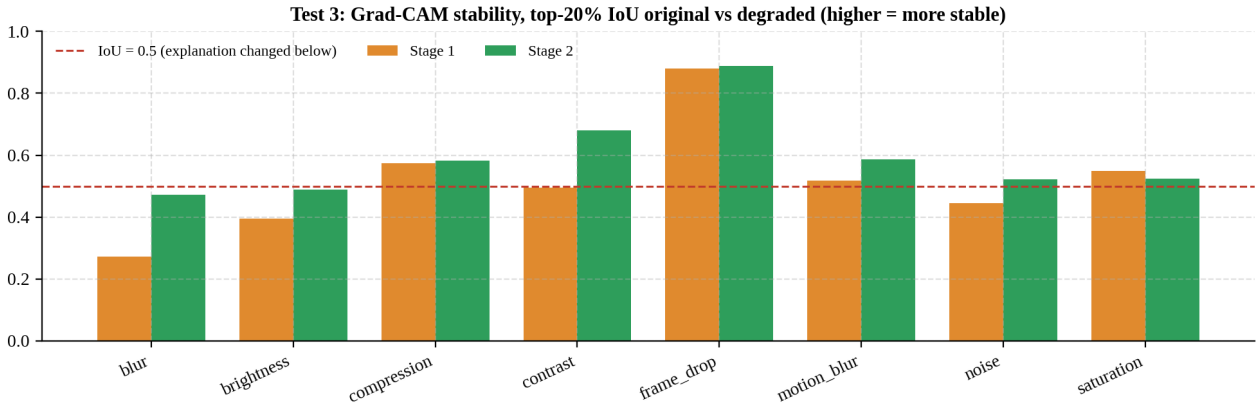

Figure 5.24: Test 3 Grad-CAM top-20% IoU per degradation type (red line: IoU 0.5).

5.4.13 Key findings

- **Face cropping and balancing fixed most of the FF++ failure.** FF++ AUC rose from 0.584 (Test 2) to 0.804, and FF++ real videos recognised rose from 1 of 200 to 80 of 200, on the same test videos. Combined AUC rose from 0.870 to 0.936 with BioDeepAV unchanged (0.996).
- **Degraded training again restores robustness.** Stage 2 cut the Combined AUC drop from 0.083 to 0.024 and the flip rate from 13.5% to 6.2%; on BioDeepAV the flip rate fell from 10.7% to 2.2%.
- **Without degraded training, noise breaks the detector:** Stage 1 AUC was 0.557 at medium and 0.471 at severe noise (below chance); severe blur gave a 52.7% flip rate. Stage 2 lifted every degradation × severity to AUC 0.86–0.94.

- **Explanations are more stable but still not reliable.** Stage 2 IoU 0.605 and r 0.788 (Test 2: 0.567, 0.741); explanation-changed-while-stable fell from 43.2% to 35.8% but remains over one in three. Frame drop barely changes maps (IoU 0.89); blur, brightness, noise and saturation stay near or below IoU 0.5.
- **Detection does not transfer between datasets.** BioDeepAV-only → FF++ AUC 0.457 and FF++-only → BioDeepAV AUC 0.165 (inverted), so each dataset's model learns cues specific to that dataset.
- **The validation threshold helps overall but is a compromise.** 0.31 gives the best Combined balanced accuracy (0.856 vs 0.826 at 0.5), but FF++ would prefer about 0.5 (0.743) and its specificity at 0.31 is only 40%.

5.4.14 Limitations and things to check

- **BioDeepAV resolution confound:** real test videos are 168×168 and fakes 512×512 (AUC of resolution alone 0.994), and crop sharpness also separates them (0.792). BioDeepAV's AUC of 0.996 may partly reflect image quality, not manipulation. Resizing all BioDeepAV videos to a common resolution before cropping would test this.
- **Audio is a potential shortcut:** only BioDeepAV has audio, p(fake) follows audio presence within each class, and masking audio flips 12.1% of BioDeepAV decisions.
- **FF++ remains weak:** AUC 0.80 and 60% of real FF++ videos still called fake at 0.31; cross-dataset results show no transferable cue yet.
- **Stage 2 epoch 4 collapsed** (val AUC 0.600) before recovering; the macro-AUC selection was unaffected, but training is not fully stable.
- **The 96×96 baseline comparison was skipped** in the script (Test 2 checkpoints not found in the run folder); the Test 2 vs Test 3 comparison in Section 5.11 uses the logged numbers on the same split. It changes face cropping, resolution, sampling and threshold at once, so gains cannot be attributed to one change.
- **Audio is still never degraded**, and the threshold was chosen on the combined validation set.

5.4.15 Suggested thesis sentence

> "With face-cropped 128×128 inputs, dataset-balanced training and a validation-selected threshold, the degradation-trained detector reached ROC-AUC 0.936 on 2,141 held-out videos and 0.913 on their degraded versions (BioDeepAV 0.996/0.992, FaceForensics++ 0.804/0.747), changing its decision for 6.2% of videos after degradation. Even so, for 35.8% of the videos whose prediction stayed the same, the most salient Grad-CAM regions of the original and the pixel-aligned degraded version overlapped by less than 50%."

5.4.16 Output files and frozen model

File	Contents
results/final/final_model.pt, FROZEN.json	Frozen Stage 2 model, threshold 0.31, 128×128; SHA-256 ab09929346da…f24de; frozen 2026-10-02 07:19
results/stage{1,2}/checkpoints, threshold/threshold.json	Best checkpoints with thresholds; validation threshold curves
metrics_long.csv	All metrics: model × split × dataset × condition, validation and fixed-0.5 thresholds
degradation_metrics.csv, prediction_stability.csv, robustness/*	Degradation × severity, flips per degradation, robustness summary
threshold_analysis.csv	Metrics at thresholds 0.30–0.70 (analysis only)
cross_dataset_metrics.csv	Cross-dataset results
diagnostics/*	Shortcut features, prediction by group, audio-masking test
xai/gradcam_stability.csv, gradcam_stability_summary.json, gradcam_examples.png	Grad-CAM stability per video and summary
logs/face_detection_*.csv, sampling_distribution_*.csv, history_*.csv	Face detection, sampling and training logs
final_experiment_summary.json, experiment_config.json, split_1-10-26.json	Full summary, configuration and split

Table 5.42: Test 3 output files (F:\Code\run_1-10-26\results).

5.5 Test 4 — Shortcut-Resistant, Explanation-Consistent Detector (Variant 4D)

5.5.1 Overview

Test 4 attacks the two problems Test 3 left open: shortcuts (resolution and audio presence) and unstable explanations. It normalises the effective face resolution, trains Stage 2 on explicit original/degraded pairs with a feature-consistency and a Grad-CAM-consistency loss, uses a mild-to-severe degradation curriculum, and trains with clean, degraded or missing audio. It uses only the degraded video files on disk (no synthetic degradations) and, unlike Test 3, trains **one model per dataset** — BioDeepAV, then FF++ — and tests each on both datasets. The script defines four variants (4A–4D); this run used the proposed full variant **4D**. Script: *6-10-26.py* (run 6-10-26, 2 runs, total 6:33:39).

Key result. The BioDeepAV-trained model kept AUC 0.992 / 0.988 on original / degraded BioDeepAV videos with a 2.2% decision flip rate, and its Grad-CAM explanations were the most stable so far: top-20% IoU 0.674, Pearson r 0.855, and only 19.6% of unchanged predictions had a changed explanation (Test 3: 35.8%). The FF++-trained model reached 0.747 / 0.680 on FF++ (IoU 0.794, 11.1%). Cross-dataset detection still failed (AUC 0.527 and 0.500), and the FF++-trained model called every BioDeepAV video fake because it had never seen audio.

5.5.2 Code pipeline

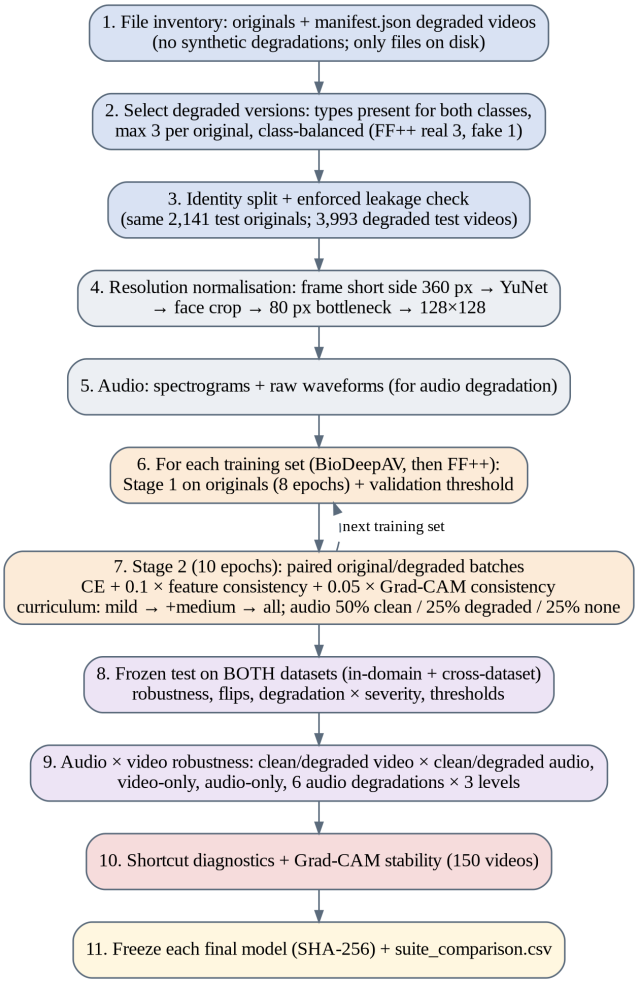

Figure 5.25: What 6-10-26.py does. Modes include inventory, selftest, suite (4A–4D), single variant, stage1/stage2, evaluate, threshold, cross, lambda_search, predict.

Variant	Adds	Question it answers	Run here
4A	Test 3 + resolution normalisation	How much of the score was the resolution shortcut?	No
4B	4A + paired training + feature consistency	Does consistency improve degradation robustness?	No
4C	4B + Grad-CAM consistency loss	Does it improve explanation robustness?	No
4D	4C + mixed audio policy + audio degradation + curriculum	Proposed model	**Yes** (both datasets)

Table 5.43: Test 4 variants defined in the script.

5.5.3 Setup

Item	Test 3	Test 4 (4D)
Training data	BioDeepAV + FF++ together	One model per dataset: BioDeepAV-trained, FF++-trained; both tested on both datasets
Input	YuNet face crop → 128×128	Frame short side resized to 360 px before detection; face crop passed through an 80×80 bottleneck, then 128×128
Stage 2	Mixed original + degraded items, cross-entropy	Paired batches (16 original + 16 matching degraded, same frames and augmentation): CE(orig) + CE(deg) + 0.1 × feature consistency (L2-normalised 256-d video embedding) + 0.05 × (1 − soft-IoU of differentiable Grad-CAM); clean branch is the anchor
Curriculum	None	Stage 2 epochs 1–3 clean + mild, 4–6 + medium, 7–10 all severities
Audio	20% dropout	Per sample: 50% clean, 25% degraded, 25% none
Epochs	8 + 8	8 + 10
Unchanged	—	Network, 16 frames / 8-frame clips, batch 32, AdamW + OneCycle (1e-3 / 3e-4), seed 42, balanced sampling, validation threshold, macro-AUC selection
Time	Stage 1 47:06, Stage 2 3:06:27	BioDeepAV: 16:33 + 1:19:59; FF++: 30:58 + 1:26:57
Caches	8.42 + 11.10 GB	8.42 GB (originals) + 15.71 GB (19,980 degraded); waveforms for 3,702 videos with audio

Table 5.44: Test 4 configuration compared with Test 3.

5.5.4 Degradations

Video. No degradation is generated in code. Of 79,582 degraded files listed in the manifests (all present on disk), 19,980 were used under three rules: (1) only types available for both classes of a dataset (BioDeepAV fake has no fps_reduction or low_resolution, so 6,072 BioDeepAV real files of those types were dropped); (2) at most 3 versions per original; (3) class balance — FF++ real 3 versions, FF++ fake 1. BioDeepAV uses 8 types and FF++ 10 types, each at mild, medium and severe.

Dataset / class	Mild	Medium	Severe	Total used
BioDeepAV fake	1,706	1,657	1,716	5,079
BioDeepAV real	2,031	2,029	1,970	6,030
FF++ fake	1,892	1,988	2,120	6,000
FF++ real	888	968	1,015	2,871

Table 5.45: Test 4 degraded videos used, by severity.

Audio. New in Test 4: six audio degradations, used in training (25% of samples) and in a dedicated audio × video test.

Audio degradation	Mild	Medium	Severe
Noise (SNR)	20 dB	10 dB	0 dB
Volume (gain + int16 quantisation)	−10 dB	−20 dB	−30 dB
Low-pass cut-off	4 kHz	2 kHz	1 kHz
Resample (down and back)	11,025 Hz	8,000 Hz	4,000 Hz
Dropout (100 ms segments zeroed)	10%	25%	50%
Silence	—	whole clip	—

Table 5.46: Test 4 audio degradation settings.

5.5.5 Data

Folder	Label	Originals	Degraded listed	Degraded used	Train orig. / deg.	Val orig. / deg.	Test orig. / deg.
BioDeepAV/real	REAL	2,010	54,312	6,030	1,407 / 4,221	201 / 603	402 / 1,206
BioDeepAV/fake	FAKE	1,693	5,079	5,079	1,185 / 3,555	169 / 507	339 / 1,017
FF++/original	REAL	1,000	2,871	2,871	700 / 2,014	100 / 287	200 / 570
FF++/Deepfakes	FAKE	1,000	2,864	1,000	700 / 700	100 / 100	200 / 200
FF++/Face2Face	FAKE	1,000	2,864	1,000	700 / 700	100 / 100	200 / 200
FF++/FaceShifter	FAKE	1,000	2,864	1,000	700 / 700	100 / 100	200 / 200
FF++/FaceSwap	FAKE	1,000	2,864	1,000	700 / 700	100 / 100	200 / 200
FF++/NeuralTextures	FAKE	1,000	2,864	1,000	700 / 700	100 / 100	200 / 200
FF++/DeepFakeDetection	FAKE	1,000	3,000	1,000	700 / 700	100 / 100	200 / 200
Total		**10,703**	**79,582**	**19,980**	7,492 / 13,990	1,070 / 1,997	**2,141 / 3,993**

Table 5.47: Test 4 file inventory and split (identity-grouped; degraded videos follow their original). Leakage check: 2,523 groups, 0 crossings.

Each model trains and selects only on its own dataset's rows: BioDeepAV-trained 2,592 train / 370 validation originals (7,776 / 1,110 degraded); FF++-trained 4,900 / 700 (6,214 / 887). Both are tested on all 2,141 test originals and 3,993 degraded test videos. Face detection found a face in 99.13% of 171,248 frames (BioDeepAV real 95.44%, 53 videos fell back to full frames; all other subsets ≥ 99.97%).

5.5.6 Training and threshold selection

Run	Epoch	Train loss	Val loss	Val AUC	Val BAcc@0.5	Val deg. AUC	Feat. loss	CAM soft-IoU	Best
BioDeepAV	S1-1	0.4672	0.5042	0.8216	0.7908	—	—	—	Yes
	S1-2	0.2595	0.1740	0.9896	0.9156	—	—	—	Yes
	S1-3	0.2122	0.1840	0.9719	0.9452	—	—	—	
	S1-4	0.1475	0.2323	0.9589	0.9274	—	—	—	
	S1-5	0.1055	0.1183	0.9948	0.9531	—	—	—	Yes
	S1-6	0.0954	0.0527	0.9984	0.9802	—	—	—	Yes
	S1-7	0.0818	0.0456	0.9982	0.9861	—	—	—	
	S1-8	0.0745	0.0543	0.9974	0.9861	—	—	—	
	S2-1	0.1766	0.0687	0.9960	0.9861	0.9599	0.182	0.651	Yes
	S2-2	0.1631	0.1076	0.9907	0.9605	0.9793	0.145	0.677	Yes
	S2-3	0.1531	0.0865	0.9944	0.9866	0.9531	0.126	0.687	
	S2-4	0.1454	0.0479	0.9994	0.9921	0.9870	0.139	0.686	Yes
	S2-5	0.1200	0.0401	0.9977	0.9921	0.9923	0.100	0.705	Yes
	S2-6	0.1096	0.0275	0.9991	0.9950	0.9947	0.088	0.718	Yes
	S2-7	0.1201	0.0607	0.9976	0.9841	0.9970	0.127	0.686	Yes
	S2-8	0.0927	0.0841	0.9990	0.9776	0.9975	0.100	0.698	Yes
	S2-9	0.0772	0.0392	0.9994	0.9950	0.9989	0.101	0.705	Yes
	S2-10	0.0866	0.0480	0.9992	0.9921	0.9986	0.090	0.707	
FF++	S1-1	0.6856	0.6026	0.5973	0.5667	—	—	—	Yes
	S1-2	0.6798	0.7559	0.6336	0.5858	—	—	—	Yes
	S1-3	0.6682	0.7096	0.6773	0.6117	—	—	—	Yes
	S1-4	0.6409	0.7947	0.7029	0.6183	—	—	—	Yes
	S1-5	0.6333	0.5893	0.7133	0.6600	—	—	—	Yes
	S1-6	0.6218	0.5634	0.7183	0.6542	—	—	—	Yes
	S1-7	0.6070	0.6180	0.7296	0.6683	—	—	—	Yes
	S1-8	0.6044	0.6061	0.7245	0.6542	—	—	—	
	S2-1	0.6725	0.6745	0.7267	0.6425	0.6862	0.298	0.663	Yes
	S2-2	0.6503	0.6504	0.7186	0.6433	0.6797	0.222	0.709	
	S2-3	0.6473	0.6848	0.7229	0.6433	0.6803	0.183	0.732	
	S2-4	0.6584	0.7000	0.7148	0.6267	0.6926	0.145	0.754	
	S2-5	0.6450	0.6308	0.7239	0.6708	0.6783	0.121	0.755	
	S2-6	0.6434	0.6157	0.7242	0.6625	0.6938	0.106	0.761	Yes
	S2-7	0.6340	0.6769	0.7246	0.6600	0.6882	0.106	0.755	
	S2-8	0.6274	0.5934	0.7387	0.6767	0.6866	0.099	0.768	Yes
	S2-9	0.6247	0.5836	0.7375	0.6842	0.6900	0.086	0.772	Yes
	S2-10	0.6098	0.6123	0.7379	0.6725	0.6844	0.084	0.777	

Table 5.48: Test 4 training (validation on the training dataset only). Best epochs: BioDeepAV S1-6 / S2-9; FF++ S1-7 / S2-9.

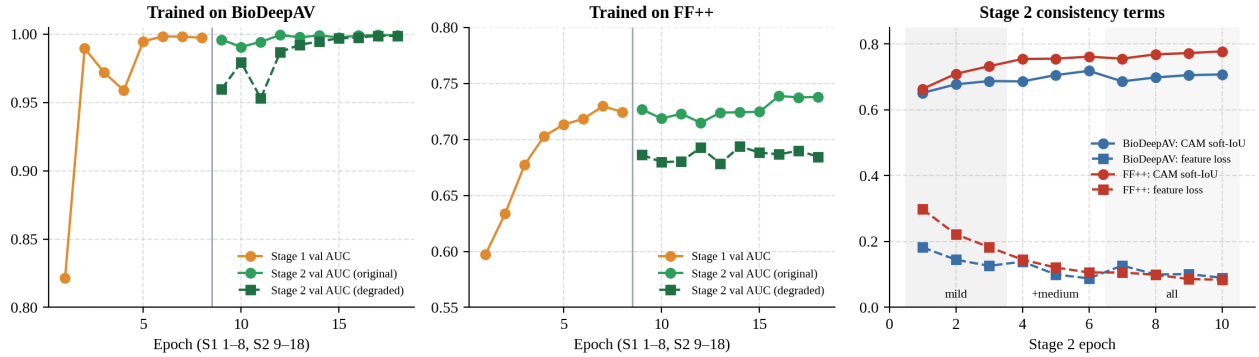

Figure 5.26: Test 4 validation AUC for both runs, and the Stage 2 consistency terms under the curriculum.

Run	Stage	Threshold	Val balanced accuracy	Val F1
BioDeepAV-trained	Stage 1	0.12	0.9950	0.9941
	Stage 2	0.50	0.9882	0.9868
FF++-trained	Stage 1	0.50	0.6683	0.6987
	Stage 2	0.50	0.6434	0.7596

Table 5.49: Test 4 validation-selected thresholds.

5.5.7 Results: original vs degraded

Run	St.	Test data	Cond.	Acc	BAcc	Prec.	Recall	Spec.	F1	AUC	TN / FP / FN / TP
Bio-trained	S1	BioDeepAV (in-domain)	orig.	93.4%	0.9379	0.884	0.985	0.891	0.932	0.9905	358/44/5/334
			degr.	86.8%	0.8689	0.838	0.882	0.856	0.859	0.9080	344/58/40/299
		FF++ (cross)	orig.	62.9%	0.5042	0.859	0.678	0.330	0.758	0.5034	66/134/386/814
			degr.	53.0%	0.4883	0.852	0.547	0.430	0.666	0.4664	86/114/544/656
		Combined	orig.	73.4%	0.7251	0.866	0.746	0.704	0.801	0.8102	424/178/391/1148
			degr.	64.7%	0.6674	0.847	0.621	0.714	0.716	0.7309	430/172/584/955
	S2	BioDeepAV (in-domain)	orig.	96.2%	0.9638	0.938	0.982	0.945	0.960	0.9919	380/22/6/333
			degr.	96.2%	0.9626	0.951	0.968	0.958	0.959	0.9876	385/17/11/328
		FF++ (cross)	orig.	53.3%	0.5254	0.869	0.536	0.515	0.663	0.5273	103/97/557/643
			degr.	47.1%	0.4979	0.856	0.461	0.535	0.599	0.5041	107/93/647/553
		Combined	orig.	68.1%	0.7183	0.891	0.634	0.802	0.741	0.8170	483/119/563/976
			degr.	64.1%	0.6949	0.889	0.572	0.817	0.696	0.7935	492/110/658/881
FF++-trained	S1	BioDeepAV (cross)	orig.	45.7%	0.5000	0.457	1.000	0.000	0.628	0.5000	0/402/0/339
			degr.	45.7%	0.5000	0.457	1.000	0.000	0.628	0.5000	0/402/0/339
		FF++ (in-domain)	orig.	57.4%	0.6558	0.934	0.542	0.770	0.686	0.7212	154/46/550/650
			degr.	65.9%	0.6029	0.896	0.681	0.525	0.774	0.6594	105/95/383/817
		Combined	orig.	53.4%	0.4492	0.688	0.643	0.256	0.665	0.3335	154/448/550/989
			degr.	58.9%	0.4628	0.699	0.751	0.174	0.724	0.3175	105/497/383/1156
	S2	BioDeepAV (cross)	orig.	45.7%	0.5000	0.457	1.000	0.000	0.628	0.5000	0/402/0/339
			degr.	45.7%	0.5000	0.457	1.000	0.000	0.628	0.5000	0/402/0/339
		FF++ (in-domain)	orig.	65.6%	0.6887	0.936	0.642	0.735	0.762	0.7475	147/53/429/771
			degr.	67.0%	0.6367	0.909	0.683	0.590	0.780	0.6800	118/82/380/820
		Combined	orig.	58.7%	0.4827	0.709	0.721	0.244	0.715	0.3404	147/455/429/1110
			degr.	59.6%	0.4745	0.705	0.753	0.196	0.728	0.3229	118/484/380/1159

Table 5.50: Test 4 test metrics at the validation thresholds (BioDeepAV n = 741, FF++ n = 1,400, Combined n = 2,141).

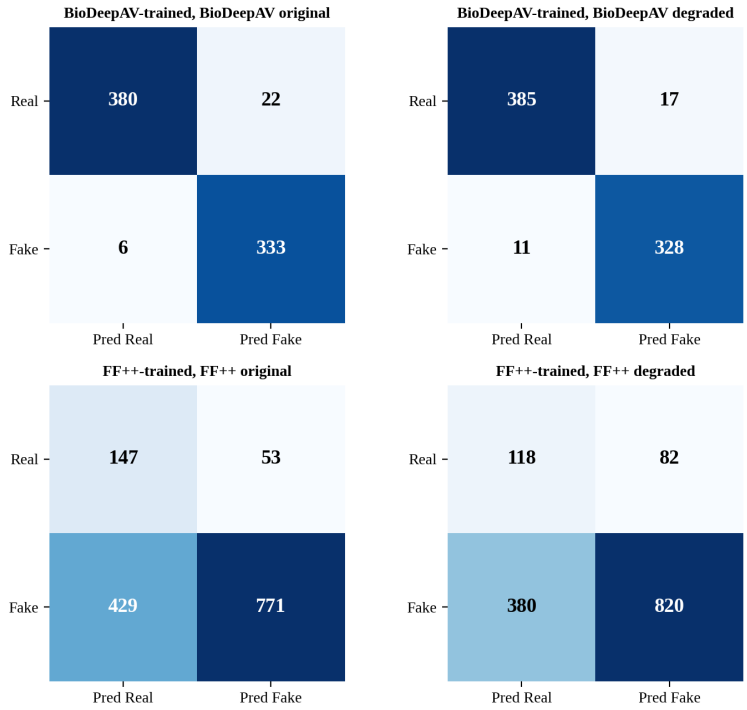

Figure 5.27: Test 4 Stage 2 confusion matrices on each model's own dataset.

Run	Model	Test data	AUC orig.	AUC deg.	AUC drop	BAcc drop	F1 drop	Flip rate
BioDeepAV-trained	Stage 1	BioDeepAV (in-domain)	0.9905	0.9080	0.0825	0.0690	0.0725	10.12%
		FF++ (cross)	0.5034	0.4664	0.0371	0.0158	0.0919	19.29%
	Stage 2	BioDeepAV (in-domain)	0.9919	0.9876	**0.0043**	0.0012	0.0006	**2.16%**
		FF++ (cross)	0.5273	0.5041	0.0232	0.0275	0.0638	16.00%
FF++-trained	Stage 1	FF++ (in-domain)	0.7212	0.6594	0.0618	0.0529	−0.0880	18.57%
		BioDeepAV (cross)	0.5000	0.5000	0.0000	0.0000	0.0000	0.00%
	Stage 2	FF++ (in-domain)	0.7475	0.6800	0.0675	0.0521	−0.0184	15.43%
		BioDeepAV (cross)	0.5000	0.5000	0.0000	0.0000	0.0000	0.00%

Table 5.51: Test 4 robustness: same videos, frozen thresholds.

5.5.8 Results by modality (audio × video)

Test 4 adds a full audio × video test on the 741 BioDeepAV test videos (the only ones with audio), using the frozen Stage 2 model and its threshold. Flip rate is measured against the clean-video + clean-audio decision.

Condition	AUC	Balanced accuracy	Mean p(fake)	Flip rate
Clean video + clean audio	0.9919	0.9638	0.478	0.00%
Degraded video + clean audio	0.9876	0.9626	0.472	2.16%
Clean video + degraded audio (random)	0.9901	0.9528	0.494	1.75%
Degraded video + degraded audio (random)	0.9851	0.9454	0.490	3.78%
Video only (audio masked)	0.9881	0.9524	0.487	2.56%
Audio only	0.7042	0.6224	0.562	38.60%

Table 5.52: Test 4 audio × video robustness, BioDeepAV-trained Stage 2 model.

Audio degradation (clean video)	AUC mild	AUC medium	AUC severe	Flip mild	Flip medium	Flip severe
noise	0.9919	0.9921	0.9899	0.14%	1.48%	2.16%
volume	0.9919	0.9915	0.9912	1.21%	1.89%	2.02%
lowpass	0.9916	0.9898	0.9892	1.48%	1.75%	2.16%
resample	0.9918	0.9915	0.9899	1.21%	1.62%	1.75%
dropout	0.9919	0.9915	0.9899	0.81%	2.56%	3.51%
silence	—	0.9880	—	—	2.43%	—

Table 5.53: Test 4 per audio degradation (BioDeepAV-trained Stage 2).

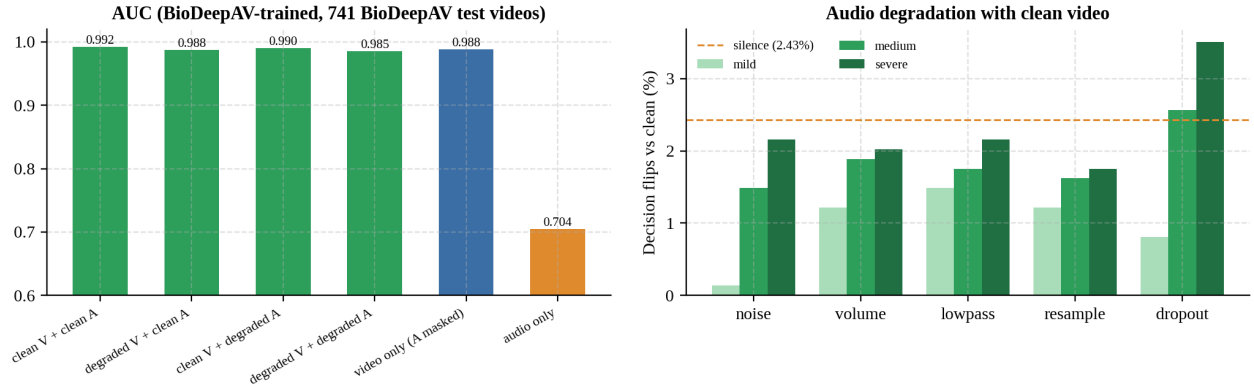

Figure 5.28: Test 4 audio × video robustness (BioDeepAV-trained Stage 2).

FF++-trained model. FF++ has no audio, so this model never saw non-zero audio features. On BioDeepAV every condition with audio gave mean p(fake) ≈ 1.000 (AUC 0.500, all 741 videos called fake); masking the audio dropped mean p(fake) to 0.530 and changed 37.4% of decisions, but AUC stayed at chance (0.490).

5.5.9 Results per degradation and severity

Degradation	Severity	n	AUC S1	AUC S2	BAcc S1	BAcc S2	Flip S1	Flip S2
blur	mild	92	0.998	1.000	0.980	0.980	4.3%	2.2%
	medium	86	0.993	0.999	0.887	0.977	15.1%	2.3%
	severe	101	0.985	0.997	0.663	0.980	35.6%	4.0%
brightness	mild	89	0.974	0.993	0.908	0.922	4.5%	5.6%
	medium	75	0.879	0.968	0.783	0.911	16.0%	6.7%
	severe	94	0.682	0.934	**0.558**	0.890	37.2%	10.6%
compression	mild	99	0.980	0.974	0.949	0.939	2.0%	1.0%
	medium	104	0.990	0.990	0.942	0.962	5.8%	3.8%
	severe	85	0.975	0.970	0.977	0.941	4.7%	2.4%
contrast	mild	91	0.984	0.995	0.947	0.988	1.1%	0.0%
	medium	103	0.959	0.999	0.884	0.981	8.7%	2.9%
	severe	84	0.914	0.995	0.775	0.980	25.0%	4.8%
frame_drop	mild	113	0.993	0.987	0.932	0.966	0.9%	0.0%
	medium	104	0.977	0.980	0.920	0.946	1.9%	0.0%
	severe	94	1.000	0.999	0.949	0.979	0.0%	2.1%
motion_blur	mild	79	0.986	0.993	0.964	0.976	5.1%	0.0%
	medium	89	0.998	0.998	0.939	0.965	0.0%	2.2%
	severe	92	0.999	1.000	0.970	0.970	1.1%	2.2%
noise	mild	105	0.990	0.999	0.911	0.972	3.8%	0.0%
	medium	85	**0.498**	0.990	**0.500**	0.965	43.5%	1.2%
	severe	87	**0.282**	0.996	**0.500**	0.989	46.0%	0.0%
saturation	mild	100	0.991	0.996	0.954	0.979	2.0%	1.0%
	medium	86	0.935	0.995	0.849	0.965	16.3%	5.8%
	severe	86	0.879	0.999	0.789	0.989	23.3%	2.3%

Table 5.54: Test 4 BioDeepAV-trained model on BioDeepAV degraded test videos (in-domain). Red: below 0.6.

Degradation	Severity	n	AUC S1	AUC S2	BAcc S1	BAcc S2	Flip S1	Flip S2
blur	mild	50	**0.528**	0.678	0.630	0.642	52.0%	40.0%
	medium	58	**0.477**	**0.485**	**0.509**	**0.431**	51.7%	46.6%
	severe	51	**0.548**	**0.593**	**0.500**	**0.464**	54.9%	31.4%
brightness	mild	76	0.670	0.719	0.652	0.637	11.8%	15.8%
	medium	72	**0.546**	0.738	**0.592**	0.692	43.1%	29.2%
	severe	84	**0.581**	**0.579**	**0.500**	**0.576**	47.6%	31.0%
compression	mild	49	0.709	0.774	0.626	0.720	6.1%	4.1%
	medium	83	0.721	0.701	0.687	0.730	6.0%	4.8%
	severe	75	0.731	0.795	0.614	0.748	21.3%	22.7%
contrast	mild	45	0.750	0.718	0.681	0.601	13.3%	8.9%
	medium	43	**0.576**	0.756	**0.559**	**0.599**	34.9%	20.9%
	severe	63	**0.525**	0.701	**0.543**	0.655	57.1%	30.2%
frame_drop	mild	72	0.724	0.759	0.656	0.731	0.0%	1.4%
	medium	100	0.831	0.871	0.729	0.801	1.0%	2.0%
	severe	65	0.653	0.691	0.675	0.720	0.0%	1.5%
motion_blur	mild	50	0.681	0.764	0.633	0.639	12.0%	22.0%
	medium	54	0.794	0.825	0.680	0.702	20.4%	7.4%
	severe	48	0.827	0.789	0.682	**0.589**	35.4%	29.2%
noise	mild	56	0.781	0.739	0.710	0.681	1.8%	3.6%
	medium	54	0.636	0.662	**0.569**	0.653	3.7%	3.7%
	severe	64	**0.573**	**0.549**	**0.562**	**0.575**	7.8%	17.2%
saturation	mild	64	0.679	0.748	**0.589**	0.637	10.9%	7.8%
	medium	77	0.656	0.723	0.601	0.660	11.7%	18.2%
	severe	59	0.749	0.748	0.633	0.679	33.9%	27.1%
fps_reduction	mild	56	0.675	**0.533**	0.610	**0.529**	0.0%	1.8%
	medium	64	0.617	**0.528**	0.603	**0.477**	3.1%	4.7%
	severe	47	0.760	0.838	0.709	0.749	2.1%	10.6%
low_resolution	mild	6	0.625	0.750	0.625	**0.500**	0.0%	16.7%
	medium	25	**0.548**	**0.421**	**0.480**	**0.421**	8.0%	16.0%
	severe	60	0.692	0.706	0.632	0.666	18.3%	15.0%

Table 5.55: Test 4 FF++-trained model on FF++ degraded test videos (in-domain). low_resolution mild has only 6 videos.

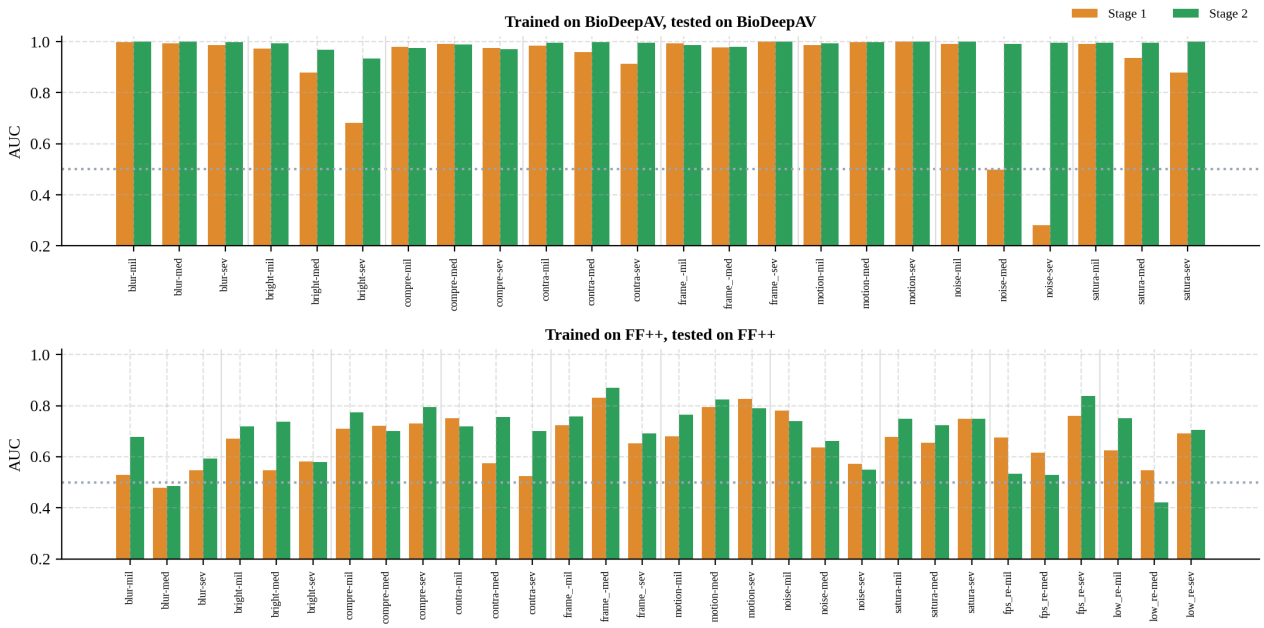

Figure 5.29: Test 4 in-domain AUC per degradation and severity, Stage 1 vs Stage 2.

Degradation	Bio-trained S1 on BioDeepAV	Bio-trained S2 on BioDeepAV	Bio-trained S2 on FF++ (cross)	FF++-trained S1 on FF++	FF++-trained S2 on FF++
blur	17.2%	4.3%	15.3%	46.6%	32.8%
brightness	14.3%	6.0%	32.8%	33.3%	25.3%
compression	1.9%	1.0%	19.9%	11.1%	10.5%
contrast	6.7%	0.0%	6.8%	37.3%	20.3%
frame_drop	0.9%	1.9%	8.6%	0.5%	1.5%
motion_blur	0.0%	2.6%	16.2%	17.9%	14.5%
noise	25.0%	0.0%	5.9%	4.4%	8.1%
saturation	15.6%	2.2%	16.7%	19.3%	20.0%
fps_reduction	—	—	8.6%	1.7%	6.0%
low_resolution	—	—	27.3%	16.7%	19.7%

Table 5.56: Test 4 prediction flip rate per degradation (one random degraded version per test video).

5.5.10 Cross-dataset results

Training data	Model	BioDeepAV orig. AUC	BioDeepAV deg. AUC	FF++ orig. AUC	FF++ deg. AUC
Test 3: BioDeepAV only	Video-only, Stage 1 protocol	0.9898	0.9098	0.4574	0.4640
Test 4: BioDeepAV	4D Stage 1	0.9905	0.9080	0.5034	0.4664
Test 4: BioDeepAV	4D Stage 2	**0.9919**	**0.9876**	0.5273	0.5041
Test 3: FF++ only	Video-only, Stage 1 protocol	0.1654	0.3015	0.7353	0.6671
Test 4: FF++	4D Stage 1	0.5000	0.5000	0.7212	0.6594
Test 4: FF++	4D Stage 2	0.5000	0.5000	**0.7475**	**0.6800**
Test 3: both datasets	Audio-visual Stage 2	0.9958	0.9915	0.8042	0.7468

Table 5.57: Test 4 in-domain vs cross-dataset AUC, with Test 3 for reference (red: chance level or below). Test 4 degraded test versions differ from Test 3.

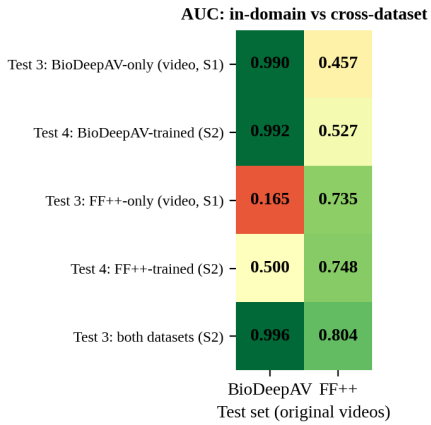

Figure 5.30: In-domain vs cross-dataset AUC (original test videos).

5.5.11 Shortcut diagnostics

Subset	Label	BioDeepAV-trained: predicted fake	FF++-trained: predicted fake	Size (px)	Sharpness T3 → T4
BioDeepAV/fake	FAKE	98.2%	100.0%	512 × 512	592.1 → 180.6
BioDeepAV/real	REAL	5.5%	**100.0%**	168 × 168	252.5 → 133.2
FF++/DeepFakeDetection	FAKE	55.0%	99.5%	1920 × 1080	311.4 → 116.8
FF++/Deepfakes	FAKE	51.5%	68.5%	854 × 480	443.9 → 155.0
FF++/Face2Face	FAKE	51.5%	59.5%	832 × 480	464.9 → 158.8
FF++/FaceShifter	FAKE	63.5%	69.5%	854 × 480	386.2 → 145.0
FF++/FaceSwap	FAKE	49.5%	48.0%	854 × 480	468.7 → 159.7
FF++/NeuralTextures	FAKE	50.5%	40.5%	832 × 480	400.7 → 150.2
FF++/original	REAL	48.5%	26.5%	854 × 480	487.9 → 167.0

Table 5.58: Test 4 prediction rate by subset (Stage 2, original test videos) and face-crop sharpness before (Test 3) and after resolution normalisation.

Run	Scope	Feature	AUC feature vs label	ρ within real	ρ within fake
BioDeepAV-trained	BioDeepAV	resolution_px	0.994	0.420	0.015
	BioDeepAV	face_px_source	0.995	0.489	0.039
	Combined	has_audio	0.276	−0.609	0.657
	FF++	face_px_source	0.522	0.515	0.463
FF++-trained	Combined	has_audio	0.276	0.974	0.722
	Combined	resolution_px	0.827	−0.790	−0.159
	FF++	fps	0.419	−0.020	−0.453

Table 5.59: Test 4 selected flagged features (ρ = Spearman correlation of the feature with p(fake), label held fixed).

Resolution normalisation narrowed the BioDeepAV sharpness gap (fake/real ratio 2.34 in Test 3 → 1.36), but BioDeepAV resolution still separates the labels (AUC 0.994), and within real videos p(fake) still rises with source resolution (ρ 0.42, Test 3: 0.21), so the resolution shortcut is reduced but not ruled out. Masking audio changed 2.6% of BioDeepAV decisions (Test 3: 12.1%), so the mixed audio policy cut audio dependence for the in-domain model.

5.5.12 Explainability: Grad-CAM stability

Grad-CAM stability was measured as in Test 3 on 150 test videos (both datasets, aligned original/degraded crops) for each frozen model. Note that the Stage 2 training loss itself rewards similar Grad-CAM maps (soft-IoU), so stability is partly an optimised property here; the evaluation metric (top-20% IoU) is different from the training term.

Run	Model	Pearson r	Top-20% IoU	Centre shift	Temporal CAM r	Prediction stable	Explanation changed while stable
BioDeepAV-trained	Stage 1	0.783	0.603	0.081	0.504	88.0%	31.1%
	Stage 2	0.855	0.674	0.045	0.518	92.0%	**19.6%**
FF++-trained	Stage 1	0.721	0.597	0.048	0.577	86.7%	31.5%
	Stage 2	0.932	0.794	0.028	0.717	90.0%	**11.1%**

Table 5.60: Test 4 Grad-CAM stability, original vs degraded version of the same video.

Degradation	Bio S1 IoU	Bio S2 IoU	Bio S2 r	FF++ S1 IoU	FF++ S2 IoU	FF++ S2 r
blur	0.477	0.650	0.854	0.399	0.742	0.913
brightness	0.332	0.441	0.629	0.255	0.627	0.820
compression	0.614	0.662	0.887	0.622	0.843	0.974
contrast	0.421	0.548	0.745	0.401	0.689	0.911
fps_reduction	0.890	0.835	0.981	0.790	0.939	0.990
frame_drop	0.840	0.854	0.981	0.875	0.948	0.998
low_resolution	0.660	0.718	0.930	0.779	0.892	0.992
motion_blur	0.791	0.862	0.982	0.772	0.918	0.994
noise	0.766	0.859	0.979	0.789	0.886	0.991
saturation	0.468	0.533	0.778	0.534	0.612	0.843
BioDeepAV videos	0.538	0.646	0.836	0.603	0.754	0.912
FF++ videos	0.638	0.689	0.866	0.595	0.816	0.942

Table 5.61: Test 4 Grad-CAM stability per degradation type and per dataset.

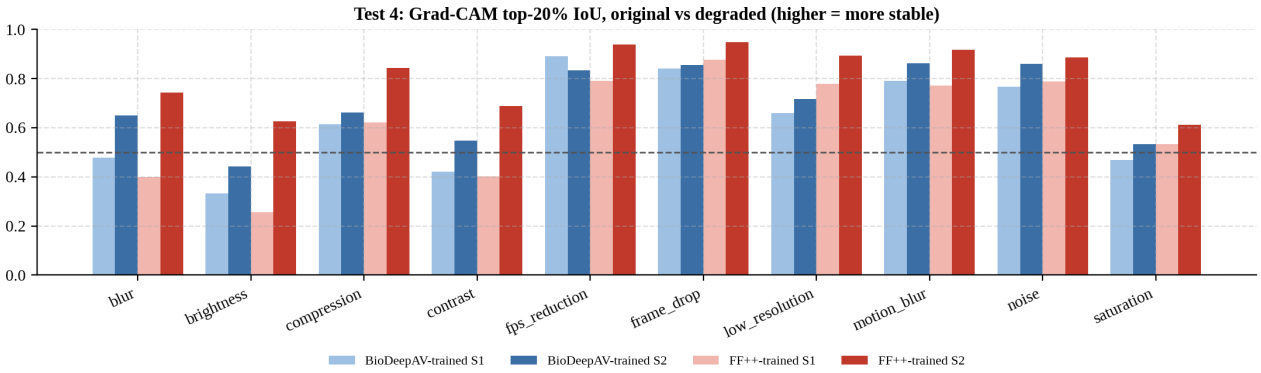

Figure 5.31: Test 4 Grad-CAM top-20% IoU per degradation type (dashed line: IoU 0.5).

5.5.13 Key findings

- **Explanation consistency training works.** With the Grad-CAM consistency loss, the share of unchanged predictions whose explanation changed fell to 19.6% (BioDeepAV-trained) and 11.1% (FF++-trained), from 35.8% in Test 3 and 43.2% in Test 2; IoU rose to 0.674 and 0.794. Brightness and saturation remain the least stable (IoU 0.44–0.63).
- **In-domain robustness is the best so far.** The BioDeepAV-trained Stage 2 model lost only 0.004 AUC under degradation (0.992 → 0.988) with a 2.2% flip rate, and every BioDeepAV degradation × severity kept AUC ≥ 0.934.
- **Without degraded training, noise again breaks detection:** BioDeepAV-trained Stage 1 AUC 0.498 (medium) and 0.282 (severe noise); Stage 2 restored them to 0.990 and 0.996.
- **Audio dependence fell in-domain:** masking audio flipped 2.6% of decisions (Test 3: 12.1%); audio degradations flipped at most 3.5%, and audio alone gave AUC 0.704.
- **FF++ remains hard:** in-domain AUC 0.747 / 0.680, about the same as Test 3's FF++-only model (0.735 / 0.667); several FF++ degradations stay near chance (e.g. blur medium 0.485).
- **No cross-dataset transfer:** BioDeepAV → FF++ AUC 0.527 and FF++ → BioDeepAV 0.500; the FF++-trained model labelled every BioDeepAV video fake because it reacted to audio it had never seen.

5.5.14 Limitations and things to check

- **Only variant 4D was run.** Without 4A–4C, the separate effects of resolution normalisation, feature consistency, Grad-CAM consistency and the audio policy cannot be isolated (run *--mode suite*).
- **Not directly comparable with Test 3's main model:** Test 4 trains one model per dataset, uses only the provided degraded files (different degraded test versions, 10 FF++ types) and 10 Stage 2 epochs.
- **The explanation metric is partly optimised.** Stage 2 trains towards similar Grad-CAM maps; report this alongside the stability gain, and check that explanations still point to plausible regions (e.g. with ExDDV-style annotations).
- **Audio presence is a decisive shortcut across datasets:** a model trained without audio fails completely on audio-bearing videos; the mixed audio policy cannot help when the training set has no audio.
- **BioDeepAV resolution confound is reduced, not removed** (labels still separable by source resolution; within-real ρ 0.42).
- **Small cells:** low_resolution mild has 6 FF++ test videos; FF++ fps_reduction and low_resolution exist only in FF++. The 96×96 baseline comparison was skipped (no Test 2 checkpoints in the run folder).

5.5.15 Suggested thesis sentence

> “Adding paired original/degraded training with feature- and Grad-CAM-consistency losses, a degradation curriculum and mixed audio kept the in-domain detector’s ROC-AUC at 0.992 on original and 0.988 on degraded BioDeepAV videos (2.2% decision flips), and reduced the share of unchanged predictions whose Grad-CAM explanation changed substantially from 35.8% to 19.6% (11.1% for the FF++-trained model). However, neither model transferred to the other dataset (AUC 0.527 and 0.500), showing that robustness to degradation does not imply generalisation across datasets.”

5.5.16 Output files and frozen models

File	Contents
4D_train-BioDeepAV/results/final/final_model.pt, FROZEN.json	BioDeepAV-trained Stage 2, threshold 0.50; SHA-256 bcd9b29bd416…68f6a; frozen 2026-10-08 02:59
4D_train-FFpp/results/final/final_model.pt, FROZEN.json	FF++-trained Stage 2, threshold 0.50; SHA-256 b94ffd5924bd…a22f; frozen 2026-10-08 05:08
suite_comparison.csv	One row per run: in-domain / cross metrics, flips, Grad-CAM, audio and shortcut summaries
metrics_long.csv, degradation_metrics.csv, prediction_stability.csv, threshold_analysis.csv	Per run: all metrics, degradation × severity, flips, threshold sensitivity
final_experiment_summary.json, experiment_config.json, file_inventory.json, split_6-10-26.json	Per run: full summary (incl. audio robustness, Grad-CAM), configuration, file inventory, split

Table 5.62: Test 4 output files (E:\Code\run_6-10-26).

5.6 Test 5 — Independent Replication with Hard-Degradation-Aware Training (Variants 5A–5D, 3 Seeds)

5.6.1 Overview

Test 5 asks whether training aimed at the degradation types whose explanations were least stable in Test 4 (brightness, saturation, contrast and blur — the **hard types**) improves detection robustness and Grad-CAM explanation stability. Unlike Tests 1–4, it was run as an **independent replication**: a new run folder, a fresh file inventory with full-file SHA-256, new duplicate-aware grouping, a new split (seed 20261009), its own caches and freshly initialised weights; no Test 4 code, cache or checkpoint was loaded (0 blocked reads). It trains four Stage 2 variants (5A–5D) that add one component at a time, one model per dataset (BioDeepAV, FF++) and with **three seeds** (42, 123, 2026): 6 Stage 1 and 24 Stage 2 runs, all completed. Variant differences are tested with a paired cluster bootstrap and count as supported only if the 95% CI excludes zero and all seeds agree in sign. Script: *9-10-26_test5_independent.py* (run 10–16 Oct 2026).

Key result. No variant reliably beat the independent baseline 5A on detection. 5A reached AUC 0.966 / 0.939 (original / degraded) on BioDeepAV and 0.770 / 0.705 on FF++ (mean of 3 seeds). Hard-degradation sampling (5B) gave the best BioDeepAV values (0.981 / 0.962), but the gain was not statistically supported, and on FF++ it lowered brightness AUC (supported). The Grad-CAM consistency loss (5C) was the only component with a supported benefit, and only on FF++: hard-type Grad-CAM IoU rose from 0.398 to 0.629 and unchanged predictions with a changed explanation fell from 23.4% to 9.5%. The full method (5D, closest to Test 4) was worse than 5A on BioDeepAV degraded AUC (−0.036) and worst-type AUC (−0.099; severe-noise AUC 0.665). Cross-dataset AUC stayed at chance (0.49–0.54 and 0.500).

5.6.2 Code pipeline

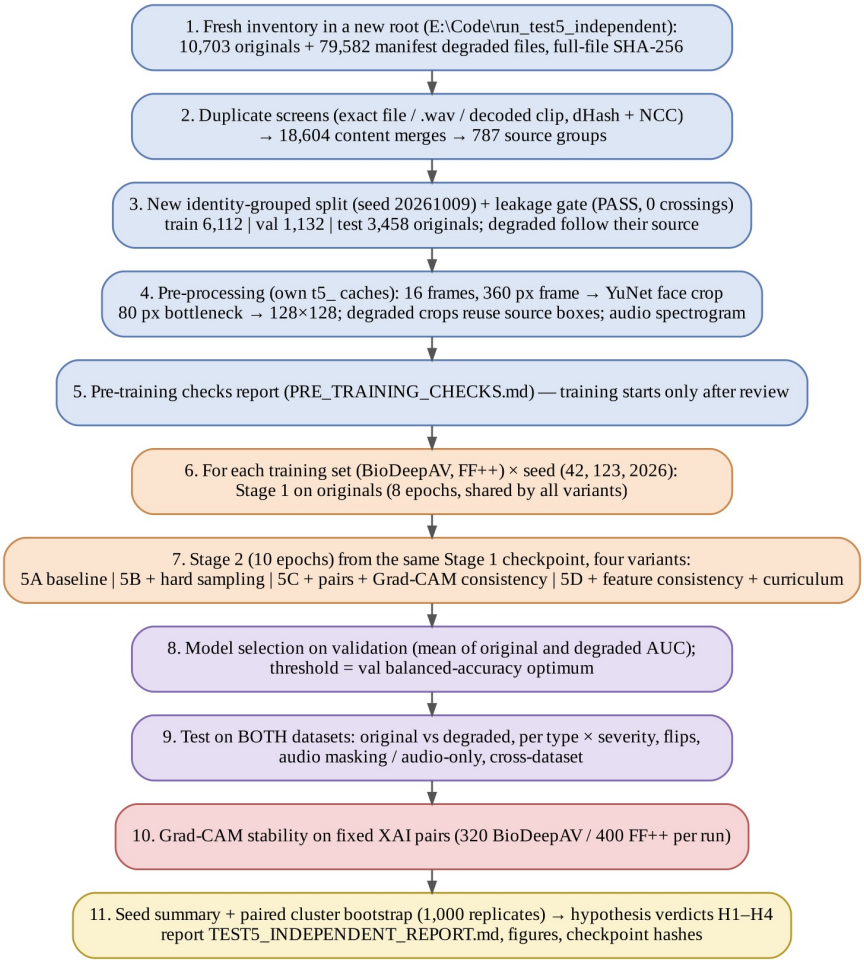

Figure 5.32: What 9-10-26_test5_independent.py does. Phases: inventory and audits, split and leakage gate, pre-processing, pre-training checks (training starts only after review), training, evaluation, Grad-CAM, statistics, report.

Variant	Stage 2 adds	Question it answers
S1	Stage 1 only: originals, 8 epochs (shared starting point of all variants)	Reference without degraded training
5A	Unpaired clips, 16 original + 16 degraded per step, natural degradation mixture, cross-entropy only	Independent baseline
5B	5A + hard-degradation sampling (half of the degraded draws from brightness 0.3, saturation 0.3, contrast 0.2, blur 0.2)	H1: does hard sampling improve detection on hard types? (5B − 5A)
5C	5B + original/degraded pairs + Grad-CAM consistency loss (λ_{cam} 0.05)	H2: does Grad-CAM consistency improve explanation stability? (5C − 5B)
5D	5C + feature consistency (λ_{feat} 0.1) + curriculum (mild → +medium → all)	H3: do they add beyond 5C? (5D − 5C) H4: full method vs baseline (5D − 5A)

Table 5.63: Test 5 variants (each run on both training sets with 3 seeds). Unavoidable confounds: 5C − 5B measures pairing and Grad-CAM consistency together; 5D − 5C measures feature consistency and the curriculum together.

5.6.3 Setup

Item	Test 4 (4D)	Test 5
Independence	Built on the Test 3 data handling	New root, fresh inventory (SHA-256), new grouping and split, own *t5_* caches with provenance, fresh weights; Test 4 never loaded
Split	Identity-grouped, seed 42	Identity + content-duplicate grouping (18,604 merges, 787 groups), seed 20261009; DFD grouped by actor
Training sets	One model per dataset	Same (BioDeepAV-trained, FF++-trained), each tested on both datasets
Variants / seeds	Only 4D; 1 seed	5A–5D (+ Stage 1 reference); 3 seeds (42, 123, 2026)
Input	360 px frame, 80 px face bottleneck, YuNet crop 128×128	Same; 16 cached frames, 8-frame clips (evaluation averages the 2 frame offsets)
Stage 2 batches	Paired original/degraded	Unpaired (5A, 5B) or paired (5C, 5D); 16 original + 16 degraded clips per step; hard sampling in 5B–5D
Losses	CE + 0.1 × feature + 0.05 × Grad-CAM consistency	CE (5A, 5B); + 0.05 × Grad-CAM (5C); + 0.1 × feature (5D); clean anchor with stop-gradient
Curriculum	Mild → +medium → all	Only 5D (epochs 1–3 mild, 4–6 +medium, 7–10 all)
Audio	50% clean / 25% degraded / 25% none	Same
Epochs, optimiser	8 + 10	8 + 10; AdamW (wd 1e-4) + OneCycle (lr 1e-3 / 3e-4), 32 clips per step
Model selection	Validation AUC	Mean of validation original and degraded AUC; threshold = validation balanced-accuracy optimum
Statistics	Single run	Seed mean ± SD; paired cluster bootstrap over source groups (1,000 replicates) + sign agreement across seeds
Hardware, time	BioDeepAV 1:19:59, FF++ 1:26:57	RTX 3050 (PyTorch 2.14, CUDA 12.6); Stage 2 ≈ 56–72 min (BioDeepAV) / 73–90 min (FF++) per run; 10–16 Oct 2026

Table 5.64: Test 5 configuration compared with Test 4.

5.6.4 Degradations

Only the degraded files produced with the tool (Section 5.0) and listed in the manifests were used; no degradation is generated in code. Of the 79,582 manifest files, 19,791 were selected (class-balanced, about three per original; BioDeepAV real: 6,000 of 54,312 available) and 62 were then excluded because their content link to the source could not be verified, leaving **19,729**. BioDeepAV has 8 types (no fps_reduction or low_resolution) and FF++ 10, each at mild, medium and severe.

Dataset / class	Mild	Medium	Severe	Total used
BioDeepAV fake	1,649	1,591	1,637	4,877
BioDeepAV real	2,009	2,019	1,972	6,000
FF++ fake	1,883	2,031	2,075	5,989
FF++ real	888	968	1,007	2,863
Total	**6,429**	**6,609**	**6,691**	**19,729**

Table 5.65: Test 5 degraded videos used, by severity (all splits). Test split: BioDeepAV 4,076 (1,242 real / 2,834 fake), FF++ 2,411 (574 / 1,837); hard types: 2,083 and 1,062 of them.

Hard types. Brightness, saturation, contrast and blur were chosen because they had the lowest Stage 2 Grad-CAM IoU in Test 4 (BioDeepAV-trained: 0.44, 0.53, 0.55 and 0.65; Table 5.61). In 5B–5D, half of the degraded draws (hard share 0.5) are a hard-type version from the same dataset × class cell, weighted brightness 0.3, saturation 0.3, contrast 0.2, blur 0.2; the other half follow the natural mixture (uniform over the source's verified versions). Audio degradation follows the mixed policy (50% clean, 25% degraded, 25% none).

5.6.5 Data

Dataset / subset	Videos	Train	Val	Test	Median size	Audio
BioDeepAV fake	1,693*	453	233	1,006	512 × 512	Yes
BioDeepAV real	2,010	1,396	200	414	168 × 168	Yes
FF++ DeepFakeDetection	1,000	63	99	838	1920 × 1080	No
FF++ Deepfakes, Face2Face, FaceShifter, FaceSwap, NeuralTextures (each)	1,000	700	100	200	832–854 × 480	No
FF++ original	1,000	700	100	200	854 × 480	No
Total	**10,703**	**6,112**	**1,132**	**3,458**		

*Table 5.66: Test 5 inventory and split of the original videos (identity- and content-grouped; degraded videos follow their source). *One BioDeepAV fake video was excluded (face-clip extraction failed); 10,702 used.*

Identity tokens (BioDeepAV real: YouTube id; FF++: source and target sequence ids; DFD: individual actors) and content duplicates (identical file, .wav, decoded face clip or audio, or strong near-duplicates of face crops and spectrograms) were merged with union-find before splitting, giving 787 groups. The groups are very uneven — one BioDeepAV group holds 910 videos and the 838 DFD test videos form one group — so the split differs from the 70/10/20 target: most BioDeepAV fakes and almost all DFD videos are in the test set, and the test sets are not class-balanced (BioDeepAV 414 real / 1,006 fake; FF++ 200 real / 1,838 fake).

Check	Result
Leakage gate (9 checks, incl. source isolation, original/degraded grouping, ≥ 20 real and fake per dataset in val and test)	**PASS**
Groups / identity tokens in more than one split	0 of 787 / 0 of 2,793
Degraded videos with split or group mismatch	0 of 19,729
Exact file / decoded-clip duplicate sets across splits	0 / 0 (within split: 344 / 109)
Near-duplicate candidates across splits (below the strong criterion; documented, not merged)	2,474
Reads of Test 4 or older Test 5 artifacts; cache files without provenance	0; 0
Input properties that still separate BioDeepAV labels (pre-processing self-test)	Native face size AUC 0.993 (real 158 px, fake 510 px); width / height 0.990 / 0.992

Table 5.67: Test 5 leakage audit and data self-tests (PRE_TRAINING_CHECKS.md, leakage_audit_report.json).

5.6.6 Training and threshold selection

Stage 1 trains on clean originals for 8 epochs and is shared by all four variants of the same training set and seed; each Stage 2 variant then starts from that checkpoint for 10 epochs. The best epoch is chosen by the validation score (mean of validation original and degraded AUC of the training dataset), and the decision threshold is the validation balanced-accuracy optimum.

Training set	Variant	Best epoch (seed 42 / 123 / 2026)	Threshold	Validation score
BioDeepAV	S1	2 / 6 / 3	0.01 / 0.05 / 0.01	0.989 / 0.989 / 0.992
	5A	1 / 7 / 8	0.02 / 0.01 / 0.01	0.977 / 0.984 / 0.992
	5B	4 / 8 / 4	0.01 / 0.01 / 0.01	0.990 / 0.982 / 0.986
	5C	1 / 5 / 5	0.06 / 0.06 / 0.01	0.984 / 0.977 / 0.989
	5D	2 / 1 / 5	0.01 / 0.12 / 0.04	0.968 / 0.964 / 0.983
FF++	S1	8 / 8 / 8	0.45 / 0.51 / 0.45	0.689 / 0.569 / 0.676
	5A	6 / 10 / 9	0.67 / 0.75 / 0.67	0.696 / 0.699 / 0.692
	5B	8 / 10 / 6	0.70 / 0.71 / 0.66	0.695 / 0.640 / 0.678
	5C	10 / 9 / 7	0.69 / 0.77 / 0.68	0.687 / 0.632 / 0.681
	5D	4 / 10 / 7	0.68 / 0.72 / 0.74	0.672 / 0.568 / 0.673

Table 5.68: Test 5 model selection per run (S1 validation score = validation original AUC).

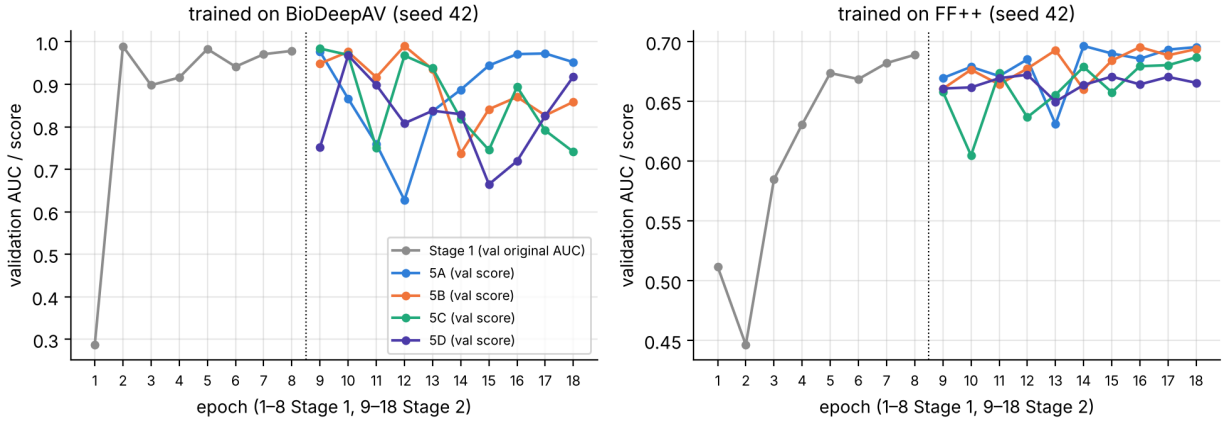

Figure 5.33: Test 5 validation curves, seed 42 (Stage 1: validation original AUC; Stage 2: validation score).

Two points matter for interpreting the results. First, the BioDeepAV thresholds are extremely low (0.01–0.12): the balanced-accuracy optimum lies where p(fake) is close to zero, so the BioDeepAV models are strongly over-confident and their accuracy figures depend heavily on the threshold (FF++: 0.66–0.77). Second, the BioDeepAV validation score swings by up to 0.35 between Stage 2 epochs (Figure 5.33) and several best epochs are 1 or 2, so which checkpoint is selected is partly a matter of chance — one reason why results vary between seeds.

5.6.7 Results: original vs degraded

Training / test	Measure	S1	5A	5B	5C	5D
BioDeepAV	AUC original	0.880 ± 0.069	0.966 ± 0.032	**0.981 ± 0.002**	0.952 ± 0.038	0.967 ± 0.008
	AUC degraded	0.795 ± 0.064	0.939 ± 0.038	**0.962 ± 0.007**	0.922 ± 0.047	0.903 ± 0.041
	AUC hard types	0.800 ± 0.072	0.919 ± 0.040	**0.947 ± 0.008**	0.911 ± 0.045	0.904 ± 0.026
	Worst-type AUC	0.558 ± 0.092	0.858 ± 0.053	**0.894 ± 0.022**	0.840 ± 0.051	0.759 ± 0.110
	Flip rate	14.25 ± 1.55%	**6.46 ± 0.72%**	6.58 ± 0.30%	8.70 ± 1.47%	11.12 ± 1.75%
FF++	AUC original	0.704 ± 0.054	**0.770 ± 0.014**	0.755 ± 0.020	0.744 ± 0.037	0.736 ± 0.046
	AUC degraded	0.631 ± 0.036	**0.705 ± 0.008**	0.688 ± 0.027	0.694 ± 0.027	0.704 ± 0.024
	AUC hard types	0.548 ± 0.037	0.647 ± 0.018	0.621 ± 0.039	0.625 ± 0.023	**0.682 ± 0.024**
	Worst-type AUC	0.441 ± 0.043	**0.570 ± 0.009**	0.565 ± 0.020	0.559 ± 0.041	0.533 ± 0.115
	Flip rate	13.37 ± 2.63%	15.10 ± 2.55%	16.13 ± 0.76%	15.66 ± 1.52%	**13.04 ± 3.75%**

Table 5.69: Test 5 in-domain results, mean ± SD over 3 seeds (BioDeepAV n = 1,420 original / 4,076 degraded; FF++ n = 2,038 / 2,411). Hard types = brightness, saturation, contrast, blur. Flip = decision change between a test video and its degraded version at the frozen threshold. Bold: best Stage 2 value (not necessarily a supported difference, see Table 5.78).

Training / test	Condition	Measure	S1	5A	5B	5C	5D
BioDeepAV	Original	Balanced acc.	77.1%	80.5%	88.5%	84.0%	87.2%
		Real recognised	61.0%	62.1%	79.0%	69.0%	75.0%
		Fake detected	93.3%	99.0%	98.0%	99.0%	99.3%
	Degraded	Balanced acc.	71.9%	78.9%	85.8%	80.9%	83.5%
		Real recognised	61.2%	61.6%	76.8%	67.3%	77.5%
		Fake detected	82.7%	96.3%	94.9%	94.5%	89.6%
FF++	Original	Balanced acc.	64.9%	70.1%	68.6%	68.7%	67.3%
		Real recognised	54.2%	65.7%	66.7%	79.0%	71.2%
		Fake detected	75.6%	74.5%	70.5%	58.4%	63.5%
	Degraded	Balanced acc.	59.0%	64.0%	63.1%	64.4%	63.7%
		Real recognised	38.3%	50.0%	50.8%	62.2%	56.3%
		Fake detected	79.7%	78.0%	75.5%	66.5%	71.2%

Table 5.70: Test 5 test metrics at each run's validation threshold (seed means).

Every Stage 2 variant clearly beats Stage 1 on both datasets, so degraded training again restores most of the robustness lost without it. Between the Stage 2 variants the differences are small compared with the spread across seeds. Real videos remain the main source of error: only 62–79% of BioDeepAV real and 50–79% of FF++ real test videos are recognised, while 90–99% of BioDeepAV fakes are caught.

Training	Variant	Original AUC (seed 42 / 123 / 2026)	Degraded AUC (seed 42 / 123 / 2026)	Range (original)
BioDeepAV	S1	0.818 / 0.954 / 0.868	0.749 / 0.867 / 0.768	0.136
	5A	0.929 / 0.981 / 0.987	0.896 / 0.950 / 0.970	0.059
	5B	0.983 / 0.982 / 0.979	0.968 / 0.963 / 0.954	0.005
	5C	0.909 / 0.981 / 0.966	0.868 / 0.957 / 0.940	0.072
	5D	0.966 / 0.960 / 0.976	0.880 / 0.878 / 0.950	0.016
FF++	S1	0.729 / 0.642 / 0.741	0.637 / 0.592 / 0.664	0.099
	5A	0.755 / 0.782 / 0.775	0.707 / 0.695 / 0.711	0.027
	5B	0.770 / 0.732 / 0.763	0.713 / 0.660 / 0.689	0.038
	5C	0.748 / 0.705 / 0.778	0.686 / 0.672 / 0.723	0.074
	5D	0.758 / 0.682 / 0.767	0.703 / 0.681 / 0.729	0.084

Table 5.71: Test 5 in-domain AUC per seed. Seed-to-seed differences (up to 0.14 for BioDeepAV Stage 1 and 0.07 for 5C) are as large as most differences between variants.

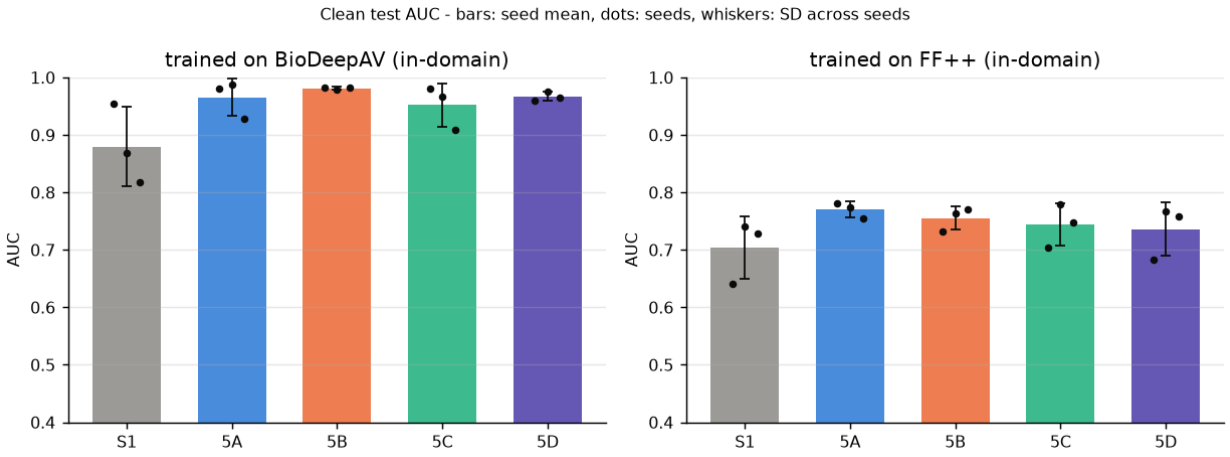

Figure 5.34: Test 5 in-domain AUC on original test videos (bars: seed mean, dots: seeds, whiskers: SD).

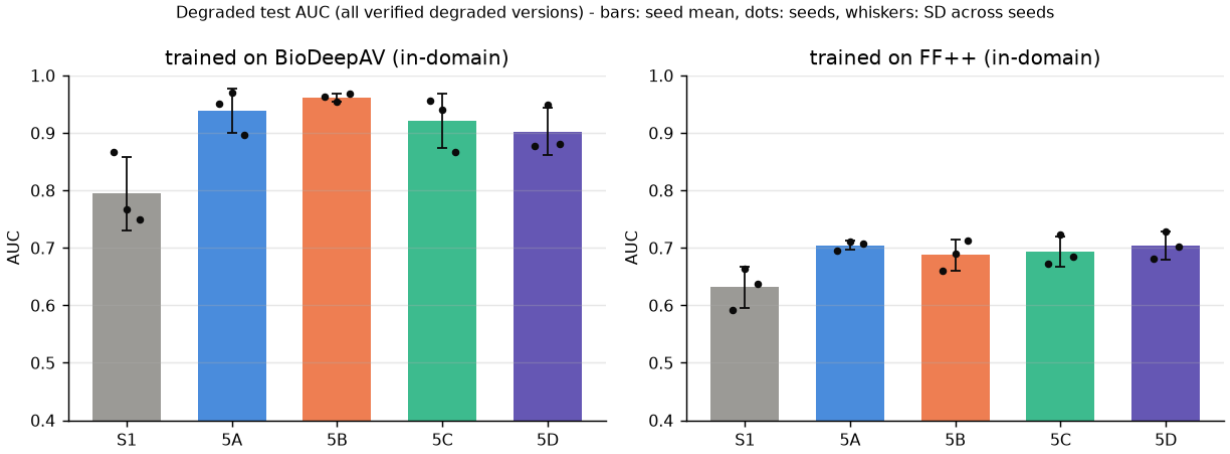

Figure 5.35: Test 5 in-domain AUC on degraded test videos (pooled over all types and severities).

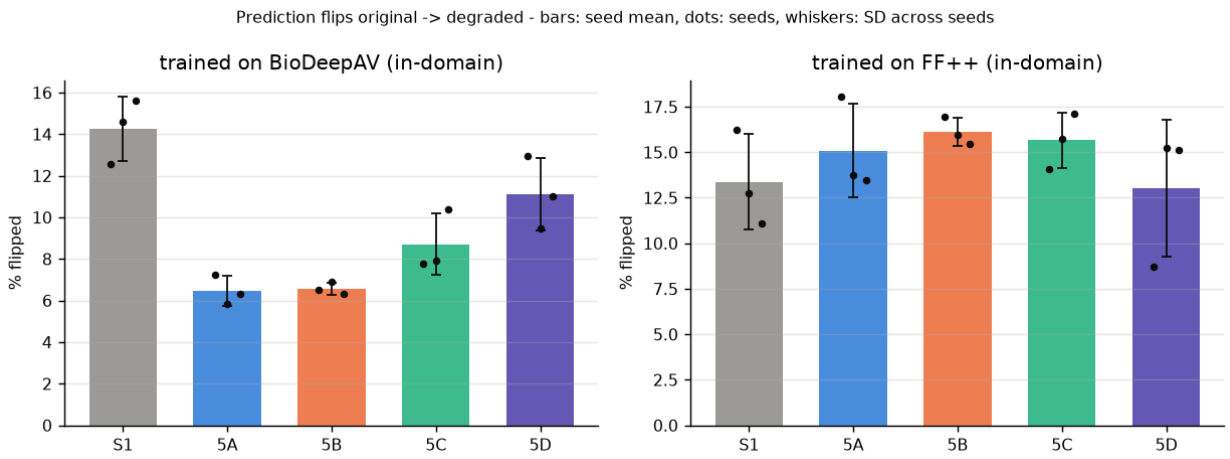

Figure 5.36: Test 5 prediction flip rate, original vs degraded version of the same test video.

5.6.8 Results by modality (audio)

As in Test 4, audio was tested by masking it (zero spectrogram) or keeping only audio, on the 1,420 BioDeepAV test videos (FF++ has no audio). Flips are measured against the normal audio-visual decision.

Model	Measure	S1	5A	5B	5C	5D
BioDeepAV-trained	Normal AUC	0.880	0.966	0.981	0.952	0.967
	Audio masked AUC	0.867	0.948	0.976	0.935	0.931
	Flips when audio masked	9.6%	5.5%	5.4%	6.8%	3.4%
	Audio-only AUC	0.540	0.588	0.661	0.595	0.595
	Flips audio-only	21.8%	15.2%	20.6%	18.7%	20.9%
FF++-trained (cross)	Normal: predicted fake	73.7%	100.0%	100.0%	100.0%	100.0%
	Audio masked AUC	0.501	0.550	0.490	0.479	0.415
	Flips when audio masked	36.0%	49.5%	50.2%	70.1%	58.3%

Table 5.72: Test 5 audio ablation on BioDeepAV test videos (seed means).

Audio dependence of the BioDeepAV-trained models is low: masking audio changes 3.4–6.8% of Stage 2 decisions (Test 4: 2.6%) and lowers AUC by at most 0.036, and audio alone gives only AUC 0.59–0.66. The FF++-trained models, which never saw audio, call every BioDeepAV video fake (AUC 0.500); masking the audio changes 50–70% of their decisions but does not make them useful (AUC 0.42–0.55). Audio presence therefore remains a decisive shortcut across datasets.

5.6.9 Results per degradation

Training	Degradation	S1	5A	5B	5C	5D
BioDeepAV	**blur**	0.898	0.956	0.971	0.948	0.951
	brightness	0.750	0.858	0.894	0.840	0.806
	compression	0.867	0.957	0.971	0.940	0.951
	contrast	0.842	0.952	0.980	0.935	0.945
	frame_drop	0.873	0.958	0.981	0.947	0.966
	motion_blur	0.864	0.952	0.976	0.945	0.955
	noise	0.558	0.961	0.972	0.927	0.793
	saturation	0.792	0.926	0.947	0.931	0.939
FF++	**blur**	0.485	0.642	0.588	0.615	0.686
	brightness	0.561	0.674	0.603	0.628	0.678
	compression	0.657	0.678	0.703	0.717	0.693
	contrast	0.498	0.570	0.602	0.597	0.662
	fps_reduction	0.626	0.670	0.633	0.627	0.609
	frame_drop	0.726	0.788	0.768	0.776	0.741
	low_resolution	0.583	0.665	0.672	0.680	0.578
	motion_blur	0.736	0.809	0.795	0.775	0.805
	noise	0.690	0.756	0.724	0.740	0.720
	saturation	0.680	0.755	0.726	0.717	0.740

Table 5.73: Test 5 in-domain AUC per degradation type (all severities, seed means). Bold: hard types. Red: below 0.6.

Training	Type	Severity	n	S1	5A	5B	5C	5D
BioDeepAV	brightness	mild	175	0.859	0.935	0.947	0.905	0.937
		medium	167	0.791	0.844	0.885	0.834	0.880
		severe	159	0.661	0.776	0.829	0.776	0.737
	saturation	mild	186	0.819	0.958	0.980	0.960	0.954
		medium	178	0.822	0.934	0.946	0.945	0.940
		severe	190	0.765	0.879	0.919	0.904	0.929
	contrast	mild	169	0.903	0.982	0.994	0.972	0.982
		medium	164	0.845	0.952	0.980	0.939	0.948
		severe	169	0.823	0.922	0.966	0.902	0.928
	blur	mild	164	0.949	0.980	0.994	0.981	0.994
		medium	172	0.907	0.959	0.964	0.939	0.954
		severe	190	0.921	0.968	0.975	0.957	0.975
	noise	mild	158	0.885	0.971	0.989	0.959	0.981
		medium	188	0.516	0.944	0.961	0.874	0.796
		severe	148	0.329	0.977	0.976	0.965	0.665
FF++	brightness	mild	74	0.589	0.720	0.704	0.688	0.705
		medium	88	0.614	0.739	0.655	0.714	0.714
		severe	90	0.447	0.527	0.453	0.448	0.574
	saturation	mild	89	0.641	0.656	0.625	0.633	0.652
		medium	98	0.752	0.871	0.851	0.829	0.812
		severe	94	0.682	0.762	0.727	0.718	0.764
	contrast	mild	94	0.482	0.587	0.586	0.581	0.654
		medium	91	0.502	0.548	0.581	0.607	0.656
		severe	94	0.500	0.547	0.649	0.614	0.657
	blur	mild	77	0.537	0.633	0.658	0.621	0.677
		medium	87	0.496	0.642	0.596	0.657	0.732
		severe	86	0.424	0.634	0.528	0.576	0.671
	noise	mild	81	0.736	0.861	0.826	0.828	0.794
		medium	74	0.620	0.685	0.638	0.651	0.648
		severe	91	0.711	0.743	0.704	0.746	0.712

Table 5.74: Test 5 in-domain AUC by severity for the hard types and noise (seed means; n = test videos per cell). Red: below 0.6.

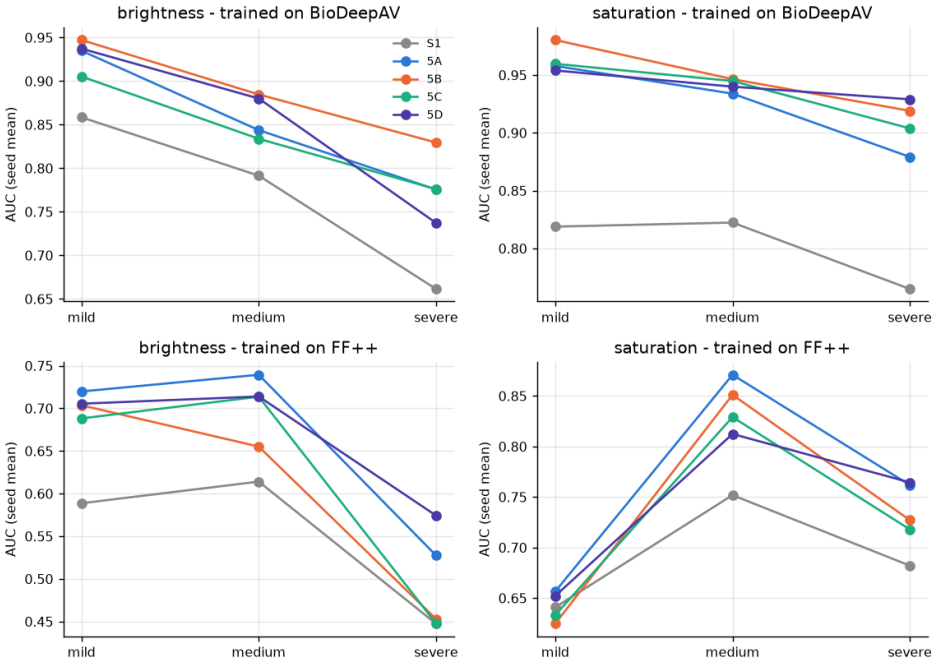

Figure 5.37: Test 5 brightness and saturation AUC by severity (seed means).

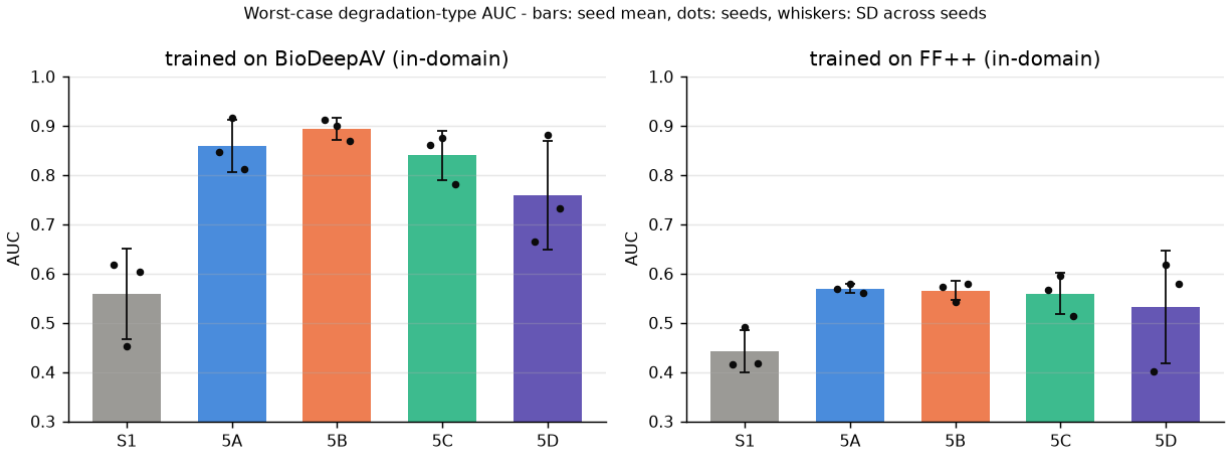

Figure 5.38: Test 5 worst-case degradation-type AUC per variant.

On BioDeepAV, all Stage 2 variants keep most types above 0.93, but brightness stays the hardest type for 5A–5C (0.84–0.89) and severe brightness their hardest cell (0.78–0.83). 5D introduced a new failure: severe-noise AUC fell to 0.665 (medium 0.796), from 0.977 in 5A — with half of the degraded draws spent on the four hard types and severe versions only allowed from epoch 7, the model saw less severe noise. Without degraded training (S1) noise again breaks detection (severe 0.329, below chance). On FF++, cells range from 0.45 to 0.87; contrast and blur stay near chance for 5A–5C (0.53–0.66). 5D raised contrast (0.662) and blur (0.686) but lost on low_resolution medium (0.448, only 19 videos) and fps_reduction mild (0.510).

5.6.10 Cross-dataset results

Train → test	Measure	S1	5A	5B	5C	5D
BioDeepAV → FF++	AUC original	0.477 ± 0.023	0.542 ± 0.027	0.514 ± 0.037	0.490 ± 0.048	0.498 ± 0.066
	AUC degraded	0.466 ± 0.013	0.517 ± 0.021	0.491 ± 0.033	0.475 ± 0.042	0.477 ± 0.045
	Real called fake	85.8%	81.5%	71.5%	76.7%	75.2%
	Predicted fake	83.4%	84.0%	71.4%	76.2%	75.9%
FF++ → BioDeepAV	AUC original	0.500 ± 0.001	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
	AUC degraded	0.501 ± 0.002	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
	Predicted fake	73.7%	100.0%	100.0%	100.0%	100.0%

Table 5.75: Test 5 cross-dataset results (mean ± SD over seeds; FF++ test n = 2,038 original / 2,411 degraded; BioDeepAV n = 1,420 / 4,076). Red: chance level (AUC ≤ 0.55).

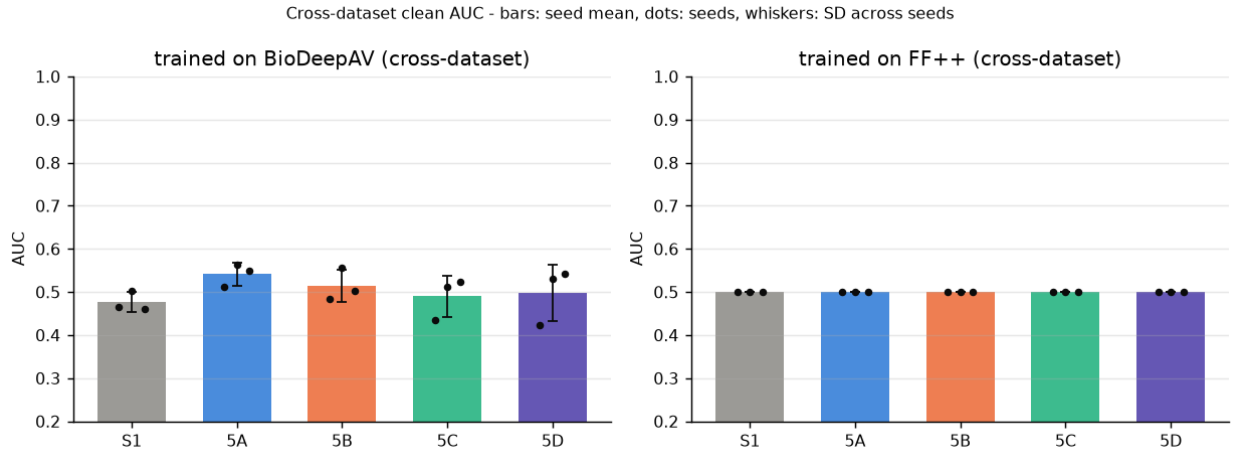

Figure 5.39: Test 5 cross-dataset AUC on original test videos.

Neither direction transfers. BioDeepAV-trained models reach AUC 0.49–0.54 on FF++ and call 71–84% of all FF++ videos fake, including 72–82% of the real ones; FF++-trained Stage 2 models call every BioDeepAV video fake. The full method is also worse than the baseline across datasets: BioDeepAV → FF++ AUC 5D − 5A = −0.044 (supported).

5.6.11 Explainability: Grad-CAM stability

Grad-CAM stability was measured on a fixed list of pairs (manifests/xai_pairs.csv), the same for every run: 320 BioDeepAV and 400 FF++ original/degraded pairs per run, half of them hard types. The measures are those of Tests 3–4 (Pearson r, top-20% IoU, centre shift, temporal CAM r, share of unchanged predictions whose explanation changed) plus a **clean reference IoU**: the IoU between two different frame subsets of the same original, i.e. the stability expected without any degradation. 5C and 5D train towards similar Grad-CAM maps, so their stability is partly optimised.

Training	Var.	Pearson r	Top-20% IoU	Centre shift	Temp. CAM r	Pred. stable	Expl. changed | stable	IoU hard types	Expl. ch. (hard)	Clean ref. IoU
BioDeepAV	S1	0.767	0.625	0.115	0.512	85.3%	26.6%	0.470	47.1%	0.829
	5A	0.780	0.602	0.065	0.523	92.4%	34.1%	0.436	60.1%	0.786
	5B	0.777	0.604	0.065	0.508	91.8%	30.4%	0.457	50.4%	0.771
	5C	0.804	0.614	0.050	0.525	88.8%	30.3%	0.459	54.7%	0.786
	5D	0.795	0.620	0.082	0.544	87.2%	27.5%	0.464	49.9%	0.807
FF++	S1	0.717	0.574	0.043	0.599	83.5%	31.0%	0.337	65.3%	0.560
	5A	0.759	0.614	0.033	0.610	83.2%	26.8%	0.379	60.6%	0.604
	5B	0.767	0.627	0.033	0.624	82.0%	23.4%	0.398	56.3%	0.622
	5C	0.916	0.775	0.024	0.702	83.2%	9.5%	0.629	24.7%	0.696
	5D	0.908	0.750	0.016	0.758	85.4%	11.2%	0.599	28.8%	0.641

Table 5.76: Test 5 Grad-CAM stability, original vs degraded version of the same video (in-domain, seed means; 320 BioDeepAV / 400 FF++ pairs per run, 160 of them hard types).

Degradation	Bio S1	Bio 5A	Bio 5C	Bio 5D	FF++ S1	FF++ 5A	FF++ 5C	FF++ 5D
blur	0.460	0.505	0.537	0.451	0.396	0.345	0.653	0.629
brightness	0.371	0.255	0.289	0.355	0.181	0.245	0.408	0.429
compression	0.727	0.726	0.716	0.722	0.700	0.716	0.856	0.819
contrast	0.509	0.524	0.526	0.532	0.374	0.400	0.716	0.646
fps_reduction	—	—	—	—	0.791	0.829	0.905	0.892
frame_drop	0.938	0.889	0.893	0.907	0.817	0.828	0.907	0.897
low_resolution	—	—	—	—	0.789	0.807	0.904	0.886
motion_blur	0.797	0.804	0.793	0.797	0.638	0.684	0.809	0.784
noise	0.660	0.652	0.674	0.676	0.657	0.756	0.857	0.824
saturation	0.541	0.458	0.483	0.518	0.396	0.527	0.738	0.692

Table 5.77: Test 5 Grad-CAM top-20% IoU per degradation type (seed means). Bold: hard types. Red: IoU below 0.5.

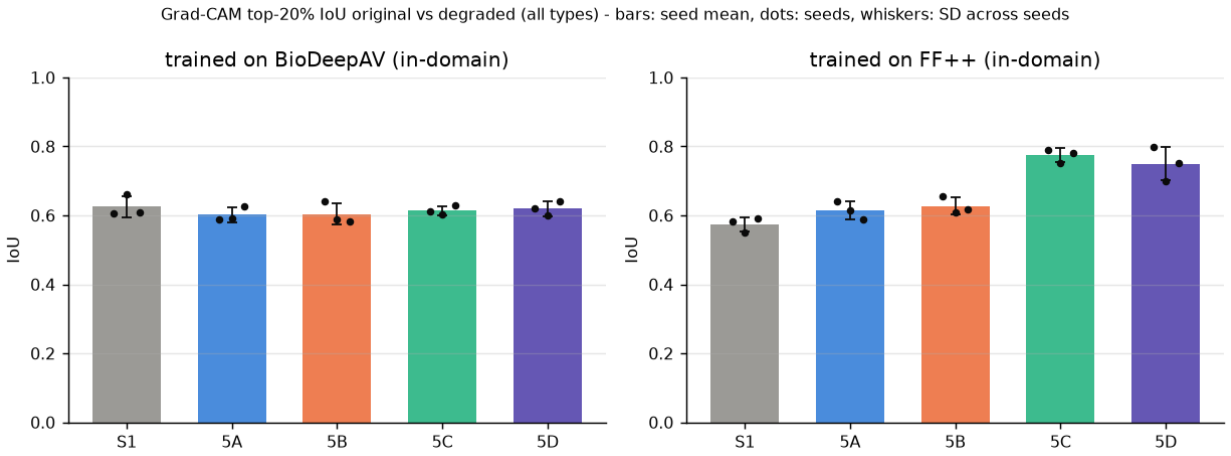

Figure 5.40: Test 5 Grad-CAM top-20% IoU, all degradation types.

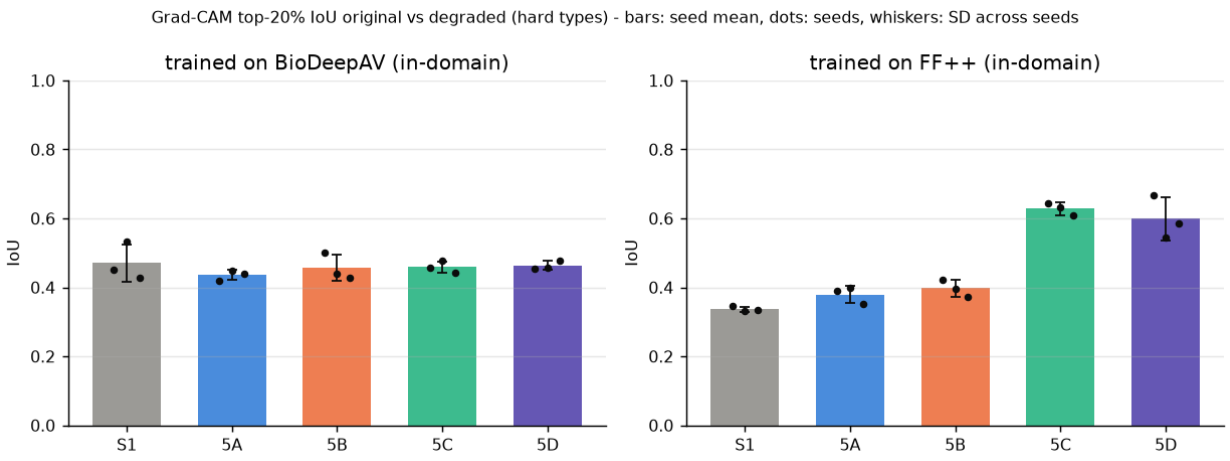

Figure 5.41: Test 5 Grad-CAM top-20% IoU, hard types only.

Figure 5.42: Test 5 Grad-CAM examples (FF++-trained 5D, seed 42): original, its CAM, degraded version, its CAM. Stability is not correctness.

The Grad-CAM consistency loss helps strongly on FF++: from 5B to 5C, IoU rose from 0.627 to 0.775 (hard types 0.398 → 0.629) and the share of unchanged predictions with a changed explanation fell from 23.4% to 9.5%. On BioDeepAV the gains are small and not supported (IoU 0.604 → 0.614; hard 0.457 → 0.459), about one in three unchanged predictions still has a changed explanation (27.5–34.1%), and the Stage 1 model is as stable as any Stage 2 variant. Brightness (IoU 0.18–0.43) and saturation remain the least stable types in every variant. For FF++ 5C, the original-vs-degraded IoU (0.775) is higher than the clean reference IoU (0.696), so consistency training may partly make the maps less specific to the input rather than more faithful.

5.6.12 Statistical tests and hypothesis verdicts

Each comparison uses a paired cluster bootstrap over source groups (1,000 replicates; 53 BioDeepAV and 101 FF++ groups) of the seed-averaged difference on the same test videos. A difference is **supported** only if the 95% CI excludes zero and all three seeds agree in sign. Of the 96 in-domain comparisons (4 comparisons × 12 metrics × 2 training sets), 12 were supported and 84 inconclusive; all supported effects, including cross-dataset ones, are listed below.

Training → test	Compar.	Metric	Mean diff. [95% CI]	Per seed (42; 123; 2026)	Result
BioDeepAV → BioDeepAV	5C − 5B	AUC degraded	-0.040 [-0.064, -0.001]	-0.100; -0.006; -0.014	worse
BioDeepAV → BioDeepAV	5D − 5A	AUC degraded	-0.036 [-0.057, -0.003]	-0.016; -0.073; -0.020	worse
BioDeepAV → BioDeepAV	5D − 5A	Worst-type AUC	-0.099 [-0.140, -0.007]	-0.081; -0.181; -0.036	worse
BioDeepAV → BioDeepAV	5D − 5C	Grad-CAM r (hard)	-0.033 [-0.072, -0.002]	-0.002; -0.021; -0.075	worse
FF++ → FF++	5B − 5A	Brightness AUC	-0.071 [-0.118, -0.020]	-0.058; -0.060; -0.095	worse
FF++ → FF++	5C − 5B	Grad-CAM IoU (all)	+0.148 [+0.131, +0.163]	+0.163; +0.135; +0.145	better
FF++ → FF++	5C − 5B	Grad-CAM IoU (hard)	+0.231 [+0.208, +0.256]	+0.235; +0.222; +0.236	better
FF++ → FF++	5C − 5B	Grad-CAM r (hard)	+0.310 [+0.276, +0.370]	+0.278; +0.293; +0.359	better
FF++ → FF++	5D − 5A	Grad-CAM IoU (all)	+0.136 [+0.123, +0.154]	+0.162; +0.158; +0.087	better
FF++ → FF++	5D − 5A	Grad-CAM IoU (hard)	+0.220 [+0.196, +0.243]	+0.234; +0.278; +0.147	better
FF++ → FF++	5D − 5A	Grad-CAM r (hard)	+0.317 [+0.278, +0.352]	+0.342; +0.339; +0.269	better
FF++ → FF++	5D − 5C	Flip rate (pts)	-2.627 [-3.781, -0.458]	-1.991; -5.392; -0.498	better
BioDeepAV → FF++ (cross)	5D − 5A	AUC original	-0.044 [-0.067, -0.010]	-0.007; -0.089; -0.035	worse
BioDeepAV → FF++ (cross)	5D − 5A	AUC degraded	-0.040 [-0.059, -0.006]	-0.019; -0.070; -0.031	worse
BioDeepAV → FF++ (cross)	5D − 5A	Flip rate (pts)	+8.226 [+4.462, +10.345]	+9.789; +8.337; +6.553	worse
BioDeepAV → FF++ (cross)	5D − 5C	Flip rate (pts)	+4.936 [+3.030, +6.070]	+4.396; +5.475; +4.936	worse
FF++ → BioDeepAV (cross)	5C − 5B	Grad-CAM IoU (all)	+0.091 [+0.072, +0.105]	+0.127; +0.070; +0.077	better
FF++ → BioDeepAV (cross)	5C − 5B	Grad-CAM IoU (hard)	+0.123 [+0.100, +0.143]	+0.158; +0.100; +0.112	better
FF++ → BioDeepAV (cross)	5C − 5B	Grad-CAM r (hard)	+0.122 [+0.094, +0.146]	+0.175; +0.090; +0.102	better
FF++ → BioDeepAV (cross)	5D − 5A	Grad-CAM IoU (all)	+0.118 [+0.099, +0.139]	+0.130; +0.164; +0.060	better
FF++ → BioDeepAV (cross)	5D − 5A	Grad-CAM IoU (hard)	+0.168 [+0.141, +0.186]	+0.168; +0.228; +0.108	better
FF++ → BioDeepAV (cross)	5D − 5A	Grad-CAM r (hard)	+0.206 [+0.187, +0.259]	+0.195; +0.267; +0.157	better

Table 5.78: All statistically supported differences in Test 5 (statistics_test5.csv). For the flip rate, lower is better.

Hypothesis	Comparison	Verdict	Evidence
H1: hard sampling improves detection on hard types	5B − 5A	Not supported	BioDeepAV hard AUC +0.028 [−0.002, 0.048] and brightness +0.036 (seeds disagree); FF++ hard AUC −0.026, brightness −0.071 (supported deterioration)
H2: Grad-CAM consistency improves explanation stability	5C − 5B	Partially supported (FF++ only)	FF++ IoU all +0.148, hard +0.231, r hard +0.310 (all supported); BioDeepAV IoU +0.010 / +0.002 (inconclusive)
H3: feature consistency + curriculum add beyond 5C	5D − 5C	Not supported	Only FF++ flip rate −2.6 pts (supported); BioDeepAV Grad-CAM r (hard) −0.033 (supported deterioration)
H4: full method beats the baseline	5D − 5A	Not supported (deterioration on BioDeepAV)	BioDeepAV degraded AUC −0.036, worst-type AUC −0.099 (supported deterioration); FF++ Grad-CAM IoU hard +0.220 (supported), detection inconclusive

Table 5.79: Test 5 hypothesis verdicts.

5.6.13 Key findings

- **Degraded training works — replicated.** Every Stage 2 variant beat Stage 1 on both datasets (BioDeepAV degraded AUC 0.795 → 0.90–0.96, worst-type 0.558 → 0.76–0.89; FF++ degraded 0.631 → 0.69–0.71), and noise again breaks a model trained without degraded data (S1 severe-noise AUC 0.329).
- **Hard-degradation sampling did not help reliably (H1).** 5B gave the highest BioDeepAV values (degraded AUC 0.962, brightness 0.894), but the seeds disagreed in sign; on FF++ it lowered brightness AUC by 0.071 (supported).
- **Grad-CAM consistency stabilises explanations on FF++ only (H2).** Hard-type IoU 0.398 → 0.629 and explanation changed while prediction stable 23.4% → 9.5%; no supported effect on BioDeepAV.
- **The full method did not beat the baseline (H3, H4).** 5D was worse than 5A on BioDeepAV degraded AUC (−0.036) and worst-type AUC (−0.099), driven by severe noise (0.665), and also worse across datasets.
- **Results depend on the seed.** BioDeepAV original AUC varied by up to 0.07 between seeds of the same Stage 2 variant (5C: 0.909–0.981); single-run differences of this size, as in Tests 1–4, are therefore not evidence of a method effect.
- **Shortcuts and missing transfer persist.** Cross-dataset AUC 0.49–0.54 / 0.500; FF++-trained models call every BioDeepAV video fake (audio presence); BioDeepAV labels are still separable by native face size (AUC 0.993).

5.6.14 Limitations and things to check

- **Different split from Tests 2–4** (seed 20261009, content-duplicate merging, DFD grouped by actor): Test 4 vs Test 5 differences mix method and split changes, and the Test 4 model was never evaluated on the Test 5 test set (its training data overlaps it).
- **Uneven split.** Large groups put most BioDeepAV fakes (1,006 of 1,692) and almost all DFD videos (838) in the test set, so training sees only 453 BioDeepAV fakes and 63 DFD videos; the test sets are class-imbalanced (AUC is unaffected, accuracy is not).
- **Confounded steps:** 5C − 5B = pairing + Grad-CAM consistency; 5D − 5C = feature consistency + curriculum.
- **Limited statistical power:** 3 seeds and 53 BioDeepAV source groups; most differences are inconclusive rather than absent.
- **Calibration:** BioDeepAV thresholds 0.01–0.12 and near-0/1 probabilities; accuracy-type metrics depend on the threshold.
- **Stability is not correctness:** no manipulation-region annotations exist; for FF++ 5C the degraded IoU exceeds the clean reference IoU, so check that consistency training is not just making the maps less specific.
- **Residual leakage risk:** 2,474 near-duplicate candidate pairs across splits are documented but unresolved; the screens cannot detect the same person in different videos, and BioDeepAV fake identities are unverifiable.
- **Script changed during the run:** 27 checkpoints were trained with script SHA-256 d40fd034…, the last three (FF++ seed 2026, 5B–5D) and the final evaluation with e789a2a2…; record both. Logged training times of resumed runs are inflated (e.g. FF++ seed 2026 5A: 23.5 h).

5.6.15 Suggested thesis sentence

> “In an independent replication with a new duplicate-aware split, four training variants and three seeds, degraded (Stage 2) training again restored robustness (BioDeepAV degraded ROC-AUC 0.795 → 0.939 for the baseline), but targeting the hardest degradation types did not reliably improve detection: no hard-sampling gain was statistically supported, and the full method lowered degraded AUC on BioDeepAV by 0.036. A Grad-CAM consistency loss was the only component with a supported benefit, raising hard-degradation Grad-CAM IoU on FF++ from 0.398 to 0.629 and cutting the share of unchanged predictions with a changed explanation from 23.4% to 9.5%, while cross-dataset detection remained at chance (AUC ≤ 0.54).”

5.6.16 Output files and checkpoints

File	Contents
reports/TEST5_INDEPENDENT_REPORT.md	Full report: inventory, split, results, statistics, verdicts (generated 2026-10-16 02:52)
reports/PRE_TRAINING_CHECKS.md, gate_status.json, leakage_audit_report.json	Leakage gate (PASS) and audits, reviewed before training
reports/final_experiment_summary.json	Headline numbers, hypothesis verdicts, 30 checkpoint records
metrics/metrics_test5.csv, robustness_test5.csv, degradation_metrics_test5.csv	Per run: metrics per condition, flips, type × severity
metrics/cross_dataset_test5.csv	Cross-dataset results and audio ablations
metrics/gradcam_summary_test5.csv, gradcam_stability_test5.csv, gradcam_runs/	Grad-CAM stability (summary and per pair)
metrics/seed_comparison.csv, statistics_test5.csv	Seed summaries and paired bootstrap tests
metrics/predictions/*.csv.gz	Per-video predictions of all 30 runs
checkpoints/train-{BioDeepAV, FFpp}/seed{42, 123, 2026}/stage1.pt, 5A–5D_stage2.pt; checkpoint_hashes.csv	30 checkpoints with file and weights SHA-256, threshold, best epoch
configs/experiment_config_5A–5D.json, data_sources.json, preprocessing_config.json	Configuration
splits/split_manifest.csv, split_meta.json, split_integrity_report.json	Split (originals hash 6ae9fe14…), groups, integrity
data_audit/, manifests/, logs/	Inventory, duplicates, exclusions, face detection, XAI pairs, training histories, run logs
figures/	11 result figures + Grad-CAM examples

Table 5.80: Test 5 output files (E:\Code\run_test5_independent).

5.7 Basic Retest — Real / Fake Video Detector with Explainable AI (151026.py)

5.7.1 Overview

After five tests that focused on degradation, the basic retest goes back to the core question: **can the detector tell a real video from a fake one, and can it explain its decision?** It is a fresh, standalone experiment: it uses only the **original** real and fake videos of BioDeepAV and FaceForensics++ C23 (no degraded videos) and reads nothing from Tests 1–5 (no caches, splits or checkpoints). Its design follows the lessons of the earlier tests: an ImageNet-pretrained backbone instead of a small CNN trained from scratch, video only (audio presence was a shortcut), a fixed 112 px face bottleneck against the resolution shortcut, joint training on both datasets, an identity- and duplicate-aware split, and a calibrated decision with an "uncertain" band. The trained model can be applied to any video (*--predict* or a small GUI) and returns REAL, FAKE or UNCERTAIN with Grad-CAM++ heat-maps, temporal attention and an occlusion map. Script: *151026.py* (run 16–17 Oct 2026, one seed, 38 min training on an RTX 3050).

Key result. On 1,602 held-out test videos the detector reached AUC **0.979** on BioDeepAV and **0.874** on FF++ (0.927 overall). At the validation threshold it recognised 90.7% of BioDeepAV real and 87.0% of BioDeepAV fake videos, and 83.5% of FF++ real and 75.4% of FF++ fake videos; 12.7% of the videos were flagged as uncertain, and on the remaining 87.3% the accuracy was 85.9%. NeuralTextures stayed near chance (52.4%). The explanations are faithful for REAL decisions (hiding the Grad-CAM region changed the decision more than hiding a random region in 82–84% of videos) but not for FAKE decisions (8–26%): the fake evidence is spread over the face texture rather than localised.

5.7.2 Code pipeline

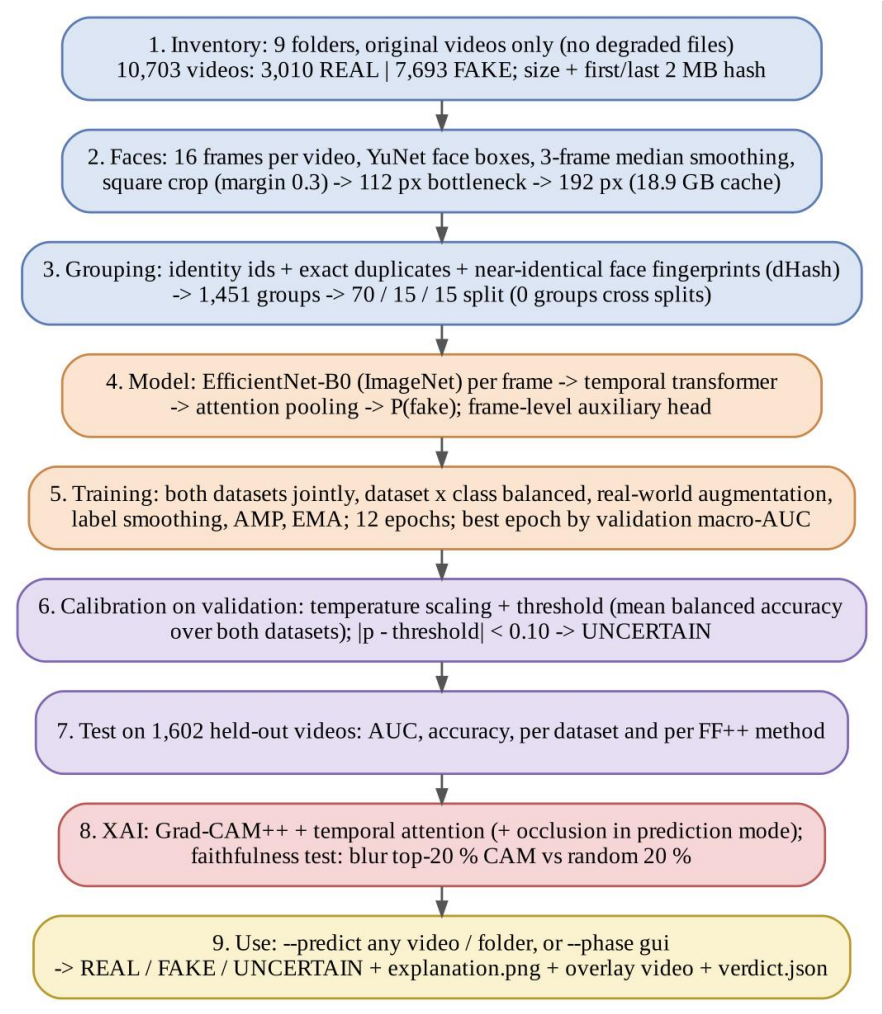

Figure 5.43: What 151026.py does. Phases: selftest, inventory, faces, split, train, evaluate, xai, report; plus predict and gui for new videos.

Item	Basic retest (151026.py)	Reason (lesson from Tests 1–5)
Data	Original real and fake videos only, both datasets, 10,703 videos	Check real/fake detection before robustness
Backbone	EfficientNet-B0, ImageNet-pretrained (4.87 M parameters with the temporal head)	Small CNNs from scratch reached a ceiling (Test 5)
Temporal model	16 frames; 1-layer transformer (4 heads) + attention pooling; per-frame auxiliary loss (0.3)	Attention shows which frames mattered
Modality	Face video only	Audio presence was a cross-dataset shortcut (Tests 4–5)
Face crop	YuNet, 3-frame median box smoothing, margin 0.3, 112 px bottleneck → 192 px	Removes the BioDeepAV resolution shortcut
Training	Joint, dataset × class balanced; JPEG, blur, down/up-scaling, colour jitter, random erasing, flips, crops; label smoothing 0.05; EMA; AMP	Single-dataset models did not transfer (Tests 3–5)
Optimiser	AdamW (backbone 1e-4, head 1e-3), warm-up + cosine, 12 epochs, 8 clips × 8 frames × accumulation 2	
Selection	Best epoch by validation macro-AUC over the two datasets	Neither dataset dominates
Decision	Temperature scaling; threshold maximising mean balanced accuracy over both datasets; ±0.10 band = uncertain	Test 5 thresholds were 0.01–0.12
XAI	Grad-CAM++, temporal attention, occlusion (prediction mode), faithfulness test	Stability alone is not correctness

Table 5.81: Basic retest design.

5.7.3 Data and split

All 10,703 original videos (3,010 real, 7,693 fake) were decoded without failure. YuNet found a face in 99.0% of the sampled frames on average; 56 videos had no detectable face and were analysed with a centre crop. Videos were grouped by identity (FF++ source and target sequence ids, DFD actor ids, BioDeepAV YouTube ids) and by duplicate content (170 exact duplicate files, 15,516 near-identical face-fingerprint pairs), giving 1,451 groups; no group crosses the train / validation / test split.

Dataset	Folder	Label	Train	Validation	Test
BioDeepAV	real	REAL	1,408	301	301
BioDeepAV	fake	FAKE	1,185	254	254
FF++	original	REAL	678	158	164
FF++	Deepfakes, Face2Face, FaceShifter, FaceSwap, NeuralTextures (each)	FAKE	678	158	164
FF++	DeepFakeDetection	FAKE	838	99	63
Total			**7,499**	**1,602**	**1,602**

Table 5.82: Basic retest split (identity- and duplicate-grouped). Test: BioDeepAV 301 real / 254 fake; FF++ 164 real / 883 fake.

5.7.4 Training

Epoch	Train loss	Val AUC BioDeepAV	Val AUC FF++	Val macro-AUC	Epoch time
1	0.690	0.968	0.598	0.7829	3.6 min
2	0.567	0.984	0.710	0.8472	2.8 min
3	0.527	0.987	0.773	0.8802	2.8 min
4	0.492	0.986	0.831	0.9089	2.8 min
5	0.473	0.979	0.856	0.9175	2.8 min
6	0.458	0.985	0.870	0.9276	2.8 min
7	0.431	0.991	0.877	0.9339	2.8 min
8	0.422	0.983	0.894	0.9385	2.8 min
9	0.404	0.985	0.900	0.9425	2.8 min
10	0.392	0.989	0.899	0.9440	2.8 min
11	**0.383**	**0.988**	**0.903**	**0.9455**	2.8 min
12	0.374	0.990	0.897	0.9438	2.8 min

Table 5.83: Basic retest training (seed 42). Bold: selected epoch 11 (validation macro-AUC 0.9455). Temperature 1.45, threshold 0.38; validation ECE 0.156 → 0.159 (temperature scaling did not improve calibration).

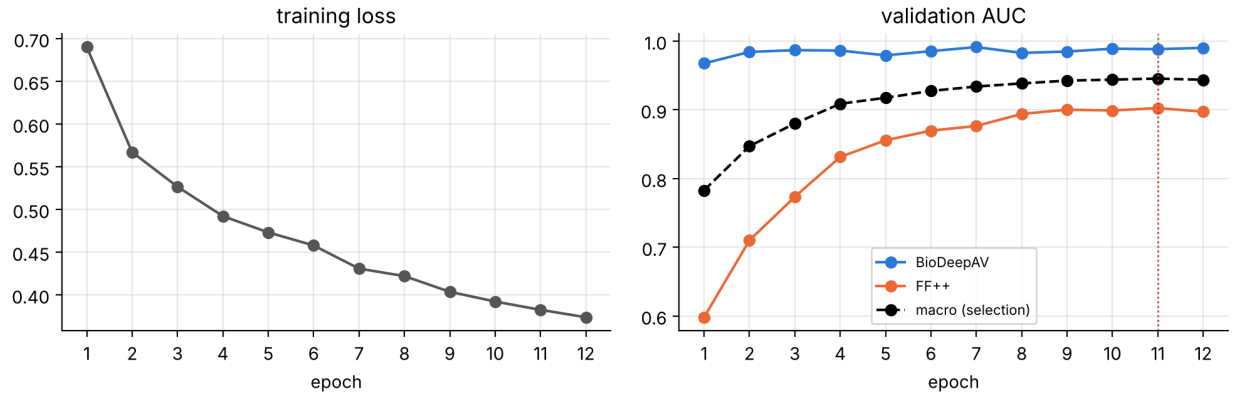

Figure 5.44: Basic retest training loss and validation AUC per dataset (dotted line: selected epoch).

BioDeepAV was learned almost at once (validation AUC 0.968 after one epoch), while FF++ improved steadily from 0.598 to 0.903 and was still close to its best at epoch 12, so a longer schedule may help FF++.

5.7.5 Results: real vs fake

Test set	n (real / fake)	AUC	Accuracy	Balanced acc.	Real recognised	Fake detected	Precision (fake)	F1	ECE	Uncertain
ALL (both datasets)	1,602 (465 / 1137)	0.927	81.0%	83.1%	88.2%	78.0%	94.2%	0.853	0.179	12.7%
BioDeepAV	555 (301 / 254)	0.979	89.0%	88.9%	90.7%	87.0%	88.8%	0.879	0.120	6.3%
FF++	1,047 (164 / 883)	0.874	76.7%	79.5%	83.5%	75.4%	96.1%	0.845	0.271	16.1%

Table 5.84: Basic retest test results at the validation threshold 0.38 (accuracy at 0.5: 75.7% overall, 87.7% BioDeepAV, 69.3% FF++). Confusion counts: BioDeepAV TN 273, FP 28, FN 33, TP 221; FF++ TN 137, FP 27, FN 217, TP 666.

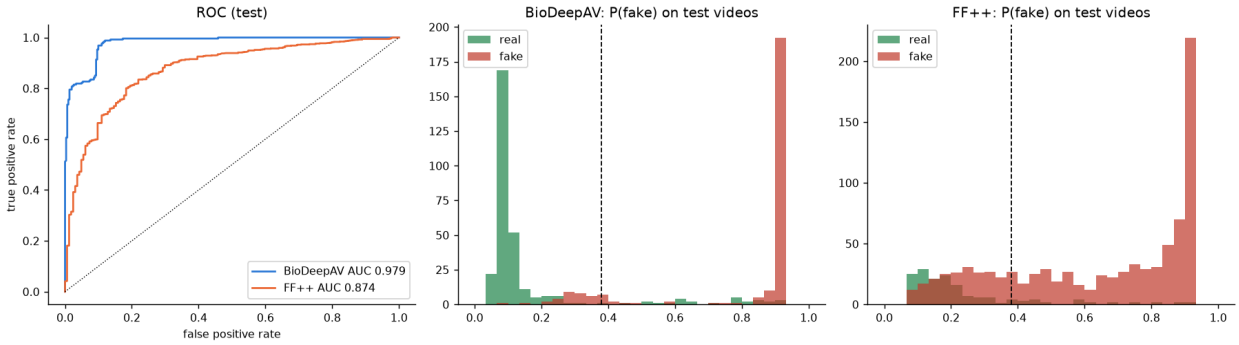

Figure 5.45: Basic retest ROC curves and P(fake) distributions of the test videos (dashed line: threshold 0.38).

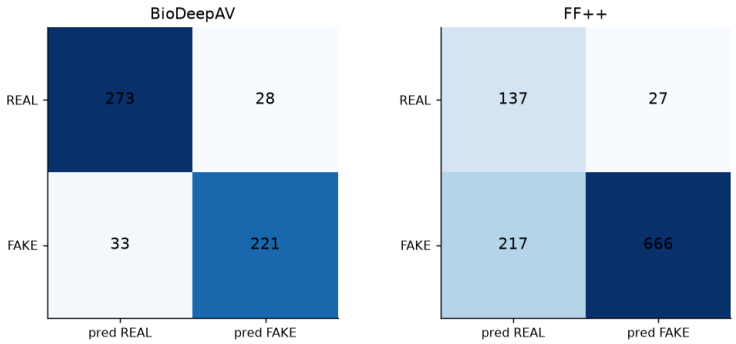

Figure 5.46: Basic retest confusion matrices (test videos, threshold 0.38).

Dataset	Folder	True label	n	Correct	Mean P(fake)	Uncertain	Correct when decided
BioDeepAV	fake	FAKE	254	87.0%	0.815	12.2%	95.1%
BioDeepAV	real	REAL	301	90.7%	0.154	1.3%	91.2%
FF++	DeepFakeDetection	FAKE	63	100.0%	0.923	0.0%	100.0%
FF++	Deepfakes	FAKE	164	90.2%	0.756	6.7%	91.5%
FF++	Face2Face	FAKE	164	72.6%	0.582	23.8%	79.2%
FF++	FaceShifter	FAKE	164	75.0%	0.628	20.1%	82.4%
FF++	FaceSwap	FAKE	164	77.4%	0.641	17.1%	83.1%
FF++	NeuralTextures	FAKE	164	52.4%	0.451	22.0%	55.5%
FF++	original	REAL	164	83.5%	0.239	13.4%	87.3%

Table 5.85: Basic retest per folder (test videos). Red: below 60%.

Test set	Decided videos	Accuracy when decided	Uncertain	Accuracy inside the uncertain band
ALL	1,398 of 1,602	85.9%	12.7%	47.1%
BioDeepAV	520 of 555	92.9%	6.3%	31.4%
FF++	878 of 1,047	81.8%	16.1%	50.3%

Table 5.86: Effect of the uncertain band (|P(fake) − 0.38| < 0.10). Inside the band the decision is close to a coin flip, so reporting it as UNCERTAIN is justified.

BioDeepAV is separated well: real videos cluster near P(fake) 0.1 and fake videos near 0.9. FF++ is harder: its fake scores are spread over the whole range, and almost all FF++ errors are missed fakes (217 false negatives vs 27 false positives). By method, DeepFakeDetection (100%, only 63 test videos) and Deepfakes (90.2%) are detected best, FaceSwap, FaceShifter and Face2Face reach 73–77%, and NeuralTextures — which changes only the mouth region with subtle texture edits — stays at chance (52.4%). Compared with the earlier tests on original videos (different splits, so not a paired comparison), this is the highest FF++ AUC so far (Test 3: 0.804, Test 4: 0.747, Test 5 5A: 0.770) and real BioDeepAV videos are recognised far more often (90.7% vs 62.1% for Test 5 5A), while BioDeepAV AUC (0.979) is just below the shortcut-affected 0.99 of Tests 3–4.

5.7.6 Explainability (XAI)

Every decision comes with three explanations: (1) **Grad-CAM++** on the last EfficientNet block, for the predicted class; (2) the **temporal attention** weights of the 16 frames; and (3) in prediction mode an **occlusion map** (grey patches, drop in P(fake)). To check whether the heat-maps can be trusted, a **faithfulness test** was run on 200 test videos (50 per dataset × class): the top-20% Grad-CAM pixels of every frame were blurred, and so was a random 20% of pixels; a faithful explanation must lower the confidence of the predicted class more than the random mask.

Dataset	True class	n	Correct	Confidence drop: CAM top-20%	Confidence drop: random 20%	CAM beats random	CAM energy in face centre
BioDeepAV	REAL	50	94%	0.133	0.033	84%	0.65
BioDeepAV	FAKE	50	92%	0.042	0.697	8%	0.71
FF++	REAL	50	82%	0.521	0.006	82%	0.88
FF++	FAKE	50	74%	0.049	0.379	26%	0.87

Table 5.87: Basic retest faithfulness test of Grad-CAM++ (drop in the confidence of the predicted class). Red: the explanation is less informative than a random region.

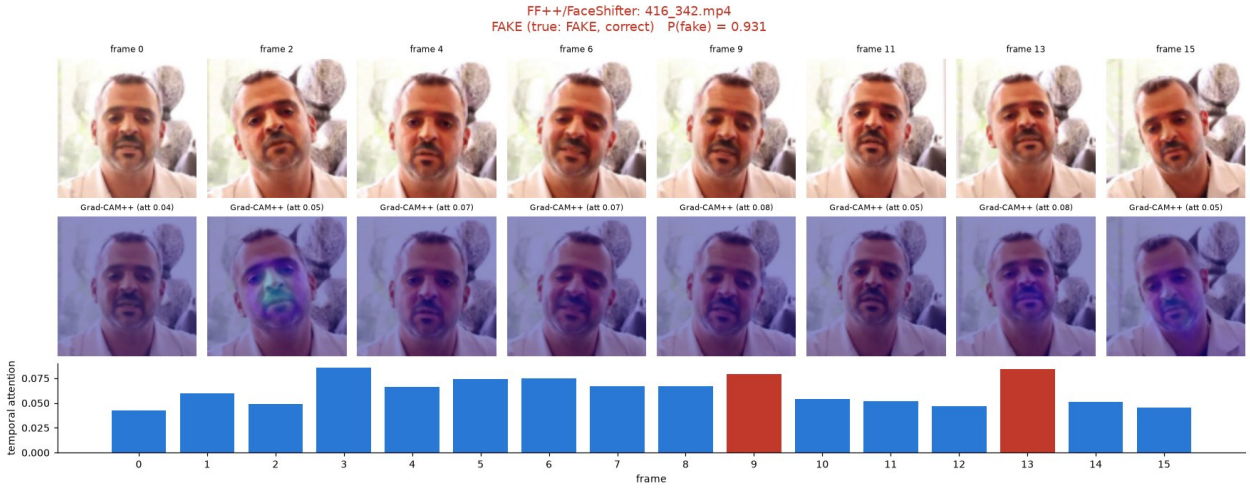

Figure 5.47: Basic retest explanation, correct FAKE decision, FF++ FaceShifter (P(fake) 0.931). Top: face crops; middle: Grad-CAM++ (normalised per video); bottom: temporal attention.

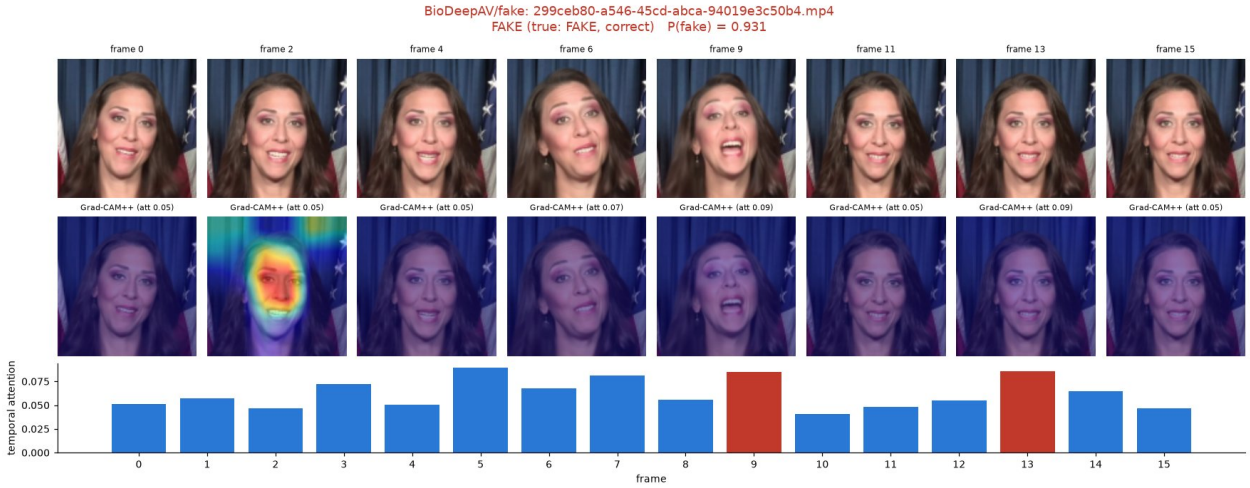

Figure 5.48: Basic retest explanation, correct FAKE decision, BioDeepAV fake (P(fake) 0.931). Top: face crops; middle: Grad-CAM++ (normalised per video); bottom: temporal attention.

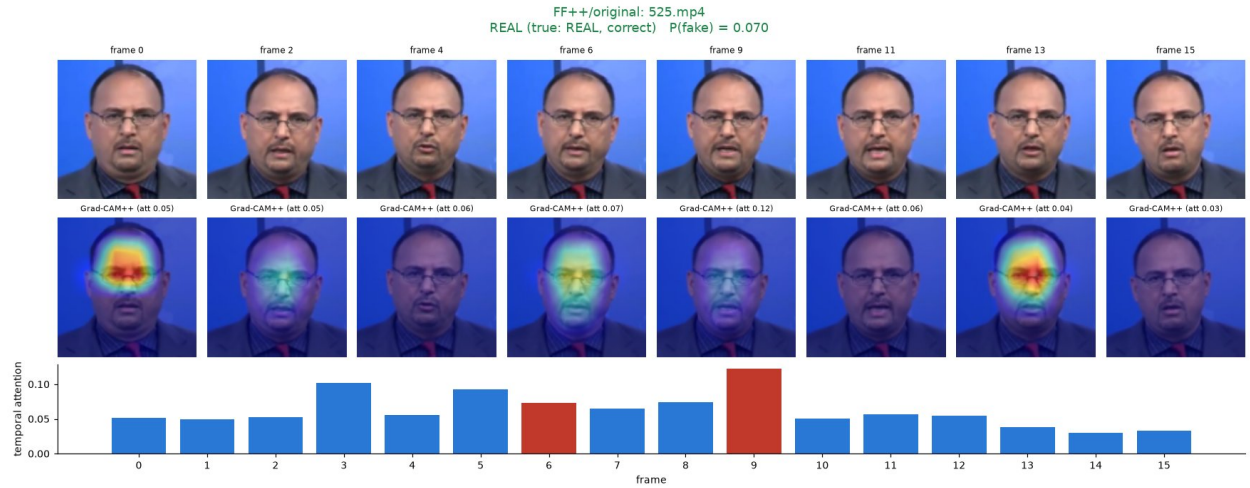

Figure 5.49: Basic retest explanation, correct REAL decision, FF++ original. Top: face crops; middle: Grad-CAM++ (normalised per video); bottom: temporal attention.

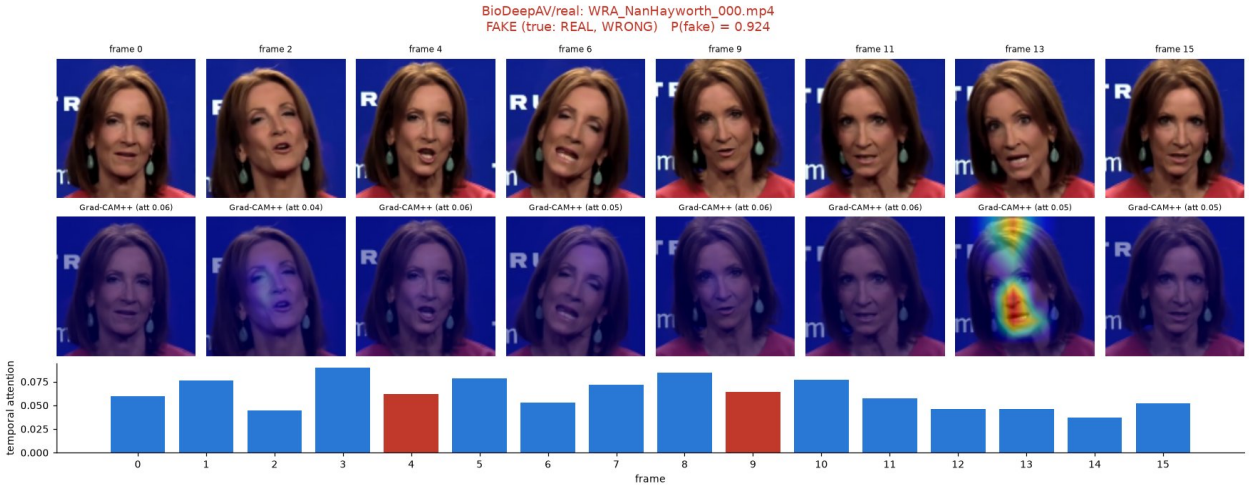

Figure 5.50: Basic retest explanation, WRONG decision: BioDeepAV real video called FAKE (P(fake) 0.924). Top: face crops; middle: Grad-CAM++ (normalised per video); bottom: temporal attention.

The test separates two cases. For **REAL** decisions the explanation is faithful: blurring the Grad-CAM region lowered the confidence far more than a random region (0.13 vs 0.03 on BioDeepAV, 0.52 vs 0.01 on FF++; CAM beats random in 84% and 82% of videos). For **FAKE** decisions it is not: a random 20% blur lowered P(fake) much more (0.70 vs 0.04 on BioDeepAV, 0.38 vs 0.05 on FF++; CAM better in only 8% and 26%). The model therefore relies on manipulation traces spread over the whole face texture, which any blur removes, rather than on one region; the Grad-CAM++ map of a FAKE decision should not be read as "the manipulated area". The heat lies mostly on the face (68% of the CAM energy in the central face region on BioDeepAV, 88% on FF++). Because the maps are normalised over the whole video, usually only one or two frames show strong heat (Figures 5.47–5.50).

5.7.7 Key findings

- **The detector works on original videos.** Test AUC 0.979 (BioDeepAV) and 0.874 (FF++), 0.927 overall; 85.9% accuracy on the 87% of videos it does not flag as uncertain.
- **Pretraining + resolution bottleneck + joint training fixed the earlier weaknesses:** best FF++ AUC of all tests, and real videos are no longer mostly called fake (BioDeepAV 90.7%, FF++ 83.5% recognised).
- **Remaining weakness: subtle manipulations.** NeuralTextures 52.4% (chance), Face2Face 72.6%; most FF++ errors are missed fakes.
- **The uncertain band is useful:** inside it accuracy is 31% (BioDeepAV) and 50% (FF++), so those videos really are undecidable.
- **Explanations are trustworthy for REAL decisions, not for FAKE decisions** (faithfulness 82–84% vs 8–26%): fake evidence is distributed over the face texture.

5.7.8 Limitations and things to check

- **One seed only** (42); Test 5 showed seed-to-seed differences of up to 0.07 AUC, so train 2–3 seeds (*--seeds 42,123,2026*) before final claims.
- **Original videos only:** robustness to degradation is not tested here by design (that is Tests 1–5).
- **Different split** from Tests 2–5 (stronger near-duplicate merging), so the comparison with earlier tests is indicative, not paired.
- **Calibration:** temperature scaling did not reduce ECE (FF++ ECE 0.271); the threshold 0.38 was tuned on validation.
- **Small cells:** only 63 DeepFakeDetection test videos (identity grouping placed most DFD actors in training).
- **XAI:** the deletion test uses blur, which also removes high-frequency texture, and heat-maps are normalised per video. In the figures, the red attention bars mark the two highest-attention frames among the eight frames shown, not across all 16.
- **Scope of "any video":** trained on face-swap / re-enactment fakes of these two datasets; fully synthetic (diffusion) videos or videos without a visible face are outside its training data (56 such videos here were analysed with a centre crop).

5.7.9 Suggested thesis sentence

> “A basic retest on the original videos only, with an ImageNet-pretrained EfficientNet-B0 and a temporal transformer trained jointly on both datasets with a fixed face-resolution bottleneck, distinguished real from fake videos with a test ROC-AUC of 0.979 on BioDeepAV and 0.874 on FaceForensics++ (85.9% accuracy on the 87% of videos not flagged as uncertain). NeuralTextures remained undetected (52.4%), and a faithfulness test showed that Grad-CAM++ explanations are reliable for real decisions but not for fake decisions, whose evidence is spread across the face texture.”

5.7.10 Output files and model

File	Contents
151026.py	Script: selftest, inventory, faces, split, train, evaluate, xai, report, predict, gui
checkpoints/best_model.pt (seed42_best.pt, seed42_last.pt)	Selected model (seed 42, epoch 11) with temperature 1.45, threshold 0.38 and preprocessing settings
reports/REPORT_151026.md, model_selection.json	Run report and model selection
metrics/test_metrics.csv, test_per_subset.csv, predictions.csv	Test metrics, per folder, per-video predictions (validation + test)
metrics/xai_faithfulness.csv, xai_faithfulness_summary.csv	Faithfulness test per video and summary
figures/roc_and_scores.png, confusion_matrices.png; xai/*.png	Figures and 15 example explanations (correct and wrong)
data/inventory.csv, splits/split.csv, split_summary.json	Inventory, groups and split
cache/faces_n10703_T16_S192_F112_*.npy (18.9 GB)	Face crops of all videos
predictions/<video>_<time>/	For each new video: explanation.png, explanation_overlay.mp4, verdict.json

Table 5.88: Basic retest output files (E:\Code\run_151026).

5.8 Test 5.7 Improvement Suite — Reproduction, Seeds, Sampling, Training Length, Calibration and XAI Faithfulness (171026.py)

5.8.1 Overview

The basic retest (Section 5.7) produced a usable real / fake detector, but it rested on **one training run**. Before any of its numbers can be used as thesis evidence, six questions must be answered: Can the run be reproduced? How much do the results move between seeds? Does a different training sampler or a longer schedule really help the hard FF++ methods? Is the decision threshold well calibrated? And are the heat-maps faithful to what the model uses? The improvement suite *171026.py* answers these questions in six experiments (E1–E6). It re-uses the inventory, split and face cache of the basic retest **read-only**, so every run sees exactly the same 1,602 test videos and all differences between configurations are paired. 15 models were trained (5 configurations × seeds 42, 123 and 2026) on an RTX 3050; the run finished on 17 Oct 2026.

Key result. (1) The split and the checkpoint selection of the basic retest reproduce exactly (identical split hash, same best epoch 11), but GPU training is not bit-wise repeatable: test AUC moved by +0.013 and 8.1% of the test decisions changed. (2) Over three seeds the 151026 settings give test AUC **0.933 ± 0.006** (BioDeepAV 0.982, FF++ 0.880). (3) Changing only the sampler to adaptive group-DRO (*harddro12*) gave the only **supported** overall gains: AUC +0.015, balanced accuracy +0.031, fake detected +0.055, with NeuralTextures recall +0.161 and better recall on all four hard FF++ methods, while BioDeepAV stayed unchanged. (4) Training for 20 or 30 epochs improved FF++ (AUC up to 0.916) but not BioDeepAV, which fell slightly. (5) Isotonic calibration fitted on validation lowered ECE from 0.19 to 0.035; abstaining on the 10% least certain videos raised balanced accuracy from 0.87 to 0.89. (6) No explanation method is faithful in every test: Grad-CAM++ and occlusion pass the randomisation check, Integrated Gradients does not, and deletion results depend on the perturbation used.

5.8.2 Experiment design

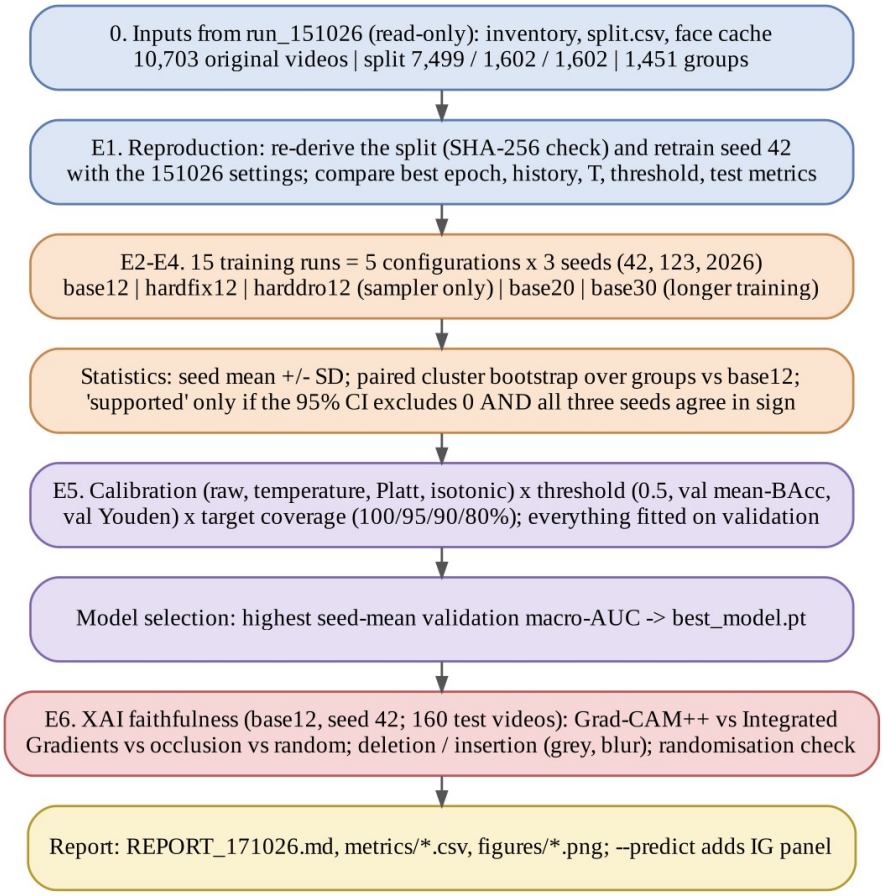

Figure 5.51: What 171026.py does. The basic-retest data are used read-only; only the factor named in each experiment changes.

#	Question	What changes	What is measured	Decision rule
E1	Is the basic retest reproducible?	Nothing: seed 42, 151026 settings	Split hash, best epoch, history, T, threshold, test metrics, prediction agreement	Split identical; AUC within 0.01, accuracy within 1.5 points
E2	How large is the seed variation?	Seed (42, 123, 2026)	Mean ± SD of AUC, accuracy, balanced accuracy, ECE	Report SD next to every result
E3	Does hard-type sampling help?	Only the training sampler (hardfix12, harddro12)	Paired differences vs base12; recall per FF++ method	Supported = 95% CI excludes 0 and all seeds agree
E4	Does longer training help?	Epochs 12 / 20 / 30 (early stopping, patience 4 / 5 / 5)	As E3; validation curves	As E3
E5	Are threshold and probabilities calibrated?	Calibration method, threshold rule, target coverage (fitted on validation)	ECE, Brier, coverage, mean balanced accuracy, AURC	Lower ECE / AURC; bacc at matched coverage
E6	Are the explanations faithful?	XAI method: Grad-CAM++, Integrated Gradients, occlusion, random	Deletion / insertion AUC (grey and blur), face-centre energy, randomisation check	Better than random with CI excluding 0

Table 5.89: The six experiments of the improvement suite (171026.py).

Configuration	Sampler	Epochs (patience)	Everything else
base12	Dataset × class balanced (as 151026)	12 (4)	Identical to 151026: EfficientNet-B0 + temporal transformer, 16 frames, 112 px face → 192 px, AdamW, warm-up + cosine, label smoothing 0.05, EMA, AMP
hardfix12	Fixed weights: folders with high validation error in 151026 (e.g. NeuralTextures) drawn more often	12 (4)	Same
harddro12	Adaptive group-DRO: each epoch, groups (dataset × folder) with higher training loss are drawn more often; uniform floor keeps every group	12 (4)	Same
base20	As base12	20 (5)	Same; cosine schedule stretched to 20 epochs
base30	As base12	30 (5)	Same; cosine schedule stretched to 30 epochs

Table 5.90: The five training configurations. Only the sampler (E3) or the schedule length (E4) differs from base12.

5.8.3 E1 — Baseline reproduction

The split was re-derived from the inventory with the 151026 code and compared with the stored split; seed 42 was then trained again with exactly the 151026 settings.

Check	Basic retest (151026)	Re-run (171026)	Result
Split assignment (SHA-256)	c0b494ad05426e2f…	c0b494ad05426e2f…	identical
Groups	1,451	1,451	identical
Selected (best) epoch	11	11	identical
Validation macro-AUC	0.9455	0.9464	max difference per epoch 0.007
Temperature / threshold	1.45 / 0.38	1.49 / 0.36	close
Same test videos	1,602	1,602	yes
Test decisions that agree	—	91.9%	Spearman ρ of P(fake) 0.963
Bit-wise identical predictions	—	no	GPU non-determinism
Metrics within tolerance	—	no	not reproduced (strict)

Table 5.91: E1 reproduction checks.

Test set	n	AUC 151026	AUC re-run	Δ AUC	Accuracy 151026	Accuracy re-run	Δ accuracy	Within tolerance
ALL	1,602	0.9266	0.9394	+0.0128	81.0%	82.7%	+1.7	no
BioDeepAV	555	0.9787	0.9855	+0.0068	89.0%	95.3%	+6.3	no
FF++	1,047	0.8741	0.8832	+0.0091	76.7%	76.0%	-0.7	yes

Table 5.92: E1 test metrics of the original run and the re-run (tolerance: AUC 0.01, accuracy 1.5 points).

Interpretation. The data pipeline is fully deterministic: the same split, the same groups and the same checkpoint epoch were obtained. The model weights are not: cuDNN autotuning and atomic GPU operations make each run slightly different, and because the threshold is tuned on validation, small probability shifts move the threshold (0.38 → 0.36) and change 8% of the decisions. The difference in AUC (+0.013) is about twice the seed SD of E2, so it is ordinary run-to-run noise rather than a bug. Consequence for the thesis: **a single run is not enough evidence**; all following comparisons use three seeds and paired statistics. A slower, more repeatable run is possible with *--deterministic*.

5.8.4 E2 — Three-seed evaluation

Configuration	Test set	AUC	Accuracy	Balanced acc.	Real recognised	Fake detected	ECE
base12	ALL	0.933 ± 0.006	0.815 ± 0.012	0.841 ± 0.008	0.902 ± 0.003	0.780 ± 0.017	0.197 ± 0.020
	BioDeepAV	0.982 ± 0.004	0.933 ± 0.019	0.934 ± 0.021	0.921 ± 0.007	0.948 ± 0.040	0.102 ± 0.028
	FF++	0.880 ± 0.003	0.752 ± 0.008	0.799 ± 0.009	0.866 ± 0.022	0.731 ± 0.011	0.293 ± 0.012
hardfix12	ALL	0.931 ± 0.009	0.826 ± 0.034	0.846 ± 0.020	0.893 ± 0.025	0.799 ± 0.054	0.199 ± 0.011
	BioDeepAV	0.984 ± 0.002	0.943 ± 0.011	0.945 ± 0.011	0.921 ± 0.020	0.969 ± 0.026	0.088 ± 0.009
	FF++	0.877 ± 0.010	0.764 ± 0.048	0.796 ± 0.017	0.841 ± 0.037	0.750 ± 0.063	0.295 ± 0.012
harddro12	ALL	0.948 ± 0.002	0.856 ± 0.024	0.871 ± 0.014	0.908 ± 0.010	0.835 ± 0.038	0.123 ± 0.020
	BioDeepAV	0.983 ± 0.001	0.944 ± 0.009	0.944 ± 0.009	0.939 ± 0.005	0.949 ± 0.014	0.051 ± 0.016
	FF++	0.902 ± 0.004	0.810 ± 0.034	0.826 ± 0.009	0.850 ± 0.034	0.802 ± 0.046	0.186 ± 0.020
base20	ALL	0.942 ± 0.001	0.834 ± 0.015	0.852 ± 0.005	0.895 ± 0.020	0.810 ± 0.029	0.164 ± 0.006
	BioDeepAV	0.974 ± 0.010	0.899 ± 0.020	0.897 ± 0.024	0.921 ± 0.022	0.873 ± 0.071	0.104 ± 0.010
	FF++	0.902 ± 0.002	0.800 ± 0.014	0.819 ± 0.008	0.846 ± 0.034	0.792 ± 0.023	0.229 ± 0.009
base30	ALL	0.938 ± 0.004	0.836 ± 0.007	0.853 ± 0.010	0.893 ± 0.020	0.813 ± 0.004	0.124 ± 0.012
	BioDeepAV	0.960 ± 0.010	0.886 ± 0.024	0.884 ± 0.024	0.905 ± 0.028	0.864 ± 0.033	0.083 ± 0.003
	FF++	0.916 ± 0.002	0.810 ± 0.011	0.835 ± 0.005	0.872 ± 0.006	0.798 ± 0.014	0.181 ± 0.016

Table 5.93: Test results per configuration, mean ± SD over seeds 42, 123 and 2026 (1,602 test videos; validation threshold and temperature of each run).

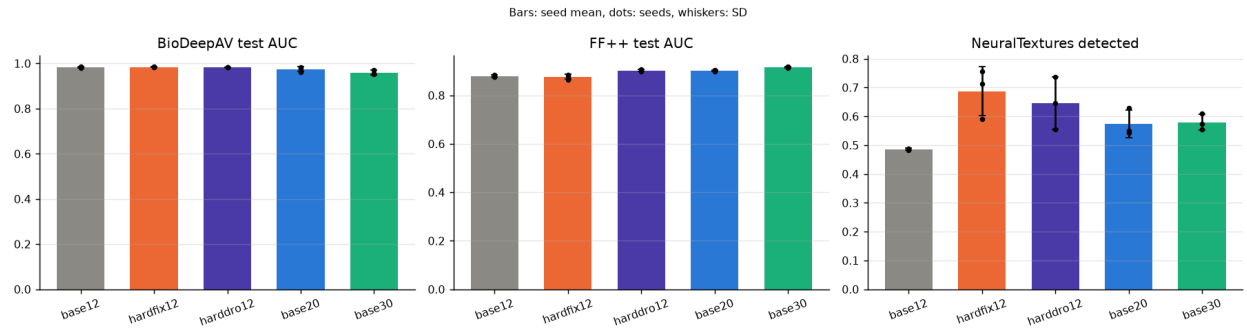

Figure 5.52: BioDeepAV AUC, FF++ AUC and NeuralTextures detection per configuration (bars: seed mean; dots: seeds; whiskers: SD).

Interpretation. The basic-retest result is stable: over three seeds the 151026 settings (base12) reach AUC 0.933 ± 0.006, BioDeepAV 0.982 ± 0.004 and FF++ 0.880 ± 0.003, close to the single run of Section 5.7 (0.927 / 0.979 / 0.874). The SD of the AUC is small (≤ 0.01), but threshold-dependent values move more: fake detected on BioDeepAV varies by ± 4 points. NeuralTextures stays at chance

(48.6%) in every base12 seed, so this weakness is a property of the method, not of one unlucky run.

5.8.5 E3 and E4 — Hard-type sampling and longer training

Test folder	base12	hardfix12	harddro12	base20	base30
BioDeepAV/real	92.1 ± 0.7	92.1 ± 2.0	93.9 ± 0.5	92.1 ± 2.2	90.5 ± 2.8
BioDeepAV/fake	94.8 ± 4.0	96.9 ± 2.6	94.9 ± 1.4	87.3 ± 7.1	86.4 ± 3.3
FF++/original	86.6 ± 2.2	84.1 ± 3.7	85.0 ± 3.4	84.6 ± 3.4	87.2 ± 0.6
FF++/DeepFakeDetection	100.0 ± 0.0	99.5 ± 0.9	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
FF++/Deepfakes	89.2 ± 1.3	81.3 ± 6.7	91.9 ± 3.4	91.7 ± 2.1	92.7 ± 1.1
FF++/Face2Face	70.7 ± 1.6	72.2 ± 7.4	76.6 ± 4.1	74.0 ± 1.5	76.4 ± 3.1
FF++/FaceShifter	75.6 ± 1.6	78.7 ± 4.8	84.1 ± 2.8	85.0 ± 2.5	83.3 ± 2.0
FF++/FaceSwap	71.1 ± 4.7	64.8 ± 6.7	76.2 ± 5.6	79.9 ± 1.8	81.1 ± 1.1
FF++/NeuralTextures	**48.6 ± 0.4**	68.7 ± 8.5	64.6 ± 9.1	**57.3 ± 4.8**	**57.9 ± 2.8**

Table 5.94: Accuracy per test folder (%, seed mean ± SD). Green / red: at least 5 points better / worse than base12; bold: below 60%.

Test set	Comparison	Metric	Mean Δ	Seeds 42 / 123 / 2026	95% CI	Verdict
ALL	hardfix12 − base12	recall NeuralTextures	+0.201	+0.2683 / +0.1098 / +0.2256	[+0.150, +0.257]	supported improvement
ALL	hardfix12 − base12	recall FaceSwap	-0.063	-0.0488 / -0.1220 / -0.0183	[-0.102, -0.019]	supported deterioration
ALL	harddro12 − base12	AUC	+0.015	+0.0102 / +0.0154 / +0.0192	[+0.005, +0.031]	supported improvement
ALL	harddro12 − base12	fake detected	+0.055	+0.0783 / +0.0642 / +0.0237	[+0.038, +0.073]	supported improvement
ALL	harddro12 − base12	balanced acc.	+0.031	+0.0381 / +0.0364 / +0.0172	[+0.018, +0.044]	supported improvement
ALL	harddro12 − base12	recall NeuralTextures	+0.161	+0.2500 / +0.1646 / +0.0671	[+0.117, +0.204]	supported improvement
ALL	harddro12 − base12	recall Face2Face	+0.059	+0.1098 / +0.0610 / +0.0061	[+0.023, +0.096]	supported improvement
ALL	harddro12 − base12	recall FaceShifter	+0.085	+0.0976 / +0.1098 / +0.0488	[+0.052, +0.122]	supported improvement
ALL	harddro12 − base12	recall FaceSwap	+0.051	+0.0610 / +0.0488 / +0.0427	[+0.015, +0.091]	supported improvement
ALL	base20 − base12	recall NeuralTextures	+0.087	+0.0549 / +0.0671 / +0.1402	[+0.048, +0.131]	supported improvement
ALL	base20 − base12	recall Face2Face	+0.033	+0.0366 / +0.0305 / +0.0305	[+0.002, +0.064]	supported improvement
ALL	base20 − base12	recall FaceShifter	+0.093	+0.0732 / +0.0915 / +0.1159	[+0.059, +0.131]	supported improvement
ALL	base20 − base12	recall FaceSwap	+0.087	+0.0183 / +0.0976 / +0.1463	[+0.059, +0.122]	supported improvement
ALL	base30 − base12	recall NeuralTextures	+0.093	+0.0854 / +0.1280 / +0.0671	[+0.047, +0.139]	supported improvement
ALL	base30 − base12	recall Face2Face	+0.057	+0.0915 / +0.0732 / +0.0061	[+0.022, +0.095]	supported improvement

Test set	Comparison	Metric	Mean Δ	Seeds 42 / 123 / 2026	95% CI	Verdict
ALL	base30 − base12	recall FaceShifter	+0.077	+0.0732 / +0.1098 / +0.0488	[+0.032, +0.126]	supported improvement
ALL	base30 − base12	recall FaceSwap	+0.100	+0.0549 / +0.1159 / +0.1280	[+0.066, +0.140]	supported improvement
FF++	harddro12 − base12	AUC	+0.022	+0.0181 / +0.0214 / +0.0271	[+0.012, +0.033]	supported improvement
FF++	harddro12 − base12	fake detected	+0.071	+0.1065 / +0.0725 / +0.0340	[+0.051, +0.093]	supported improvement
FF++	harddro12 − base12	balanced acc.	+0.027	+0.0349 / +0.0393 / +0.0078	[+0.009, +0.045]	supported improvement
FF++	base20 − base12	AUC	+0.022	+0.0162 / +0.0269 / +0.0229	[+0.013, +0.030]	supported improvement
FF++	base20 − base12	fake detected	+0.060	+0.0328 / +0.0566 / +0.0917	[+0.045, +0.077]	supported improvement
FF++	base20 − base12	balanced acc.	+0.020	+0.0317 / +0.0222 / +0.0062	[+0.004, +0.036]	supported improvement
FF++	base30 − base12	AUC	+0.036	+0.0314 / +0.0419 / +0.0351	[+0.024, +0.048]	supported improvement
FF++	base30 − base12	fake detected	+0.067	+0.0589 / +0.0861 / +0.0566	[+0.047, +0.090]	supported improvement
FF++	base30 − base12	balanced acc.	+0.037	+0.0386 / +0.0491 / +0.0222	[+0.017, +0.056]	supported improvement

Table 5.95: Supported differences to base12 (paired cluster bootstrap over source groups, 1,000 replicates; supported = CI excludes 0 and all three seeds agree in sign). The other 41 of 80 comparisons were inconclusive, including every BioDeepAV comparison.

Comparison	Δ BioDeepAV AUC	Seeds	95% CI	Verdict
hardfix12 − base12	+0.0013	-0.0020 / +0.0067 / -0.0008	[-0.0065, +0.0087]	inconclusive
harddro12 − base12	+0.0011	-0.0027 / +0.0059 / +0.0001	[-0.0111, +0.0158]	inconclusive
base20 − base12	-0.0078	-0.0026 / -0.0156 / -0.0052	[-0.0332, +0.0080]	inconclusive
base30 − base12	-0.0226	-0.0337 / -0.0223 / -0.0118	[-0.0589, +0.0012]	inconclusive

Table 5.96: BioDeepAV AUC differences. Longer training lowered BioDeepAV AUC in all three seeds (base30: −0.023) but the confidence interval still includes 0.

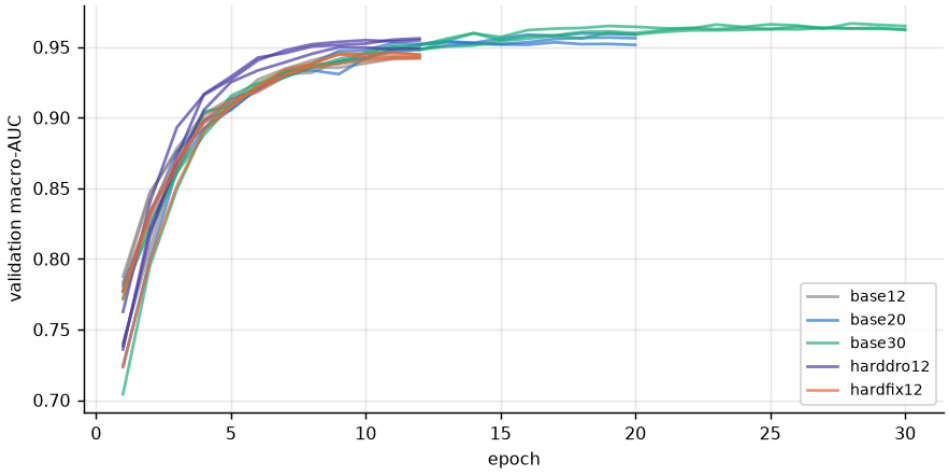

Figure 5.53: Validation macro-AUC per epoch for all 15 runs. harddro12 rises fastest; base20 / base30 keep improving slowly after epoch 12.

Configuration	Best epoch (42 / 123 / 2026)	Validation macro-AUC (seed mean)	Test AUC ALL	Test AUC FF++	Test AUC BioDeepAV	AURC (mean)
base12	11 / 12 / 11	0.9437	0.933 ± 0.006	0.880 ± 0.003	0.982 ± 0.004	0.0541
hardfix12	11 / 12 / 9	0.9443	0.931 ± 0.009	0.877 ± 0.010	0.984 ± 0.002	0.0553
harddro12	10 / 9 / 12	0.9536	0.948 ± 0.002	0.902 ± 0.004	0.983 ± 0.001	0.0466
base20	17 / 19 / 18	0.9566	0.942 ± 0.001	0.902 ± 0.002	0.974 ± 0.010	0.0416
base30	28 / 26 / 29	0.9649	0.938 ± 0.004	0.916 ± 0.002	0.960 ± 0.010	0.0483

Table 5.97: Best epochs, validation score and test AUC (seed mean ± SD); AURC = area under the risk–coverage curve (lower = better; Section 5.8.6).

Interpretation — sampling (E3). Re-weighting the folders by the old validation errors (*hardfix12*) is a trade-off, not an improvement: NeuralTextures recall rose by 20 points (48.6% → 68.7%), but FaceSwap fell by 6 points and Deepfakes by 8 points, and overall AUC did not change. The adaptive group-DRO sampler (*harddro12*) avoided this trade-off: it improved NeuralTextures (+16), FaceShifter (+9), Face2Face (+6) and FaceSwap (+5), raised overall AUC by 0.015 and balanced accuracy by 0.031, and left BioDeepAV unchanged (0.983). These are the only **supported** overall gains of the suite, and they were obtained without longer training.

Interpretation — training length (E4). Validation macro-AUC kept rising until epochs 17–29, and FF++ benefited (AUC 0.880 → 0.902 → 0.916; supported for AUC, fake detected and balanced accuracy). BioDeepAV did not: its AUC fell in all seeds (0.982 → 0.974 → 0.960), and fake BioDeepAV videos were detected less often (94.8% → 86.4%). These BioDeepAV drops are inconclusive under the bootstrap, but the pattern suggests that the extra epochs fit FF++ better at the cost of BioDeepAV. The 12-epoch schedule of the basic retest therefore stopped too early for FF++, but not for BioDeepAV.

5.8.6 E5 — Threshold, calibration and coverage

Four calibration methods were fitted on the validation predictions of each run (raw = no calibration; temperature scaling as in 151026; Platt scaling; isotonic regression), combined with three threshold rules (fixed 0.5; maximum mean balanced accuracy over the two datasets; Youden's J) and an abstention band whose width was chosen on validation to keep a target share of the videos (coverage 100%, 95%, 90% or 80%). Nothing was tuned on the test set.

Configuration	ECE raw	ECE temperature	ECE Platt	ECE isotonic	Brier raw	Brier isotonic
base12	0.192	0.197	0.065	0.035	0.166	0.087
hardfix12	0.191	0.199	0.051	0.036	0.157	0.086
harddro12	0.121	0.123	0.052	0.041	0.113	0.077
base20	0.154	0.164	0.073	0.044	0.146	0.086
base30	0.118	0.124	0.071	0.048	0.134	0.091

Table 5.98: Expected calibration error (ECE) and Brier score on the test videos, seed mean. Green: best per row.

Configuration	Calibration	Coverage 100%	Coverage 95%	Coverage 90%	Coverage 80%
base12	temperature	0.866 (100%)	0.880 (94%)	0.894 (89%)	0.920 (78%)
	isotonic	0.867 (100%)	0.868 (99%)	0.892 (91%)	0.909 (84%)
harddro12	temperature	0.885 (100%)	0.896 (95%)	0.909 (90%)	0.924 (80%)
	isotonic	0.885 (100%)	0.891 (98%)	0.908 (91%)	0.922 (82%)
base30	temperature	0.860 (100%)	0.874 (95%)	0.894 (90%)	0.926 (80%)
	isotonic	0.861 (100%)	0.862 (100%)	0.884 (92%)	0.895 (85%)

Table 5.99: Mean balanced accuracy over the two datasets on the decided test videos at the target coverage (actual test coverage in brackets); threshold = maximum mean balanced accuracy on validation; seed mean.

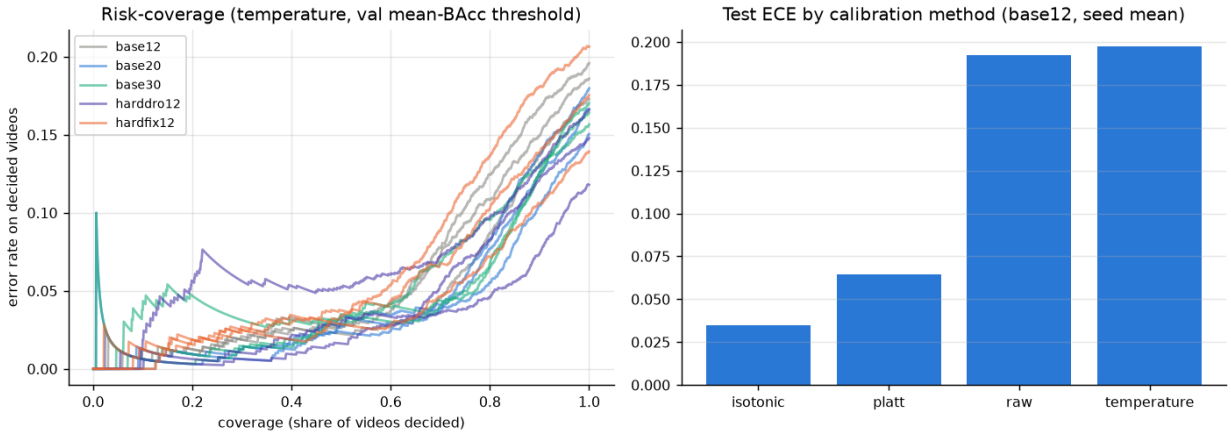

Figure 5.54: Left: risk–coverage curves of all 15 runs (temperature scaling, validation threshold); right: test ECE of base12 per calibration method.

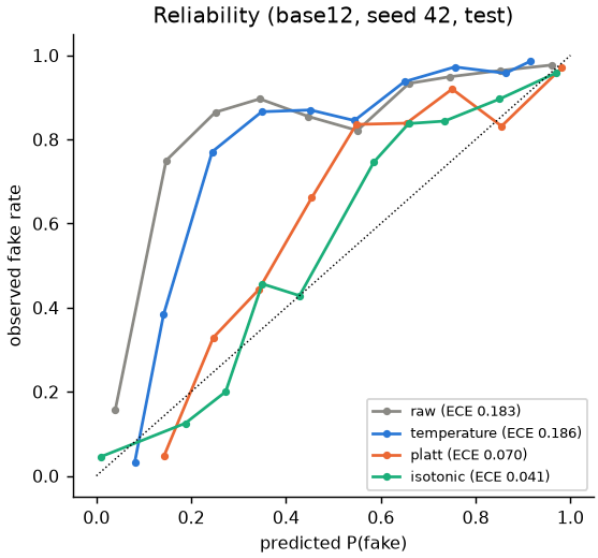

Figure 5.55: Reliability diagram (base12, seed 42, test videos). Raw and temperature-scaled probabilities are too low for fakes in the 0.1–0.4 range; isotonic regression follows the diagonal.

Interpretation. The probabilities of the basic retest are **badly calibrated** (ECE 0.19): many fake videos receive P(fake) of only 0.1–0.4. Temperature scaling, used in 151026, does not fix this because it can only stretch or compress the scale, and the problem is a shift between the validation-tuned

threshold and 0.5. Platt scaling (0.05–0.07) and especially isotonic regression (0.035–0.048) remove most of the miscalibration in every configuration. The threshold rule matters more than the calibration for balanced accuracy: with the validation threshold all methods reach 0.86–0.89 at full coverage. Allowing the detector to abstain helps steadily: at 90% coverage balanced accuracy rises to 0.89 (base12) and 0.91 (harddro12), and at 80% to 0.91–0.92. For a media-literacy interface this supports showing “uncertain” for about one video in ten and reporting calibrated (isotonic) probabilities.

5.8.7 E6 — XAI faithfulness: Grad-CAM++ vs Integrated Gradients vs occlusion

160 test videos of the base12 seed-42 model (75 predicted FAKE, 85 predicted REAL; sampled per dataset and label) were explained with four maps: Grad-CAM++ (as in 151026), Integrated Gradients (24 steps, blurred-video baseline), occlusion (grey patches) and a random map as the control. **Deletion:** the highest-ranked pixels are removed (grey or blurred) in steps of 5–75%; a faithful map makes the confidence of the predicted class fall quickly (low AUC). **Insertion:** the highest-ranked pixels are restored into a removed video; a faithful map makes the confidence rise quickly (high AUC). **Randomisation check:** the top layers are re-initialised; a map that barely changes (high Spearman ρ) reflects the image rather than the model.

Method	Pred.	n	Deletion AUC grey ↓	Deletion AUC blur ↓	Insertion AUC grey ↑	Insertion AUC blur ↑	Face-centre share	Random-model ρ ↓	Seconds per video
Grad-CAM++	FAKE	75	0.464	0.624	0.542	0.639	0.76	0.10	0.12
Grad-CAM++	REAL	85	0.759	0.589	0.858	0.848	0.85	—	0.26
Integr. Gradients	FAKE	75	0.371	0.448	0.260	0.578	0.62	0.74	1.70
Integr. Gradients	REAL	85	0.829	0.736	0.909	0.849	0.63	0.78	1.61
Occlusion	FAKE	75	0.375	0.458	0.499	0.625	0.64	—	0.71
Occlusion	REAL	85	0.662	0.727	0.742	0.866	0.70	-0.02	0.83
Random	FAKE	75	0.499	0.171	0.546	0.089	0.39	—	0.00
Random	REAL	85	0.436	0.894	0.404	0.913	0.39	—	0.00

Table 5.100: E6 faithfulness summary (mean over videos). ↓ lower is better, ↑ higher is better. Face-centre share = share of the map energy inside the central face disc. ρ: Spearman correlation between the maps of the trained and the randomised model (— = randomised map was empty). Red: fails the randomisation check.

Method	Pred.	Test (grey)	Mean gain vs random	95% CI	Videos better than random	Verdict
Grad-CAM++	REAL	deletion	-0.323	[-0.373, -0.271]	12%	worse than random
Grad-CAM++	REAL	insertion	+0.455	[+0.426, +0.484]	100%	faithful (better than random)
Grad-CAM++	FAKE	deletion	+0.035	[-0.032, +0.097]	51%	not distinguishable from random
Grad-CAM++	FAKE	insertion	-0.003	[-0.062, +0.055]	57%	not distinguishable from random
Integr. Gradients	REAL	deletion	-0.394	[-0.438, -0.348]	9%	worse than random
Integr. Gradients	REAL	insertion	+0.506	[+0.479, +0.535]	100%	faithful (better than random)
Integr. Gradients	FAKE	deletion	+0.128	[+0.069, +0.182]	75%	faithful (better than random)
Integr. Gradients	FAKE	insertion	-0.285	[-0.328, -0.242]	4%	worse than random

Method	Pred.	Test (grey)	Mean gain vs random	95% CI	Videos better than random	Verdict
Occlusion	REAL	deletion	-0.226	[-0.261, -0.193]	11%	worse than random
Occlusion	REAL	insertion	+0.338	[+0.309, +0.369]	98%	faithful (better than random)
Occlusion	FAKE	deletion	+0.124	[+0.071, +0.177]	68%	faithful (better than random)
Occlusion	FAKE	insertion	-0.047	[-0.092, +0.002]	49%	not distinguishable from random

Table 5.101: Comparison with the random map under grey perturbation (n = 85 REAL, 75 FAKE; gain = random AUC − method AUC for deletion, method − random for insertion; bootstrap over videos).

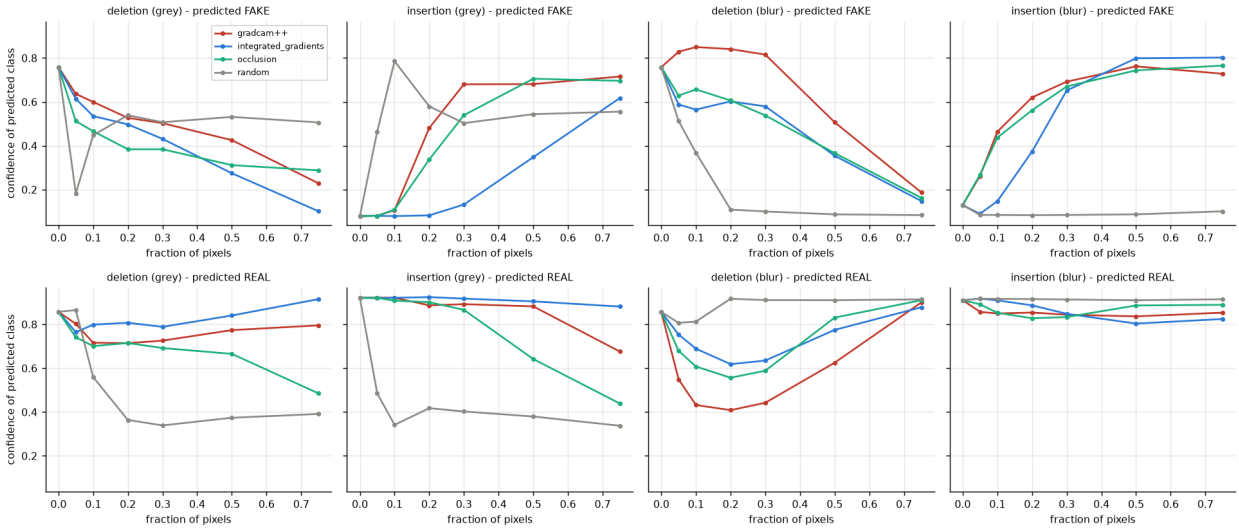

Figure 5.56: Mean deletion and insertion curves (confidence of the predicted class) for predicted FAKE (top) and predicted REAL (bottom), grey and blur perturbation.

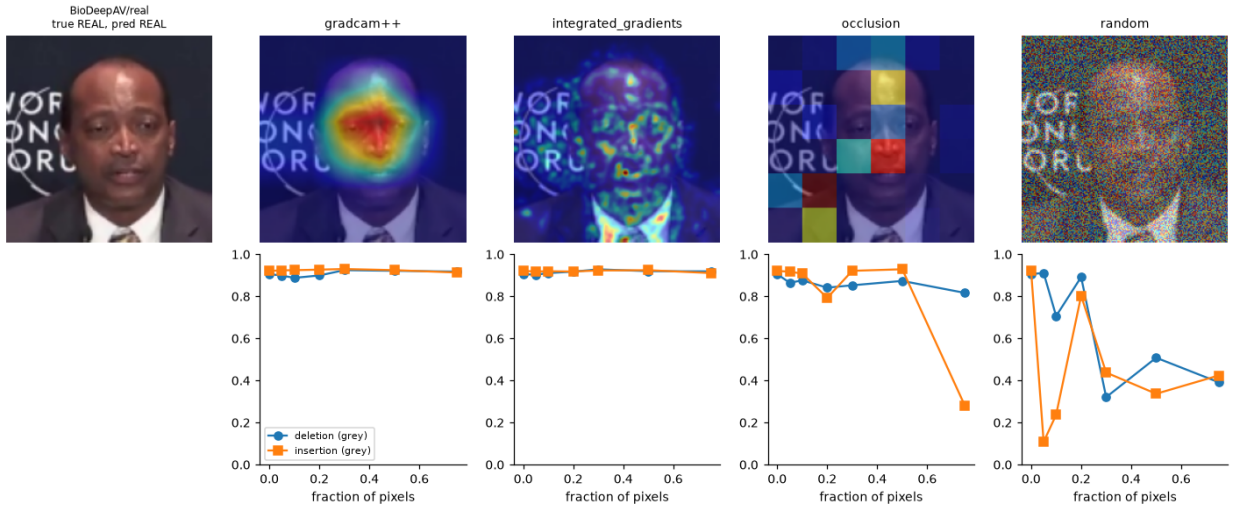

Figure 5.57: Example (BioDeepAV real video, predicted REAL): Grad-CAM++ is smooth and centred on the face; Integrated Gradients is a speckled edge-like map that also covers the background and tie; occlusion marks the mouth and eye patches.

Interpretation. The faithfulness of a map depends strongly on **how pixels are removed**. Random pixel masks are scattered, so grey removal adds salt-and-pepper noise that the model reads as a manipulation trace — the confidence of REAL decisions collapses after only 10–20% random grey

pixels. Blurring scattered pixels, in contrast, smooths the face texture everywhere and erases the fake evidence fastest. Because of these artefacts, “worse than random” results for one perturbation are reversed for the other (e.g. Grad-CAM++ REAL deletion: worse than random with grey, 0.589 vs 0.894 and clearly better with blur). The robust findings are: (1) **insertion** favours all three methods over random for REAL decisions under grey and for FAKE decisions under blur, but not in the other two combinations; (2) for FAKE decisions under grey perturbation, only Integrated Gradients and occlusion beat random in deletion (gain +0.13 and +0.12), and Grad-CAM++ is not distinguishable from random — consistent with Section 5.7, where Grad-CAM++ was faithful for REAL but not for FAKE decisions; (3) Integrated Gradients **fails the randomisation sanity check** ($\rho = 0.74$–0.78): its map hardly changes when the model is destroyed, so it mainly shows image edges; Grad-CAM++ ($\rho = 0.10$) and occlusion ($\rho = -0.02$) pass; (4) Grad-CAM++ concentrates on the face (76–85% of the energy in the central disc) and is the fastest (0.1–0.3 s per video vs 1.6–1.7 s for IG). For the user interface, Grad-CAM++ remains the main map, occlusion is the best second opinion for FAKE decisions, and Integrated Gradients should not be shown to users without the sanity warning.

5.8.8 Model selection and conclusions

The suite selects the final model by the highest seed-mean **validation** macro-AUC, without looking at the test set: *base30* (0.965), seed 42 (0.967, epoch 28), exported as *run_171026/checkpoints/best_model.pt*. On the test set, however, *harddro12* is the better all-round configuration (AUC 0.948 vs 0.938, balanced accuracy 0.871 vs 0.853, BioDeepAV 0.983 vs 0.960, ECE 0.123 vs 0.124). The validation set rewards the FF++ gains of longer training more than the BioDeepAV losses, which is a known limitation of selecting on a macro-average of two validation sets of different size.

Checklist item	Result	Status
1 Baseline reproduction	Split, groups and best epoch identical; AUC +0.013 and 8.1% changed decisions (GPU non-determinism)	Partly reproduced
2 Three-seed evaluation	base12: AUC 0.933 ± 0.006; BioDeepAV 0.982 ± 0.004; FF++ 0.880 ± 0.003; NeuralTextures 48.6% in every seed	Done
3 Hard-type balanced sampling	harddro12: supported gains (AUC +0.015, bal. acc. +0.031, NT +0.161, all hard types); hardfix12: trade-off only	Improvement (harddro12)
4 Longer training	FF++ AUC 0.880 → 0.902 → 0.916 (supported); BioDeepAV 0.982 → 0.960 (inconclusive drop)	FF++ only
5 Threshold and calibration	Isotonic ECE 0.19 → 0.035; 90% coverage: bal. acc. 0.87 → 0.89 (base12), 0.91 (harddro12)	Improvement
6 XAI faithfulness	Grad-CAM++ and occlusion pass sanity check, IG fails; only IG / occlusion beat random in FAKE deletion (grey); results depend on perturbation	Mixed

Table 5.102: Test 5.7 improvement checklist — outcome of 171026.py.

Recommendation for the final system. Train with the adaptive group-DRO sampler for 20 epochs (harddro + base20 were tested separately, their combination is the next experiment), select by a validation score that weights both datasets equally and checks that neither drops, calibrate with isotonic regression, abstain on about 10% of the videos (“uncertain”), and show Grad-CAM++ with an occlusion map as a second opinion for FAKE decisions.

Limitations. Three seeds give a rough SD only; the bootstrap is over source groups of one split, so the results do not cover a different split or other datasets; E6 used one model (base12, seed 42) and 160 videos; pixel-level deletion with random masks creates out-of-distribution inputs, so region-level tests (as in Section 5.7) remain needed; NeuralTextures is still detected in only 57–69% of the videos at best.

5.9 Test 6 — Adaptive Group-DRO Sampling with 20-Epoch Training (harddro20, 201026.py)

5.9.1 Overview

The improvement suite (Section 5.8) found two separate gains: the adaptive group-DRO sampler (*harddro12*) gave the best all-round detector, and the 20-epoch schedule (*base20*) improved FF++. Its recommendation was to test the combination. Test 6 (*201026.py*) trains exactly this combination, **harddro20** = group-DRO sampler + 20 epochs, patience 5, cosine schedule stretched to 20 epochs, with the three seeds 42, 123 and 2026. Nothing else changes: 151026.py and 171026.py are imported read-only, so model, preprocessing, augmentation, loss, optimiser, EMA, sampler, checkpoint rule, bootstrap and calibration code are byte-identical. The existing base12, harddro12 and base20 runs of Section 5.8 are re-scored without retraining, which gives a 2 × 2 design (sampler × schedule length) on the same 1,602 test videos. Four hypotheses with decision rules were fixed before evaluation. Test 6 also adds Holm-corrected p-values, a per-manipulation AUC analysis and a Grad-CAM++ evaluation (faithfulness and stability) on **every** run, not only on one model. Training took about 3 h and the XAI phase about 1.5 h on the RTX 3050 (run 20 Oct 2026).

Key result. The combination does **not** beat its parts. Pooled test AUC was **0.944 ± 0.007** for harddro20 vs 0.948 ± 0.002 for harddro12 and 0.942 ± 0.001 for base20 (both differences inconclusive). It gave the best FF++ AUC so far (**0.912**; supported vs both) and the best NeuralTextures AUC (0.861), but BioDeepAV fell to 0.967 ± 0.015 (supported deterioration vs harddro12, −0.016). It produced the lowest test ECE (0.094) but twice as many confidently wrong predictions, so abstaining helps it less (AURC 0.121 vs 0.047). Grad-CAM++ faithfulness and stability were non-inferior to harddro12. Verdicts: H1 not supported, H2 partially supported, H3 contradicted, H4 inconclusive. **harddro12 remains the recommended model.**

5.9.2 Design, hypotheses and data audit

Configuration	Source	Sampler	Epochs (patience)	Role
base12	existing (171026)	dataset × class balanced	12 (4)	Reference (151026 settings)
harddro12	existing (171026)	group-DRO over 9 dataset × folder cells	12 (4)	Sampler effect
base20	existing (171026)	dataset × class balanced	20 (5)	Schedule effect
harddro20	**new (201026)**	group-DRO (η 2.0, floor 0.5, momentum 0.5)	20 (5)	**Combination (Test 6)**

Table 5.103: The 2 × 2 design. harddro20 differs from harddro12 only in schedule length and from base20 only in the sampler.

	Hypothesis	Decision rule (fixed in advance)	Verdict
H1	harddro20 improves pooled test AUC vs harddro12 and vs base20	Supported improvement vs both	NOT SUPPORTED (inconclusive)
H2	harddro20 improves recall and AUC on NeuralTextures, FaceSwap, Face2Face, FaceShifter	Count supported improvements / deteriorations over 4 types × 2 metrics × 2 comparators	PARTIALLY SUPPORTED (4 / 16, 0 worse)
H3	BioDeepAV AUC maintained while FF++ AUC improves	BioDeepAV: CI lower bound > −0.01 and no supported deterioration; FF++: supported improvement, vs both	CONTRADICTED
H4	Grad-CAM++ faithfulness and stability are non-inferior	CI lower bound of the difference > −0.05, vs both	INCONCLUSIVE (7 / 8 non-inferior)

Table 5.104: Pre-specified hypotheses and outcome. "Supported" = 95% paired cluster-bootstrap CI excludes 0 and all three seeds agree in sign (the 171026 rule); Holm-adjusted p-values over the 56 primary comparisons are reported in addition.

Data and leakage audit. Before training, a preflight audit checked all 24 conditions and stopped the script on any failure. All passed: the split file is identical to the one used in Sections 5.7 and 5.8 (SHA-256 c0b494ad…) and was re-derived from scratch with identical groups; no group, identity token or exact duplicate occurs in more than one partition; labels and datasets are correct; the epoch-1 DRO weights are uniform over the nine training cells; and every one of the 12 runs uses the same validation and test videos, labels, configuration and calibration rule. No degraded videos are present, so robustness to degradation was not evaluated (as in Section 5.8).

5.9.3 Training

Run	Epochs completed	Best epoch	Val macro-AUC	Val AUC BioDeepAV	Val AUC FF++	Temperature	Threshold
base12_s42	12	11	0.9464	0.9876	0.9052	1.49	0.36
base12_s123	12	12	0.9420	0.9828	0.9011	1.49	0.37
base12_s2026	12	11	0.9428	0.9841	0.9015	1.49	0.39
harddro12_s42	12	10	0.9549	0.9793	0.9304	1.13	0.28
harddro12_s123	12	9	0.9498	0.9878	0.9117	1.03	0.50
harddro12_s2026	12	12	0.9561	0.9816	0.9306	1.13	0.54
base20_s42	20	17	0.9533	0.9916	0.9151	1.38	0.35
base20_s123	20	19	0.9567	0.9887	0.9248	1.32	0.37
base20_s2026	20	18	0.9596	0.9890	0.9302	1.45	0.28
harddro20_s42	20	20	**0.9642**	0.9830	0.9454	1.19	0.55
harddro20_s123	20	18	**0.9633**	0.9831	0.9436	1.19	0.67
harddro20_s2026	20	18	**0.9590**	0.9816	0.9364	1.32	0.61

Table 5.105: All 12 runs (bold: new harddro20 runs). No run stopped early. Temperature and threshold are fitted on validation.

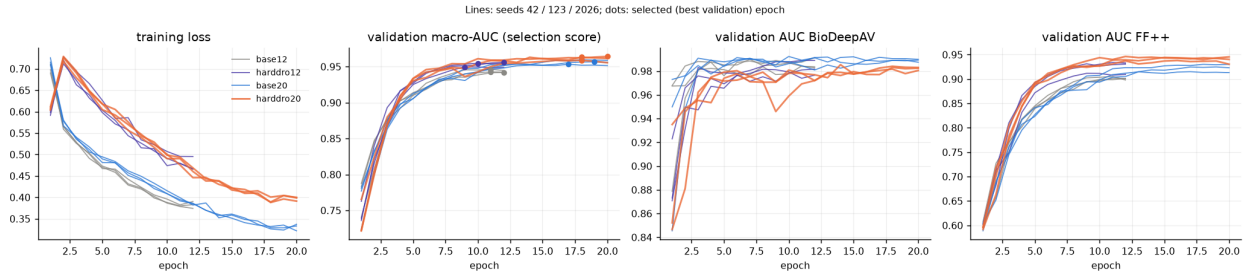

Figure 5.58: Training loss and validation AUC per epoch (lines: seeds; dots: selected epoch). The DRO runs have a higher training loss because they draw the hard cells more often.

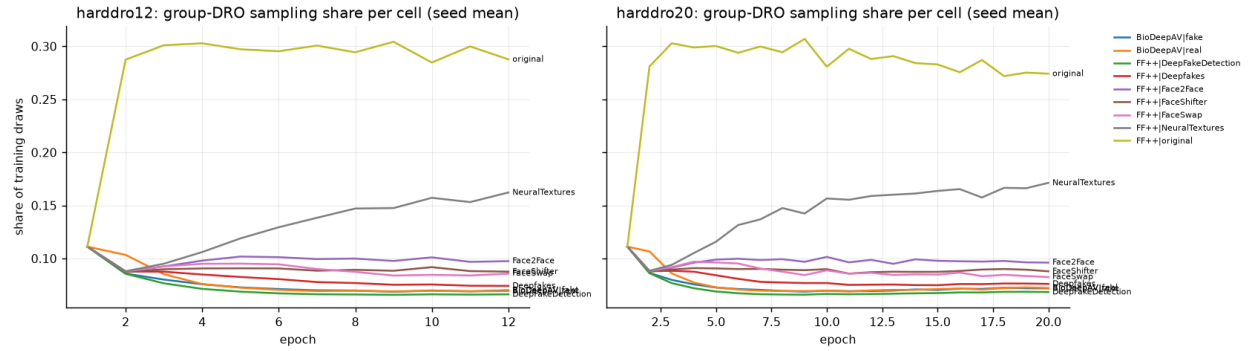

Figure 5.59: Group-DRO share of training draws per cell (seed mean). Both DRO schedules give the FF++ originals ≈ 0.28–0.30 and NeuralTextures up to 0.17 of the draws, while each BioDeepAV cell falls to ≈ 0.07.

Interpretation. harddro20 reached the highest validation macro-AUC of all configurations (0.962 ± 0.003), driven by FF++ (0.936–0.945), while its validation BioDeepAV AUC (0.982–0.983) stayed below base20 (0.989–0.992). The sampling shares explain the trade-off: the DRO sampler moves most of the training effort to FF++ originals and NeuralTextures, and with 20 epochs the two BioDeepAV cells are seen at only about 7% each for eight more epochs. The validation curves of harddro20 for BioDeepAV are also the most unstable (one seed dips to 0.946 at epoch 9).

5.9.4 Detection results

Configuration	Test set	AUC	Balanced acc.	Real recognised	Fake detected	F1	ECE
base12	ALL	0.933 ± 0.006	0.841 ± 0.008	0.902 ± 0.003	0.780 ± 0.017	0.857 ± 0.010	0.197 ± 0.020
	BioDeepAV	0.982 ± 0.004	0.934 ± 0.021	0.921 ± 0.007	0.948 ± 0.040	0.928 ± 0.022	0.102 ± 0.028
	FF++	0.880 ± 0.003	0.799 ± 0.009	0.866 ± 0.022	0.731 ± 0.011	0.833 ± 0.006	0.293 ± 0.012
harddro12	ALL	0.948 ± 0.002	0.871 ± 0.014	0.908 ± 0.010	0.835 ± 0.038	0.891 ± 0.021	0.123 ± 0.020
	BioDeepAV	0.983 ± 0.001	0.944 ± 0.009	0.939 ± 0.005	0.949 ± 0.014	0.939 ± 0.009	0.051 ± 0.016
	FF++	0.902 ± 0.004	0.826 ± 0.009	0.850 ± 0.034	0.802 ± 0.046	0.876 ± 0.025	0.186 ± 0.020
base20	ALL	0.942 ± 0.001	0.852 ± 0.005	0.895 ± 0.020	0.810 ± 0.029	0.874 ± 0.014	0.164 ± 0.006
	BioDeepAV	0.974 ± 0.010	0.897 ± 0.024	0.921 ± 0.022	0.873 ± 0.071	0.887 ± 0.029	0.104 ± 0.010
	FF++	0.902 ± 0.002	0.819 ± 0.008	0.846 ± 0.034	0.792 ± 0.023	0.870 ± 0.011	0.229 ± 0.009
harddro20	ALL	**0.944 ± 0.007**	**0.863 ± 0.026**	**0.903 ± 0.004**	**0.822 ± 0.056**	**0.882 ± 0.032**	**0.094 ± 0.025**
	BioDeepAV	**0.967 ± 0.015**	**0.901 ± 0.052**	**0.930 ± 0.003**	**0.873 ± 0.101**	**0.891 ± 0.060**	**0.069 ± 0.027**
	FF++	**0.912 ± 0.004**	**0.831 ± 0.018**	**0.854 ± 0.012**	**0.807 ± 0.046**	**0.880 ± 0.027**	**0.140 ± 0.024**

Table 5.106: Test results (mean ± SD over three seeds, validation threshold and temperature of each run).

Test set	Metric	harddro20 − harddro12	Verdict	harddro20 − base20	Verdict
ALL	AUC	-0.004 [-0.012, +0.005]	inconclusive	+0.002 [-0.006, +0.015]	inconclusive
	balanced acc.	-0.009 [-0.026, +0.008]	inconclusive	+0.010 [-0.006, +0.031]	inconclusive
	fake detected	-0.013 [-0.045, +0.017]	inconclusive	+0.012 [-0.020, +0.046]	inconclusive
	real recognised	-0.004 [-0.020, +0.013]	inconclusive	+0.009 [-0.009, +0.037]	inconclusive
BioDeep AV	AUC	-0.016 [-0.049, -0.002]	supported deterioration	-0.007 [-0.049, +0.022]	inconclusive
	balanced acc.	-0.042 [-0.094, +0.014]	inconclusive	+0.004 [-0.056, +0.070]	inconclusive
	fake detected	-0.076 [-0.177, +0.031]	inconclusive	-0.000 [-0.127, +0.126]	inconclusive
	real recognised	-0.009 [-0.033, +0.011]	inconclusive	+0.009 [-0.011, +0.072]	inconclusive
FF++	AUC	+0.010 [+0.002, +0.019]	supported improvement	+0.011 [+0.001, +0.021]	supported improvement
	balanced acc.	+0.005 [-0.012, +0.022]	inconclusive	+0.012 [-0.002, +0.026]	inconclusive
	fake detected	+0.005 [-0.016, +0.027]	inconclusive	+0.016 [-0.000, +0.034]	inconclusive
	real recognised	+0.004 [-0.022, +0.030]	inconclusive	+0.008 [-0.017, +0.034]	inconclusive

Table 5.107: Primary paired comparisons (seed-mean difference [95% cluster-bootstrap CI]). None of the 56 primary comparisons is significant after Holm correction (smallest p_Holm = 0.056).

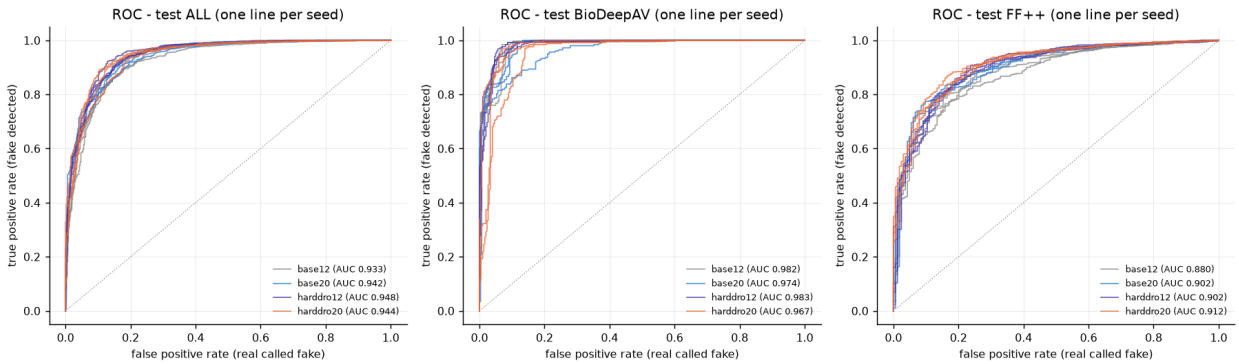

Figure 5.60: Test ROC curves per dataset (seeds pooled).

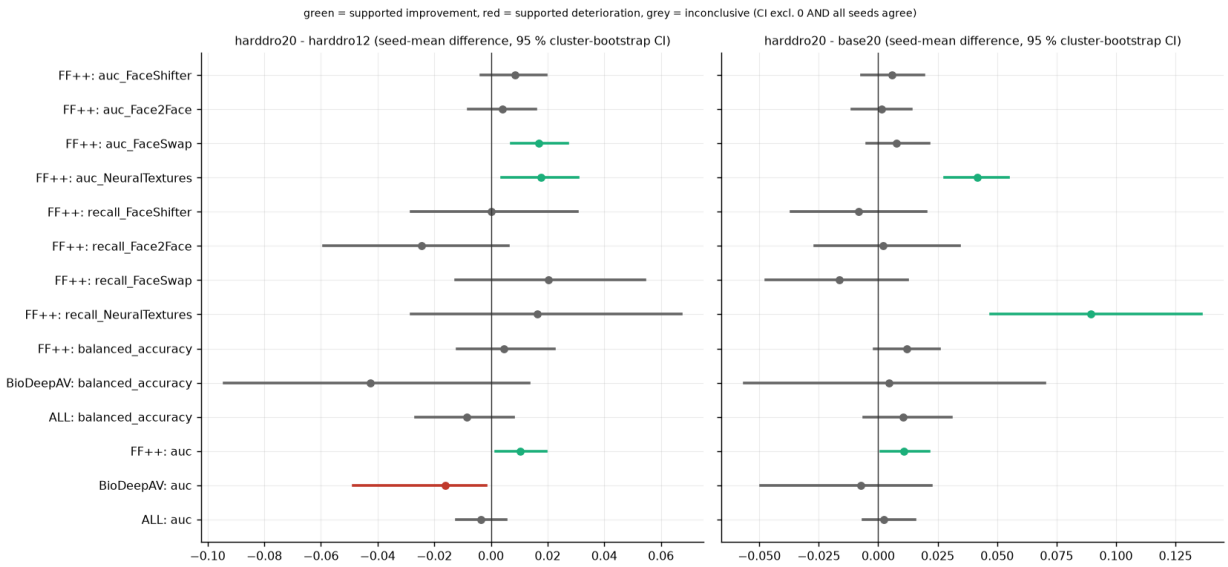

Figure 5.61: Paired differences of harddro20 to harddro12 (left) and base20 (right) with 95% cluster-bootstrap CI. Green: supported improvement; red: supported deterioration; grey: inconclusive.

Interpretation (H1, H3). The two effects did not add up. Pooled AUC (0.944) lies between base20 and harddro12 and neither difference is supported, so **H1 is not supported**. The combination does what longer training did in Section 5.8, only more strongly: FF++ AUC rose to 0.912, a supported gain over both parents (+0.010 and +0.011), but BioDeepAV AUC fell to 0.967 with a much larger seed SD than harddro12 (0.015 vs 0.001), and the drop vs harddro12 is supported (−0.016, CI −0.049 to −0.002). Fake BioDeepAV videos were detected in only 87.3% (harddro12: 94.9%). Because BioDeepAV was not maintained, **H3 is contradicted**. Against base12, harddro20 still keeps supported gains (pooled AUC +0.011, balanced accuracy +0.022), so it is a valid improvement over the basic retest, but not over harddro12.

5.9.5 Per-manipulation results (H2)

Folder (n per seed)	base12 recall	harddro12 recall	base20 recall	harddro20 recall	base12 AUC	harddro12 AUC	base20 AUC	harddro20 AUC
FF++/NeuralTextures (164)	48.6	64.6	57.3	**66.3**	0.787	0.843	0.819	**0.861**
FF++/FaceSwap (164)	71.1	76.2	**79.9**	78.3	0.869	0.889	0.898	**0.906**
FF++/Face2Face (164)	70.7	**76.6**	74.0	74.2	0.863	0.880	0.882	**0.884**
FF++/FaceShifter (164)	75.6	84.1	**85.0**	84.1	0.893	0.914	0.917	**0.923**
FF++/Deepfakes (164)	89.2	91.9	91.7	**93.5**	0.940	0.947	0.956	**0.959**
FF++/DeepFakeDetection (63)	**100.0**	**100.0**	**100.0**	**100.0**	**0.996**	0.993	0.993	0.987
BioDeepAV/fake (254)	94.8	**94.9**	87.3	87.3	0.982	**0.983**	0.974	0.967
FF++/original (164) *	86.6	85.0	84.6	85.4	—	—	—	—
BioDeepAV/real (301) *	92.1	93.9	92.1	93.0	—	—	—	—

*Table 5.108: Per manipulation (seed mean). Recall in % at the validation threshold; AUC against the real test videos of the same dataset. Bold: best per row. * real folders: specificity (real recognised).*

Hard type	Metric	vs harddro12	vs base20
NeuralTextures	recall	+0.016 [-0.028, +0.067]	+0.089 [+0.047, +0.136]
	AUC	+0.018 [+0.004, +0.031]	+0.042 [+0.028, +0.055]

Hard type	Metric	vs harddro12	vs base20
FaceSwap	recall	+0.020 [-0.013, +0.054]	-0.016 [-0.047, +0.012]
	AUC	+0.017 [+0.007, +0.027]	+0.008 [-0.005, +0.021]
Face2Face	recall	-0.024 [-0.059, +0.006]	+0.002 [-0.027, +0.034]
	AUC	+0.004 [-0.008, +0.016]	+0.001 [-0.011, +0.014]
FaceShifter	recall	+0.000 [-0.028, +0.030]	-0.008 [-0.037, +0.020]
	AUC	+0.008 [-0.004, +0.020]	+0.006 [-0.007, +0.019]

Table 5.109: H2 comparisons on the four hard FF++ types (difference [95% CI]; green = supported improvement). 4 of 16 supported, 0 worse.

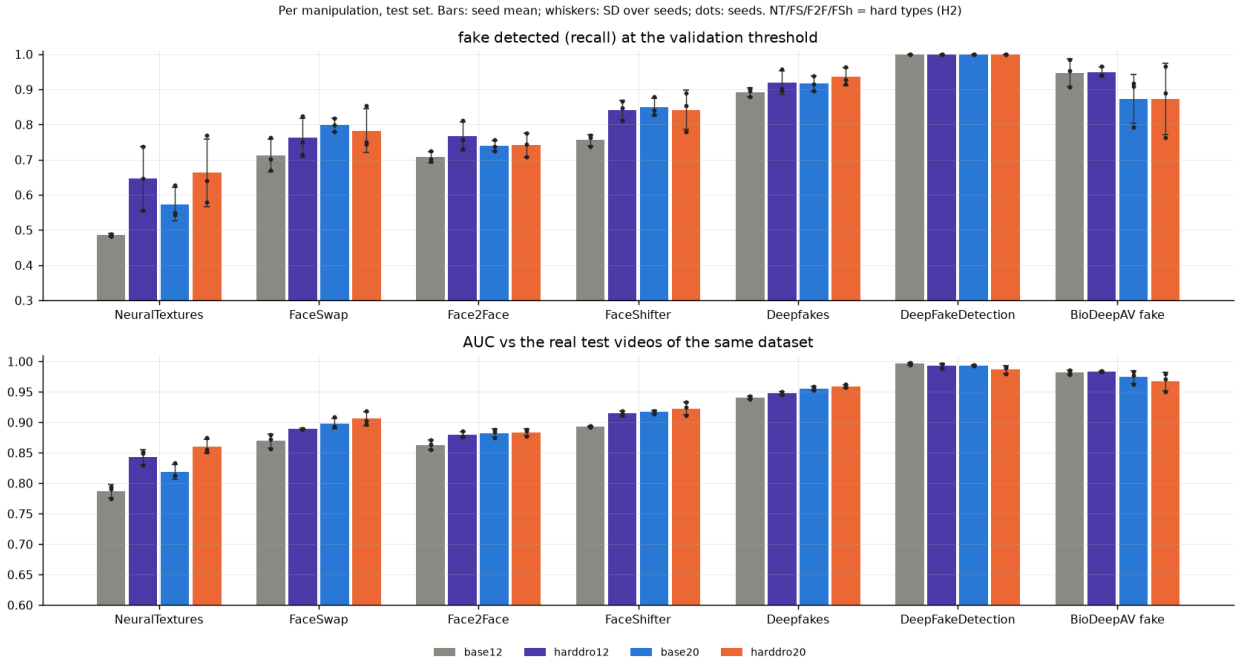

Figure 5.62: Recall (top) and AUC against real videos (bottom) per manipulation; bars: seed mean, dots: seeds, whiskers: SD.

Interpretation. harddro20 has the highest AUC for NeuralTextures (0.861), FaceSwap (0.906), Face2Face (0.884), FaceShifter (0.923) and Deepfakes (0.959), i.e. it ranks the hard fakes better than any earlier model. At the decision threshold, however, recall is not consistently higher: NeuralTextures 66.3% (harddro12 64.6%) varies between seeds from 58% to 77%. The supported gains are AUC on NeuralTextures and FaceSwap vs harddro12, and recall and AUC on NeuralTextures vs base20; there is no supported deterioration, so **H2 is partially supported**. NeuralTextures remains the weakest type in every configuration.

5.9.6 Calibration, confident errors and abstention

Configuration	ECE raw	ECE temperature	ECE isotonic	Bal. acc. 100%	Bal. acc. 90%	Bal. acc. 80%	AURC ↓
base12	0.192	0.197	0.035	0.867	0.892	0.909	0.054 ± 0.007
harddro12	0.121	0.123	0.041	0.885	0.908	0.922	0.047 ± 0.019
base20	0.154	0.164	0.044	0.860	0.889	0.906	0.042 ± 0.002
harddro20	0.091	0.094	0.048	0.867	0.899	0.924	0.121 ± 0.046

Table 5.110: Calibration and abstention (seed mean; calibrators and thresholds fitted on validation). Balanced accuracy = mean over the two datasets with isotonic calibration at 100 / 90 / 80% target coverage. AURC: area under the risk–coverage curve.

Configuration	Fake videos with P(fake) < 0.1	Real videos with P(fake) > 0.9	Per seed (42 / 123 / 2026)
harddro12	23.0 ± 9.6	12.0 ± 2.6	34/16/19 fakes; 11/10/15 reals
base20	26.0 ± 12.1	7.7 ± 0.6	13/37/28 fakes; 8/7/8 reals
harddro20	47.3 ± 8.7	23.7 ± 1.5	40/57/45 fakes; 24/25/22 reals

Table 5.111: Confidently wrong test predictions (temperature-scaled P(fake); 1,137 fake and 465 real test videos per seed). Additional analysis of the stored predictions.csv files, not part of REPORT_201026.md.

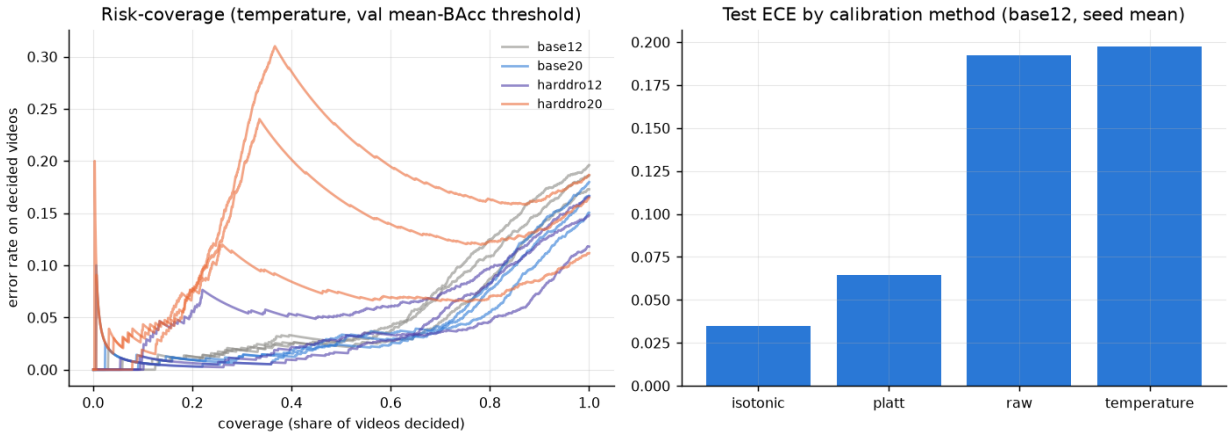

Figure 5.63: Left: risk–coverage curves of all 12 runs; harddro20 (orange) makes many errors among its most confident decisions. Right: test ECE of base12 per calibration method.

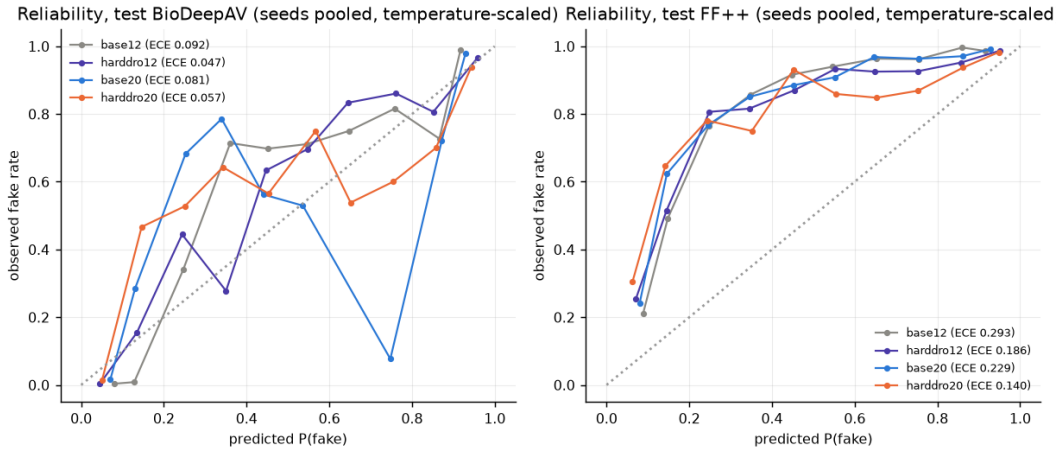

Figure 5.64: Reliability diagrams per dataset (seeds pooled, temperature-scaled).

Interpretation. harddro20 has the lowest ECE without extra calibration (0.094 after temperature scaling; FF++ 0.140 vs 0.186–0.293), because its higher thresholds (0.55–0.67) fit its probabilities better. Isotonic calibration brings all four configurations to 0.035–0.048, so this advantage disappears once calibration is applied. More important for a user interface is the **ranking** of confidence: harddro20 gives a P(fake) below 0.1 to about 47 fake videos per seed (harddro12: 23) and above 0.9 to about 24 real videos (harddro12: 12). These confident errors sit exactly where an “uncertain” band cannot catch them, so the AURC is 2.6 times worse (0.121 vs 0.047), and at 90% coverage harddro20

reaches only 0.899 balanced accuracy vs 0.908 for harddro12. Longer DRO training makes the model more extreme, not more reliable.

5.9.7 Explainability: Grad-CAM++ faithfulness and stability (H4)

Grad-CAM++ was evaluated on all 12 runs with the 160 E6 test videos of Section 5.8.7. **Faithfulness:** deletion and insertion gain over a random map on the same video (grey and blur fill; > 0 = better than random). **Stability:** the map after each of six mild photometric probes (JPEG q60, blur σ 1, downscale 0.6, brightness +0.10, contrast 1.15, saturation 0.80) is compared with the clean map (top-20% IoU, Pearson r); the probe flip rate is the share of decisions that change. Stability is not faithfulness, and no localisation ground truth exists.

Configuration	Deletion gain grey	Insertion gain grey	Deletion gain blur	Insertion gain blur	Stability IoU	Stability r	Probe flip rate
base12	-0.121 [-0.177, -0.061]	+0.206 [+0.158, +0.249]	-0.017 [-0.089, +0.051]	+0.208 [+0.144, +0.287]	0.830 [0.813, 0.847]	0.956 [0.948, 0.964]	0.067 [0.042, 0.094]
harddro12	-0.092 [-0.160, -0.025]	+0.230 [+0.165, +0.286]	-0.038 [-0.089, +0.015]	+0.217 [+0.161, +0.277]	0.839 [0.820, 0.859]	0.955 [0.945, 0.966]	0.060 [0.041, 0.085]
base20	-0.125 [-0.193, -0.045]	+0.248 [+0.181, +0.308]	-0.001 [-0.082, +0.066]	+0.235 [+0.161, +0.326]	0.814 [0.799, 0.830]	0.946 [0.937, 0.954]	0.056 [0.033, 0.079]
harddro20	-0.073 [-0.104, -0.041]	+0.235 [+0.182, +0.274]	-0.013 [-0.068, +0.037]	+0.259 [+0.211, +0.308]	0.815 [0.796, 0.837]	0.939 [0.927, 0.952]	0.047 [0.030, 0.065]

Table 5.112: Grad-CAM++ evaluation per configuration (mean over 160 videos and three seeds [95% cluster-bootstrap CI]).

Metric	harddro20 − harddro12	Verdict	harddro20 − base20	Verdict
deletion gain grey	+0.019 [-0.040, +0.071]	non-inferior	+0.051 [-0.027, +0.113]	non-inferior
insertion gain grey	+0.005 [-0.038, +0.053]	non-inferior	-0.013 [-0.078, +0.067]	inconclusive
deletion gain blur	+0.025 [-0.016, +0.061]	non-inferior	-0.012 [-0.058, +0.039]	inconclusive
insertion gain blur	+0.042 [-0.003, +0.087]	non-inferior	+0.024 [-0.053, +0.084]	inconclusive
stability iou	-0.023 [-0.030, -0.014]	non-inferior	+0.001 [-0.015, +0.018]	non-inferior
stability pearson	-0.016 [-0.021, -0.009]	non-inferior	-0.007 [-0.017, +0.004]	non-inferior

Table 5.113: H4 non-inferiority tests (margin 0.05: non-inferior if the CI lower bound is above −0.05).

Probe	base12 IoU	harddro12 IoU	base20 IoU	harddro20 IoU	base12 flips	harddro12 flips	base20 flips	harddro20 flips
blur_s1.0	0.772	0.790	0.759	0.771	9.0%	8.3%	7.7%	6.2%
brightness_+0.10	0.888	0.892	0.875	0.868	4.2%	3.5%	2.7%	3.8%
contrast_1.15	0.875	0.877	0.858	0.856	5.8%	4.8%	3.1%	3.1%
downscale_0.6	0.794	0.804	0.777	0.785	10.8%	9.2%	10.2%	6.5%
jpeg_q60	0.748	0.758	0.728	0.731	7.7%	7.5%	7.7%	6.5%
saturation_0.80	0.904	0.911	0.888	0.882	2.7%	2.7%	1.9%	2.1%

Table 5.114: Grad-CAM++ stability per photometric probe (top-20% IoU with the clean map) and share of changed decisions.

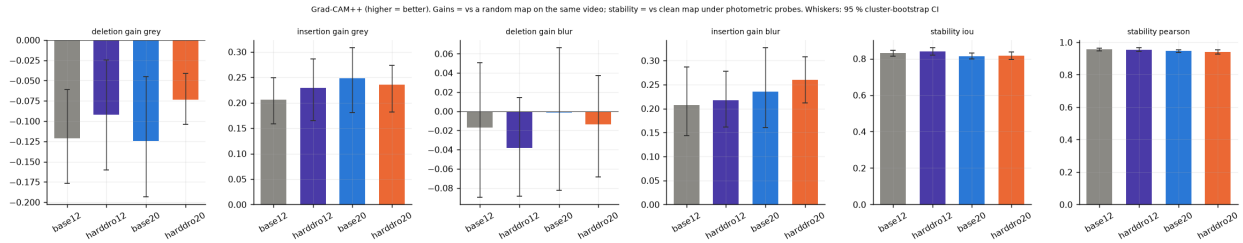

Figure 5.65: Grad-CAM++ faithfulness gains and stability per configuration (whiskers: 95% cluster-bootstrap CI).

Interpretation. Across all 12 models the picture of Section 5.8.7 is confirmed: Grad-CAM++ is clearly better than random in **insertion** (+0.21 to +0.26 for every configuration, CI above 0), but not in grey **deletion**, where all configurations are worse than random (−0.07 to −0.13) because scattered grey pixels act as artificial manipulation traces. harddro20 has the least negative grey-deletion gain (−0.073) and the highest blur-insertion gain (+0.259), and the fewest decisions that flip under the probes (4.7%). Its maps are slightly less stable than harddro12 (IoU −0.023, r −0.016; the CIs exclude 0 but stay well inside the 0.05 margin). Seven of eight H4 tests are non-inferior; insertion (grey) vs base20 is inconclusive because its CI is wide (−0.078 to +0.067), so **H4 is inconclusive**, with no sign of a real loss of explanation quality. JPEG compression and downscaling change the maps most (IoU 0.73–0.80) in every configuration.

5.9.8 Conclusions

Question	Answer from Test 6
Does DRO + 20 epochs beat DRO alone?	No. Pooled AUC 0.944 vs 0.948 (inconclusive); BioDeepAV worse (supported); FF++ better (supported).
Is it the best model for FF++?	Yes: FF++ AUC 0.912 and the best AUC on all four hard types; NeuralTextures AUC 0.861.
Is it reliable enough for an interface?	Less than harddro12: twice as many confident errors, AURC 0.121 vs 0.047, higher seed variance on BioDeepAV.
Does it harm the explanations?	No: Grad-CAM++ faithfulness and stability non-inferior; fewest decision flips under probes.
Which model to use?	**harddro12** (group-DRO, 12 epochs): best pooled AUC and balanced accuracy, BioDeepAV kept, lowest AURC.

Table 5.115: Test 6 conclusions.

Thesis statement. Combining online group-DRO sampling with a 20-epoch schedule (harddro20; three seeds) yielded a pooled test ROC-AUC of 0.944 ± 0.007, compared with 0.948 ± 0.002 for harddro12 and 0.942 ± 0.001 for base20 (paired differences −0.004 and +0.002, both inconclusive). FF++ C23 AUC improved to 0.912 ± 0.004 (supported vs both comparators), but BioDeepAV AUC decreased to 0.967 ± 0.015 (−0.016 vs harddro12, 95% CI −0.049 to −0.002, supported deterioration). Grad-CAM++ faithfulness and stability were non-inferior. The combination therefore trades BioDeepAV performance for FF++ performance and is not adopted.

Next experiment. Keep one change at a time: either keep 20 epochs but cap the DRO share of the FF++ cells (or give each dataset a fixed minimum share), or select the checkpoint with a rule that rejects any BioDeepAV drop, and re-test H3. NeuralTextures needs a targeted change (e.g. higher

face resolution or a texture-sensitive input branch), evaluated with the same split, seeds and statistics.

Limitations. Three seeds only; GPU training is not bit-wise reproducible (Section 5.8.3), so differences of about 0.01 AUC are within run-to-run noise; harddro20 vs harddro12 changes epochs, patience and schedule length together; the interaction of sampler and schedule is read from the 2 × 2 design but not tested formally; the bootstrap covers only the 1,602 test videos of one split; no comparison survives Holm correction (smallest p_Holm 0.056); 164 fakes per FF++ method (63 for DeepFakeDetection); XAI on 160 videos without localisation ground truth; no degraded videos.

5.10 Test 7 — MSTF-AV Multi-Stream Audio-Visual Detector with Consistency, Group-DRO and Shortcut-Adversarial Training (161026.py)

5.10.1 Overview

Test 5 (Section 5.6) left the AudioVisualDeepFakeCNN (584,978 parameters) at a ceiling: no training variant reliably beat the plain degraded-training baseline 5A, fixed hard-type sampling starved other degradation types (5D severe-noise AUC 0.665), Grad-CAM consistency was the only replicated benefit, and the resolution and audio-presence shortcuts remained. Test 7 (*161026.py*) therefore changes the **model**. MSTF-AV (Multi-Stream Spatio-Temporal-Frequency Audio-Visual network, 2,280,919 parameters) joins an RGB stream with an SRM noise-residual stream, models time with a transformer over frame and frame-difference tokens, and adds audio through gated cross-modal attention. Four Stage 2 variants (6A–6D) each add one training block; every variant is trained on BioDeepAV, on FF++ and on both datasets jointly, with the three Test 5 seeds (42, 123, 2026): 9 Stage 1 and 36 Stage 2 runs, all completed. No data were rebuilt: the Test 5 split, face-crop and audio caches and Grad-CAM pairs are used read-only, so Test 7 versus Test 5 (5A, 5C) is a **paired** comparison on the same 3,458 test originals and 6,487 degraded test videos. Summed training time was 28.1 h on the RTX 3050 (run 20–23 Oct 2026).

Naming. The script and all of its output files call this experiment “Test 6” (run folder *run_16-10-26_test6*, variants 6A–6D). It is numbered Test 7 in this chapter because Test 6 is the harddro20 run of Section 5.9; the variant labels 6A–6D are kept so that every value can be traced to the files.

Key result. The larger architecture does **not** improve on Test 5. With the same recipe (6A vs 5A), original / degraded AUC is 0.956 / 0.895 vs 0.966 / 0.939 on BioDeepAV and 0.726 / 0.650 vs 0.770 / 0.705 on FF++ (differences inconclusive). The full method **6D is worse than 5A** on degraded AUC for both datasets (−0.064 and −0.081, supported). Paired Grad-CAM + supervised contrastive consistency (**6B**) is again the component that works: it raised Grad-CAM top-20% IoU in all four training/test settings (FF++ 0.596 → 0.826; explanation changed while prediction stable 28.3% → 4.2%; supported). Group-DRO + EMA (6C) and the shortcut adversaries (6D) gave no supported gain and several supported losses. Cross-dataset AUC fell below chance (0.22–0.42). Verdicts: H1 not supported, H2 partially supported, H3 and H4 not supported, H5 mixed. **Test 5 baseline 5A remains the better detector**; 6B gives the most stable explanations.

5.10.2 Code pipeline

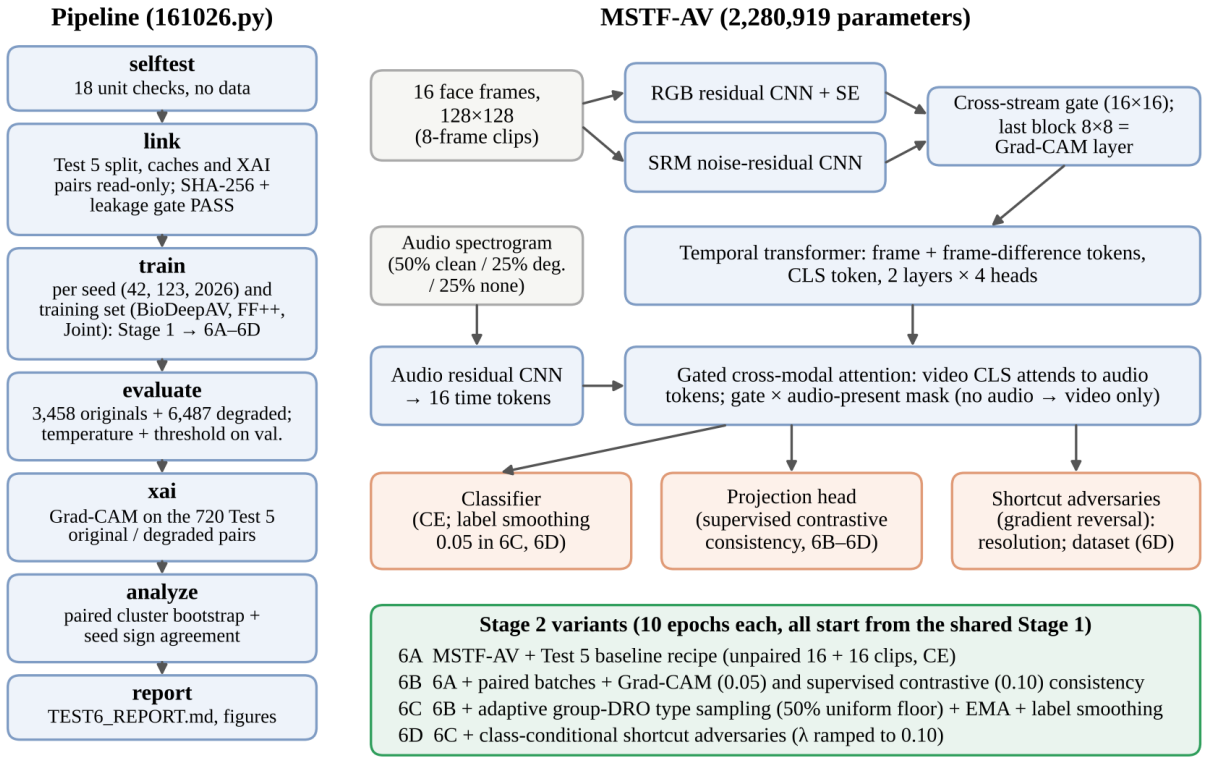

Figure 5.66: What 161026.py does (left) and the MSTF-AV architecture with the four Stage 2 variants (right). Phases: selftest, link, run (train, evaluate, xai, analyze, report per seed), each resumable; a smoke mode runs the pipeline end-to-end in a separate root.

Before any training the script ran 18 unit checks (all passed), among them: parameter count 1.5–4 M, forward shapes in audio-visual / video-only / audio-only mode, “audio mask 0 equals video only”, an 8×8 Grad-CAM layer, a differentiable soft Grad-CAM consistency term, gradient reversal, temperature scaling that reduces ECE, the DRO floor keeping every type at or above floor/K, and a guard that blocks loading any Test 5 checkpoint. The link phase then verified 21 conditions against the Test 5 run (leakage gate PASS, split-manifest and XAI-pair SHA-256 identical to the Test 5 gate, caches complete, every degraded video in its source’s partition); all passed.

5.10.3 Setup

Item	Test 5 (5A / 5C)	Test 7 (6A–6D)
Script	9-10-26_test5_independent.py	161026.py (SHA-256 b093dd7e…)
Model	AudioVisualDeepFakeCNN, 584,978 parameters; frame-wise CNN + mean / max pooling	MSTF-AV, 2,280,919 parameters: RGB residual CNN + squeeze-excitation; SRM noise-residual CNN; cross-stream gate at 16×16; 8×8 Grad-CAM layer; temporal transformer (2 layers × 4 heads, frame + frame-difference tokens); audio CNN tokens + gated cross-modal attention
Data, split, input	Own split (seed 20261009), t5_ caches; 16 cached frames, 128×128 face crops, 8-frame clips	Identical: Test 5 split, caches and XAI pairs read-only (link PASS); fresh weights, Test 5 checkpoints blocked
Training sets	BioDeepAV; FF++	BioDeepAV; FF++; Joint (both datasets, validation score = macro over datasets)
Stage 1 / Stage 2	8 + 10 epochs; AdamW (wd 1e-4) + OneCycle	8 + 10 epochs; AdamW (wd 5e-4), linear warm-up (15%) + cosine; lr 1e-3 / 3e-4; 32 clips per step; AMP; gradient clip 5
Losses	CE (5A); + 0.05 × Grad-CAM consistency (5C)	CE (6A); + 0.05 × Grad-CAM + 0.10 × supervised contrastive (τ 0.10) (6B–6D); label smoothing 0.05, EMA 0.999 (6C, 6D); adversaries λ 0.10 (6D)
Sampling	Natural mixture (5A, 5C)	Natural (6A, 6B); adaptive group-DRO over degradation types, η 2.0, 50% uniform floor (6C, 6D)
Audio	50% clean / 25% degraded / 25% none	Same (6 audio degradations, 3 levels)
Calibration, threshold	Raw probability; validation balanced-accuracy optimum	Temperature scaling on validation (NLL grid); threshold = validation balanced-accuracy optimum on the calibrated probability
Model selection	Mean of validation original and degraded AUC	Same (Joint: macro over datasets; 6C, 6D on EMA weights)
Statistics	Paired cluster bootstrap (1,000) + sign agreement	Same; comparisons within Test 7 and against T5-5A, T5-5C
Hardware, time	RTX 3050; 10–16 Oct 2026	RTX 3050 (PyTorch 2.14, CUDA 12.6); 28.1 h for 45 runs; 20–23 Oct 2026

Table 5.116: Test 7 configuration compared with Test 5.

	Comparison	Hypothesis
H1	6A − T5-5A	The MSTF-AV architecture improves robustness over the Test 5 model (same recipe)
H2	6B − 6A	Paired Grad-CAM + supervised contrastive consistency improves explanation stability
H3	6C − 6B	Group-DRO sampling + EMA improve worst-case robustness without starving types
H4	6D − 6C	Shortcut adversaries reduce shortcut use (cross-dataset AUC, calibration)
H5	6D − T5-5A	The full Test 7 method beats the Test 5 baseline

Table 5.117: Pre-specified hypotheses. A difference is "supported" only if the 95% paired cluster-bootstrap CI excludes zero and all three seeds agree in sign (the Test 5 rule).

5.10.4 Data

Dataset	Train	Val	Test	Degraded test videos	Grad-CAM pairs
BioDeepAV	1,849	433	1,420	4,076 (8 types × 3 levels)	320
FF++	4,263	699	2,038	2,411 (10 types × 3 levels)	400
Total	**6,112**	**1,132**	**3,458**	**6,487**	**720**

Table 5.118: Originals per partition (from the Test 5 split; 10,702 originals and 19,729 degraded videos in total). Clusters for the bootstrap: 53 BioDeepAV and 101 FF++ source groups.

5.10.5 Training and threshold selection

Training set	Variant	Best epoch (42 / 123 / 2026)	Threshold (42 / 123 / 2026)	Temp. (mean)	Val. score (mean)	Val. ECE raw → cal.	Time / run
BioDeepAV	S1	2 / 6 / 5	0.32 / 0.10 / 0.22	2.56	0.954	0.220 → 0.203	6 min
	6A	5 / 5 / 5	0.14 / 0.09 / 0.18	3.44	0.973	0.200 → 0.205	28 min
	6B	4 / 4 / 3	0.24 / 0.11 / 0.20	3.57	0.972	0.252 → 0.249	28 min
	6C	10 / 7 / 6	0.22 / 0.15 / 0.22	2.33	0.953	0.234 → 0.224	29 min
	6D	7 / 7 / 10	0.27 / 0.15 / 0.30	3.26	0.957	0.281 → 0.219	29 min
FF++	S1	6 / 8 / 8	0.54 / 0.51 / 0.46	0.90	0.698	0.339 → 0.336	13 min
	6A	10 / 6 / 10	0.72 / 0.81 / 0.76	0.82	0.713	0.053 → 0.040	36 min
	6B	10 / 7 / 10	0.77 / 0.71 / 0.75	0.81	0.704	0.061 → 0.035	37 min
	6C	10 / 7 / 9	0.72 / 0.65 / 0.75	0.72	0.693	0.079 → 0.041	38 min
	6D	10 / 1 / 4	0.74 / 0.51 / 0.73	0.64	0.686	0.127 → 0.073	38 min
Joint	S1	7 / 6 / 6	0.20 / 0.21 / 0.20	2.46	0.742	0.306 → 0.306	18 min
	6A	6 / 5 / 6	0.22 / 0.24 / 0.23	3.57	0.744	0.165 → 0.217	62 min
	6B	8 / 8 / 4	0.16 / 0.33 / 0.19	3.11	0.721	0.158 → 0.207	65 min
	6C	10 / 10 / 8	0.22 / 0.17 / 0.26	2.11	0.721	0.164 → 0.180	66 min
	6D	1 / 9 / 10	0.24 / 0.34 / 0.20	1.91	0.711	0.161 → 0.171	67 min

Table 5.119: Model selection per training set and variant (S1 = Stage 1, shared by 6A–6D; S1 validation score = original AUC). Thresholds are on the temperature-calibrated probability.

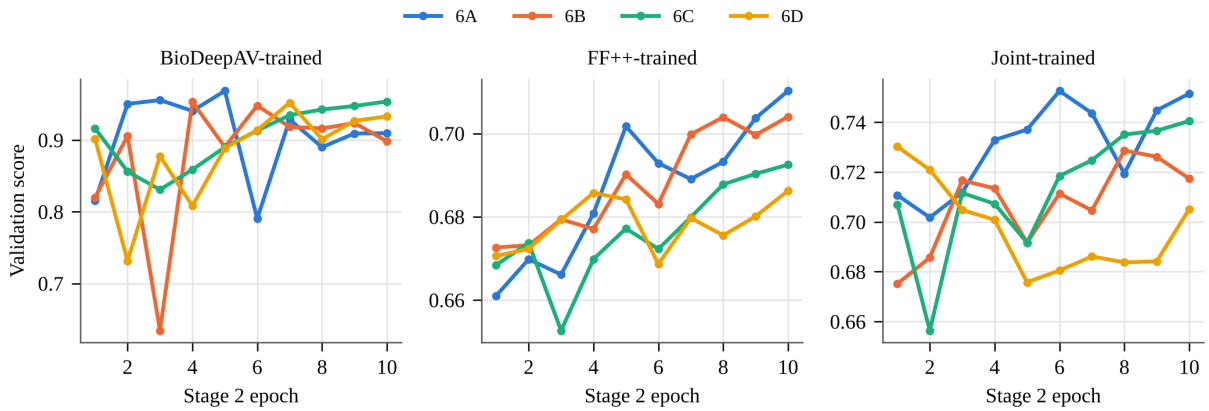

Figure 5.67: Test 7 validation score per Stage 2 epoch, seed 42 (mean of validation original and degraded AUC; Joint: macro over datasets).

Temperature scaling moved the BioDeepAV thresholds away from the extreme values of Test 5 (0.01–0.12) to 0.09–0.32, but the BioDeepAV models are still over-confident (temperature 1.4–4.8) and their validation ECE stays at 0.12–0.26 after calibration, because the calibration is fitted on a validation set in which the classes are almost perfectly separated. On FF++ the temperature is below 1 (the model is under-confident) and calibration works (Stage 2 validation ECE 0.03–0.06; one 6D run 0.16). As in Test 5, the BioDeepAV validation score swings between epochs (6B seed 42: 0.63–0.95, Figure 5.67), and the best epoch varies widely between seeds of the same variant (FF++ 6D: 10 / 1 / 4; Joint 6D: 1 / 9 / 10), so checkpoint choice is partly chance. The adaptive group-DRO weights (seed 42) stayed close to uniform on FF++ (final weights 0.09–0.11 per type) and moved most on BioDeepAV, where brightness reached 0.23–0.24.

5.10.6 Results: original vs degraded

Measure	S1	5A (T5)	5C (T5)	6A	6B	6C	6D
BioDeepAV-trained → BioDeepAV							
AUC original	0.897 (.119)	0.966 (.032)	0.952 (.038)	0.956 (.014)	0.941 (.018)	0.945 (.022)	0.928 (.053)
AUC degraded	0.735 (.098)	0.939 (.038)	0.922 (.047)	0.895 (.024)	0.873 (.028)	0.878 (.016)	0.875 (.070)
Worst-type AUC	0.544 (.061)	0.858 (.053)	0.840 (.051)	0.815 (.053)	0.797 (.054)	0.791 (.050)	0.791 (.138)
Hard-type AUC	0.694 (.106)	0.919 (.040)	0.911 (.045)	0.856 (.035)	0.847 (.040)	0.855 (.038)	0.842 (.094)
Noise AUC	0.565 (.028)	0.961 (.026)	0.927 (.067)	0.922 (.003)	0.883 (.064)	0.872 (.047)	0.911 (.016)
Flip rate (%)	20.0 (2.1)	6.5 (0.7)	8.7 (1.5)	6.7 (1.3)	10.3 (2.6)	11.5 (0.6)	12.0 (6.1)
Real recognised	0.850 (.074)	0.621 (.364)	0.690 (.173)	0.386 (.097)	0.599 (.134)	0.573 (.191)	0.530 (.206)
Bal. acc. original	0.856 (.083)	0.805 (.179)	0.840 (.088)	0.691 (.048)	0.782 (.048)	0.772 (.085)	0.758 (.105)
Bal. acc. degraded	0.736 (.061)	0.789 (.128)	0.809 (.076)	0.688 (.041)	0.737 (.029)	0.760 (.052)	0.752 (.075)
ECE (calibrated)	0.225 (.118)	0.108 (.068)	0.132 (.056)	0.177 (.065)	0.243 (.058)	0.227 (.033)	0.235 (.058)
FF++-trained → FF++							
AUC original	0.670 (.036)	0.770 (.014)	0.744 (.037)	0.726 (.018)	0.725 (.003)	0.716 (.015)	0.714 (.016)
AUC degraded	0.575 (.032)	0.705 (.008)	0.694 (.027)	0.650 (.027)	0.648 (.008)	0.633 (.016)	0.624 (.015)
Worst-type AUC	0.315 (.033)	0.570 (.009)	0.559 (.041)	0.519 (.040)	0.500 (.004)	0.435 (.042)	0.401 (.051)
Hard-type AUC	0.481 (.028)	0.647 (.018)	0.625 (.023)	0.596 (.019)	0.591 (.015)	0.554 (.024)	0.541 (.037)
Noise AUC	0.664 (.017)	0.756 (.016)	0.740 (.078)	0.709 (.012)	0.703 (.030)	0.686 (.023)	0.681 (.014)
Flip rate (%)	14.5 (1.6)	15.1 (2.6)	15.7 (1.5)	18.2 (0.4)	15.7 (2.5)	15.8 (1.4)	15.2 (3.4)
Real recognised	0.503 (.059)	0.657 (.069)	0.790 (.062)	0.667 (.110)	0.627 (.093)	0.582 (.043)	0.562 (.097)
Bal. acc. original	0.649 (.011)	0.701 (.011)	0.687 (.034)	0.665 (.005)	0.671 (.007)	0.672 (.010)	0.667 (.013)
Bal. acc. degraded	0.592 (.005)	0.640 (.013)	0.644 (.023)	0.620 (.017)	0.619 (.013)	0.611 (.005)	0.603 (.006)
ECE (calibrated)	0.353 (.011)	0.165 (.015)	0.185 (.025)	0.140 (.005)	0.139 (.002)	0.152 (.025)	0.187 (.071)

Table 5.120: Test 7 in-domain results next to the Test 5 baselines on the same test videos (seed mean, SD in parentheses; n = 1,420 / 4,076 BioDeepAV and 2,038 / 2,411 FF++ original / degraded videos). Flip rate = share of degraded videos whose decision differs from the original. Test 5 probabilities are uncalibrated.

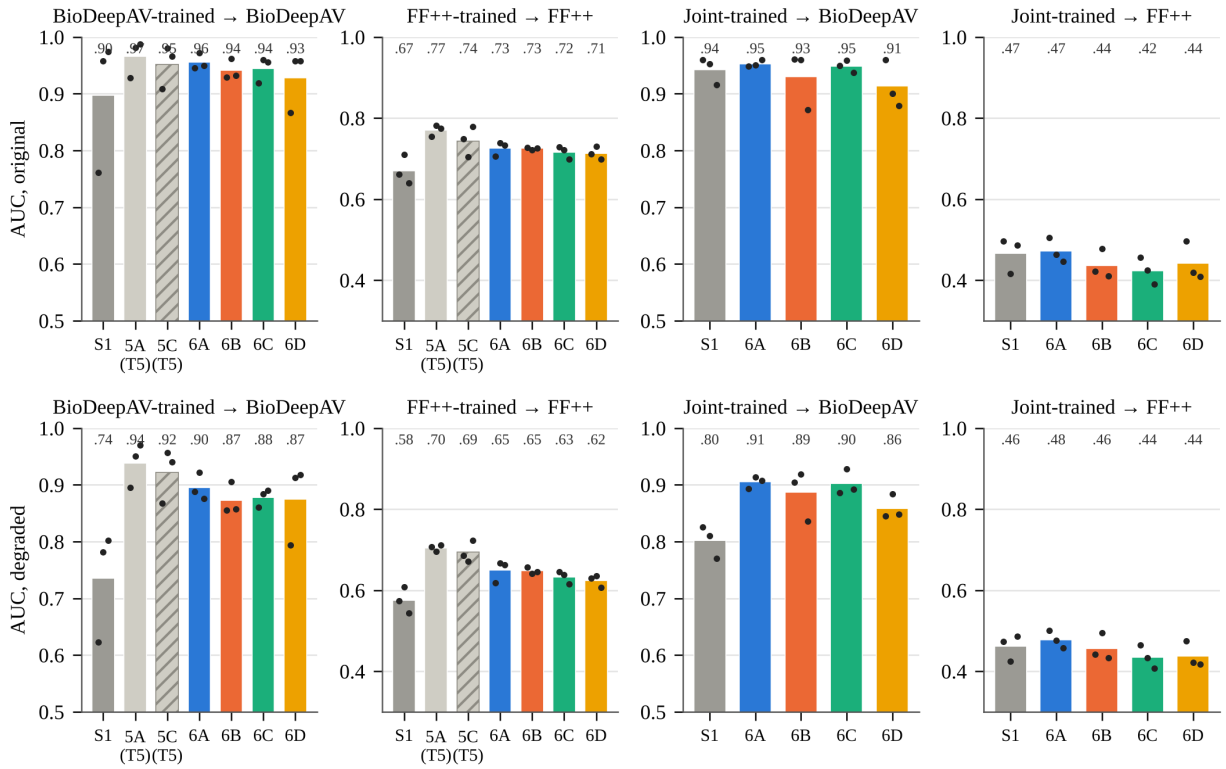

Figure 5.68: AUC on original (top) and degraded (bottom) test videos; bars: seed mean, dots: seeds. 5A (T5) and 5C (T5) are the Test 5 models on the same videos.

Measure	S1	6A	6B	6C	6D
Joint-trained → BioDeepAV					
AUC original	0.943 (.024)	0.953 (.006)	0.931 (.051)	0.949 (.011)	0.913 (.042)
AUC degraded	0.802 (.028)	0.905 (.010)	0.887 (.044)	0.902 (.023)	0.859 (.022)
Worst-type AUC	0.585 (.095)	0.850 (.028)	0.835 (.053)	0.832 (.054)	0.762 (.081)
Hard-type AUC	0.780 (.011)	0.878 (.024)	0.859 (.042)	0.870 (.041)	0.831 (.044)
Noise AUC	0.623 (.151)	0.919 (.029)	0.916 (.042)	0.933 (.005)	0.873 (.015)
Flip rate (%)	18.5 (1.9)	8.3 (2.7)	9.2 (0.3)	9.8 (2.2)	8.8 (4.8)
Real recognised	0.766 (.091)	0.746 (.194)	0.692 (.166)	0.737 (.174)	0.709 (.145)
Bal. acc. original	0.858 (.042)	0.841 (.071)	0.806 (.077)	0.832 (.064)	0.795 (.047)
Bal. acc. degraded	0.755 (.029)	0.796 (.055)	0.767 (.068)	0.790 (.063)	0.759 (.044)
ECE (calibrated)	0.182 (.075)	0.223 (.024)	0.185 (.040)	0.183 (.024)	0.158 (.070)
Joint-trained → FF++					
AUC original	0.466 (.044)	0.472 (.030)	0.437 (.036)	0.424 (.033)	0.441 (.047)
AUC degraded	0.462 (.033)	0.478 (.022)	0.457 (.034)	0.435 (.029)	0.438 (.032)
Worst-type AUC	0.379 (.045)	0.402 (.048)	0.343 (.062)	0.315 (.057)	0.369 (.030)
Hard-type AUC	0.454 (.043)	0.474 (.011)	0.467 (.031)	0.444 (.022)	0.423 (.021)
Noise AUC	0.464 (.007)	0.486 (.036)	0.485 (.019)	0.460 (.042)	0.451 (.020)
Flip rate (%)	1.3 (2.2)	0.2 (0.2)	0.2 (0.4)	0.2 (0.1)	0.5 (0.9)
Real recognised	0.000 (.000)	0.002 (.003)	0.007 (.012)	0.003 (.003)	0.015 (.026)
Bal. acc. original	0.500 (.000)	0.499 (.002)	0.502 (.004)	0.501 (.001)	0.504 (.007)
Bal. acc. degraded	0.497 (.005)	0.499 (.002)	0.503 (.005)	0.501 (.001)	0.505 (.008)
ECE (calibrated)	0.406 (.001)	0.346 (.017)	0.342 (.005)	0.317 (.030)	0.333 (.051)

Table 5.121: Jointly trained models (both datasets in training; no Test 5 counterpart). Seed mean (SD).

Three points stand out. First, degraded training works with the new model too: every Stage 2 variant beats its Stage 1 model on degraded AUC (BioDeepAV 0.735 → 0.87–0.90; FF++ 0.575 → 0.62–0.65; worst-type 0.544 → 0.79–0.82 on BioDeepAV). Second, none of the Test 7 variants reaches the Test 5 model: the best Test 7 values (6A) are 0.010–0.044 AUC lower on original and 0.044–0.055 lower on degraded videos, and on FF++ the gap grows with every added block (degraded AUC 6A 0.650 → 6D 0.624). Third, the joint models learn BioDeepAV but **not FF++**: FF++ AUC 0.42–0.47 and only 0–2% of FF++ real videos recognised, i.e. in joint training the detector separates the datasets (BioDeepAV real vs everything else) instead of real from fake in FF++.

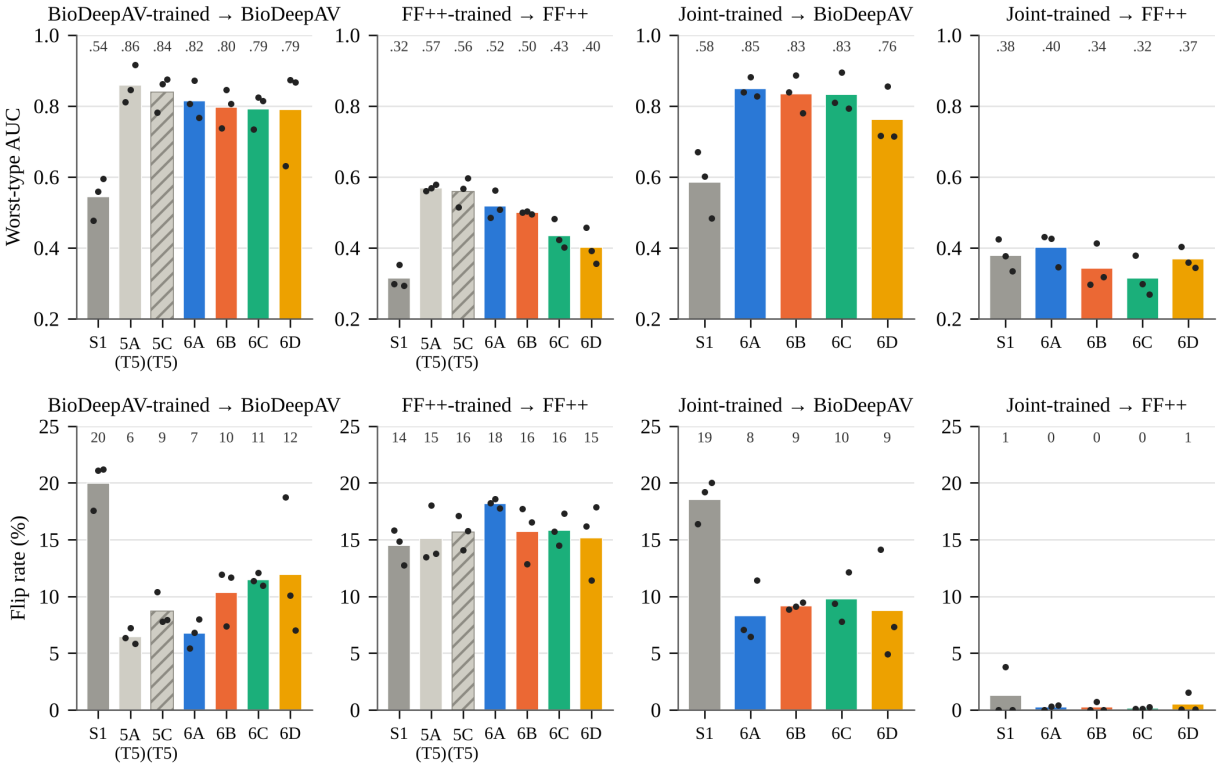

Figure 5.69: Worst degradation-type AUC (top) and prediction flip rate between original and degraded version of the same test video (bottom). Joint → FF++ flips are near zero because almost every FF++ video is called fake.

5.10.7 Results per degradation type

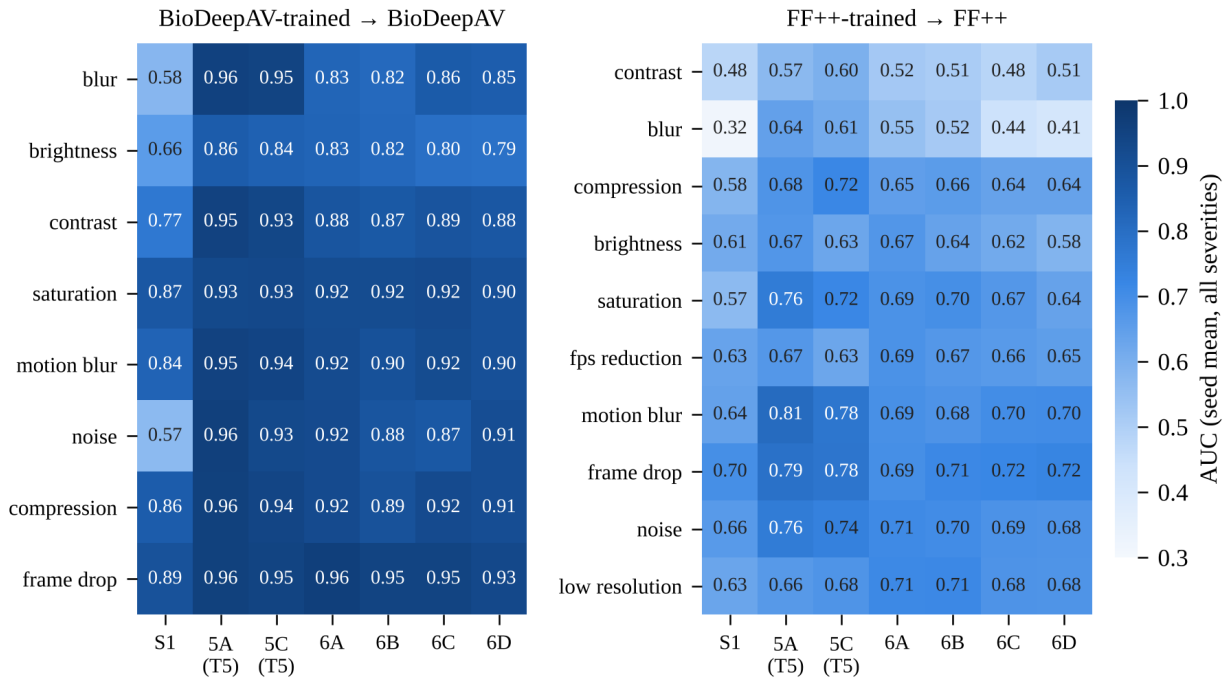

Figure 5.70: In-domain AUC per degradation type (seed mean over 3 seeds, all severities pooled).

Training / test	Type (severe)	n	S1	5A (T5)	5C (T5)	6A	6B	6C	6D
BioDeepAV	blur	190	0.438	0.968	0.957	0.874	0.873	0.889	0.877
	brightness	159	0.472	0.776	0.776	0.719	0.705	0.728	0.689
	contrast	169	0.658	0.922	0.902	0.797	0.782	0.814	0.801
	saturation	190	0.895	0.879	0.904	0.881	0.893	0.902	0.878
	noise	148	0.601	0.977	0.965	0.970	0.967	0.936	0.985
	compression	166	0.866	0.959	0.945	0.920	0.890	0.926	0.915
	motion blur	189	0.797	0.943	0.936	0.914	0.883	0.916	0.891
	frame drop	174	0.883	0.968	0.956	0.939	0.915	0.913	0.890
FF++	blur	86	0.231	0.634	0.576	0.487	0.416	0.281	0.302
	brightness	90	0.551	0.527	0.448	0.516	0.341	0.405	0.402
	contrast	94	0.491	0.547	0.614	0.354	0.345	0.377	0.456
	saturation	94	0.530	0.762	0.718	0.695	0.686	0.651	0.615
	noise	91	0.734	0.743	0.746	0.772	0.769	0.752	0.741
	compression	111	0.549	0.623	0.715	0.575	0.580	0.568	0.577
	motion blur	90	0.620	0.850	0.806	0.664	0.628	0.667	0.685
	frame drop	100	0.747	0.802	0.804	0.729	0.749	0.744	0.757
	fps reduction	42	0.567	0.707	0.640	0.604	0.580	0.588	0.585
	low resolution	50	0.593	0.651	0.666	0.703	0.698	0.662	0.643

Table 5.122: In-domain AUC at the severe level (seed mean). Hard types (brightness, saturation, contrast, blur) first.

On BioDeepAV the Test 7 models lose most against Test 5 on **blur** (severe 0.87–0.89 vs 0.97) and **contrast** (0.78–0.81 vs 0.92); brightness remains the worst type for both models (severe 0.69–0.73 vs 0.78). The group-DRO floor did what it was designed to do: no type was starved, and severe noise stays at 0.94–0.99 for all variants (Test 5 5D: 0.665). On FF++, blur is the main failure and becomes

worse with each block (severe blur 6A 0.487 → 6C 0.281 → 6D 0.302, below chance); contrast and brightness are near or below chance for all Test 7 variants.

5.10.8 Cross-dataset results and shortcut diagnostics

Measure	S1	5A (T5)	5C (T5)	6A	6B	6C	6D
BioDeepAV → FF++: AUC original	0.381 (.016)	0.542 (.027)	0.490 (.048)	0.418 (.015)	0.392 (.030)	0.411 (.008)	0.415 (.014)
BioDeepAV → FF++: AUC degraded	0.409 (.016)	0.517 (.021)	0.475 (.042)	0.445 (.009)	0.415 (.020)	0.425 (.007)	0.432 (.011)
FF++ → BioDeepAV: AUC original	0.216 (.072)	0.500 (.000)	0.500 (.000)	0.284 (.040)	0.246 (.026)	0.239 (.029)	0.219 (.047)
FF++ → BioDeepAV: AUC degraded	0.368 (.048)	0.500 (.000)	0.500 (.000)	0.400 (.040)	0.373 (.026)	0.363 (.019)	0.351 (.030)
ρ resolution, BioDeepAV → BioDeepAV	0.519 (.097)	0.518 (.084)	0.501 (.052)	0.370 (.160)	0.366 (.087)	0.425 (.076)	0.378 (.134)
ρ resolution, BioDeepAV → FF++	0.453 (.103)	0.395 (.150)	0.277 (.164)	0.293 (.117)	0.241 (.112)	0.232 (.099)	0.218 (.049)
ρ resolution, FF++ → FF++	−0.291 (.094)	−0.089 (.044)	−0.012 (.116)	−0.253 (.069)	−0.255 (.054)	−0.271 (.057)	−0.231 (.039)
ρ resolution, Joint → BioDeepAV	0.438 (.007)	—	—	0.325 (.052)	0.308 (.061)	0.279 (.090)	0.302 (.108)
ρ resolution, Joint → FF++	0.019 (.145)	—	—	−0.392 (.188)	−0.261 (.151)	−0.369 (.074)	−0.150 (.152)

Table 5.123: Cross-dataset AUC and the resolution shortcut (Spearman ρ between p(fake) and native face size within real test videos; 0 = no shortcut). Seed mean (SD).

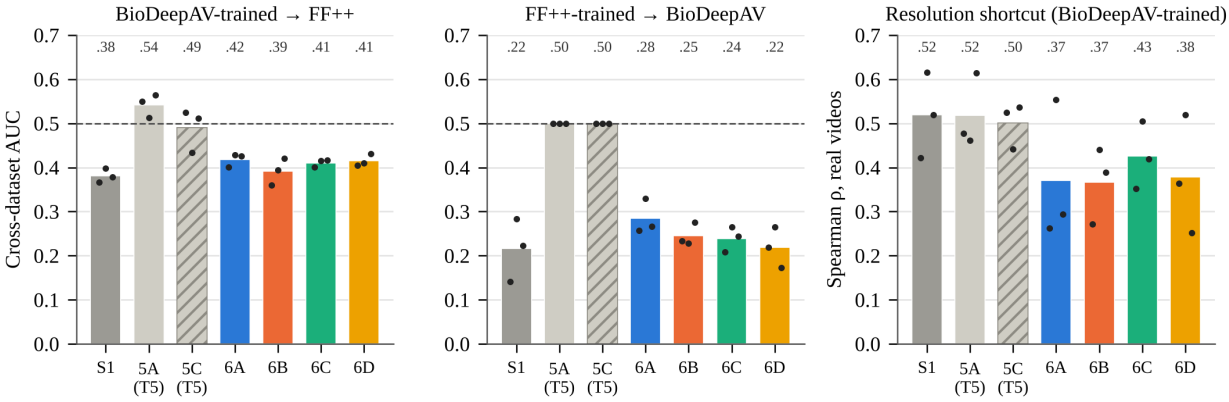

Figure 5.71: Cross-dataset AUC (dashed line: chance) and the resolution correlation of the BioDeepAV-trained models.

Condition (test on BioDeepAV)	S1	5A (T5)	5C (T5)	6A	6B	6C	6D
BioDeepAV-trained: audio masked, AUC	0.900	0.948	0.935	0.976	0.944	0.937	0.913
BioDeepAV-trained: audio masked, flips (%)	9.3 (5.9)	5.5 (2.4)	6.8 (2.3)	7.5 (0.6)	9.4 (2.5)	13.6 (3.8)	8.8 (2.9)
BioDeepAV-trained: audio only, AUC	0.503	0.588	0.595	0.581	0.576	0.514	0.510
Joint-trained: audio masked, AUC	0.874	—	—	0.903	0.901	0.926	0.865
Joint-trained: audio masked, flips (%)	25.8 (3.3)	—	—	22.9 (8.1)	23.3 (4.7)	23.8 (7.6)	26.8 (7.7)
Joint-trained: audio only, AUC	0.558	—	—	0.584	0.574	0.572	0.544

Table 5.124: Audio ablation on BioDeepAV original test videos (flips = decisions changed vs normal audio-visual input; seed mean, SD in parentheses). FF++-trained Test 7 models skip the audio branch because FF++ has no audio in training.

Cross-dataset transfer did not improve; it became **worse than chance**. BioDeepAV-trained models reach 0.39–0.42 on FF++ (Test 5: 0.49–0.54), and FF++-trained models 0.22–0.28 on BioDeepAV, i.e. their scores are systematically inverted: BioDeepAV real videos look more “fake” to them than BioDeepAV fakes. The FF++-trained Test 5 models scored exactly 0.500 because they called every BioDeepAV video fake (audio presence); the Test 7 audio mask removes that shortcut (no audio branch without audio in training), but what is left is a resolution cue with the opposite sign (ρ −0.28 to −0.35). The resolution correlation of the BioDeepAV-trained model fell from 0.52 (5A) to 0.37 (6A), so part of the shortcut is removed by the architecture itself, but the shortcut adversaries of 6D add nothing (0.378 vs 6C 0.425; inconclusive). In-domain audio dependence is modest (7–14% flips when audio is masked, 5A: 5.5%); the joint models depend far more on audio (23–27% flips).

5.10.9 Explainability: Grad-CAM stability

Grad-CAM stability was measured as in Test 5 on the same 720 original/degraded pairs (320 BioDeepAV, 400 FF++) for every model, so the Test 5 rows are paired. Because 6B–6D are trained with a Grad-CAM consistency loss, their stability is partly optimised and is not evidence of correct localisation.

Setting / measure	S1	5A (T5)	5C (T5)	6A	6B	6C	6D
BioDeepAV-trained → BioDeepAV							
Top-20% IoU, all	0.467 (.041)	0.602 (.021)	0.614 (.013)	0.523 (.043)	0.578 (.036)	0.538 (.065)	0.549 (.124)
Top-20% IoU, hard types	0.391 (.061)	0.436 (.014)	0.459 (.017)	0.439 (.035)	0.498 (.018)	0.466 (.038)	0.465 (.120)
Pearson r, all	0.639 (.078)	0.780 (.013)	0.804 (.015)	0.705 (.061)	0.774 (.027)	0.742 (.067)	0.735 (.122)
Expl. changed, pred. stable (%)	47.5 (8.0)	34.1 (1.2)	30.3 (1.2)	42.7 (9.1)	32.6 (7.1)	36.9 (8.5)	36.2 (19.1)
Clean reference IoU	0.578 (.036)	0.786 (.037)	0.786 (.045)	0.608 (.047)	0.633 (.047)	0.593 (.047)	0.599 (.116)
FF++-trained → FF++							
Top-20% IoU, all	0.587 (.029)	0.614 (.026)	0.775 (.020)	0.596 (.021)	0.826 (.021)	0.782 (.023)	0.699 (.112)
Top-20% IoU, hard types	0.447 (.063)	0.379 (.025)	0.629 (.019)	0.467 (.026)	0.756 (.024)	0.703 (.029)	0.593 (.153)
Pearson r, all	0.740 (.047)	0.759 (.024)	0.916 (.005)	0.788 (.021)	0.965 (.000)	0.946 (.009)	0.863 (.124)
Expl. changed, pred. stable (%)	29.5 (3.8)	26.8 (4.3)	9.5 (2.1)	28.3 (2.2)	4.2 (0.8)	5.9 (2.3)	15.4 (15.5)
Clean reference IoU	0.520 (.048)	0.604 (.037)	0.696 (.055)	0.501 (.030)	0.733 (.012)	0.681 (.022)	0.606 (.118)
Joint-trained → BioDeepAV							
Top-20% IoU, all	0.460 (.035)	—	—	0.462 (.017)	0.542 (.040)	0.561 (.056)	0.536 (.072)
Top-20% IoU, hard types	0.415 (.025)	—	—	0.383 (.006)	0.434 (.040)	0.464 (.058)	0.447 (.042)
Pearson r, all	0.645 (.065)	—	—	0.653 (.023)	0.746 (.030)	0.764 (.050)	0.720 (.075)
Expl. changed, pred. stable (%)	51.0 (7.4)	—	—	52.5 (5.6)	39.8 (8.9)	35.4 (9.6)	39.7 (9.7)
Clean reference IoU	0.541 (.034)	—	—	0.519 (.035)	0.596 (.038)	0.597 (.049)	0.581 (.053)
Joint-trained → FF++							
Top-20% IoU, all	0.446 (.031)	—	—	0.509 (.028)	0.657 (.068)	0.655 (.050)	0.583 (.119)
Top-20% IoU, hard types	0.351 (.025)	—	—	0.373 (.029)	0.515 (.065)	0.524 (.044)	0.428 (.116)
Pearson r, all	0.547 (.047)	—	—	0.680 (.047)	0.830 (.053)	0.836 (.041)	0.724 (.166)
Expl. changed, pred. stable (%)	52.2 (5.1)	—	—	44.7 (4.1)	25.3 (9.5)	25.2 (6.0)	34.3 (13.9)
Clean reference IoU	0.461 (.048)	—	—	0.453 (.006)	0.549 (.071)	0.530 (.041)	0.509 (.030)
FF++-trained → BioDeepAV (cross)							
Top-20% IoU, all	0.514 (.018)	0.583 (.008)	0.701 (.052)	0.530 (.043)	0.784 (.018)	0.730 (.026)	0.628 (.116)
Top-20% IoU, hard types	0.419 (.030)	0.423 (.020)	0.574 (.033)	0.443 (.051)	0.741 (.023)	0.676 (.040)	0.565 (.132)
Pearson r, all	0.679 (.032)	0.752 (.003)	0.855 (.005)	0.722 (.041)	0.943 (.025)	0.918 (.028)	0.829 (.135)
Expl. changed, pred. stable (%)	37.6 (5.8)	36.5 (0.4)	20.8 (3.9)	33.2 (8.2)	6.7 (4.9)	8.7 (2.6)	22.9 (20.7)
Clean reference IoU	0.610 (.046)	0.711 (.054)	0.785 (.067)	0.580 (.054)	0.812 (.008)	0.769 (.008)	0.679 (.123)

Table 5.125: Grad-CAM stability between each original and its degraded version (seed mean, SD in parentheses). Clean reference IoU = IoU between two clean views of the same video (upper bound for stability). Expl. changed = IoU below 0.5 while the decision is unchanged.

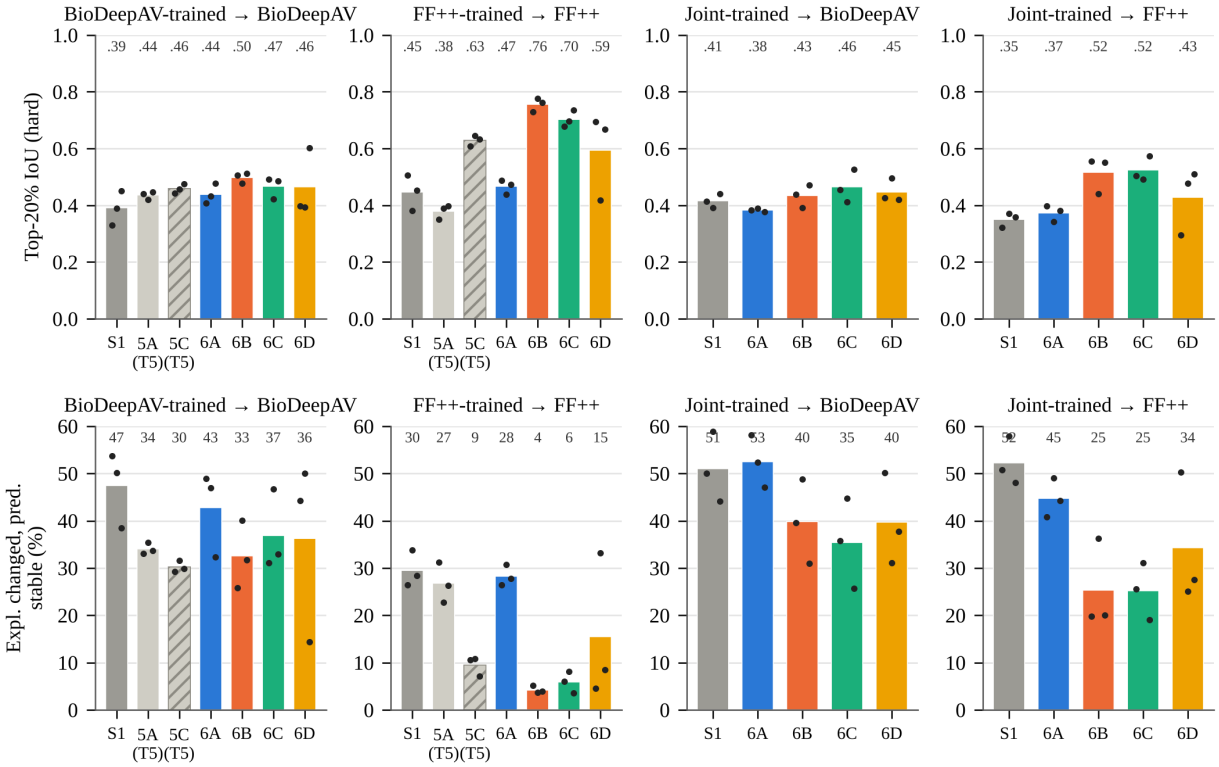

Figure 5.72: Grad-CAM top-20% IoU on hard degradation types (top) and share of unchanged decisions with a changed explanation (bottom).

Figure 5.73: Test 7 Grad-CAM examples (FF++-trained 6D, seed 42): original, its CAM, degraded version, its CAM, with IoU and Pearson r.

The consistency block (6B) is the one Test 7 component with a clear, replicated effect: IoU rose in every setting (BioDeepAV +0.055, FF++ +0.230, Joint → BioDeepAV +0.080, Joint → FF++ +0.147; all supported) and on FF++ only 4.2% of unchanged decisions came with a changed explanation (6A: 28.3%; Test 5 5C: 9.5%). It is also the first time a supported IoU gain appears on BioDeepAV, which Test 5 did not achieve. Adding DRO + EMA (6C) cost part of this gain (FF++ IoU −0.044, BioDeepAV −0.040, supported), and the adversaries (6D) made stability seed-dependent (FF++ hard-type IoU 0.42–0.70 across seeds). On BioDeepAV, Test 5 5C stays as stable as 6B (IoU 0.614 vs 0.578) with a much higher clean reference IoU (0.786 vs 0.633): the MSTF-AV maps are noisier even between two clean views.

5.10.10 Statistical tests and hypothesis verdicts

Each comparison uses a paired cluster bootstrap over source groups (1,000 replicates) of the seed-averaged difference on the same test videos. Of the 270 comparisons, 20 were supported improvements, 54 supported deteriorations and 196 inconclusive. All supported in-domain and joint effects are listed below; the 33 supported cross-dataset effects are summarised after the table.

Training → test	Comparison	Metric	Mean diff. [95% CI]	Per seed (42; 123; 2026)	Result
BioDeepAV → BioDeepAV	6A − T5-5A	Hard-type AUC	−0.063 [−0.088, −0.001]	−0.029; −0.028; −0.131	worse
BioDeepAV → BioDeepAV	6B − 6A	AUC degraded	−0.022 [−0.045, −0.001]	−0.033; −0.016; −0.018	worse
BioDeepAV → BioDeepAV	6B − 6A	Grad-CAM IoU (all)	+0.055 [+0.019, +0.079]	+0.132; +0.027; +0.006	better
BioDeepAV → BioDeepAV	6B − 6A	Grad-CAM IoU (hard)	+0.059 [+0.014, +0.091]	+0.097; +0.046; +0.035	better
BioDeepAV → BioDeepAV	6C − 6B	Grad-CAM IoU (all)	−0.040 [−0.060, −0.006]	−0.032; −0.079; −0.008	worse
BioDeepAV → BioDeepAV	6D − T5-5A	AUC original	−0.038 [−0.054, −0.003]	−0.061; −0.023; −0.029	worse
BioDeepAV → BioDeepAV	6D − T5-5A	AUC degraded	−0.064 [−0.075, −0.011]	−0.102; −0.038; −0.052	worse
BioDeepAV → BioDeepAV	6D − T5-5A	Hard-type AUC	−0.077 [−0.100, −0.008]	−0.142; −0.022; −0.066	worse
BioDeepAV → BioDeepAV	6D − T5-5A	Noise AUC	−0.050 [−0.066, −0.002]	−0.033; −0.064; −0.053	worse
BioDeepAV → BioDeepAV	6D − T5-5A	Flip rate (pts)	+5.50 [+2.01, +6.41]	+12.44; +2.85; +1.20	worse
BioDeepAV → BioDeepAV	6D − T5-5A	ECE original	+0.127 [+0.088, +0.215]	+0.132; +0.031; +0.219	worse
BioDeepAV → BioDeepAV	6B − T5-5C	Hard-type AUC	−0.063 [−0.103, −0.000]	−0.055; −0.060; −0.076	worse
BioDeepAV → BioDeepAV	6B − T5-5C	ECE original	+0.111 [+0.017, +0.162]	+0.095; +0.111; +0.127	worse
FF++ → FF++	6A − T5-5A	Flip rate (pts)	+3.10 [+0.56, +4.39]	+4.77; +0.54; +3.98	worse
FF++ → FF++	6A − T5-5A	ECE original	−0.025 [−0.047, −0.016]	−0.030; −0.009; −0.037	better
FF++ → FF++	6A − T5-5A	Grad-CAM IoU (hard)	+0.087 [+0.055, +0.116]	+0.137; +0.049; +0.076	better
FF++ → FF++	6B − 6A	Flip rate (pts)	−2.49 [−3.32, −1.03]	−0.54; −5.72; −1.20	better
FF++ → FF++	6B − 6A	Grad-CAM IoU (all)	+0.230 [+0.215, +0.251]	+0.188; +0.271; +0.232	better
FF++ → FF++	6B − 6A	Grad-CAM IoU (hard)	+0.290 [+0.265, +0.330]	+0.242; +0.339; +0.288	better
FF++ → FF++	6C − 6B	AUC degraded	−0.015 [−0.026, −0.004]	−0.012; −0.004; −0.030	worse
FF++ → FF++	6C − 6B	Hard-type AUC	−0.036 [−0.058, −0.009]	−0.029; −0.033; −0.048	worse
FF++ → FF++	6C − 6B	Noise AUC	−0.017 [−0.033, −0.002]	−0.019; −0.026; −0.007	worse
FF++ → FF++	6C − 6B	Grad-CAM IoU (all)	−0.044 [−0.054, −0.037]	−0.043; −0.066; −0.023	worse
FF++ → FF++	6C − 6B	Grad-CAM IoU (hard)	−0.053 [−0.074, −0.040]	−0.052; −0.081; −0.027	worse
FF++ → FF++	6D − 6C	AUC degraded	−0.009 [−0.017, −0.002]	−0.016; −0.003; −0.008	worse
FF++ → FF++	6D − T5-5A	AUC original	−0.057 [−0.088, −0.006]	−0.043; −0.051; −0.076	worse
FF++ → FF++	6D − T5-5A	AUC degraded	−0.081 [−0.126, −0.008]	−0.077; −0.060; −0.105	worse
FF++ → FF++	6D − T5-5A	Noise AUC	−0.075 [−0.173, −0.006]	−0.066; −0.052; −0.106	worse
FF++ → FF++	6D − T5-5A	Grad-CAM IoU (hard)	+0.214 [+0.186, +0.246]	+0.344; +0.028; +0.270	better
FF++ → FF++	6B − T5-5C	ECE original	−0.046 [−0.063, −0.038]	−0.058; −0.016; −0.065	better
FF++ → FF++	6B − T5-5C	Grad-CAM IoU (all)	+0.051 [+0.038, +0.066]	+0.021; +0.054; +0.077	better

Training → test	Comparison	Metric	Mean diff. [95% CI]	Per seed (42; 123; 2026)	Result
FF++ → FF++	6B − T5-5C	Grad-CAM IoU (hard)	+0.128 [+0.098, +0.175]	+0.122; +0.132; +0.130	better
Joint → BioDeepAV	6B − 6A	Grad-CAM IoU (all)	+0.080 [+0.070, +0.092]	+0.091; +0.041; +0.108	better
Joint → BioDeepAV	6B − 6A	Grad-CAM IoU (hard)	+0.051 [+0.041, +0.088]	+0.056; +0.003; +0.095	better
Joint → BioDeepAV	6D − 6C	AUC degraded	−0.043 [−0.071, −0.008]	−0.041; −0.044; −0.045	worse
Joint → BioDeepAV	6D − 6C	Hard-type AUC	−0.040 [−0.065, −0.012]	−0.043; −0.036; −0.040	worse
Joint → BioDeepAV	6D − 6C	Worst-type AUC	−0.070 [−0.127, −0.020]	−0.093; −0.039; −0.079	worse
Joint → FF++	6B − 6A	Grad-CAM IoU (all)	+0.147 [+0.134, +0.162]	+0.177; +0.101; +0.165	better
Joint → FF++	6B − 6A	Grad-CAM IoU (hard)	+0.142 [+0.124, +0.171]	+0.158; +0.098; +0.171	better
Joint → FF++	6D − 6C	ECE original	+0.016 [+0.013, +0.019]	+0.045; +0.002; +0.001	worse
Joint → FF++	6D − 6C	Grad-CAM IoU (hard)	−0.096 [−0.132, −0.073]	−0.210; −0.014; −0.063	worse

Table 5.126: All statistically supported in-domain and joint differences in Test 7 (statistics_test6.csv). For flip rate and ECE, lower is better.

Cross-dataset. All supported cross-dataset effects on detection are deteriorations against Test 5: BioDeepAV → FF++ AUC original 6A − 5A −0.124 and 6D − 5A −0.127; FF++ → BioDeepAV AUC original 6A − 5A −0.216, 6D − 5A −0.281 and worst-type AUC −0.237 / −0.294, with flip rates about 24–25 points higher. The only supported cross-dataset gains are explanation stability of 6B (FF++ → BioDeepAV IoU +0.254 vs 6A, +0.083 vs 5C) and the ECE of 6C (BioDeepAV → FF++ −0.063 vs 6B).

	Comparison	Verdict	Evidence
H1	6A − T5-5A	NOT SUPPORTED (inconclusive, all differences negative)	AUC original −0.010 (BioDeepAV), −0.044 (FF++); degraded −0.044 / −0.055; worst-type −0.043 / −0.051; all CIs include 0; hard-type AUC −0.063 on BioDeepAV is a supported deterioration
H2	6B − 6A	PARTIALLY SUPPORTED	Grad-CAM IoU all / hard better in all 4 settings (+0.051 to +0.290, supported); FF++ flip rate −2.5 pts (supported); BioDeepAV degraded AUC −0.022 (supported deterioration)
H3	6C − 6B	NOT SUPPORTED (deterioration)	No supported worst-case gain; FF++ degraded AUC −0.015, hard-type −0.036, noise −0.017 (supported deteriorations); Grad-CAM IoU lower; noise not starved
H4	6D − 6C	NOT SUPPORTED (deterioration)	No supported gain in cross-dataset AUC or calibration; Joint → FF++ ECE +0.016 and Joint → BioDeepAV degraded −0.043, worst-type −0.070 (supported deteriorations)
H5	6D − T5-5A	MIXED	Degraded AUC −0.064 (BioDeepAV) and −0.081 (FF++), BioDeepAV flip rate +5.5 pts, ECE +0.127 (supported deteriorations); FF++ hard-type Grad-CAM IoU +0.214 (supported improvement)

Table 5.127: Test 7 hypothesis verdicts (final_summary_test6.json).

5.10.11 Key findings

- **A larger, multi-stream model did not help (H1).** MSTF-AV has 3.9 times the parameters of the Test 5 model but is not better on any in-domain detection metric with the same recipe; all differences point the other way and BioDeepAV hard-type AUC is significantly lower (−0.063). The bottleneck of Tests 5–7 is therefore not model capacity.
- **Degraded training remains the main source of robustness.** Stage 2 lifts degraded AUC over Stage 1 for every variant and dataset (BioDeepAV +0.14–0.16, FF++ +0.05–0.08), as in Tests 2–5.
- **Explanation-consistency training is replicated and now generalises (H2).** 6B improves Grad-CAM stability on BioDeepAV, FF++ and in joint training; on FF++ the explanation changes for only 4% of unchanged decisions. It costs a little detection on BioDeepAV (degraded AUC

−0.022).

- **More training machinery made things worse (H3, H4).** Group-DRO + EMA + label smoothing and the shortcut adversaries reduced FF++ robustness and part of the stability gain. The DRO floor did prevent type starvation (severe noise ≥ 0.94), the failure mode of Test 5 5D.
- **Shortcuts persist in a different form.** Without the audio-presence cue, FF++-trained models are inverted on BioDeepAV (AUC 0.22–0.28), joint models separate datasets instead of real and fake in FF++ (FF++ AUC 0.42–0.47, real recognised ≤ 2%), and the resolution correlation stays at 0.37–0.43.
- **Calibration helps on FF++ only.** Temperature scaling gave FF++ ECE 0.14 (Test 5: 0.17) but not on BioDeepAV (0.18–0.24 vs 0.11), where only 39–60% of real videos are recognised at the selected threshold.

5.10.12 Limitations and things to check

- Three seeds only; seed SD is up to 0.05 AUC (6D on BioDeepAV) and 0.11–0.15 in Grad-CAM IoU for 6D, so small differences are not interpretable.
- Training settings were not tuned for the larger model: the Test 5 schedule (8 + 10 epochs, lr 3e-4) was kept; many FF++ and Joint best epochs are the last one (10), which suggests under-training, while others are 1–4.
- Test 7 changes the architecture and the optimiser settings (weight decay 5e-4, warm-up + cosine instead of OneCycle, temperature scaling) together, so 6A − 5A is not a pure architecture comparison.
- The first evaluation of the FF++-trained seed-42 models produced NaN outputs (audio tokens overflowed in fp16 for a model trained without audio). The script was fixed to skip the audio branch when a batch has no audio; those models were re-evaluated without retraining and the stale files kept in *stale_ffpp_seed42_audio_nan/*. 30 of 45 checkpoints were written by the final script version (b093dd7e…), 14 by cc108bc4… and one by b4459fa6…; the architecture signature is identical for all.
- The paired bootstrap covers one split; Joint models have no held-out dataset; 196 of 270 comparisons are inconclusive; no multiple-comparison correction was applied.
- Grad-CAM stability is partly optimised in 6B–6D and there is no manipulation-region ground truth, so stability is not correctness.

5.10.13 Suggested thesis sentence

“Replacing the Test 5 detector by a 2.3 M-parameter multi-stream spatio-temporal-frequency audio-visual network (MSTF-AV) did not improve robustness on the same split and test videos (degraded AUC 0.895 vs 0.939 on BioDeepAV and 0.650 vs 0.705 on FF++, seed means); paired Grad-CAM and supervised contrastive consistency training was the only component with a replicated benefit, raising Grad-CAM top-20% IoU under degradation on both datasets (FF++ 0.596 → 0.826), whereas group-DRO sampling and shortcut adversaries reduced robustness and cross-dataset AUC remained below chance.”

5.10.14 Output files and checkpoints

File (run folder E:\Code\run_16-10-26_test6)	Contents
161026.py	Script: selftest, link, train, evaluate, xai, analyze, report, status; smoke mode
reports/TEST6_REPORT.md, final_summary_test6.json	Full report, hypothesis verdicts, checkpoint records (SHA-256 of files, weights, script, config, split)
reports/selftest_report.json, test5_input_link.json	18 unit checks; 21 link checks against the Test 5 run (all PASS)
configs/experiment_config_6A–6D.json	Variant, model, optimiser, calibration, data hashes, environment
metrics/metrics_test6.csv, robustness_test6.csv, degradation_metrics_test6.csv	Per-run metrics on original and degraded videos; per type and severity
metrics/seed_summary_test6.csv, statistics_test6.csv	Seed means / SD; 270 paired bootstrap comparisons
metrics/gradcam_summary_test6.csv, gradcam_stability_test6.csv, gradcam_runs/	Grad-CAM stability per run, scope and pair
metrics/audio_ablation_test6.csv; metrics/predictions/*.csv.gz	Audio masking / audio-only; per-video predictions of all 45 + 12 Test 5 baseline runs
logs/training/history_*.csv, sampling_*.csv; logs/run_status.csv	Training histories, DRO weights, run status (45 / 45 completed)
checkpoints/train-{BioDeepAV,FFpp,Joint}/seed{42,123, 2026}/	stage1.pt and 6A–6D_stage2.pt (about 9.2 MB each); checkpoint_hashes.csv
figures/*.png	Report figures and Grad-CAM examples (Figure 5.73)

Table 5.128: Test 7 output files. Figures 5.67–5.72 were redrawn for this chapter from seed_summary_test6.csv, robustness_test6.csv, degradation_metrics_test6.csv and the training histories without changing any value.

5.11 Comparison of Tests

Tests 2 and 3 share the same split, test videos and degraded versions. Test 4 keeps the same 2,141 test originals but uses only the provided degraded files (3,993 degraded test videos) and trains one model per dataset, so its Combined numbers mix in-domain and cross-dataset results; compare its in-domain rows. Test 1 uses a BioDeepAV-only split. All values: audio-visual model.

Measure	Test 1	T2 S1	T2 S2	T3 S1	T3 S2	T4 Bio-trained S2	T4 FF++-trained S2
Combined AUC original	0.9541*	0.8772	0.8700	0.9143	**0.9362**	0.8170†	0.3404†
Combined AUC degraded	0.9593*	0.8207	0.8656	0.8316	**0.9127**	0.7935†	0.3229†
BioDeepAV AUC original	0.9541	0.9931	0.9923	**0.9960**	0.9958	0.9919	0.5000 (cross)
BioDeepAV AUC degraded	0.9593	0.9507	0.9900	0.9149	**0.9915**	0.9876	0.5000 (cross)
BioDeepAV real recognised	62.6%	**97.5%**	93.8%	96.8%	97.0%	94.5%	0.0%
FF++ AUC original	—	0.5844	0.5838	0.7278	**0.8042**	0.5273 (cross)	0.7475
FF++ AUC degraded	—	0.5469	0.5842	0.6310	**0.7468**	0.5041 (cross)	0.6800
FF++ real recognised	—	2.5%	0.5%	36.5%	40.0%	51.5%	**73.5%**
Flip rate (in-domain)	—	12.42%	**1.45%**	13.55%	6.21%	2.16% (Bio)	15.43% (FF++)
Threshold	0.5	0.5	0.5	0.31	0.31	0.50	0.50
Grad-CAM Pearson r	—	0.678	0.741	0.685	0.788	0.855	**0.932**
Grad-CAM top-20% IoU	—	0.557	0.567	0.531	0.605	0.674	**0.794**
Explanation changed while prediction stable	—	39.7%	43.2%	40.3%	35.8%	19.6%	**11.1%**
Audio masking flips (BioDeepAV)	—	—	—	—	12.1%	**2.6%**	37.4%
Training time	2:34:41	26:31	1:44:46	47:06	3:06:27	1:19:59	1:26:57

*Table 5.129: All tests side by side. *Test 1 is BioDeepAV only. †Test 4 Combined mixes in-domain and cross-dataset predictions. Bold: best.*

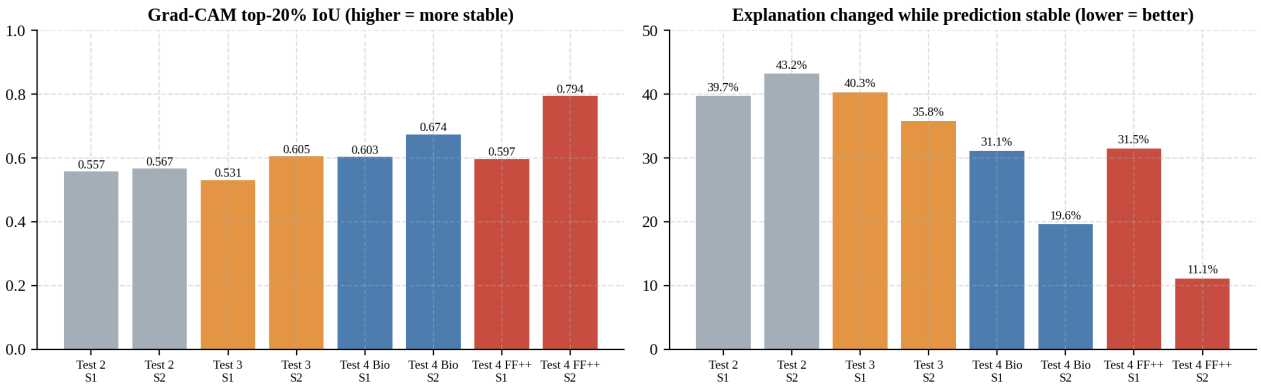

Figure 5.75: Grad-CAM explanation stability across tests (Stage 1 and Stage 2 of each test).

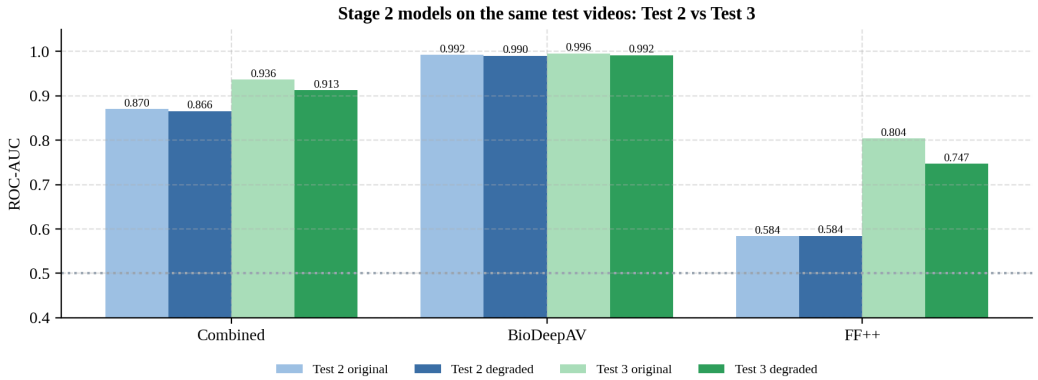

Figure 5.76: Stage 2 models of Test 2 and Test 3 on the same test videos.

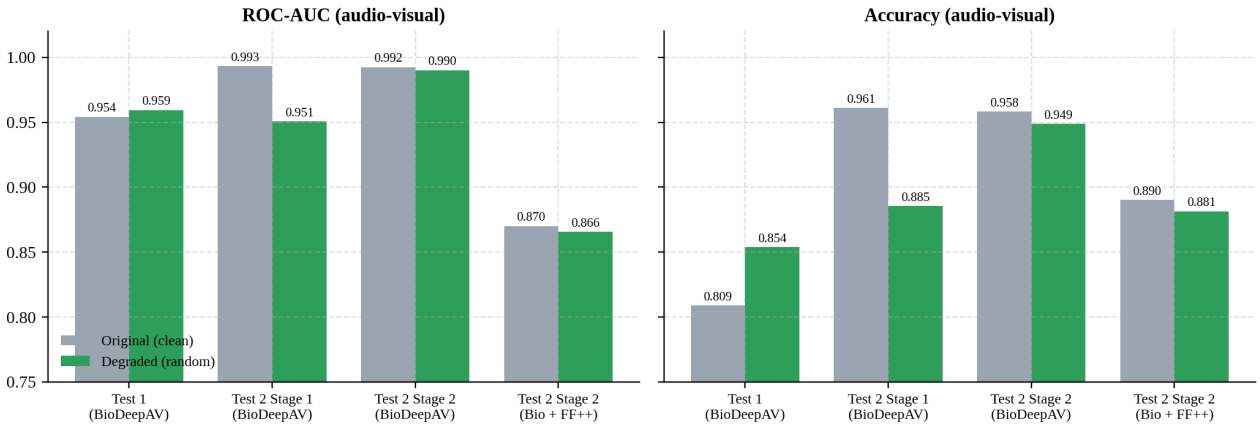

Figure 5.77: Original vs degraded performance on BioDeepAV across Tests 1 and 2 (Test 3: 0.996 / 0.992 AUC; Test 4 BioDeepAV-trained: 0.992 / 0.988).

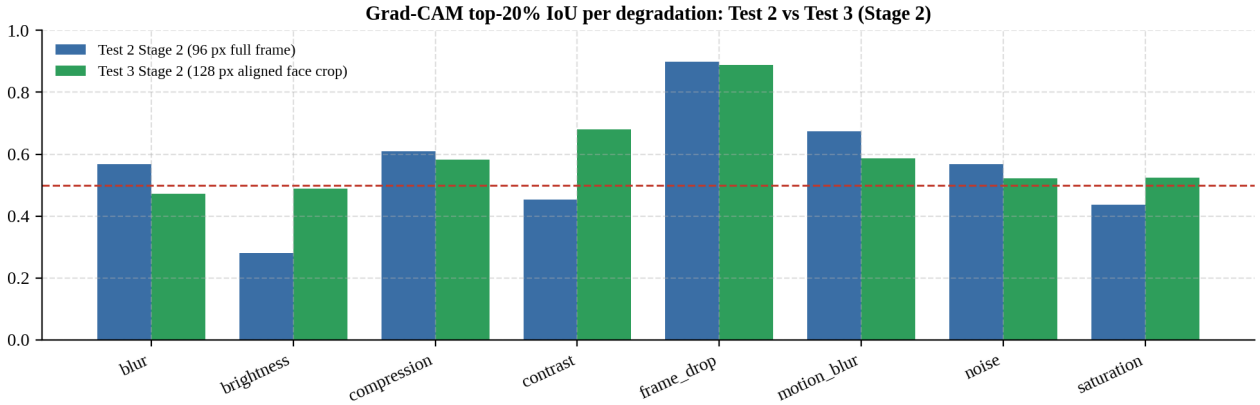

Figure 5.78: Grad-CAM stability per degradation type, Test 2 vs Test 3 (Stage 2); Test 4 per-degradation values are in Figure 5.31.

What the comparison shows

- **Each test fixed the main problem of the one before.** Test 1 called many real videos fake; Test 2 fixed that on BioDeepAV but collapsed on FF++; Test 3 lifted FF++ AUC to 0.80; Test 4 roughly halved the explanation instability (35.8% → 19.6%) and cut audio dependence (12.1% → 2.6% flips).

- **Degraded training works in every test.** Stage 2 removes most of the damage seen in Stage 1 (BioDeepAV degraded AUC: Test 2 0.951 → 0.990; Test 3 0.915 → 0.992; Test 4 0.908 → 0.988). Noise is consistently the degradation that breaks a model trained without degraded data.
- **The explanation gap shrinks only when it is trained for.** Face crops (Test 3) helped a little; adding the Grad-CAM consistency loss (Test 4) helped most. Even then, one in five (BioDeepAV) unchanged predictions has a substantially changed explanation, and brightness and saturation remain the hardest.
- **Generalisation across datasets has not improved.** Single-dataset models fail on the other dataset in both Test 3 and Test 4 (AUC 0.17–0.53); only joint training (Test 3) gives good results on both.
- **Shortcuts persist in reduced form:** the BioDeepAV resolution confound (narrowed but not removed by Test 4) and audio presence (decisive when a model has never seen audio).

Adding Test 5, the basic retest, the improvement suite and Test 6 to the comparison

Test 5 (Section 5.6), the basic retest (Section 5.7) and the improvement suite (Section 5.8) and Test 6 (Section 5.9) were added after Table 5.129 was compiled. Test 5 uses its own, stricter split, so its values are not paired with Tests 2–4; each Test 5 value is the mean of three seeds. Test 5 variant 5D uses the same components as Test 4 (4D) plus hard-degradation sampling; 5A is the plain degraded-training baseline. All values: in-domain Stage 2 model unless marked “cross”.

Measure	T4 Bio-trained S2	T4 FF++-trained S2	T5 5A Bio-trained	T5 5D Bio-trained	T5 5A FF++-trained	T5 5D FF++-trained
BioDeepAV AUC original	0.9919	—	0.966	0.967	—	—
BioDeepAV AUC degraded	0.9876	—	0.939	0.903	—	—
FF++ AUC original	—	0.7475	—	—	0.770	0.736
FF++ AUC degraded	—	0.6800	—	—	0.705	0.704
Cross-dataset AUC (original)	0.5273	0.5000	0.542	0.498	0.500	0.500
Real recognised (original)	94.5%	73.5%	62.1%	75.0%	65.7%	71.2%
Flip rate (in-domain)	2.16%	15.43%	6.46%	11.12%	15.10%	13.04%
Threshold	0.50	0.50	0.01–0.02	0.01–0.12	0.67–0.75	0.68–0.74
Grad-CAM Pearson r	0.855	0.932	0.780	0.795	0.759	0.908
Grad-CAM top-20% IoU	0.674	0.794	0.602	0.620	0.614	0.750
Explanation changed while prediction stable	19.6%	11.1%	34.1%	27.5%	26.8%	11.2%
Audio masking flips (BioDeepAV)	2.6%	37.4%	5.5%	3.4%	49.5%	58.3%
Seeds / Grad-CAM pairs per run	1 / 150	1 / 150	3 / 320	3 / 320	3 / 400	3 / 400

Table 5.130: Test 5 next to Test 4 (Test 5: seed means, own split; Tests 4 and 5 are not a paired comparison). Thresholds are ranges over seeds.

Item	Test 5	Basic retest	Improvement suite
Script	9-10-26_test5_independent.py	151026.py	171026.py
Training data	One dataset per model (BioDeepAV; FF++) × variants 5A–5D × 3 seeds, tested on both	Both datasets jointly, original videos only; 1 seed	As basic retest (same split and cache); 5 configurations × 3 seeds
Model / input	AudioVisualDeepFakeCNN; face crop 128×128, resolution-normalised (80 px face)	EfficientNet-B0 (ImageNet) + temporal transformer, video only; 192 px crop, 112 px face bottleneck	As basic retest
Degradations	Only the provided files: 8 / 10 types, 3 levels; hard-type sampling in 5B–5D	None (augmentation only during training)	None
Training	Stage 1 → Stage 2; 5A CE; 5B hard sampling; 5C pairs + Grad-CAM consistency; 5D + feature consistency + curriculum	Single stage, 12 epochs, balanced sampling, label smoothing, EMA	base12; hard-type samplers (hardfix12, harddro12); 20 and 30 epochs
Threshold	Validation (BioDeepAV 0.01–0.12; FF++ 0.66–0.77)	Validation, calibrated: 0.38; ±0.10 = uncertain	Raw / temperature / Platt / isotonic × 3 threshold rules × coverage 100–80%
Test set	3,458 originals + 6,487 degraded (own split)	1,602 originals (own split)	Same 1,602 originals (paired)
Main result	BioDeepAV 0.966 / 0.939; FF++ 0.770 / 0.705 (5A, original / degraded AUC)	BioDeepAV AUC 0.979; FF++ 0.874 (original videos)	base12 0.933 ± 0.006; harddro12 0.948 ± 0.002 (BioDeepAV 0.983, FF++ 0.902); base30 FF++ 0.916
New	Independent re-run, variant ablation, 3 seeds, paired bootstrap, clean reference IoU	Predict any video + GUI; Grad-CAM++, temporal attention, occlusion; faithfulness test	Reproduction check, seed SD, calibration and abstention, Integrated Gradients, deletion / insertion and randomisation tests

Table 5.131: Test 5, basic retest and improvement-suite entries for the overview (Table 5.3) and implementation status (Table 5.4).

What Test 5, the basic retest, the improvement suite and Test 6 add to the comparison

- **Single-run gains need replication.** The Test 4 recipe (≈ 5D) did not beat a plain degraded-training baseline (5A) when repeated with three seeds on a new split; on BioDeepAV it was worse (degraded AUC −0.036, supported).
- **Grad-CAM consistency is the one replicated component**, with supported gains in explanation stability on FF++ (IoU 0.627 → 0.775; changed-while-stable 23.4% → 9.5%), but none on BioDeepAV.
- **In-domain BioDeepAV values are lower than in Tests 3–4** (0.97 vs 0.99) under the stricter, duplicate-aware split; this may mean the earlier test sets contained easier near-duplicates, but it was not tested directly.
- **Cross-dataset failure and the audio-presence shortcut replicate** (AUC 0.49–0.54 and 0.500).
- **A pretrained, video-only, jointly trained detector works best on original videos** (basic retest): FF++ AUC 0.874, the highest of all tests, and BioDeepAV 0.979 without the resolution shortcut; real videos are recognised in 84–91% of cases. Different splits, so indicative only.
- **Seeded, paired tests** (5.8–5.9): group-DRO gave supported gains (AUC +0.015); 20 epochs helped FF++ only; both together (Test 6) lost BioDeepAV.

Adding Test 7 to the comparison

Test 7 (Section 5.10) uses the Test 5 split, test videos, Grad-CAM pairs and seeds, so — unlike all other rows of this section — its comparison with Test 5 is **paired**: the same 3,458 test originals and 6,487 degraded videos, the same three seeds and the same bootstrap clusters. Each cell gives BioDeepAV / FF++ for the model trained on that dataset (seed means).

Measure (in-domain)	T5 5A	T5 5C	T7 6A	T7 6B	T7 6C	T7 6D
AUC original	0.966 / 0.770	0.952 / 0.744	0.956 / 0.726	0.941 / 0.725	0.945 / 0.716	0.928 / 0.714
AUC degraded	0.939 / 0.705	0.922 / 0.694	0.895 / 0.650	0.873 / 0.648	0.878 / 0.633	0.875 / 0.624
Worst-type AUC	0.858 / 0.570	0.840 / 0.559	0.815 / 0.519	0.797 / 0.500	0.791 / 0.435	0.791 / 0.401
Noise AUC	0.961 / 0.756	0.927 / 0.740	0.922 / 0.709	0.883 / 0.703	0.872 / 0.686	0.911 / 0.681
Flip rate (%)	6.5 / 15.1	8.7 / 15.7	6.7 / 18.2	10.3 / 15.7	11.5 / 15.8	12.0 / 15.2
Real recognised	0.621 / 0.657	0.690 / 0.790	0.386 / 0.667	0.599 / 0.627	0.573 / 0.582	0.530 / 0.562
ECE (original)	0.108 / 0.165	0.132 / 0.185	0.177 / 0.140	0.243 / 0.139	0.227 / 0.152	0.235 / 0.187
Grad-CAM top-20% IoU	0.602 / 0.614	0.614 / 0.775	0.523 / 0.596	0.578 / 0.826	0.538 / 0.782	0.549 / 0.699
Grad-CAM IoU, hard types	0.436 / 0.379	0.459 / 0.629	0.439 / 0.467	0.498 / 0.756	0.466 / 0.703	0.465 / 0.593
Grad-CAM Pearson r	0.780 / 0.759	0.804 / 0.916	0.705 / 0.788	0.774 / 0.965	0.742 / 0.946	0.735 / 0.863
Expl. changed, pred. stable (%)	34.1 / 26.8	30.3 / 9.5	42.7 / 28.3	32.6 / 4.2	36.9 / 5.9	36.2 / 15.4
Cross-dataset AUC (Bio → FF++ / FF++ → Bio)	0.542 / 0.500	0.490 / 0.500	0.418 / 0.284	0.392 / 0.246	0.411 / 0.239	0.415 / 0.219
ρ resolution (BioDeepAV-trained, real)	0.518	0.501	0.370	0.366	0.425	0.378
Parameters	584,978	584,978	2,280,919	2,280,919	2,280,919	2,280,919

Table 5.132: Test 7 next to the Test 5 models on the same split and test videos (paired; seed means; BioDeepAV / FF++). Test 5 values were re-scored by 161026.py from the Test 5 prediction and Grad-CAM files and agree with Section 5.6.

What Test 7 adds to the comparison

- **Capacity is not the bottleneck.** A 3.9× larger multi-stream model with a noise-residual stream and a temporal transformer did not beat the Test 5 model on any in-domain detection metric on the same videos.
- **Explanation-consistency training is the most robust finding across Tests 4, 5 and 7.** In Test 7 it improves Grad-CAM stability on both datasets and in joint training (FF++ changed-while-stable 28.3% → 4.2%), extending the FF++-only effect of Test 5 to BioDeepAV.
- **Extra robustness machinery keeps failing to add value** (Test 5 5D; Test 7 6C, 6D): group-DRO with a floor avoids type starvation but lowers FF++ robustness; adversarial shortcut removal did not improve transfer.
- **Cross-dataset generalisation is still absent** and in Test 7 even inverted (0.22–0.42), and jointly trained models separate datasets rather than real and fake on FF++.

5.12 Using the Results in the Thesis

Two frozen detectors are available for the XAI phase: the Test 3 Stage 2 model (trained on both datasets; best overall detection) and the Test 4 BioDeepAV-trained Stage 2 model (best in-domain robustness and most stable explanations on BioDeepAV). Choose according to the scope of the XAI study, and state the choice. Recommended headline results:

Claim	Evidence	Source
Without degraded training, degradation hurts	Stage 1 flip rates 10–16%; severe noise AUC 0.28–0.47 (Tests 3–4)	5.4.9, 5.5.9
Degraded training restores robustness	BioDeepAV AUC under degradation 0.988–0.992 after Stage 2 (Tests 2–4)	5.3.7, 5.4.7, 5.5.7
Prediction stability ≠ explanation stability	Unchanged predictions with changed Grad-CAM: 43.2% (T2), 35.8% (T3), 19.6% / 11.1% (T4)	5.3.11, 5.4.12, 5.5.12
Explanation-consistency training reduces the gap	35.8% → 19.6%; IoU 0.605 → 0.674 (BioDeepAV-trained)	5.5.12
Explanation stability depends on degradation type	Frame drop / fps reduction IoU 0.83–0.95; brightness and saturation 0.44–0.63 (T4 Stage 2)	5.5.12
Face crops + balancing fix the FF++ collapse	FF++ AUC 0.584 → 0.804 on the same test videos	5.11
Robustness does not imply cross-dataset generalisation	Cross-dataset AUC 0.165–0.527 (Tests 3–4)	5.4.10, 5.5.10
Audio dependence can be reduced in-domain	Audio-masking flips 12.1% → 2.6%; audio degradations ≤ 3.5% flips	5.5.8

Table 5.133: Recommended thesis claims.

Do not claim

- That BioDeepAV's AUC above 0.99 shows the model detects manipulations — BioDeepAV real and fake differ in resolution (AUC 0.994 from resolution alone); Test 4 only narrowed this.
- That the detector works across datasets — cross-dataset AUC is at or below chance in Tests 3 and 4.
- That Test 4's Grad-CAM maps are more *correct* — only that they are more *stable*; stability was partly optimised by the training loss.
- That any single Test 4 component caused its gains — only variant 4D was run (no 4A–4C ablation).
- That degradation improves detection (Test 1 degraded > clean is a training-mix effect), or the Mid report figures as Test 1's audio-visual result.
- That audio makes the detector robust — audio-presence is a shortcut across datasets (FF++-trained model on BioDeepAV).

State in the method section

- Test 4: resolution normalisation (360 px frame, 80 px face bottleneck); paired Stage 2 with λ_{feat} 0.1 and λ_{cam} 0.05 (clean anchor); curriculum; audio 50/25/25; only provided degraded files (max 3 per original, class-balanced); one model per dataset; model selection by validation AUC, never by explanation metrics.

- Test 3: YuNet face crops (99% of frames), 128×128, source face boxes for degraded copies, balanced sampling, validation threshold 0.31.
- Tests 1–2: 96×96 full frames with 16 sampled frames; degraded files re-encoded while originals are not.
- All splits are identity-disjoint (0 crossings); Tests 2–4 share the same test originals; models frozen (SHA-256) before XAI.

Test 5, basic retest, improvement-suite and Test 6 additions to the recommended claims

Claim	Evidence	Source
Degraded training restores robustness (replicated with 3 seeds on a new split)	BioDeepAV degraded AUC 0.795 (Stage 1) → 0.939 (5A); FF++ 0.631 → 0.705	5.6.7
Hard-degradation sampling does not reliably improve robustness	No supported gain for 5B − 5A; FF++ brightness AUC −0.071 (supported)	5.6.9, 5.6.12
Grad-CAM consistency training improves explanation stability (FF++)	Hard-type IoU 0.398 → 0.629; changed-while-stable 23.4% → 9.5% (5B → 5C), all seeds agree	5.6.11, 5.6.12
Single-run differences are not method effects	Original AUC varies by up to 0.07 between seeds of the same variant	5.6.7
Robustness does not imply cross-dataset generalisation (replicated)	Cross-dataset AUC 0.49–0.54 / 0.500 for all variants and seeds	5.6.10
The detector separates real from fake original videos (basic retest)	Test AUC 0.979 BioDeepAV, 0.874 FF++; 85.9% accuracy on non-uncertain videos	5.7.5
Grad-CAM++ is faithful for REAL decisions, not for FAKE decisions	CAM beats random region in 82–84% (real) vs 8–26% (fake) of videos	5.7.6
The basic-retest result is stable across seeds	AUC 0.933 ± 0.006 (3 seeds); same split and best epoch on re-run	5.8.3, 5.8.4
Adaptive hard-type sampling (group-DRO) improves detection of hard FF++ methods	AUC +0.015, bal. acc. +0.031, NeuralTextures +0.161; supported, BioDeepAV unchanged	5.8.5
Isotonic calibration and a 10% abstention band make the decision more reliable	ECE 0.19 → 0.035; bal. acc. 0.87 → 0.89 at 90% coverage	5.8.6
Grad-CAM++ and occlusion pass the model-randomisation check; Integrated Gradients does not	Spearman ρ 0.10 / −0.02 vs 0.74–0.78	5.8.7
Combining group-DRO with 20 epochs trades BioDeepAV for FF++ performance	FF++ AUC +0.010 (supported); BioDeepAV −0.016 (supported); pooled AUC inconclusive	5.9.4
Grad-CAM++ quality is preserved across training variants	Faithfulness and stability non-inferior (7 / 8 tests, margin 0.05) on all 12 runs	5.9.7

Table 5.134: Recommended thesis claims added by Test 5, the basic retest, the improvement suite and Test 6.

Do not claim (Test 5, basic retest, improvement suite and Test 6)

- That the Test 4 method (paired consistency training + curriculum) improves detection robustness — in Test 5 the full method (5D) was worse than the baseline on BioDeepAV.
- That Grad-CAM consistency stabilises explanations on every dataset — there was no supported effect on BioDeepAV, and stability is not correctness.
- That differences between Test 4 and Test 5 are caused by the method — the splits differ.
- That the basic-retest heat-maps show the manipulated region of a fake video — the faithfulness test does not support this.
- That the basic-retest detector works on any deepfake — NeuralTextures stayed at chance and other generators were not tested.

- That longer training improves the detector in general — in the improvement suite BioDeepAV AUC fell in all seeds with 30 epochs.
- That a single explanation method is faithful in all tests — deletion results reverse between grey and blur perturbation (Section 5.8.7).
- That harddro20 is the best overall detector — its pooled AUC did not beat harddro12 and no Test 6 comparison survives Holm correction.

State in the method section (Test 5)

- Independent re-run: new inventory with full-file SHA-256, identity and content-duplicate grouping (787 groups), split seed 20261009, own caches, freshly initialised weights; leakage gate PASS with 0 confirmed crossings.
- Four Stage 2 variants (5A–5D) × two training sets × three seeds (42, 123, 2026); model selection by mean validation original/degraded AUC; validation balanced-accuracy threshold.
- Differences reported as supported only if the paired cluster-bootstrap 95% CI (1,000 replicates over source groups) excludes zero and all seeds agree in sign.

Test 7 additions to the recommended claims

Claim	Evidence	Source
A larger multi-stream architecture does not improve robustness on the same data	6A − 5A: degraded AUC −0.044 (BioDeepAV), −0.055 (FF++), inconclusive; 6D − 5A −0.064 / −0.081 (supported deterioration)	5.10.6, 5.10.10
Degraded training restores robustness (replicated with a new architecture)	BioDeepAV degraded AUC 0.735 (Stage 1) → 0.873–0.895; FF++ 0.575 → 0.624–0.650	5.10.6
Paired Grad-CAM + contrastive consistency training improves explanation stability on both datasets	6B − 6A IoU +0.055 (BioDeepAV), +0.230 (FF++), +0.080 / +0.147 (Joint), all supported; FF++ changed-while-stable 28.3% → 4.2%	5.10.9, 5.10.10
A uniform floor in group-DRO sampling prevents degradation-type starvation	Severe-noise AUC 0.94–0.99 for 6C / 6D (Test 5 5D: 0.665)	5.10.7
Shortcut adversaries do not reduce shortcut use	6D − 6C: no supported gain in cross-dataset AUC, resolution ρ or ECE; three supported deteriorations	5.10.8, 5.10.10
Joint training on both datasets learns dataset identity, not FF++ manipulations	Joint → FF++ AUC 0.42–0.47; FF++ real videos recognised 0–2%	5.10.6

Table 5.135: Recommended thesis claims added by Test 7.

Do not claim (Test 7)

- That the MSTF-AV architecture is a better deepfake detector — it was not better than the Test 5 model on any in-domain metric, and the architecture and optimiser settings changed together.
- That 6B's stable Grad-CAM maps show where the manipulation is — stability was optimised by the training loss and there is no localisation ground truth.
- That removing the audio-presence shortcut improved cross-dataset transfer — FF++-trained Test 7 models are below chance on BioDeepAV (AUC 0.22–0.28).
- That temperature scaling calibrates the BioDeepAV detector — calibrated ECE stayed at 0.18–0.24 (Test 5 uncalibrated: 0.11).

5.13 Final Test — Pretrained, Degradation-Aware and Explanation-Consistent Video Detector with Calibrated Decisions (FINALTEST.py)

5.13.1 Overview

The final test brings together what Tests 1–7, the basic retest and the improvement suite established. Degraded training restores robustness (all tests); paired Grad-CAM consistency training is the only component that replicated as an improvement of explanation stability (Test 5 5C, Test 7 6B); a larger model trained from scratch did not help (Test 7); an ImageNet-pretrained EfficientNet-B0 with a temporal transformer gave the best detection on original videos (Sections 5.7–5.8) but had never been trained or tested on degraded videos; audio presence is a dataset shortcut; and isotonic calibration with an abstention band made decisions more reliable (Section 5.8). *FINALTEST.py* combines these elements in one pre-registered experiment and ends by freezing **one final detector** that can be applied to any video.

Three variants are compared. **F0** fine-tunes the pretrained detector on clean originals for 12 epochs with the group-DRO dataset × folder sampler of the improvement suite (harddro12). **F1** continues F0 with unpaired 50/50 original/degraded clips. **F2**, pre-registered as the final system, continues F0 with paired original/degraded clips, a Grad-CAM consistency term ($\lambda = 0.05$) and a supervised contrastive term ($\lambda = 0.10$). Each variant was trained on BioDeepAV, on FF++ and on both datasets jointly, with seeds 42, 123 and 2026 (27 runs, all completed; 79.0 h on the RTX 3050, 24–28 Oct 2026). The Test 5 split, face-crop caches, degraded videos and Grad-CAM pairs were re-used read-only, so every comparison with Test 5 (5A, 5C) and Test 7 (6A, 6B) is **paired** on the same 3,458 test originals, 6,487 degraded test videos, 720 Grad-CAM pairs and seeds.

Key result. The pretrained, degradation-trained detector is the **best detector of the thesis**. In-domain degraded AUC rises from 0.939 to 0.982 on BioDeepAV and from 0.705 to 0.899 on FF++ compared with the Test 5 baseline 5A (F1 − 5A, supported; FF++ significant after Holm). Degraded training again helps the pretrained model (F1 − F0: degraded AUC +0.031 to +0.040, supported in all four settings). Explanation-consistency training (F2) improved Grad-CAM stability on FF++ and in joint training (hard-type IoU +0.067 / +0.077, explanation changed while prediction stable 27.3% → 18.8% on FF++), but it did not keep detection non-inferior on BioDeepAV, and the joint F2 models were unstable across seeds. On validation, joint F2 fell below joint F1 (0.757 vs 0.777), so the pre-registered fallback froze **joint F1 (seed 42)** as the final model: test AUC 0.958 / 0.942 (BioDeepAV original / degraded) and 0.929 / 0.860 (FF++), isotonic ECE 0.05–0.08, and a prediction pipeline that reproduces the evaluation input exactly. Cross-dataset generalisation is still absent (0.35–0.59). Verdicts: H1, H2, H3, H4 partially supported; H5 not supported; H6 supported.

5.13.2 Code pipeline

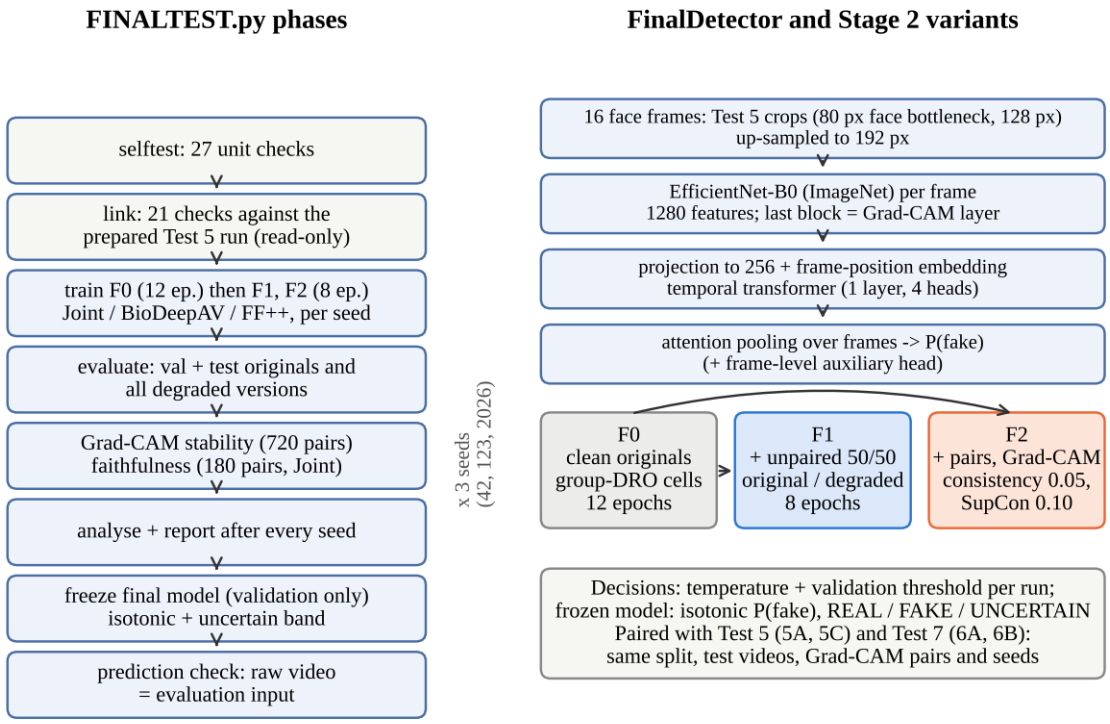

Figure 5.79: What FINALTEST.py does (left) and the detector with its three variants (right). Every phase is resumable; a report is written after each seed. A smoke mode runs the whole pipeline on a small subset in a separate folder.

Before training, the script ran 27 unit checks (all passed), among them: model output shapes, Grad-CAM and Grad-CAM++ ranges and sign handling, the second-order gradient needed by the Grad-CAM consistency loss on the GPU with mixed precision, temperature and isotonic calibration, the abstention band, the weighted bootstrap AUC and ECE (equal to the resampled-video values), Holm correction, the group-DRO sampler, paired augmentation, EMA, and a guard that blocks loading any checkpoint of an earlier test. The link phase then verified 21 conditions against the Test 5 run (leakage gate PASS, split-manifest and Grad-CAM-pair SHA-256 equal to the Test 5 gate, caches complete, every degraded video in its source's partition); all passed. The protocol (all settings, comparisons and hypotheses) was hashed and written before the first training step.

5.13.3 Setup

Item	Test 5 (5A / 5C) and Test 7 (6A / 6B)	Final test (F0 / F1 / F2)
Script	9-10-26_test5_independent.py; 161026.py	FINALTEST.py (4,357 lines)
Model	AudioVisualDeepFakeCNN (584,978 parameters); MSTF-AV (2,280,919)	EfficientNet-B0 (ImageNet) per frame + temporal transformer (1 layer, 4 heads) + attention pooling; 4,967,167 parameters; video only
Data and input	Test 5 split and caches; 16 cached frames, 128 px face crops (80 px face bottleneck); 8-frame clips	Identical caches (read-only), up-sampled to 192 px on the GPU; 8 random sorted frames for training, all 16 frames for evaluation
Training sets, seeds	5A/5C: BioDeepAV, FF++; 6A/6B: + Joint; seeds 42, 123, 2026	BioDeepAV, FF++, Joint; seeds 42, 123, 2026
Stage 1	8 epochs on clean originals	F0: 12 epochs, lr 1e-4 (backbone) / 1e-3 (head), group-DRO over dataset × folder cells (η 2.0, 50% floor), label smoothing 0.05, frame loss 0.3, EMA 0.998, real-world augmentation
Stage 2	10 epochs; 5A/6A unpaired; 5C/6B paired + Grad-CAM consistency	8 epochs from the best F0 (EMA) weights, lr 5e-5 / 5e-4; F1 unpaired 50/50; F2 paired + Grad-CAM consistency 0.05 + supervised contrastive 0.10
Batch	32 clips	8 clips × gradient accumulation 2 (151026 setting); AMP; warm-up + cosine; patience 4
Model selection	Mean of validation original and degraded AUC	F0: validation original AUC; F1/F2: mean of validation original and degraded AUC (macro over datasets)
Calibration, decision	Raw (T5) / temperature (T7); validation balanced-accuracy threshold	Temperature + threshold per run; frozen model: isotonic calibration, threshold and an UNCERTAIN band tuned for 90% validation coverage
XAI	Grad-CAM stability on 720 pairs	Same pairs and metrics (+ Grad-CAM++, attention); faithfulness: deletion / insertion vs a random map on 180 pairs (Joint models)
Statistics	Paired cluster bootstrap (1,000) + sign agreement	Same; Holm over 8 primary comparisons; non-inferiority margins 0.01 AUC / 0.05 IoU

Table 5.139: Final-test configuration compared with Test 5 and Test 7.

	Comparison	Hypothesis
H1	F1 − 5A (in-domain)	An ImageNet-pretrained backbone improves robustness over the Test 5 baseline
H2	F1 − F0	Degraded training also helps the pretrained detector
H3	F2 − F1	Explanation-consistency training improves Grad-CAM stability without losing detection (non-inferiority 0.01 AUC)
H4	F2 (Joint) − 5A (dataset-specific)	The final joint detector beats the best earlier detector on the same test videos
H5	F2 cross-dataset AUC	Dataset-specific final detectors generalise (95% CI above 0.5)
H6	Joint F2: isotonic vs temperature; 90% vs 100% coverage	Isotonic calibration and abstention make decisions reliable on original and degraded videos
RQ	Joint F0–F2: degraded − original	Exploratory: does degradation change Grad-CAM faithfulness?

Table 5.140: Pre-registered hypotheses. "Supported" requires the 95% paired cluster-bootstrap CI to exclude zero and all three seeds to agree in sign. Primary family (Holm): F1 − 5A and F1 − F0 (degraded AUC), F2 − F1 (hard-type IoU) and F2 (Joint) − 5A (degraded AUC), each on both datasets.

5.13.4 Data

Dataset	Train	Val	Test	Degraded test videos	Grad-CAM / faithfulness pairs
BioDeepAV	1,849	433	1,420	4,076 (8 types × 3 levels)	320 / 80
FF++	4,263	699	2,038	2,411 (10 types × 3 levels)	400 / 100
Total	**6,112**	**1,132**	**3,458**	**6,487**	**720 / 180**

Table 5.141: Originals per partition (Test 5 split; 10,702 originals and 19,729 degraded videos). Faithfulness used every fourth Grad-CAM pair. Bootstrap clusters: 53 BioDeepAV and 101 FF++ source groups.

5.13.5 Training and model selection

Training set	Variant	Best / run epochs (42 · 123 · 2026)	Val. score (42 · 123 · 2026)	Threshold (mean)	Temp. (mean)	Val. ECE raw → temp.	Time / run
Joint	F0	11/12 · 4/8 · 2/6	0.853 · 0.837 · 0.782	0.56	1.27	0.127 → 0.125	186 min
	F1	2/6 · 1/5 · 1/5	0.803 · 0.778 · 0.750	0.63	1.33	0.092 → 0.093	230 min
	F2	2/6 · 1/5 · 5/8	0.818 · 0.748 · 0.704	0.61	1.49	0.120 → 0.115	279 min
BioDeepAV	F0	2/6 · 6/10 · 2/6	0.950 · 0.992 · 0.874	0.40	0.87	0.137 → 0.110	50 min
	F1	2/6 · 1/5 · 3/7	0.900 · 0.960 · 0.931	0.49	1.20	0.075 → 0.076	81 min
	F2	1/5 · 1/5 · 4/8	0.902 · 0.953 · 0.941	0.38	1.02	0.090 → 0.083	80 min
FF++	F0	12/12 · 12/12 · 11/12	0.899 · 0.881 · 0.895	0.51	1.17	0.144 → 0.148	183 min
	F1	7/8 · 8/8 · 8/8	0.895 · 0.892 · 0.890	0.57	1.20	0.103 → 0.104	245 min
	F2	6/8 · 8/8 · 8/8	0.881 · 0.886 · 0.883	0.62	1.04	0.083 → 0.086	247 min

Table 5.142: Training of the 27 runs (validation data only). Stage 2 often stopped early (best epoch 1–2 for BioDeepAV and Joint), while FF++ F0 used all 12 epochs. Summed training time 79.0 h (Joint 34.8 h, FF++ 33.7 h, BioDeepAV 10.5 h).

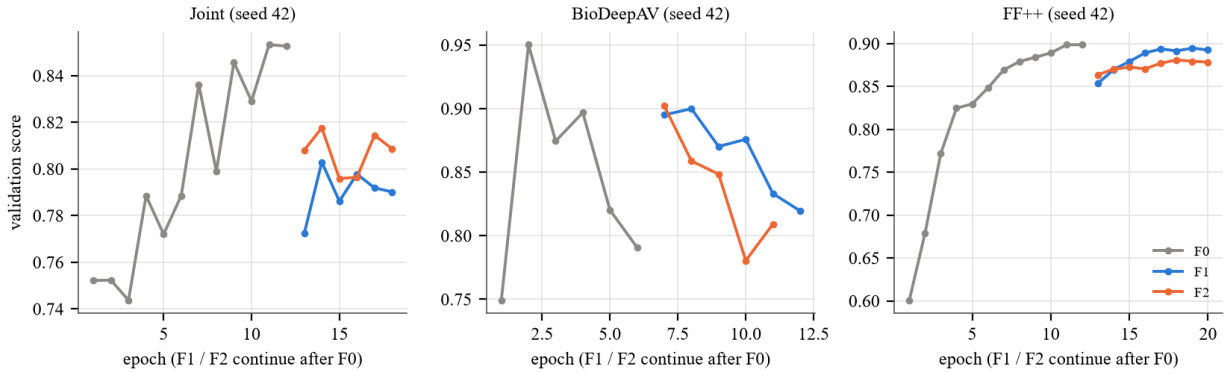

Figure 5.80: Validation score per epoch, seed 42 (F1/F2 continue after F0).

Joint training was the hardest setting: validation scores of 0.70–0.85 against 0.87–0.99 for the dataset-specific runs, with large differences between seeds (Joint F2 0.818 / 0.748 / 0.704). Seed 2026 was the weakest seed for every Joint variant.

5.13.6 Detection on original and degraded test videos

BioDeepAV-trained → BioDeepAV

Model	AUC orig.	AUC degr.	Worst-type AUC	Flips %	Real rec.	Fake det.	ECE orig.
5A (T5)	0.966 (.032)	0.939 (.038)	0.858 (.053)	6.5 (.7)	0.621 (.364)	0.990 (.011)	0.108 (.068)
5C (T5)	0.952 (.038)	0.922 (.047)	0.840 (.051)	8.7 (1.5)	0.690 (.173)	0.990 (.005)	0.132 (.056)
6A (T7)	0.956 (.014)	0.895 (.024)	0.815 (.053)	6.7 (1.3)	0.386 (.097)	0.996 (.001)	0.177 (.065)
6B (T7)	0.941 (.018)	0.873 (.028)	0.797 (.054)	10.3 (2.6)	0.599 (.134)	0.966 (.038)	0.243 (.058)
F0	0.975 (.015)	0.946 (.025)	0.847 (.027)	9.3 (4.1)	0.833 (.069)	0.940 (.093)	0.093 (.056)
F1	0.990 (.003)	0.982 (.005)	0.921 (.030)	4.5 (1.6)	0.872 (.044)	0.996 (.001)	0.062 (.010)
F2	0.986 (.005)	0.973 (.009)	0.916 (.033)	5.6 (1.9)	0.828 (.007)	0.994 (.003)	0.078 (.034)

Table 5.143: BioDeepAV-trained → BioDeepAV: seed mean (SD over seeds). Test 5 stored uncalibrated probabilities, so its ECE is not temperature-scaled; AUC does not depend on calibration.

FF++-trained → FF++

Model	AUC orig.	AUC degr.	Worst-type AUC	Flips %	Real rec.	Fake det.	ECE orig.
5A (T5)	0.770 (.014)	0.705 (.008)	0.570 (.009)	15.1 (2.6)	0.657 (.069)	0.745 (.054)	0.165 (.015)
5C (T5)	0.744 (.037)	0.694 (.027)	0.559 (.041)	15.7 (1.5)	0.790 (.062)	0.584 (.124)	0.185 (.025)
6A (T7)	0.726 (.018)	0.650 (.027)	0.519 (.040)	18.2 (.4)	0.667 (.110)	0.663 (.113)	0.140 (.005)
6B (T7)	0.725 (.003)	0.648 (.008)	0.500 (.004)	15.7 (2.5)	0.627 (.093)	0.715 (.085)	0.139 (.002)
F0	0.919 (.002)	0.859 (.001)	0.734 (.016)	11.3 (1.0)	0.893 (.037)	0.791 (.043)	0.192 (.012)
F1	0.934 (.003)	0.899 (.003)	0.785 (.007)	9.4 (1.1)	0.897 (.023)	0.821 (.014)	0.156 (.019)
F2	0.931 (.001)	0.893 (.006)	0.805 (.008)	8.8 (.4)	0.902 (.033)	0.813 (.027)	0.141 (.019)

Table 5.144: FF++-trained → FF++: seed mean (SD over seeds). Test 5 stored uncalibrated probabilities, so its ECE is not temperature-scaled; AUC does not depend on calibration.

Joint → BioDeepAV

Model	AUC orig.	AUC degr.	Worst-type AUC	Flips %	Real rec.	Fake det.	ECE orig.
6A (T7)	0.953 (.006)	0.905 (.010)	0.850 (.028)	8.3 (2.7)	0.746 (.194)	0.935 (.066)	0.223 (.024)
6B (T7)	0.931 (.051)	0.887 (.044)	0.835 (.053)	9.2 (.3)	0.692 (.166)	0.920 (.056)	0.185 (.040)
F0	0.933 (.021)	0.910 (.008)	0.804 (.014)	12.3 (2.2)	0.861 (.023)	0.854 (.190)	0.168 (.040)
F1	0.954 (.005)	0.940 (.003)	0.903 (.007)	9.0 (2.0)	0.886 (.013)	0.967 (.014)	0.132 (.032)
F2	0.938 (.032)	0.906 (.029)	0.809 (.050)	12.0 (6.4)	0.923 (.048)	0.715 (.237)	0.191 (.057)

Table 5.145: Joint → BioDeepAV: seed mean (SD over seeds). Test 5 stored uncalibrated probabilities, so its ECE is not temperature-scaled; AUC does not depend on calibration.

Joint → FF++

Model	AUC orig.	AUC degr.	Worst-type AUC	Flips %	Real rec.	Fake det.	ECE orig.
6A (T7)	0.472 (.030)	0.478 (.022)	0.402 (.048)	0.2 (.2)	0.002 (.003)	0.997 (.002)	0.346 (.017)
6B (T7)	0.437 (.036)	0.457 (.034)	0.343 (.062)	0.2 (.4)	0.007 (.012)	0.998 (.004)	0.342 (.005)
F0	0.855 (.084)	0.753 (.089)	0.596 (.073)	15.7 (1.2)	0.690 (.203)	0.812 (.035)	0.197 (.008)
F1	0.868 (.064)	0.784 (.076)	0.672 (.104)	15.2 (1.2)	0.717 (.217)	0.818 (.035)	0.161 (.017)
F2	0.878 (.036)	0.815 (.037)	0.680 (.109)	13.1 (1.1)	0.875 (.122)	0.701 (.118)	0.222 (.046)

Table 5.146: Joint → FF++: seed mean (SD over seeds). Test 5 stored uncalibrated probabilities, so its ECE is not temperature-scaled; AUC does not depend on calibration.

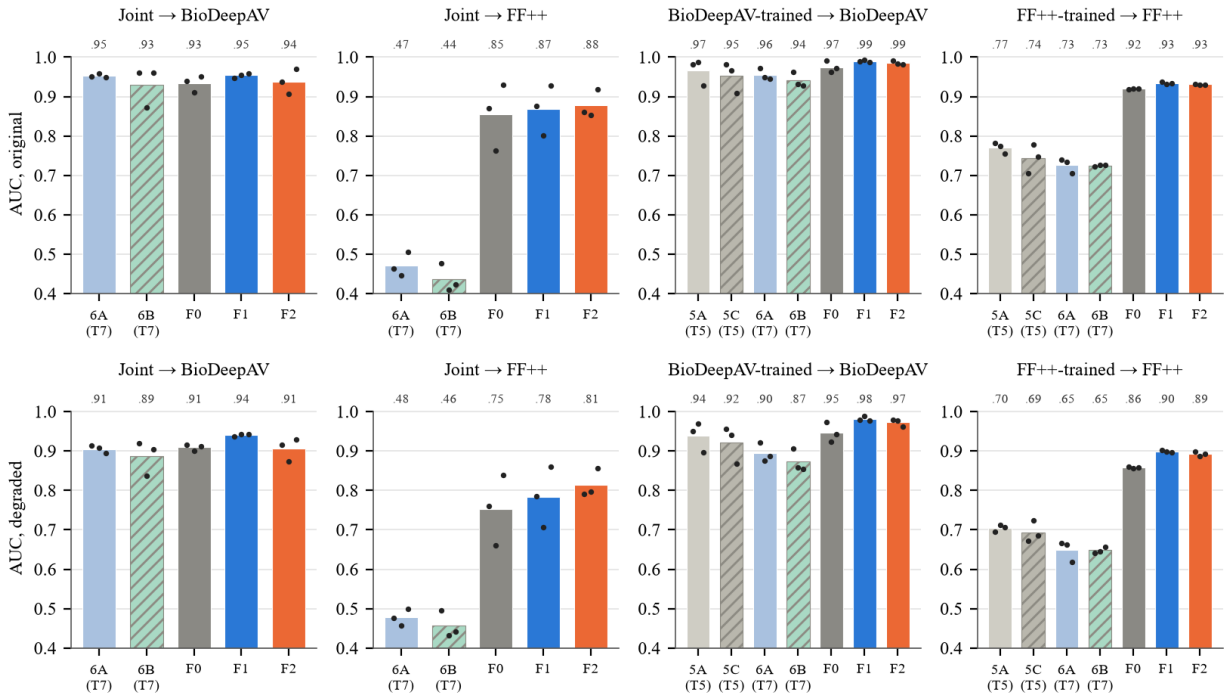

Figure 5.81: AUC on original (top) and degraded (bottom) test videos; bars = seed mean, dots = seeds, label = mean.

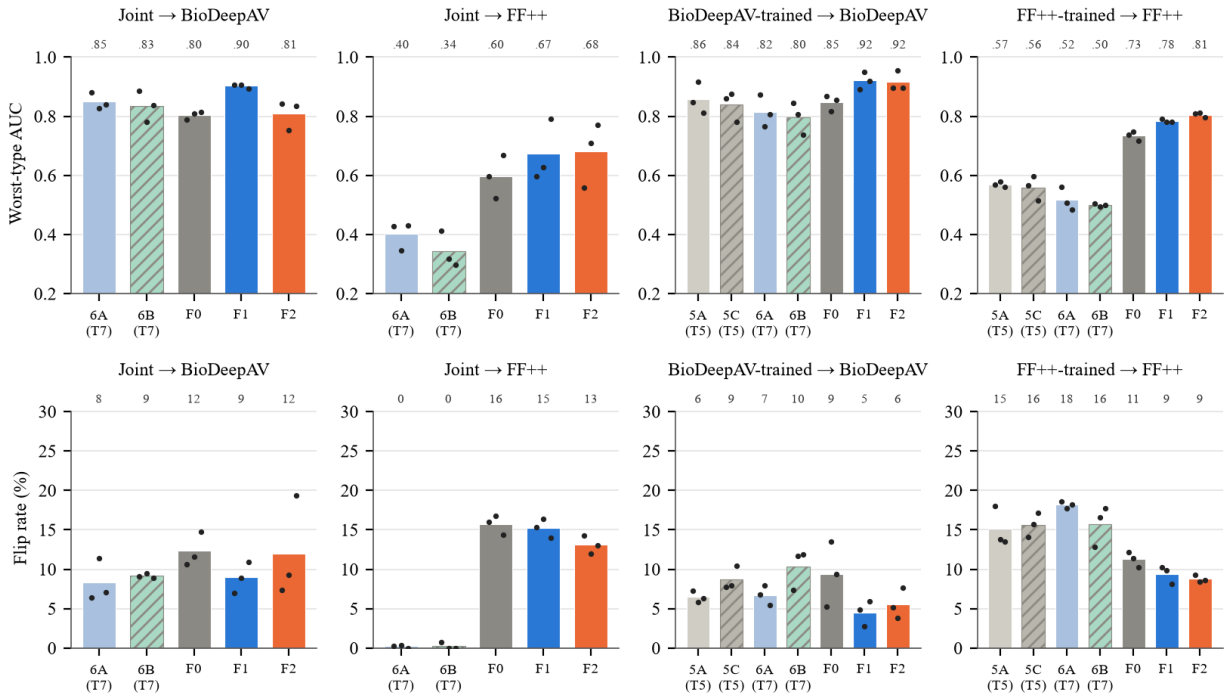

Figure 5.82: Worst degradation-type AUC (top) and prediction flips between each original and its degraded version (bottom, %).

The pretrained detector is a large step forward on **FF++**: in-domain degraded AUC 0.899 (F1) and 0.893 (F2) against 0.705 (5A) and 0.650 (6A), worst-type AUC 0.785–0.805 against 0.500–0.570, and flips 8.8–9.4% against 15–18%. On **BioDeepAV** the gain is smaller because the earlier models were already strong (degraded AUC 0.982 vs 0.939; real videos recognised 0.87 vs 0.62). The **Joint** models are the first joint detectors of the thesis that separate real and fake FF++ videos (AUC 0.855–0.878; Test 7 joint models 0.44–0.47, recognising almost no real FF++ video), but they remain below the

dataset-specific models (Joint F2 → FF++ degraded AUC 0.815 vs 0.893). Within the Joint F2 runs the single threshold traded fake detection for real recognition (BioDeepAV fake detected 0.67 / 0.97 / 0.51 for the three seeds at AUC 0.97 / 0.94 / 0.91).

5.13.7 Per degradation type and per manipulation method

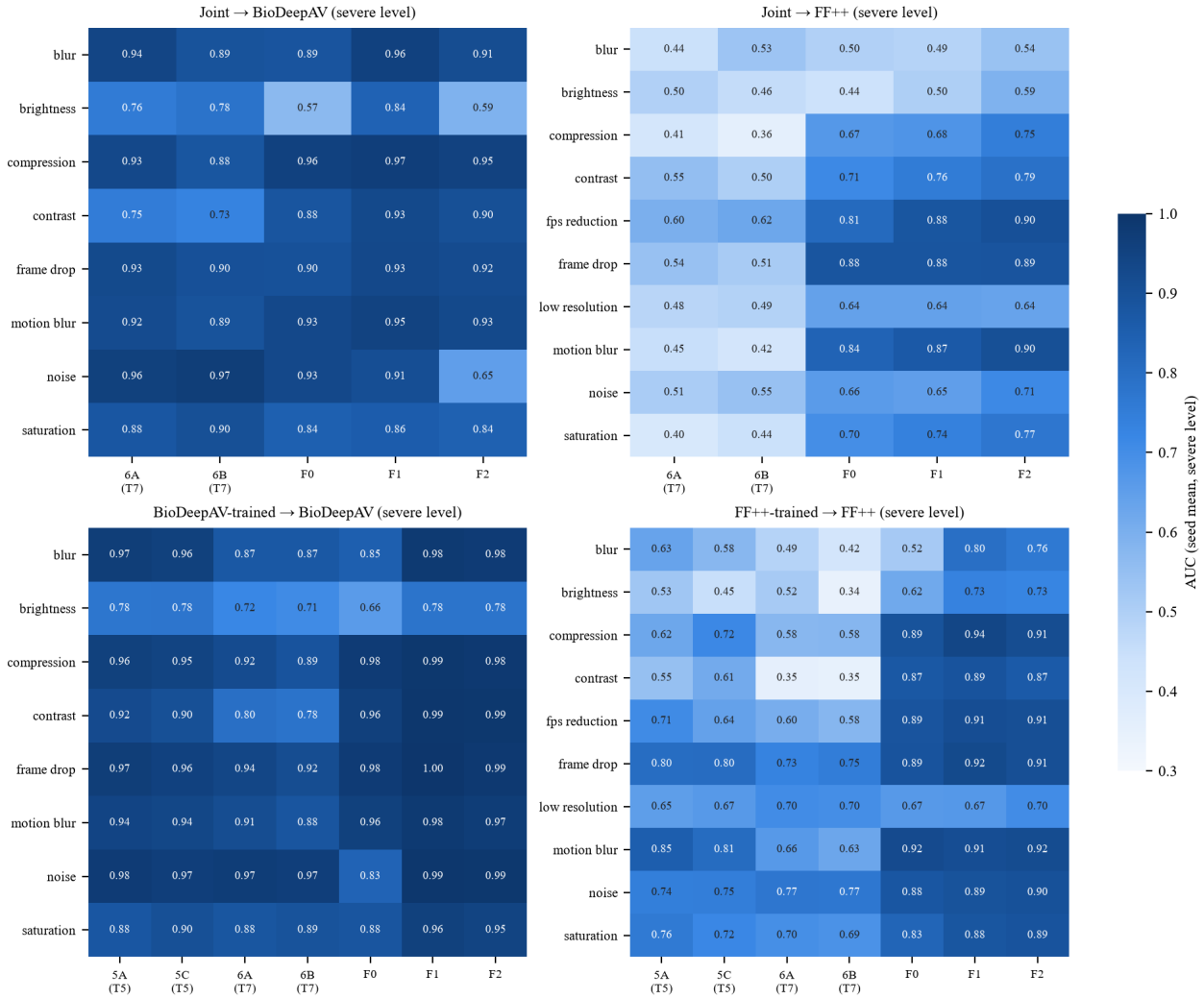

Figure 5.83: AUC per degradation type at the severe level (seed mean).

Training → test	Model	blur	bright.	compr.	contrast	fps red.	frame dr.	low res.	motion bl.	noise	satur.
BioDeepAV → BioDeepAV	5A (T5)	0.96	0.86	0.96	0.95	—	0.96	—	0.95	0.96	0.93
	F0	0.95	0.85	0.97	0.97	—	0.98	—	0.97	0.94	0.94
	F1	0.99	0.92	0.99	0.99	—	0.99	—	0.99	0.99	0.98
	F2	0.98	0.92	0.98	0.99	—	0.99	—	0.98	0.98	0.97
FF++ → FF++	5A (T5)	0.64	0.67	0.68	0.57	0.67	0.79	0.66	0.81	0.76	0.76
	F0	0.73	0.84	0.90	0.87	0.84	0.90	0.77	0.91	0.89	0.87
	F1	0.88	0.89	0.93	0.90	0.88	0.93	0.78	0.91	0.91	0.90
	F2	0.85	0.89	0.92	0.88	0.88	0.92	0.81	0.91	0.90	0.90
Joint → FF++	6A (T7)	0.46	0.51	0.40	0.46	0.58	0.52	0.53	0.46	0.49	0.45
	F0	0.70	0.61	0.77	0.79	0.78	0.84	0.66	0.84	0.78	0.75
	F1	0.71	0.73	0.77	0.82	0.82	0.85	0.67	0.86	0.78	0.78
	F2	0.75	0.80	0.81	0.82	0.83	0.86	0.68	0.88	0.82	0.80

Table 5.147: AUC per degradation type (all severities, seed mean). Brightness and blur remain the hardest types on BioDeepAV and FF++; low resolution is now the weakest FF++ type for F1/F2.

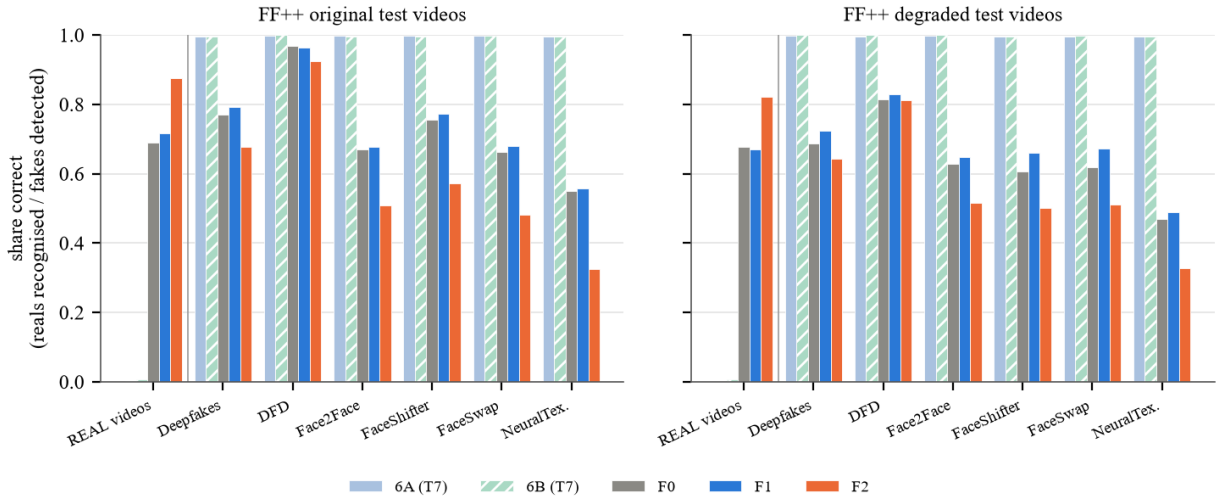

Figure 5.84: FF++ test videos: share of real videos recognised and of each manipulation detected (Joint models). Joint F2 recognises more real FF++ videos (0.875) than F1 (0.717) at the cost of fake detection; NeuralTextures remains the hardest manipulation (detected 0.33–0.56 by the Joint models, 0.43–0.50 by the FF++ models).

5.13.8 Cross-dataset generalisation

Model	Bio → FF++ orig.	Bio → FF++ degr.	FF++ → Bio orig.	FF++ → Bio degr.
5A (T5)	0.542 (.027)	0.517 (.021)	0.500 (.000)	0.500 (.000)
6A (T7)	0.418 (.015)	0.445 (.009)	0.284 (.040)	0.400 (.040)
F0	0.571 (.047)	0.518 (.039)	0.443 (.046)	0.342 (.031)
F1	0.590 (.022)	0.535 (.020)	0.376 (.034)	0.344 (.041)
F2	0.579 (.033)	0.528 (.031)	0.393 (.028)	0.351 (.041)

Table 5.148: Cross-dataset AUC (seed mean, SD). F2 95% CIs: Bio → FF++ 0.499–0.642 (original) and 0.493–0.565 (degraded); FF++ → Bio 0.088–0.528 and 0.095–0.464.

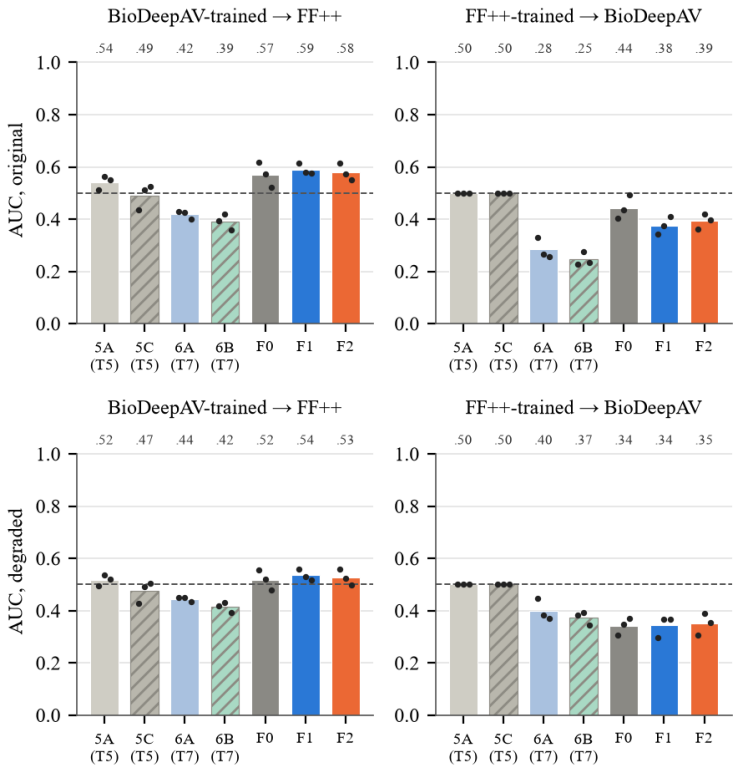

Figure 5.85: Cross-dataset AUC of the dataset-specific models (dashed line = chance).

No model generalises from one dataset to the other. BioDeepAV-trained models reach 0.57–0.59 on FF++ originals, close to chance; FF++-trained models are below chance on BioDeepAV (0.38–0.44 original, 0.34–0.35 degraded), as in Test 7. The pretrained backbone therefore improves in-domain robustness but does not remove the dataset-specific cues.

5.13.9 Calibration and selective decisions

Training → test	Model	Videos	ECE raw	ECE temperature	ECE isotonic	Bal. acc. 100%	Bal. acc. 90% band	Realised coverage
Joint → BioDeepAV	F1	original	0.088	0.132	0.105	0.925	0.931	0.970
		degraded	0.078	0.118	0.099	0.880	0.887	0.965
	F2	original	0.143	0.191	0.139	0.818	0.848	0.931
		degraded	0.125	0.167	0.118	0.777	0.798	0.943
Joint → FF++	F1	original	0.129	0.161	0.087	0.770	0.783	0.956
		degraded	0.083	0.101	0.070	0.699	0.707	0.953
	F2	original	0.190	0.222	0.116	0.789	0.802	0.965
		degraded	0.127	0.148	0.049	0.734	0.748	0.959
BioDeepAV → BioDeepAV	F1	original	0.040	0.062	0.042	0.934	0.940	0.982
		degraded	0.025	0.043	0.032	0.927	0.933	0.979
	F2	original	0.081	0.078	0.058	0.909	0.918	0.986
		degraded	0.073	0.073	0.049	0.903	0.914	0.975
FF++ → FF++	F1	original	0.143	0.156	0.075	0.859	0.870	0.970
		degraded	0.092	0.099	0.035	0.819	0.832	0.952
	F2	original	0.138	0.141	0.077	0.856	0.867	0.968
		degraded	0.077	0.080	0.030	0.811	0.823	0.952

Table 5.149: Calibration of each run fitted on its own validation videos and evaluated on test videos (seed means). The band is tuned for 90% validation coverage; because the isotonic map is a step function, the realised test coverage was 93–99%.

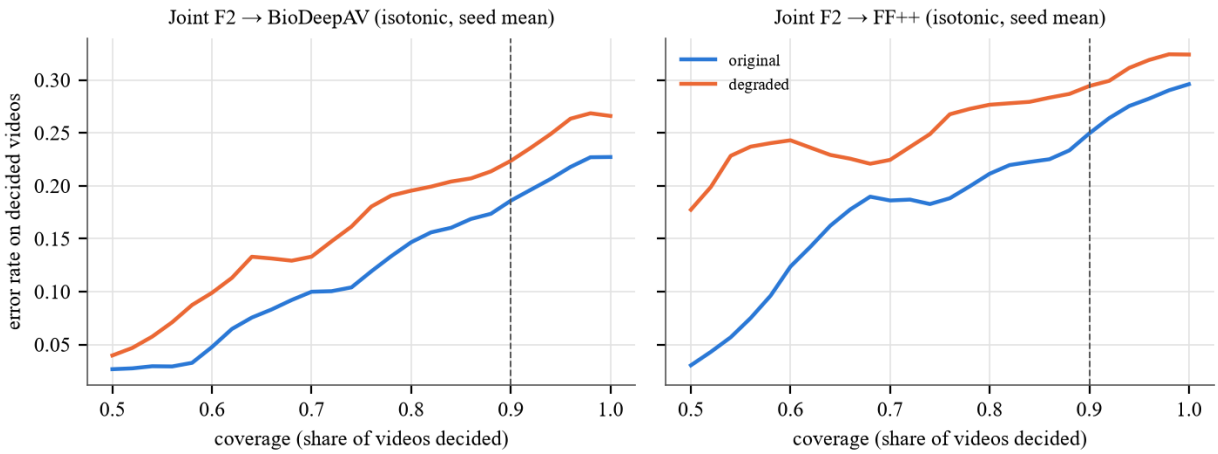

Figure 5.86: Risk–coverage of the Joint F2 models (isotonic; dashed line = 90% coverage). Error falls steadily as the least confident videos are withheld, on original and degraded videos.

Isotonic calibration gave the lowest ECE in every Joint F2 seed, test set and condition (Joint F2 seed means: original 0.206 → 0.128, degraded 0.157 → 0.083 against temperature scaling), and withholding the least confident videos raised balanced accuracy everywhere (Joint F2: 0.803 → 0.825 on originals, 0.755 → 0.773 on degraded videos). H6 is supported, with the caveat that the realised abstention rate was smaller than intended.

5.13.10 Grad-CAM stability under degradation

Training → test	Model	IoU all	IoU hard	Pearson r	Changed while stable %	Clean-ref. IoU	Grad-CAM++ IoU
BioDeepAV → BioDeepAV	5A (T5)	0.602 (.021)	0.436 (.014)	0.780 (.013)	34.1 (1.2)	0.786 (.037)	—
	5C (T5)	0.614 (.013)	0.459 (.017)	0.804 (.015)	30.3 (1.2)	0.786 (.045)	—
	6A (T7)	0.523 (.043)	0.439 (.035)	0.705 (.061)	42.7 (9.1)	0.608 (.047)	—
	6B (T7)	0.578 (.036)	0.498 (.018)	0.774 (.027)	32.6 (7.1)	0.633 (.047)	—
	F0	0.530 (.030)	0.455 (.049)	0.725 (.028)	38.8 (7.1)	0.656 (.042)	0.651 (.020)
	F1	0.535 (.027)	0.455 (.021)	0.729 (.023)	40.7 (4.1)	0.661 (.039)	0.680 (.029)
	F2	0.564 (.050)	0.487 (.059)	0.754 (.045)	36.2 (7.0)	0.699 (.061)	0.661 (.039)
FF++ → FF++	5A (T5)	0.614 (.026)	0.379 (.025)	0.759 (.024)	26.8 (4.3)	0.604 (.037)	—
	5C (T5)	0.775 (.020)	0.629 (.019)	0.916 (.005)	9.5 (2.1)	0.696 (.055)	—
	6A (T7)	0.596 (.021)	0.467 (.026)	0.788 (.021)	28.3 (2.2)	0.501 (.030)	—
	6B (T7)	0.826 (.021)	0.756 (.024)	0.965 (.000)	4.2 (.8)	0.733 (.012)	—
	F0	0.645 (.020)	0.553 (.033)	0.759 (.032)	23.9 (2.4)	0.628 (.015)	0.850 (.009)
	F1	0.626 (.010)	0.543 (.007)	0.756 (.019)	27.3 (.8)	0.610 (.020)	0.839 (.006)
	F2	0.692 (.034)	0.609 (.051)	0.823 (.035)	18.8 (3.9)	0.670 (.029)	0.857 (.015)
Joint → BioDeepAV	6A (T7)	0.462 (.017)	0.383 (.006)	0.653 (.023)	52.5 (5.6)	0.519 (.035)	—
	6B (T7)	0.542 (.040)	0.434 (.040)	0.746 (.030)	39.8 (8.9)	0.596 (.038)	—
	F0	0.554 (.042)	0.477 (.046)	0.711 (.064)	37.2 (6.6)	0.668 (.040)	0.695 (.044)
	F1	0.568 (.026)	0.486 (.036)	0.738 (.029)	35.6 (5.0)	0.689 (.020)	0.714 (.034)
	F2	0.626 (.085)	0.554 (.101)	0.778 (.080)	28.6 (9.4)	0.719 (.101)	0.740 (.017)
Joint → FF++	6A (T7)	0.509 (.028)	0.373 (.029)	0.680 (.047)	44.7 (4.1)	0.453 (.006)	—
	6B (T7)	0.657 (.068)	0.515 (.065)	0.830 (.053)	25.3 (9.5)	0.549 (.071)	—
	F0	0.648 (.034)	0.539 (.054)	0.797 (.030)	22.5 (3.9)	0.660 (.025)	0.802 (.061)
	F1	0.663 (.019)	0.551 (.038)	0.809 (.016)	23.4 (1.5)	0.664 (.033)	0.831 (.022)
	F2	0.716 (.040)	0.628 (.053)	0.856 (.044)	15.3 (4.6)	0.703 (.038)	0.835 (.024)

Table 5.150: Grad-CAM stability between each original and its degraded version on the 720 pre-registered pairs (seed mean, SD). Clean-reference IoU compares the same original on its other eight frames (natural variability). Stability is not correctness.

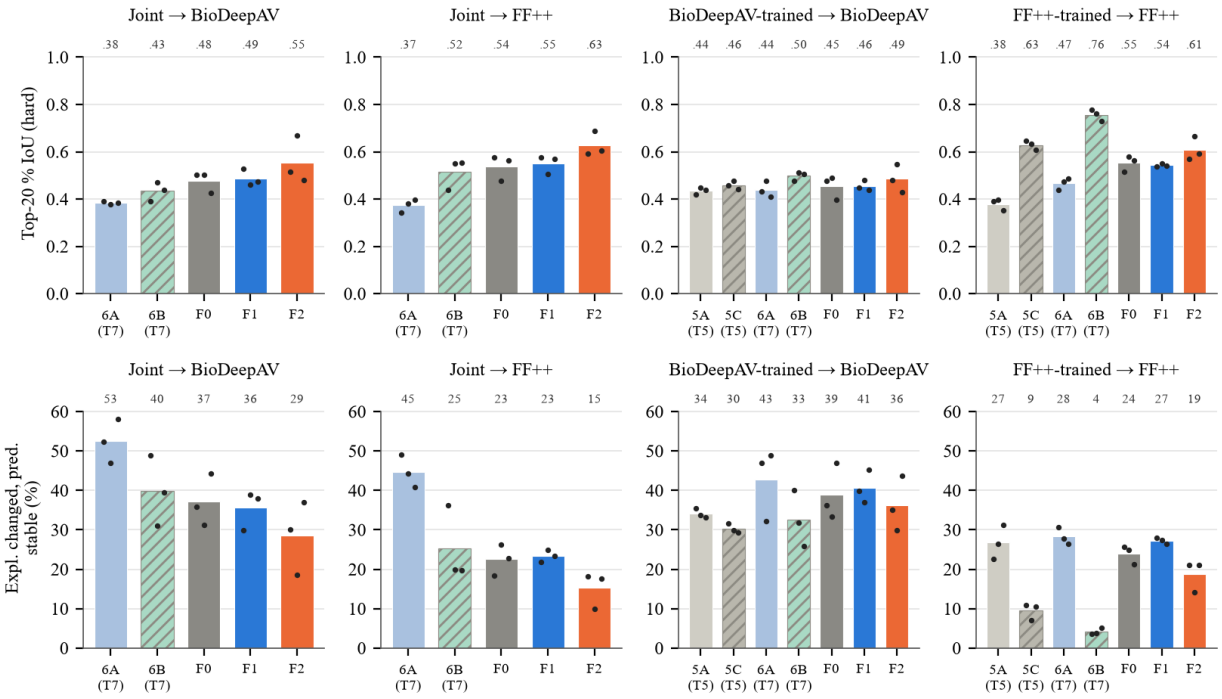

Figure 5.87: Grad-CAM top-20% IoU on the hard degradation types (top) and explanations that changed while the prediction stayed the same (bottom, %).

Explanation-consistency training again made explanations more stable where it was strongest before, on FF++: F2 − F1 hard-type IoU +0.067 (in-domain) and +0.077 (Joint → FF++), changed-while-stable −8.5 and −8.1 points, all supported. On BioDeepAV the gain was smaller (IoU +0.030 in-domain, supported; Joint → BioDeepAV +0.068, inconclusive because seeds 42 and 123 gained almost nothing and seed 2026 gained 0.21). The best Test 7 model 6B remains more stable on FF++ in-domain (IoU 0.826 vs 0.692), but the final models start from much better detection: 6B detected FF++ fakes at AUC 0.725, the final models at 0.93.

5.13.11 Faithfulness under degradation (exploratory)

For every fourth Grad-CAM pair (180 pairs), the Joint models' Grad-CAM map for the predicted class was used to delete (or insert) the top 5–75% of pixels, replaced by grey or by a strong blur, on the original and on the degraded version of the same video. The gain is the area under the confidence curve compared with a random map of the same clip (deletion: random − map; insertion: map − random), so a positive gain means the map points at evidence the model uses. For 22% of the grey-perturbed curves the network returned a non-finite output under mixed precision (mostly F2 seed 2026: 90%; F1 seed 2026: 34%); these curves were excluded, and the summaries below were recomputed from the per-video file with the non-finite values removed (FINALTEST.py has been corrected accordingly).

Model	Metric	All decisions (orig. / degr.)	Predicted FAKE (orig. / degr.)	Predicted REAL (orig. / degr.)
F0	insertion, grey	+0.22* / +0.24*	−0.26† / −0.24†	+0.35* / +0.35*
	deletion, grey	−0.15† / −0.17†	+0.25* / +0.24*	−0.24† / −0.25†
	insertion, blur	+0.10* / +0.11*	+0.32* / +0.30*	−0.02 / +0.01
	deletion, blur	+0.01 / −0.00	−0.13† / −0.13†	+0.09* / +0.07
F1	insertion, grey	+0.30* / +0.26*	−0.26† / −0.23†	+0.41* / +0.38*
	deletion, grey	−0.23† / −0.21†	+0.18* / +0.19*	−0.32† / −0.29†
	insertion, blur	+0.10* / +0.12*	+0.33* / +0.30*	−0.02 / +0.01
	deletion, blur	+0.00 / −0.02	−0.21† / −0.17†	+0.10* / +0.08
F2	insertion, grey	+0.26* / +0.23*	−0.16† / −0.13†	+0.30* / +0.27*
	deletion, grey	−0.23† / −0.19†	+0.15* / +0.13*	−0.26† / −0.22†
	insertion, blur	+0.07 / +0.09*	+0.20* / +0.19*	+0.02 / +0.04
	deletion, blur	+0.02 / +0.00	−0.13† / −0.11†	+0.07 / +0.05

*Table 5.151: Grad-CAM faithfulness gain over a random map (Joint models, seeds pooled, cluster-bootstrap CI). * CI above +0.005 (better than random); † CI below −0.005 (worse than random).*

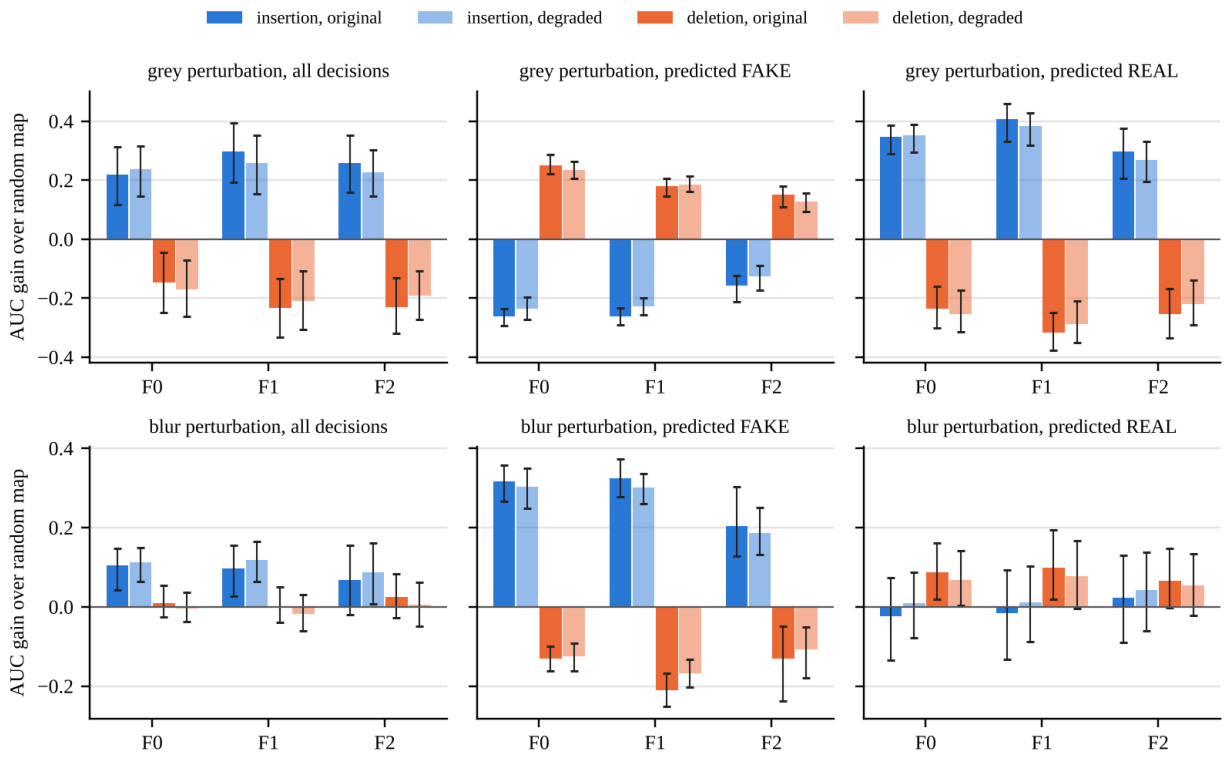

Figure 5.88: Faithfulness gains with 95% CI for original (dark) and degraded (light) versions.

Degraded − original	Insertion, grey	Deletion, grey	Insertion, blur	Deletion, blur	Grad-CAM++ insertion, grey
F0	+0.011 [−0.028, +0.043]	−0.018 [−0.039, +0.005]	+0.009 [−0.009, +0.037]	−0.012 [−0.032, +0.002]	−0.000 [−0.032, +0.035]
F1	−0.032 [−0.053, −0.008]	+0.027 [+0.003, +0.056]	+0.023 [+0.001, +0.053]	−0.019 [−0.040, −0.001]	−0.053 [−0.078, −0.022]
F2	−0.024 [−0.042, −0.002]	+0.036 [+0.010, +0.056]	+0.019 [+0.003, +0.040]	−0.019 [−0.037, +0.001]	−0.044 [−0.062, −0.020]

Table 5.152: Change of the faithfulness gain from the original to the degraded version of the same video (paired by pair and seed).

Two findings are robust. First, **faithfulness depends on the perturbation and on the predicted class**, as in the improvement suite: with grey replacement Grad-CAM is faithful by insertion for REAL decisions (+0.27 to +0.41) and by deletion for FAKE decisions (+0.13 to +0.25), but the opposite curve is worse than random; with blur replacement the pattern reverses for FAKE decisions. No single deletion/insertion score therefore certifies the maps. Second, **degradation changes faithfulness only a little**: every Grad-CAM change between the original and degraded version lies within ±0.04 (e.g. F2 grey insertion −0.024, deletion +0.036), well inside the 0.05 margin, while Grad-CAM++ insertion fell by 0.04–0.05 for F1 and F2. Explanation-consistency training did not make the maps more faithful (F2 − F1 grey insertion −0.039 on originals).

5.13.12 Statistical comparisons and hypotheses

Comparison	Training	Test	Metric	Δ (seed mean)	95% CI	Seeds agree	p (Holm)	Verdict
F1 − T5-5A	BioDeepAV	BioDeepAV	degraded auc	+0.0430	[+0.010, +0.064]	yes	0.068	supported improvement
F1 − F0	Joint	BioDeepAV	degraded auc	+0.0306	[+0.004, +0.063]	yes	0.068	supported improvement
F2 − F1	Joint	BioDeepAV	gradcam iou hard	+0.0679	[−0.008, +0.091]	no	0.158	inconclusive
F2 @ Joint − T5-5A @ BioDeepAV	Joint	BioDeepAV	degraded auc	−0.0332	[−0.100, +0.028]	no	0.279	inconclusive
F1 − T5-5A	FF++	FF++	degraded auc	+0.1939	[+0.167, +0.237]	yes	0.008	supported improvement
F1 − F0	Joint	FF++	degraded auc	+0.0312	[+0.022, +0.047]	yes	0.008	supported improvement
F2 − F1	Joint	FF++	gradcam iou hard	+0.0772	[+0.044, +0.099]	yes	0.008	supported improvement
F2 @ Joint − T5-5A @ FF++	Joint	FF++	degraded auc	+0.1100	[+0.087, +0.141]	yes	0.008	supported improvement

Table 5.153: The eight primary comparisons. Four remain significant after Holm correction (all on FF++); F1 − 5A and F1 − F0 on BioDeepAV meet the CI and sign rule but not Holm (p = 0.068).

Comparison (training → test)	AUC original	AUC degraded	Worst-type AUC	Flip rate (pp)	Grad-CAM IoU hard
F1 − F0 (BioDeepAV → BioDeepAV)	+0.015 [+0.004, +0.066] ns	+0.036 [+0.016, +0.082] S+	+0.074 [+0.049, +0.201] S+	−4.84 [−6.53, −0.75] S+	+0.001 [−0.028, +0.026] ns
F1 − F0 (FF++ → FF++)	+0.014 [+0.004, +0.035] S+	+0.040 [+0.027, +0.064] S+	+0.051 [−0.015, +0.163] ns	−1.87 [−3.09, −1.17] S+	−0.010 [−0.033, +0.006] ns
F1 − F0 (Joint → BioDeepAV)	+0.021 [−0.005, +0.060] ns	+0.031 [+0.004, +0.063] S+	+0.099 [−0.004, +0.135] ns	−3.35 [−4.71, +0.08] ns	+0.009 [−0.002, +0.056] ns
F1 − F0 (Joint → FF++)	+0.013 [+0.008, +0.026] ns	+0.031 [+0.022, +0.047] S+	+0.077 [−0.002, +0.161] ns	−0.48 [−1.36, +0.90] ns	+0.012 [−0.000, +0.031] ns
F2 − F1 (BioDeepAV → BioDeepAV)	−0.005 [−0.024, −0.001] ns	−0.009 [−0.029, −0.003] ns	−0.005 [−0.052, +0.002] ns	+1.05 [−0.24, +2.01] ns	+0.031 [+0.016, +0.050] ns
F2 − F1 (FF++ → FF++)	−0.003 [−0.013, +0.002] ns	−0.006 [−0.016, −0.001] S−	+0.021 [−0.063, +0.070] ns	−0.61 [−1.26, +0.55] ns	+0.067 [+0.046, +0.085] S+
F2 − F1 (Joint → BioDeepAV)	−0.016 [−0.041, +0.006] ns	−0.035 [−0.042, +0.001] ns	−0.094 [−0.125, +0.026] ns	+3.05 [−0.31, +3.67] ns	+0.068 [−0.008, +0.091] ns
F2 − F1 (Joint → FF++)	+0.010 [−0.001, +0.031] ns	+0.030 [+0.023, +0.039] ns	+0.008 [−0.035, +0.068] ns	−2.14 [−3.56, −1.29] S+	+0.077 [+0.044, +0.099] S+
F1 − T5-5A (BioDeepAV → BioDeepAV)	+0.025 [−0.002, +0.057] ns	+0.043 [+0.010, +0.064] S+	+0.062 [−0.030, +0.116] ns	−1.95 [−10.45, +4.93] ns	+0.020 [−0.035, +0.033] ns
F1 − T5-5A (FF++ → FF++)	+0.163 [+0.130, +0.220] S+	+0.194 [+0.167, +0.237] S+	+0.215 [+0.085, +0.343] S+	−5.70 [−8.79, −4.03] S+	+0.164 [+0.099, +0.245] S+
F2 − T7-6B (Joint → BioDeepAV)	+0.008 [−0.071, +0.074] ns	+0.019 [−0.065, +0.075] ns	−0.026 [−0.121, +0.078] ns	+2.85 [−5.55, +5.57] ns	+0.120 [+0.035, +0.156] S+
F2 − T7-6B (Joint → FF++)	+0.441 [+0.274, +0.545] S+	+0.358 [+0.206, +0.448] S+	+0.338 [+0.181, +0.444] S+	+12.86 [+11.66, +13.73] S−	+0.113 [+0.074, +0.169] S+
F2 @ Joint − F2 @ BioDeepAV (Joint → BioDeepAV)	−0.047 [−0.135, +0.027] ns	−0.067 [−0.125, +0.002] ns	−0.107 [−0.132, +0.006] ns	+6.46 [−4.16, +9.17] ns	+0.068 [+0.025, +0.089] S+
F2 @ Joint − F2 @ FF++ (Joint → FF++)	−0.053 [−0.094, −0.032] S−	−0.078 [−0.101, −0.067] S−	−0.125 [−0.188, −0.049] S−	+4.30 [+1.15, +6.28] S−	+0.019 [−0.003, +0.042] ns

Table 5.154: Key paired comparisons: difference, 95% CI and verdict (S+ supported improvement, S− supported deterioration, ns inconclusive; for flips lower is better). Full table: statistics_final.csv (598 comparisons).

	Hypothesis	Verdict	Evidence
H1	Pretrained backbone improves robustness over 5A	partially supported	F1 − 5A degraded AUC +0.043 (BioDeepAV), +0.194 (FF++); worst-type AUC +0.215 on FF++ (supported), +0.063 on BioDeepAV (inconclusive)
H2	Degraded training helps the pretrained detector	partially supported	Degraded AUC +0.031 to +0.040 supported in all four settings; flips −4.8 / −1.9 pp in-domain (supported); worst-type gains supported on BioDeepAV only
H3	Explanation consistency improves stability without losing detection	partially supported (stability), detection not non-inferior	Stability improved on FF++ and Joint → FF++ (supported) and BioDeepAV all-pair IoU; detection not shown non-inferior on BioDeepAV (degraded AUC −0.009, CI to −0.029) and FF++ in-domain (−0.006, supported deterioration)
H4	Final joint detector beats 5A on the same videos	partially supported	Joint F2 − 5A: FF++ degraded AUC +0.110 (supported, Holm-significant); BioDeepAV −0.033 (inconclusive)
H5	Cross-dataset generalisation	not supported	F2 cross-dataset AUC 0.35–0.58; no CI above 0.5
H6	Isotonic calibration + abstention are reliable	supported	Isotonic ECE lower than temperature in every seed, test set and condition; abstention raised balanced accuracy everywhere (realised coverage 93–99%)
RQ	Faithfulness under degradation	small change	Degradation changed Grad-CAM faithfulness by at most ±0.04 (Table above); faithfulness itself depends on perturbation and class

Table 5.155: Verdicts of the pre-registered hypotheses.

5.13.13 The frozen final model

The final model was chosen on validation data only by the pre-registered rule: Joint F2, unless its mean validation score is more than 0.005 below Joint F1. Joint F2 scored 0.7566 and Joint F1 0.7772, so the fallback applied and the best-validation seed of **Joint F1 (seed 42, validation score 0.803)** was frozen (FINAL_MODEL.pt, 20.2 MB, weights SHA-256 82f0e71d2a84…). Its logit is mapped to P(fake) by isotonic calibration fitted on 3,296 validation videos (originals and degraded versions); the decision is FAKE if P ≥ 0.560, REAL otherwise, and UNCERTAIN if no face is found or P lies inside the band ±0.006. Because the isotonic map is a step function with a plateau next to the threshold, the band tuned for 90% coverage collapsed to ±0.006 (validation coverage 100%), so in practice the model withholds only videos without a detectable face.

Test set	Videos	n	No face	AUC	Bal. acc. (all)	Real recognised	Fake detected	Coverage	ECE temperature	ECE isotonic
BioDeepAV	original	1,420	39	0.958	0.937	0.899	0.976	0.973	0.105	0.054
BioDeepAV	degraded	4,076	117	0.942	0.874	0.889	0.859	0.971	0.088	0.058
FF++	original	2,038	0	0.929	0.847	0.905	0.789	1.000	0.162	0.078
FF++	degraded	2,411	0	0.860	0.776	0.878	0.674	1.000	0.116	0.053

Table 5.156: Test performance of the frozen final model (Test 5 test split, never used for training, selection or calibration).

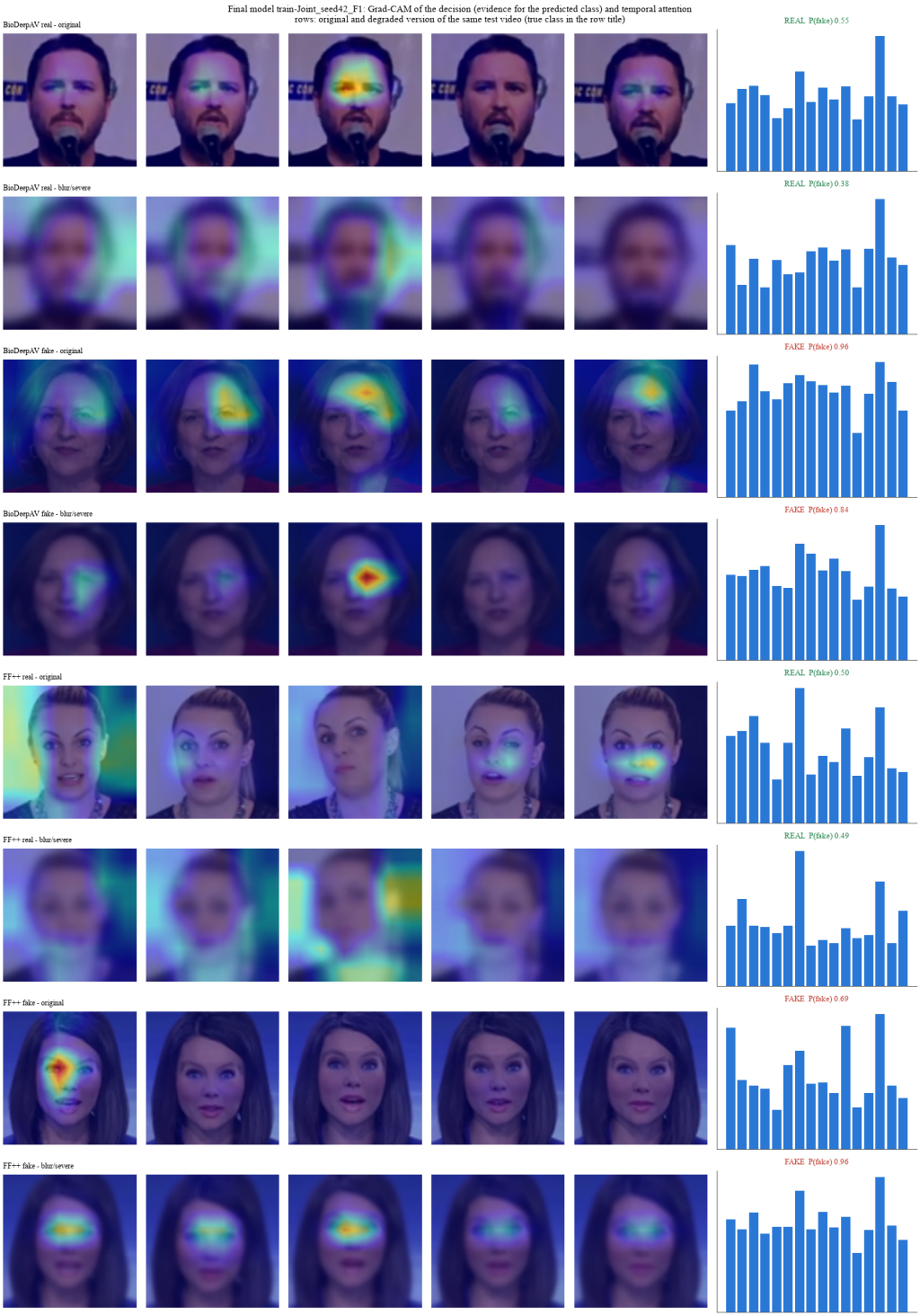

Figure 5.89: Final model: Grad-CAM of the decision (evidence for the predicted class) on five frames and temporal attention, for an original and its severely blurred version of one real and one fake test video per dataset (true class in the row title).

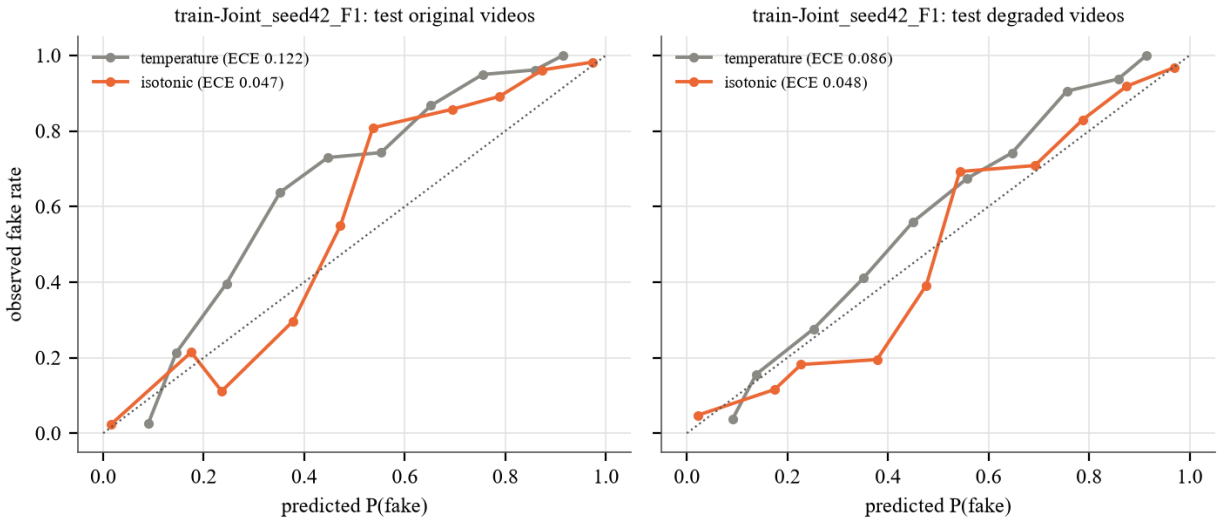

Figure 5.90: Reliability of the final model on test videos: temperature scaling vs isotonic calibration.

Prediction-pipeline check. For 24 test videos (6 per dataset and class) the raw original and one degraded version were re-processed exactly as *--predict* does (fresh YuNet face detection with the imported Test 5 preprocessing, hash ce57c0c8… unchanged). All 48 clips were pixel-identical to the cached evaluation input (median pixel difference 0.0) and gave identical probabilities (maximum |ΔP| = 0.0) and identical verdicts: status PASS. The final model can therefore be applied to any video with *python FINALTEST.py --predict video.mp4* (or the small window of *--phase gui*) and gives the same result as in this evaluation; each prediction writes a verdict file and a Grad-CAM explanation figure.

5.13.14 Comparison with the earlier tests

Measure (in-domain: BioDeepAV / FF++)	T5 5A	T5 5C	T7 6A	T7 6B	Final F0	Final F1	Final F2
AUC original	0.966 / 0.770	0.952 / 0.744	0.956 / 0.726	0.941 / 0.725	0.975 / 0.919	0.990 / 0.934	0.986 / 0.931
AUC degraded	0.939 / 0.705	0.922 / 0.694	0.895 / 0.650	0.873 / 0.648	0.946 / 0.859	0.982 / 0.899	0.973 / 0.893
Worst-type AUC	0.858 / 0.570	0.840 / 0.559	0.815 / 0.519	0.797 / 0.500	0.847 / 0.734	0.921 / 0.785	0.916 / 0.805
Flip rate (%)	6.5 / 15.1	8.7 / 15.7	6.7 / 18.2	10.3 / 15.7	9.3 / 11.3	4.5 / 9.4	5.6 / 8.8
Real recognised	0.621 / 0.657	0.690 / 0.790	0.386 / 0.667	0.599 / 0.627	0.833 / 0.893	0.872 / 0.897	0.828 / 0.902
ECE (original)	0.108 / 0.165	0.132 / 0.185	0.177 / 0.140	0.243 / 0.139	0.093 / 0.192	0.062 / 0.156	0.078 / 0.141
Grad-CAM top-20% IoU	0.602 / 0.614	0.614 / 0.775	0.523 / 0.596	0.578 / 0.826	0.530 / 0.645	0.535 / 0.626	0.564 / 0.692
Grad-CAM IoU, hard types	0.436 / 0.379	0.459 / 0.629	0.439 / 0.467	0.498 / 0.756	0.455 / 0.553	0.455 / 0.543	0.487 / 0.609
Expl. changed, pred. stable (%)	34.1 / 26.8	30.3 / 9.5	42.7 / 28.3	32.6 / 4.2	38.8 / 23.9	40.7 / 27.3	36.2 / 18.8
Cross-dataset AUC (Bio → FF++ / FF++ → Bio)	0.542 / 0.500	0.490 / 0.500	0.418 / 0.284	0.392 / 0.246	0.571 / 0.443	0.590 / 0.376	0.579 / 0.393
Joint → BioDeepAV / FF++ AUC degraded	—	—	0.905 / 0.478	0.887 / 0.457	0.910 / 0.753	0.940 / 0.784	0.906 / 0.815
Parameters	584,978	584,978	2,280,919	2,280,919	4,967,167	4,967,167	4,967,167

Table 5.157: The final test next to Test 5 and Test 7 on the same split, test videos, Grad-CAM pairs and seeds (paired; seed means; BioDeepAV / FF++).

5.13.15 What the final test shows

- **The pretrained detector is the best detector of the thesis.** On identical test videos it improves degraded-video AUC over every earlier model, most clearly on FF++ (0.705 → 0.899), and it is the first jointly trained model that works on FF++.

- **Degraded training still matters with a pretrained backbone.** F1 improved degraded AUC over F0 in all four settings; the pretrained backbone and degraded training add up.
- **Explanation-consistency training is the most replicated XAI result** (Tests 4, 5, 7 and the final test): it increased Grad-CAM stability on FF++ in every test. In the final test it cost a little detection on BioDeepAV and made the joint models less stable across seeds, so it was not frozen as the final model.
- **Stable explanations are not necessarily faithful ones.** Grad-CAM was better than a random map only for some combinations of perturbation and predicted class; degradation hardly changed faithfulness, and consistency training did not improve it.
- **Calibrated decisions are possible**: isotonic calibration roughly halves the ECE on original and degraded videos, and confidence-based abstention reduces errors.
- **Cross-dataset generalisation remains the open problem** (AUC 0.35–0.59), as in every earlier test.

5.13.16 Limitations

- Three seeds only; the Joint models varied strongly between seeds (Joint → FF++ original AUC 0.76–0.93), so small differences are not interpretable.
- The final model is F1, not the pre-registered F2: the selection rule worked as designed, but the explanation-consistency variant is not the deployed detector.
- The isotonic step function collapsed the abstention band of the frozen model (±0.006); a smoother calibrator or a band defined on the temperature-scaled probability would be needed for real selective prediction.
- 22% of the grey-perturbation faithfulness curves produced non-finite outputs under mixed precision and were excluded; the remaining pairs still cover all 180 pairs, but seed 2026 contributes little to F2 grey results.
- Only two datasets, ten degradation types and three severities; the test split is the Test 5 split, so all paired comparisons share its source groups (53 BioDeepAV, 101 FF++).
- Grad-CAM stability and faithfulness have no localisation ground truth; neither shows where a manipulation is.

5.13.17 Output files

File (run folder E:\Code\run_FINALTEST)	Contents
FINALTEST.py	Script: selftest, link, train, evaluate, xai, faith, analyze, report, freeze, predictcheck, predict, gui, status; smoke mode
reports/FINAL_TEST_REPORT.md, final_summary.json	Full report and machine-readable summary
reports/selftest_report.json, data_link.json, predict_pipeline_check.json	27 unit checks, 21 link checks, prediction-pipeline check (all PASS)
configs/final_test_protocol.json, experiment_config_F0–F2.json	Pre-registered protocol (SHA-256 3291c1a2…) and variant configurations
metrics/*_final.csv, hypotheses_final.json	Metrics per run, degradation type, manipulation; calibration; Grad-CAM; faithfulness; seed summary; 598 paired comparisons
metrics/predictions, gradcam_runs, faithfulness_runs	Per-video predictions (27 runs), Grad-CAM rows (720 pairs × 27), faithfulness curves (9 Joint runs)
checkpoints/train-{Joint,BioDeepAV,FFpp}/seed*/F0–F2.pt; checkpoint_hashes.csv	27 checkpoints (20.2 MB each) with file, weight, config and protocol hashes
final_model/FINAL_MODEL.pt, MODEL_CARD.md, model_card.json, final_model_examples.png	Frozen final detector, model card and example explanations
figures/*.png	Report figures (Figures in this section were taken from here, except the pipeline and faithfulness figures)

Table 5.158: Final-test output files.

Appendix A: Source Records

All results in Chapter 5 were carried over from the source documents below without changing any value. Raw source code is not reproduced; the pipeline diagrams in Sections 5.2.2, 5.3.2, 5.4.2 and 5.5.2 summarise each script.

Source document	Content	Used in
Literature review draft (1.pdf)	Sections 1–4	Chapters 1–4 (unchanged)
Degradation tool screenshots (4)	DeepFake Bulk Video Degradation Tool interface	5.0
Report_Mid.pdf (3 pages)	Test 1 mid report (video-only figures)	5.2.1, 5.2.8
Team01 av_mixed.py listing (30 pages)	Test 1 status, degradation tool, walkthrough, results, checks; 1,495 lines	5.1, 5.2
30-9-26test.py + console log	Test 2 code and results	5.3
1-10-26.py + console log + 6 result files	Test 3 code and results	5.4, 5.11
6-10-26.py (3,496 lines) + console log (2 runs)	Test 4 code and console results	5.5
suite_comparison.csv; per run (4D_train-BioDeepAV, 4d = FF++): metrics_long, degradation_metrics, prediction_stability, threshold_analysis, final_experiment_summary, experiment_config, file_inventory, split_6-10-26	Test 4 result files	5.5, 5.11

Table 5.136: Source documents.

Appendix A (continued): Test 5, basic retest, improvement-suite and Test 6 source records

Source document	Content	Used in
9-10-26_test5_independent.py (SHA-256 e789a2a2…; 27 of 30 checkpoints trained with d40fd034…)	Test 5 code	5.6
reports/TEST5_INDEPENDENT_REPORT.md, PRE_TRAINING_CHECKS.md, gate_status.json, leakage_audit_report.json, final_experiment_summary.json	Test 5 report, leakage gate, headline results, checkpoint records	5.6, 5.11, 5.12
metrics/metrics_test5.csv, robustness_test5.csv, degradation_metrics_test5.csv, cross_dataset_test5.csv, gradcam_summary_test5.csv, seed_comparison.csv, statistics_test5.csv	Test 5 result files	5.6, 5.11
configs/experiment_config_5A–5D.json, splits/split_manifest.csv, split_meta.json, data_audit/face_detection_summary.csv, logs/training/history_*.csv	Configuration, split, data audit, training histories	5.6
figures/*.png (run folder E:\Code\run_test5_independent)	Figures 5.34–5.42	5.6
151026.py; run_151026: REPORT_151026.md, test_metrics.csv, test_per_subset.csv, predictions.csv, xai_faithfulness*.csv, history_seed42.csv, split_summary.json	Basic retest code and results	5.7
run_151026 figures/*.png and xai/*.png	Figures 5.45–5.50	5.7
171026.py; run_171026: REPORT_171026.md, E1_reproduction.json, model_selection.json, metrics/E1–E6 *.csv, runs/*/history.csv	Improvement suite code and results	5.8
run_171026 figures/*.png and xai/E6_example_0_BioDeepAV_real.png	Figures 5.52–5.57	5.8
201026.py; run_201026: REPORT_201026.md, summary.json, preflight_audit.json, metrics/T6_*.csv and E5_*.csv, runs/harddro20_*/predictions.csv	Test 6 code and results	5.9
run_201026 figures/T6_*.png and E5_risk_coverage_ece.png	Figures 5.58–5.65	5.9

Table 5.137: Source documents for Test 5, the basic retest, the improvement suite and Test 6. Values were carried over from these files without change (seed means rounded to three decimals); Figures 5.32, 5.33, 5.43 and 5.51 were drawn from the script descriptions and the training histories.

Appendix A (continued): Test 7 source records

Source document	Content	Used in
161026.py (SHA-256 b093dd7e…; 30 of 45 checkpoints written by this version, 14 by cc108bc4…, 1 by b4459fa6…; identical architecture signature 4943176…)	Test 7 code	5.10
run_16-10-26_test6/reports/TEST6_REPORT.md, final_summary_test6.json, selftest_report.json, test5_input_link.json	Test 7 report, hypotheses, unit and link checks, checkpoint records	5.10, 5.11, 5.12
metrics/metrics_test6.csv, robustness_test6.csv, degradation_metrics_test6.csv, seed_summary_test6.csv, statistics_test6.csv, audio_ablation_test6.csv	Test 7 result files	5.10, 5.11
metrics/gradcam_summary_test6.csv, gradcam_stability_test6.csv	Grad-CAM stability	5.10.9
configs/experiment_config_6A–6D.json, logs/training/history_*.csv and sampling_*.csv, logs/run_status.csv, logs/environment_run.json	Configuration, training histories, DRO weights, environment	5.10.3, 5.10.5
figures/gradcam_examples_train-FFpp_seed42_6D.png	Figure 5.73	5.10.9
Figures 5.67–5.72 (redrawn from the files above); Figure 5.66 (drawn from the script description)	Figures	5.10

Table 5.138: Source documents for Test 7 (run folder E:\Code\run_16-10-26_test6). Values were carried over from these files without change (seed means rounded to three decimals). The script and files label this experiment “Test 6” (variants 6A–6D).

Appendix A (continued): Final test source records

Source document	Content	Used in
FINALTEST.py (4,357 lines; run with SHA-256 94be0af5…; protocol SHA-256 3291c1a2…)	Final-test code	5.13
run_FINALTEST/reports/FINAL_TEST_REPORT.md, final_summary.json, selftest_report.json, data_link.json, predict_pipeline_check.json	Final-test report, unit, link and prediction-pipeline checks	5.13
configs/final_test_protocol.json, experiment_config_F0–F2.json; logs/run_status.csv, logs/training/history_*.csv, sampling_*.csv	Pre-registered protocol, configurations, run status, training histories and DRO weights	5.13.3, 5.13.5
checkpoints/checkpoint_hashes.csv	Best epochs, validation scores, thresholds, temperatures, training times, hashes	5.13.5
metrics/robustness_final.csv, degradation_metrics_final.csv, per_manipulation_final.csv, levels_final.csv, seed_summary_final.csv	Detection results	5.13.6–5.13.8, 5.13.14
metrics/calibration_summary_final.csv, risk_coverage_final.csv	Calibration and selective decisions	5.13.9
metrics/gradcam_summary_final.csv, gradcam_stability_final.csv (incl. Test 5 / Test 7 rows)	Grad-CAM stability	5.13.10
metrics/faithfulness_per_video_final.csv (summaries recomputed with non-finite curves excluded)	Faithfulness	5.13.11
metrics/statistics_final.csv, hypotheses_final.json	Paired comparisons and hypothesis verdicts	5.13.12
final_model/FINAL_MODEL.pt, MODEL_CARD.md, model_card.json, final_model_examples.png	Frozen final model	5.13.13
figures/*.png (run folder E:\Code\run_FINALTEST)	Figures 5.80–5.87, 5.89, 5.90	5.13
Figures 5.79 and 5.88 (drawn for this chapter from the script and the per-video faithfulness file)	Figures	5.13

Table 5.159: Source documents for the final test (run folder E:\Code\run_FINALTEST, 24–28 Oct 2026). Values were carried over from these files without change (seed means rounded to three decimals), except the faithfulness summaries, which were recomputed from the per-video file.